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ORIGINAL SCIENTIFIC PAPER

Assessing Personal Experiences and Preferences in Physical Exercise for Multiple Sclerosis

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Abstract

The effectiveness of any physical exercise (PE) program is elementary based on one's interest, while that interest depends on personal experience. This study explored the PE-preferences and -experiences of individuals with Multiple Sclerosis (MS). Study included 79 individuals (13 males, and 66 females) diagnosed with MS, aged 38.99±10.72 years. The PE-preferences and -experiences were evaluated using the newly developed Physical Exercise Preferences in Multiple Sclerosis Questionnaire (PEPMSQ). The reliability of the PEPMSQ was checked with Intraclass Correlation Coefficients (ICC), while the validity was checked by determining the differences in PEPMSQ scores between males and females and between individuals who are exercising and those who are not exercising, all using the Mann-Whittney U test (MW) and Receiver Operating Characteristics curves (Area Under the Curve – AUC). The PEPMSQ demonstrated acceptable test-retest reliability (ICC values of 0.71 to 0.95 for most of the items). Analysis of differences evidenced no gender differences in PEPMSQ scores, while exercising-participants perceive: (i) greater health benefits (MW=-2.57, p=0.01), and (ii) benefits from strength exercises (MW=-2.75, p=0.01; AUC=0.71), compared to non-exercising participants. It seems that strength training should be a key component of exercise programs tailored for MS patients. Future studies are encouraged to refine the PEPMSQ and explore the dynamic relationship between MS symptoms, exercise preferences over time, and various health-indices.

Keywords: *personalized exercise, chronic illness, adapted physical activity, rehabilitation, physical fitness*

Introduction

Multiple Sclerosis (MS) is a chronic neurological condition that affects the central nervous system (CNS), which includes the brain and spinal cord. It is an autoimmune disorder (i.e. the body's immune system mistakenly attacks healthy tissue) (Dalgas, Stenager, & Ingemann-Hansen, 2008; Hunter, 2016). Specifically, the immune system targets the protective myelin which covers nerve fibers, leading to inflammation and subsequent damage. The exact cause of MS is unknown, but it is generally believed to result from a combination of genetic predisposition and environmental factors. So far it is known that certain infections (i.e., Epstein-Barr Virus, Human Herpesvirus 6, Cytomegalovirus), low vitamin D levels, and smoking may increase the risk. Four types of MS are specified so far including: relapsing-remitting MS (RRMS), secondary progressive MS (SPMS), primary progressive MS, and progressive-relapsing MS (PRMS) (Hunter, 2016).

Symptoms of MS vary widely depending on the location and extent of CNS damage, but most commonly include fatigue, numbness or tingling in limbs, muscle weakness or spasms, vision problems (e.g., double vision, partial or complete loss of vision), difficulty with coordination and balance, cognitive impairments (e.g., memory problems, difficulty concentrating), bladder and bowel dysfunction (Hunter, 2016). Diagnosing MS involves a combination of neurological examinations, magnetic resonance imaging to detect lesions in the CNS, and lumbar puncture to analyze cerebrospinal fluid. Currently, there is no cure for MS, but various treatments can help manage symptoms and slow disease progression, including disease-modifying therapies, steroids, phys-

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ical therapy and rehabilitation, and lifestyle modifications (i.e., healthy diet, stress management and physical exercise). Physical exercise (PE) plays a crucial role in managing MS, and it has been shown to have numerous benefits for individuals with the condition. Despite the challenges posed by MS symptoms, engaging in regular PE can help improve physical and mental well-being, as well as overall quality of life (Motl & Sandroff, 2015).

Specifically, resistance-based training (strength training) was shown to be beneficial for improving muscle strength, functional capacity, and overall quality of life in people with MS (Dalgas et al., 2008; Schulz et al., 2004). Further, studies collectively show that endurance training (aerobic exercise) can have profound benefits for individuals with MS. For example, in one of the early studies, Petajan et al. established the positive effects of aerobic exercise on MS patients, with the improvements of cardiovascular fitness, muscle strength, and overall quality of life in individuals with MS (Petajan et al., 1996). Supportively, studies confirmed that aerobic exercise significantly improves fitness, mobility, fatigue, and quality of life in adults with MS (Latimer-Cheung et al., 2013). More recent review highlights the various benefits of aerobic exercise in MS patients, including improvements in cardiovascular fitness, walking speed, and quality of life, emphasizing that regular aerobic exercise can reduce the severity of common MS symptoms such as fatigue and depression (Motl & Sandroff, 2015). Last, but not least, the effects of flexibility exercising (including yoga as a form of flexibility training) in MS patients were also examined. In brief, it has been reported that such exercising can improve flexibility, reduce spasticity, and enhance overall quality of life, while participants reported less fatigue and better physical function after engaging in regular yoga practice (Cramer, Lauche, Azizi, Dobos, & Langhorst, 2014). Supportively, results showed that consistent stretching helped maintain or improve flexibility, reduced muscle stiffness, and had a positive impact on spasticity, contributing to better overall mobility and comfort in MS patients (Halabchi, Alizadeh, Sahraian, & Abolhasani, 2017).

It is well known that effectiveness of any PE program is elementary based on one's interest, while it is repeatedly confirmed that interest depends on personal experience (Kastrati & Georgiev, 2020). First, when individuals have direct experiences with a subject or activity, they are more likely to develop an interest in it, while positive emotions associated with an experience (i.e. enjoyment, excitement, or satisfaction), can reinforce interest, while negative experiences may diminish interest (Versic, Idrizovic, Ahmeti, Sekulic, & Majeric, 2021). Further, repeated exposure to a particular subject can lead to familiarity, which often fosters interest, while familiarity reduces the cognitive load associated with engaging in an activity, making it more enjoyable. Off course, the importance of personal experience on interest is additionally aggravated by different influences (i.e. social and cultural, community, peers), but also with feedback and reinforcement (Hofstetter, Hovell, & Sallis, 1990; Rodrigues, Teixeira, Cid, & Monteiro, 2021). Therefore, it is of particular importance to identify most accepted types of PE for specific participants, considering personal interest, opinions and experiences on certain types of PE. There is no doubt that it is of utmost importance for participants with certain health problems, including MS patients.

Studies repeatedly identified importance of PE and emphasized the clear benefits of various PE types in MS patients. Collectively, regular PE is found to be a vital component of managing MS, can help improve physical strength, mobility, and mental well-being, as well as reduce fatigue and enhance quality of life. However, there is an evident lack of research where personal experiences about different types of PE were established among MS patients. This study examined the personal opinions about effectiveness and benefits of various types of PE in MS patients. For this purpose, we aimed to evaluate reliability and validity of the newly developed tool aimed at evaluation of the personal experience and interest for different forms of PE in MS patients. Initially, we hypothesized that newly designed tool will be: (i) reliable and (ii) valid in evaluation of PE preferences in MS patients.

Materials and methods

Participants

This study included 79 individuals (13 males, and 66 females) diagnosed with MS, aged 38.99±10.72 years. Illness was diagnosed 10.00±8.44 years ago. The majority (83.54%) of participants had the relapsing-remitting type of MS, 10.13% had a primarily progressive form, 2.53% had secondary progressive, and 1.26% had benevolent form of MS. The written consent was taken before the start of the study. The Ethical board of Faculty of Kinesiology, University of Split approved the study.

Variables and procedures

This study included demographic characteristics (age, gender), training experience (years of involvement in organized exercise programs/sports), and the newly designed questionnaire on exercise preferences in MS patients. According to the training experience, participants were divided into two groups; individuals who were exercising, and individuals who did not participate in any of the organized exercise programs/sports.

The newly designed questionnaire (Physical Exercise Preferences in Multiple Sclerosis Questionnaire – PEPMSQ) was sent to the target group via email and other social media and answers were collected on the online survey tool (SurveyMonkey). The PEPMSQ consisted of 9 items.

First 4 items had responses on the 5-point Likert scale (Strongly negative, Negative, Neutral, Positive, Strongly positive). Item 1: "Considering my diagnosis, physical activity affects my general state of health"; Item 2: "Considering my diagnosis, prolonged physical inactivity affects my general state of health"; Item 3: "Given my condition, stretching exercises work on muscle spasm"; Item 4. "Given my condition, weight training and strength training work for me". The fifth item was "Sudden movements in training make me dizzy" had responses on the 5-point scale (1-5): I totally agree, I agree, Neutrally, I don tot agree, I completely do not agree. The sixth item stated: "Please rate on the scale below how much physical activity you think would be optimal for you personally", with answers weighting from 1 to 5: Physical activity should be avoided; Minimal physical activity is required; Moderate physical activity is required; Increased physical activity is required; Emphasized physical activity is required. The last 3 items included the question: Please evaluate how suitable you think certain types of exercise are in terms of your health condition: Strength exercises (Item 7); Stretching exercises (Item 8); aerobic exercises (Item 9), with answers on the following 5-point Likert scale from 1 to 5: Extremely unfavorable, Unfavorable, Neutral, Favorable, Extremely favorable.

Statistical analysis

The normality of the distribution was checked with the Kolmogorov-Smirnov test. The descriptive statistics included means and standard deviations for continuous variables, and percentages were displayed for categorial variables. Differences between gender and exercising categories were calculated by Mann-Whitney U test. Additionally, to determine the differences in exercise preferences between exercising/non-exercising groups we used the Receivers Operating Characteristics Curve (ROC), with the area under the curve values (AUC) of above 0.70 indicating good discriminative power. To determine the test-retest reliability we used intraclass correlation coefficients (ICC), with two-Way Random Effects Model with Absolute Agreement. ICC values are interpreted as follows: less than 0.5 indicates poor reliability, 0.5 to 0.75 suggests moderate reliability, 0.75 to 0.9 indicates good reliability, and greater than 0.9 signifies excellent reliability. The statistical package Statistica (TIBCO, CA) was used for all analysis, with the p-level of 0.05.

Results

Sample characteristics

The test-retest reliability is displayed in Table 1. Items 2 (physical inactivity) and 8 (stretching suitability) had poor reliability and were not included in further analysis. The rest of the items had moderate to excellent reliability values.

Table 1. Test-retest reliability of the newly constructed questionnaire (ICC – Intraclass Correlation Coefficient; CI – Confidence interval)

	ICC	Lower 95% CI	Upper 95% CI
Item 1	0.71	0.14	0.89
Item 2	0.38	-0.72	0.78
Item 3	0.75	0.19	0.92
Item 4	0.93	0.79	0.97
Item 5	0.77	0.33	0.92
Item 6	0.81	0.47	0.93
Item 7	0.95	0.84	0.98
Item 8	0.3	-1.1	0.76
Item 9	0.75	0.29	0.91

Note. Item 1: Considering my diagnosis, physical activity affects my general state of health; Item 2: Considering my diagnosis, prolonged physical inactivity affects my general state of health; Item 3: Given my condition, stretching exercises work on muscle spasm; Item 4: Given my condition, weight training and strength training work for me; Item 5: Sudden movements in training make me dizzy; Item 6: Please rate on the scale below how much physical activity you think would be optimal for you personally; Item 7: Please evaluate how suitable you think strength exercises are in terms of your health condition; Item 8: Please evaluate how suitable you think stretching exercises are in terms of your health condition; Item 9: Please evaluate how suitable you think aerobic exercises are in terms of your health condition; please see Materials and methods for more details on scoring.

The normality of the distribution is shown in Table 2. All variables except for age did not reach the normality of the distribution, therefore, non-parametric statistical tests were used in further analysis.

Table 2. Normality of the distribution of study variables

Variable	N	max D	K-S	Lilliefors
Age	79	0.14	$p < 0.10$	$p < 0.01$
Years of training	79	0.30	$p < 0.01$	$p < 0.01$
Physical activity perceived influence	78	0.26	$p < 0.01$	$p < 0.01$
Stretching exercises perceived influence	77	0.25	$p < 0.01$	$p < 0.01$
Strength exercises perceived influence	75	0.26	$p < 0.01$	$p < 0.01$
Sudden moves and dizziness	78	0.25	$p < 0.01$	$p < 0.01$
Adequate physical activity perceived	78	0.37	$p < 0.01$	$p < 0.01$
Strength exercises suitability	78	0.31	$p < 0.01$	$p < 0.01$
Aerobic exercises suitability	78	0.30	$p < 0.01$	$p < 0.01$

Note. N – number of participants, K-S – Kolmogorov-Smirnov test

Differences according to gender are displayed in Table 3. There were no significant differences in any of the variables included in this study.

Descriptive statistics and differences according to exercising status are presented in Table 4. Participants who are involved in physical exercises consider that physical activity

Table 3. Gender differences in study variables

	Female (n=66)		Male (n=13)		Mann-Whitney U test		
Variables	Mean	SD	Mean	SD	U	Z	p-value
Age (years)	39.61	10.94	35.85	9.29	331.00	1.29	0.20
Years of training (years)	3.71	6.68	5.31	7.83	380.50	-0.63	0.53
Physical activity influence (score)	3.85	0.88	4.00	0.85	368.00	-0.38	0.70
Stretching exercises influence (score)	3.55	0.85	3.00	0.85	259.00	1.83	0.07
Strength exercises influence (score)	3.25	1.06	3.75	0.87	283.00	-1.37	0.17
Sudden moves and dizziness (score)	2.53	1.21	2.83	1.27	341.00	-0.75	0.45
Adequate Physical activity (score)	3.38	0.63	3.17	0.72	348.00	0.66	0.51
Strength exercises suitability (score)	3.64	0.89	3.58	0.79	371.50	0.33	0.74
Aerobic exercises suitability (score)	3.61	1.15	3.67	1.23	381.00	-0.20	0.84

positively influences their condition more than the non-exercising group. Furthermore, active group consider that strength exercises are more beneficial for their health condition more than does the non-exercising group.

Figure 1 shows the differences between non-exercising and exercising group by the Receiver operating characteristics curve (ROC). The most significant variable is Strength exercises influence (Item 4), with Area Under the Curve (AUC) of 0.71 (0.59-0.83 of 95% Confidence interval). Stretching exercises influence (Item 3) and aerobic exercises suitability (Item 9) did not reach statistical significance with AUC of 0.58 and 0.61, respectively.

Table 4. Descriptive statistics and differences according to exercising status

	Total sample (n=79)		**Not-exercising (n=39)**		**Exercising (n=40)**		**Mann-Whitney U test**		
Variable	Mean	SD	Mean	SD	Mean	SD	U	Z	p
Age (years)	38.99	10.72	38.85	10.70	39.13	10.88	769.50	-0.10	0.92
Years of training (years)	3.97	6.86	0.00	0.00	7.85	7.93	0.00	-7.64	0.001
Years of being diagnosed (score)	10.00	8.44	8.95	6.29	11.06	10.12	705.50	-0.17	0.87
Physical activity influence (score)	3.87	0.87	3.61	0.82	4.13	0.85	502.50	-2.57	0.01
Stretching exercises influence (score)	3.47	0.87	3.29	0.90	3.64	0.81	612.00	-1.31	0.19
Strength exercises influence (score)	3.33	1.04	2.95	0.98	3.73	0.96	408.50	-3.12	0.01
Sudden moves cause dizziness (score)	2.58	1.21	2.53	1.18	2.63	1.25	734.50	-0.25	0.80
Adequate Physical activity (score)	3.35	0.64	3.21	0.53	3.48	0.72	594.00	-1.65	0.10
Strength exercises suitability (score)	3.63	0.87	3.34	0.81	3.90	0.84	474.00	-2.85	0.01
Aerobic exercises suitability (score)	3.62	1.15	3.45	1.11	3.78	1.19	604.00	-1.55	0.12

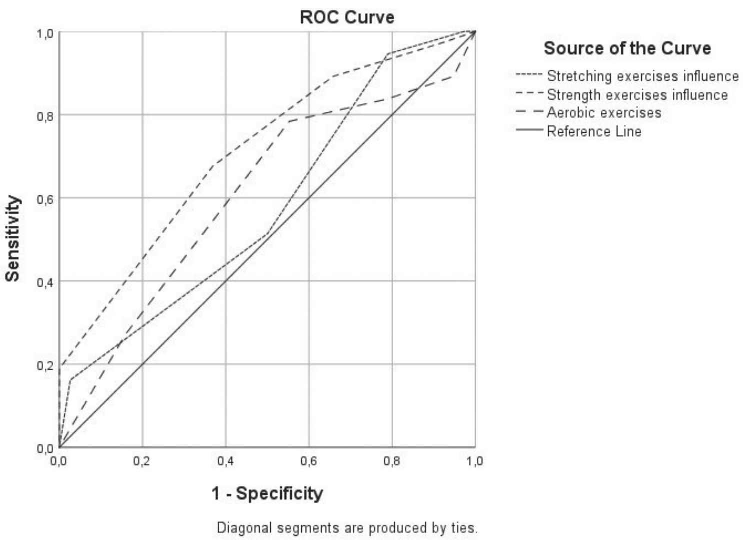

FIGURE 1. Receiver operating characteristics curve for differences between non-exercising and exercising group

Discussion

There are several main findings of the study. First, the newly developed questionnaire had acceptable test-retest reliability, with only two items not reaching satisfactory reliability level. Second, there are no gender differences in the opinions about effectiveness and benefits of various types of PE in MS patients. Third, participants who are involved in physical exercises are considering that physical activity has positive influence on their condition more than the non-exercising group does. Finally, participants who are involved in physical exercises perceive strength exercises as having the best effect on their health condition more than the non-exercising participants.

One of the aims of this research was to check the test-retest reliability of the PEPMSQ. The evaluation of test-retest reliability guarantees the stability and temporal consistency of the questionnaire in assessing the advantages and efficacy of PE among MS patients, hence contributing to more precise research conclusions, enhanced clinical decision-making, and better patient outcomes (Akbiyik et al., 2009; Mathiowetz, 2003). Generally, good test-retest reliability means that the same outcomes are reliably measured by the questionnaire after several administrations (Mathiowetz, 2003). In other words, a questionnaire must produce similar responses when filled out by the same patients at different times and under the same circumstances in order to be deemed reliable. Nevertheless, as MS is a progressive, chronic illness, its symptoms might vary from patient to patient, but answers to the questionnaire should be consistent within the individual (Hosseini, Homayuni, & Etemadifar, 2022). Notably, majority of the items included in the PEPMSQ had satisfactory test-retest reliability values, which means that this questionnaire has temporal stability.

However, two items did not reach the satisfactory level of test-retest reliability (i.e., two items had ICC values lower than 0.5), which could be explained as follows. MS symptoms tend to vary within the individual, which can influence how patients view the effects of inactivity or the benefits of stretching exercises (Motl, Mullen, Suh, & McAuley, 2014; Saedmocheshi, Yousfi, & Chamari, 2024). For instance, on a day when a patient feels less fatigued, they might find stretching helpful. However, on a day when their symptoms are more intense, their perspective might change, leading to different answers each time they're asked. Such variability can result in inconsistent responses over time in these two items. On the other side, these items might not be formed precise enough which can lead to subjective interpretations of the question (Choi & Pak, 2005). For example, a patient might see "inactivity" as a routine rest period during the first survey administration time but view it as something negative during another, depending on their mood, energy levels, or recent experiences. Moreover, there is a possibility that some external factors influence the responses to certain items of the questionnaire. Namely, changes in a patient's environment, stress levels or recent physical activity experiences can influence how they view inactivity and stretching exercises (Stults-Kolehmainen & Sinha, 2014). Nevertheless, these items should further be refined for greater clarity which could lead to adequate reliability, but the greater deal of questionnaire is reliable and can be used as such.

One of the approaches in studying validity of the newly developed tool (PEPMSQ) was the identification of the eventual differences between genders in evaluated variables/questionnaire items. In the realm of assessment and measurement, validity evaluation aims to determine whether a test or assessment tool truly measures what it claims to measure. The concept of "groups of interest" becomes crucial in this process, as validity is often examined in relation to how well the assessment performs for different groups of people. In our case, the most logical scenario was evaluation of the differences between males and females, since differences in gender might affect how certain questions or tasks are interpreted or performed, impacting the validity of an assessment for both males and females. Indeed, such an approach is frequently applied in studies evaluating validity of the measurement tools intended to be used in both genders, irrespective of the field of science (Carragher et al., 2016; Pesce, Masci, Marchetti, Vannozzi, & Schmidt, 2018). Evidently, no significant gender differences were observed, indicating that PE preferences and experiences are generally consistent across male and female MS patients. While it is generally expected that existence of significant differences between groups indicate proper validity of the measurement tool, in our study this is not the case. Quite contrary, lack of differences between males and females indicates appropriate validity of the PEPMSQ. There are several reasons for such conclusion.

It's possible that similar healthcare advice, shared experiences with the condition, and shared aspirations between male and female patients account for the lack of gender variations in perceptions regarding the efficacy and advantages of PE in MS patients. Namely, MS affects both sexes equally in terms of the physical difficulties and symptoms that both men and women face, such as exhaustion, muscle weakness, and problems with mobility (Pau et al., 2020). Regardless of gender, such similar experiences may result in comparable opinions about the efficacy and advantages of PE. Moreover, another potential reason could be the fact that healthcare professionals frequently prescribe the same PE regimens to all MS patients, regardless of gender; emphasizing exercises that enhance total strength, mobility, and endurance (Kalb et al., 2020). Therefore, male and female MS patients may have similar priorities for health. This could result in divergent opinions about the kinds of exercises that are most helpful and this common approach to PE recommendations may lead to similar results and perceptions of efficacy between genders (Kalb et al., 2020; Learmonth & Motl, 2021). Finally, another potential reason is that men and women with MS may be motivated to participate in PE for similar reasons, such as enhancing their mobility and quality of life, which may result in similar evaluations of the activity's preferences and effectiveness (McCarty, Sayer, & Kasser, 2022). While we are not able to define which of the previously stated reasons contributed to our findings (e.g., similar results in both males and females), it is clear that there are no valid reason for genders being different in PEPMSQ scores, which consequently highlights the proper validity of the measurement tool.

The results of this study reported that physically active MS patients (i.e., exercising group) are more likely to recognize and report the benefits of physical activity on their health condition compared to individuals who are not exercising. This can be explained by several factors. The most important one is the fact that the familiarity with PE seems to reinforce its perceived value, suggesting the importance of encouraging physical activity participation among MS patients (Peralta et al., 2021). Namely, active individuals get to experience the benefits of PE firsthand. By staying active, they might notice improvements in their symptoms, mood, or overall well-being, which encourages them to keep exercising and reinforces the value of staying active. Moreover, active individuals might feel the benefits of PE such as reduced stiffness, improved mobility, improved strength and higher energy levels, which can all lead to increased awareness about the benefits of physical activity (Dalgas et al., 2008).

One important thing which is crucial for long involvement in PE and perceptions of its benefits is psychological improvement. Namely, physical activity offers numerous psychological benefits, which improve the well-being of individuals dealing with chronic conditions like MS. One of the most significant psychological benefits is increasing self-efficacy (i.e., the belief in one's ability to successfully carry out tasks and achieve goals). When MS patients make exercise a regular part of their lives, they often feel more in control of their health and physical abilities which helps them stay committed to their exercise routines, even when facing the difficulties that come with their condition. Indeed, study found that MS patients who were regularly active had higher levels of self-confidence and reported a better quality of life compared to the less active ones (Motl & McAuley, 2014). Likewise, it was proven that exercise can reduce feelings of depression and anxiety and lift mood, largely by boosting self-efficacy and getting a stronger sense of control over health (Imayama et al., 2013).

Strength exercises are perceived as particularly beneficial for health condition by MS patients, compared to other exercise types. The perception that strength exercises are most beneficial when compared to other types of exercise is probably related to some specific physiological background. Due to nerve damage, MS leads to numerous motor symptoms including muscle stiffness, and weakness which lead to difficulties with mobility, coordination and balance (Hunter, 2016). All of these factors can be improved by targeted strength training. Indeed, 12-week strength training intervention on lower extremities in MS patients led to improvements in strength level and functional capacities (Dalgas et al., 2009). Supportively, experimental evidence confirmed that 12-week intervention of high-intensity interval training along with resistance training led to improvements in muscle strength and quality of life in MS patients (Zaenker et al., 2018). Moreover, an 8-week maximal strength training intervention led to improved balance in MS patients (Karpatkin et al., 2016). Collectively, the positive effects of strength training in MS patients has been synthesized in one study which reported that individuals with MS experienced notable improvements, including an 8.2% reduction in fatigue, a 21.5% increase in functional capacity, an 8.3% boost in quality of life, a 17.6% gain in muscle power, and a 24.4% increase in electromyography activity after strength training altogether supporting previous discussion (Cruickshank, Reyes, & Ziman, 2015).

It is also important to note that strength training has been reported to have significant influence on improving gait performance through improving the strength level of the muscles included in walking (Mañago, Glick, Hebert, Coote, & Schenkman, 2019). This is of particular importance as it is well known that one of the most common issues that people with MS have is difficulty walking; up to 93% of MS patients report having trouble walking ten years after diagnosis (van Asch, 2011). Therefore, strength training improves walking performance and reduces the risks of falls which is high among MS patients. Notably, guidelines recommend that adults with MS should aim for at least 30 minutes of moderate aerobic exercise twice a week, along with strength training for major muscle groups twice a week, indicating that following this routine can help reduce fatigue, improve mobility, and boost overall quality of life (Latimer-Cheung et al., 2013). Therefore, given the evidence-based facts that strength exercises have significant impact on MS patient's quality of life, it is not surprising that patients who were exercising experienced these benefits and afterwards reported positive perception of the strength exercises on their health condition.

It is important to recognize the limitations of this study. Because of the small sample size, it may be harder to extrapolate the results to the entire MS population. Furthermore, the study used self-reported data, which is prone to bias especially when it comes to symptom intensity and frequency of exercise. It is also impossible to determine a causal relationship between exercise preferences and reported benefits because of the study's cross-sectional nature. Furthermore, additional validation is required to establish the PEPMSQ's sensitivity and applicability across varied MS populations, including those with varying degrees of disability, even though it displayed adequate test-retest reliability.

Notwithstanding these limitations, the study has several strengths. An important development in determining exercise preferences among MS patients is the creation and preliminary validation of the PEPMSQ. The study has the potential to enhance the quality of life and management outcomes for people with multiple sclerosis (MS) due to the utilisation of a dependable questionnaire and the focus on an individualised approach to exercise.

Conclusions

The findings suggest that the PEPMSQ is a reliable tool, with most items demonstrating acceptable test-retest reliability, which supports its use in both clinical and research settings. Importantly, the study highlights that individuals with MS who regularly engage in PE perceive greater benefits from PE, particularly strength exercises, compared to those who are less active. This emphasizes the role of personal experience in shaping exercise preferences and underlines the necessity of tailoring exercise programs to the specific needs and interests of MS patients to enhance adherence and improve outcomes. Overall, this research contributes valuable insights into the personal exercise preferences of MS patients and underscores the importance of personalized exercises. Future studies are encouraged to refine the PEPMSQ and explore the dynamic relationship between MS symptoms and exercise preferences over time.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Relationship between Hand-Eye Coordination and Hand Grip Strength in Elite Taekwondo Athletes

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Abstract

Hand-Eye Coordination (HEC) and Hand Grip Strength (HGS) are among the most important skills in Taekwondo. On a technical level, HEC reflects neuromuscular coordination ability which enables quick responses, and HGS reflects muscle strength which indicates the strength and overall health of the athletes. Combined together, HEC and HGS likely allow Taekwondo athletes to perform better. The aim of this study was to identify if there is a relationship between HEC and HGS in elite Taekwondo athletes. 166 elite Taekwondo athletes (141 males, 25 females) voluntarily participated in this study and were divided into two categories: seniors (age >17 years) and juniors (age ≤17 years). HEC was assessed by hand-eye coordination manual dexterity test (Lafayette, IN, US) and HGS was measured using a hand dynamometer (Takei, Niigata, Japan). The results revealed that no significant relationship was observed between HEC and HGS ($p>0.05$) in elite Taekwondo athletes. Despite the several strength and neuromuscular adaptations acquired during Taekwondo practice, HEC and HGS are not significantly correlated, and hence there is no relationship between HEC and HGS in elite Taekwondo athletes. The results obtained may benefit Taekwondo coaches and athletes in improving their training strategies.

Keywords: *muscle force, muscle coordination, motor control, neuromuscular function*

Introduction

Taekwondo is a renowned Korean martial art involving the application of fast, powerful and complex kicks and punches with a great coordination (Kazemi et al., 2006). Being able to synchronize several extremities at the same time is one of the most important factors to succeed in Taekwondo. Multiple extremity coordination is the ability to use more than one extremity at the same time in a compatible and efficient manner. Hand-Eye Coordination (HEC) and Hand Grip Strength (HGS) are among the most important skills in this sport. On a technical level, HEC reflects neuromuscular coordination ability enabling quick and fine responses, however, HGS reflects the general muscle strength which plays a particularly role in strong punches (Chiodo et al., 2011; Kons et al., 2020; Pieter & Heijmans, 2000). Having a high level of HEC and HGS abilities might allow Taekwondo athletes to perform stronger and precisely controlled movements.

Neuromuscular coordination is an ability allowing the combination of several movements into an effective and efficient movement pattern (Dharma et al., 2020), by refining

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to the recruitment or inhibition of targeted motor units when visual accuracy is required (Khasawneh et al., 2009). In coordinative performance specifically, good timing is needed for fine neuromuscular activation (Cortis et al., 2009). HEC, a visual-motor function that facilitates the use of the hand towards a target, reflects the ability of both central and peripheral nervous systems (Wong et al., 2019). HEC is complex ability since several sensorimotor systems (e.g. visual system, vestibular system, proprioception, supraspinal, spinal cord and skeletal muscle system, etc.) work synergistically (Crawford et al., 2004) in order to perform different types of movement. HEC was found to be correlated with increasing age, suggesting that coordination decreases as an individual gets older (Getchell & Whitall, 2003).

At the same time, strength is a primary component of human motor skills and one of the most frequently trained motor abilities (Rdzanek et al., 2019). The development of muscle strength is promoted by a combination of several morphological and neural factors such as recruitment and synchronization of motor units (Suchomel et al., 2018). HGS, which involves the flexor muscles of the forearm and hand, is an important parameter to be assessed, since it is a marker of the upper-body muscular strength (Abaraogu et al., 2019). HGS plays a key role in injury prevention (Budoff, 2004) and in overall strength development (Tietjen-Smith et al., 2006). More recent evidence indicated that the HGS test is considered a simple and economic test which can give practical information about muscles, nerves, bones, and joints (Wind et al., 2010). In martial arts, HGS can be considered as an indicator to predict a better performance (Iermakov et al., 2016). Additionally, HGS has been related to body movements in Judo where a correlation was observed between HGS and balance control (Dias et al., 2011). When evaluating training programs, various studies have recommended martial arts training to enhance force and HEC (Kwok, 2012; Tsang et al., 2013; Wong et al., 2019). For instance, another study indicated that four weeks of Ving Tsun Chinese martial art training can improve elbow extensor isometric peak force and the time to reach peak force, although it did not enhance HEC in middle-aged and older adults living in the community (Fong et al., 2016). In contrast, research on basketball players has shown that trained athletes have significantly shorter reaction times in HEC tasks with a decrease of 23.3% compared to an increase of 8.1% of the control group (Tsang, 2014). However, existing studies have typically examined HEC and HGS separately, and there has been no investigation into the relationship between HEC and HGS in taekwondo athletes. Therefore, this study aims to determine if there is a correlation between HEC and HGS in elite Taekwondo athletes.

Materials and Methods

Participants

166 elite Taekwondo athletes (141 males, 25 females) voluntarily participated in this study and were divided into two categories: the juniors (aged ≤17 years; males: 15.48±1.12 and females: 15.19±1.01) and the seniors (aged >17 years; males: 23.34±5.09 and females: 20.50±1.13). The procedures were approved by the Scientific Research Ethics Committee of Al-Ahliyya Amman University (3/1-2020/2021). Prior to conducting the two non-invasive tests, the purpose of the study was explained to the participants, and following their consent, the necessary instructions were given. Participants were given instructions, such as not consuming caffeine before the test or any products considered stimulating.

Measurements

HEC measurement (reflecting neuromuscular coordination)

Neuromuscular coordination was evaluated by a hand-eye coordination manual dexterity test using Steadiness Tester (Lafayette Hole Type, Model 32011 Replacement Stylus, Model 32100; Lafayette, IN, US). This test consists of holding a metal-tipped stylus (with the dominant hand) in the middle of 9 consecutive holes becoming progressively smaller in diameter and stabilizing it in each hole for a period of 10 s while trying to avoid touching the perimeter of the holes. In fact, as the hole diameter becomes narrower, a higher coordination ability is required. The errors were detected by Silent Impulse Counter Model 58024C (Lafayette, IN, US). Each participant performed two trials using their dominant hand (preferred hand), the first one for stylus familiarization, and the second one for the assessment part. Factors affecting vision, such as lighting, were taken into account when applying the test.

HGS measurement (reflecting muscular strength)

A hand dynamometer (Takei Scientific Instruments Co., Ltd, Niigata, Japan) following the commonly accepted method was used to measure the HGS (kg). Following the recommendations of the American Society of Hand Therapy (Mgbemena et al., 2019), the grip bar was adjusted so that the second joints of the fingers were bent to grip the handle of the dynamometer. Concerning body position, the player stood up with the shoulder adducted and neutrally rotated. The elbow was straight, the forearm and wrist were in neutral positions, the arm was vertical, and the hand dynamometer was held close to the body. When ready, the athlete squeezed the dynamometer with a maximum isometric effort maintained for about 5 seconds (Franchini et al., 2018) without body movement. After the familiarization test with their dominant hand, the athletes performed a test for each hand.

Statistical analyses

Quantitative results were reported as mean ± SD and statistical significance was set at $p<0.05$. Pearson's correlation coefficient test was used for the correlations between HEC and HGS measurements. A two-way repeated measures ANOVA was used to identify HEC and HGS group differences (differences between males and females, as well as differences between juniors and seniors). All statistical analyses were performed using the software R (3.6.2) (foundation for statistical computing, Vienna, Austria).

Results

HEC measurements

The results concerning the HEC assessment according to sex and age are represented in Table 1. The coordination performance is expressed by the number of touches recorded between the stylus and the hole. Poor coordination is reflected when the number of touches increases, and a lower number of touches indicates a higher coordination (the value 0 is equivalent to absence of touches). The best coordination performance was obtained in the largest hole diameter (1.156 inch) since no touches (0.00) have been recorded by any of the Taekwondo athletes (Table 2). In contrast, the poorest coordination obtained was for the smallest hole diameter (0.0625 inch) with an average of 24.77 touches and a maximum of 100 touches recorded (Table 2). As the hole diameter got narrower, the number of touches increased, hence an inverse relationship was observed. A significant sex effect was observed in holes 7 and 8. A significant age effect was observed in holes 7, 8 and 9.

HGS measurements

The average HGS obtained from both hands when all participants were considered was 32.19±7.95 kg (Table 1). Significant sex differences (p<0.001) were obtained suggesting that HGS measurements were significantly higher for men (32.51±7.65 kg), compared to women (25.70±3.47 kg). Furthermore, significant age differences (p<0.001) were revealed when comparing HGS measurements between juniors (29.84±6.94 kg) and seniors (37.39±6.86 kg), where seniors recorded a greater muscular strength (37.39±6.86 kg), compared to juniors (29.84±6.94 kg).

Relationship between HEC and HGS measurements

HGS of the dominant hand. The correlation analysis between HEC and HGS from the same hand used for the HEC (dominant hand) revealed no statistical significance (p>0.05) over any of the different holes (Table 2). This result suggests that no correlation was found between these two abilities when the HGS of the dominant hand was considered.

HGS mean of the two hands. The correlation analysis between HEC (for the dominant hand) and HGS mean of the two hands revealed no statistical significance (p>0.05) over any of the different holes (Table 2). This result suggests that no significant correlation was found between these two abilities when the HGS mean of the two hands was considered.

Table 1. Hand-Eye Coordination (HEC) performance as well as Hand Grip Strength (HGS) measurements according to sex and age for Taekwondo athletes.

HEC and HGS	Sex		Age		p-value	
	Males (n=141)	Females (n=25)	Juniors (n=130)	Seniors (n=36)	Sex effect	Age effect
			HEC Performance			
Hole 1 (1.156)	0.01 ± 0.08	0.00 ± 0.00	0.01 ± 0.09	0.00 ± 0.00	0.99	0.99
Hole 2 (1.125)	0.13 ± 0.76	0.00 ± 0.00	0.13 ± 0.78	0.06 ± 0.23	0.97	0.99
Hole 3 (0.500)	0.43 ± 1.36	0.00 ± 0.00	0.40 ± 1.35	0.25 ± 0.91	0.89	0.97
Hole 4 (0.312)	0.51 ± 1.27	0.44 ± 1.64	0.49 ± 1.14	0.53 ± 1.87	0.82	0.77
Hole 5 (0.187)	1.23 ± 2.87	2.12 ± 5.97	1.33 ± 3.00	1.47 ± 4.99	0.28	0.35
Hole 6 (0.109)	4.43 ± 7.50	3.12 ± 10.09	4.03 ± 6.48	4.97 ± 11.86	0.37	0.06
Hole 7 (0.093)	7.34 ± 11.26	7.44 ± 17.06	7.00 ± 10.29	8.64 ± 17.72	0.01	<0.001
Hole 8 (0.078)	13.24 ± 14.43	13.24 ± 22.96	12.81 ± 14.06	14.81 ± 21.49	<0.01	<0.001
Hole 9 (0.063)	24.87 ± 18.91	24.20 ± 25.76	23.88 ± 19.85	27.97 ± 20.52	0.48	0.02
			HGS Measurements			
HGS min	16.70	16.50	16.50	22.00	-	-
HGS max	51.15	33.10	48.65	51.15	-	-
HGS (dominant)	32.51 ± 7.65	25.70 ± 3.47	29.84 ± 6.94	37.39 ± 6.86	<0.001	<0.001

Note: Data are presented as: Hole (diameter in inch) and mean of touches ± SD. Sex effect, as well as age effects were analyzed using two-way repeated measures ANOVA. The value of 0 is equivalent to absence of touches.

Table 2. Correlation between Hand-Eye Coordination (HEC) performance and Hand Grip Strength (HGS) measurements for the Taekwondo athletes.

	Touches recorded for the Coordination Performance			Correlation with the HGS (dominant hand)		Correlation with the HGS (average of both hands)	
Hole diameter (inch)	Minimum	Maximum	Mean ± SD	r	p-value	r	p-value
Hole 1 (1.156)	0.00	0.00	0.00 ± 0.00	0.00	0.99	0.00	0.97
Hole 2 (1.125)	0.00	8.00	0.11 ± 0.70	-0.01	0.87	0.02	0.80
Hole 3 (0.500)	0.00	11.00	0.37 ± 1.27	-0.08	0.32	-0.10	0.19
Hole 4 (0.312)	0.00	8.00	0.50 ± 1.33	-0.04	0.58	0.09	0.24
Hole 5 (0.187)	0.00	28.00	1.36 ± 3.51	-0.07	0.34	-0.10	0.19
Hole 6 (0.109)	0.00	50.00	4.23 ± 7.92	-0.11	0.15	-0.15	0.05
Hole 7 (0.093)	0.00	82.00	7.36 ± 12.24	-0.08	0.28	-0.13	0.10
Hole 8 (0.078)	0.00	99.00	13.24 ± 15.91	-0.09	0.21	-0.13	0.09
Hole 9 (0.063)	0.00	100.00	24.77 ± 20.00	0.02	0.74	0.00	0.98

Note: Data are presented as: mean of touches ± SD. Correlation between HEC and HGS were evaluated using Pearson correlation analysis. r: Pearson's correlation coefficient; significance level: p-value <0.05.

Discussion

To the best of our knowledge, this is the first study evaluating the relationship between HEC and HGS in elite Taekwondo athletes. Several recent studies have focused on martial arts since many important and interesting skills are involved. Analyzing the possible presence of a relationship between HEC and HGS in Taekwondo would allow us to understand whether these two important parameters are related or not in this sport. Even though quick responses and strong kicks are essential when performing Taekwondo, the results obtained indicated that HEC and HGS were not significantly correlated. Hence, there is no relationship between HEC and HGS in elite Taekwondo athletes.

Following the HEC performance, an inverse relationship was observed in all groups (higher number of touches when the hole diameter decreases). In fact, as the hole diameter decreases, a higher coordination ability is required. However, no significant difference was highlighted between men and women nor between juniors and seniors. For the HGS measurements, significant age and sex differences were revealed, where men recorded a significantly higher HGS than women, and seniors recorded a significantly higher HGS than juniors. This is explained by the fact that men and seniors have a higher muscular strength than women and juniors respectively (Doherty, 2001). Since Taekwondo athletes have stronger punches with their dominant hand, the analysis of the correlation between HEC and HGS was first performed considering only the dominant hand. However, no significant correlation was obtained, the correlation analysis considering both hands was carried out (average of the obtained HGS), knowing both arms are used in Taekwondo. Even though both HEC and HGS are used in Taekwondo while performing/playing, the correlation analysis results revealed that HEC and HGS (for the dominant hand and for the two hands averaged) were not significantly related in elite Taekwondo athletes.

Some studies evaluated the relationship between HEC and HGS in several sports, or relationships involving HEC particularly. In concordance to our results, similar findings were observed in some studies. It was recently reported that although a significant correlation was observed between HEC and performance, no correlation was observed between HGS and HEC in adolescent tennis players (Pipal et al., 2017). Since tennis athletes use both HEC and HGS for precision and shots respectively while performing, this result is similar to our results. Additionally, badminton athletes had no better HEC than non-badminton athletes even though their performance levels were higher (Wong et al., 2019). However, in baseball, a relationship was observed between the hand-eye visual-motor reaction time and batting performance (Laby et al., 2018).

When evaluating a different coordination in Taekwondo which is the inter-joint coordination of the roundhouse kick, it was demonstrated that this coordination was advantageous in producing both fast kicking velocity and quick kicking time (Kim et al., 2011), and hence was related to those two parameters. It would then be interesting to evaluate different types of coordination in Taekwondo and hence evaluate which coordinations are the most related to HGS, which would be useful and beneficial to build specific training strategies. When comparing several skills in martial sports, it was observed that Karate athletes have a better HEC and Taekwondo athletes have a faster perceptual processing, stating that perceptual and motor performance differs between martial sports athletes according to the specific demands of the discipline (Chen et al., 2017).

The coordination is usually practiced in any type of sports. Since both HEC and HGS are important to perform in Taekwondo specifically, the study wanted to see if there is a relationship between these two parameters. The lack of a relationship between HEC and HGS obtained in elite Taekwondo athletes might be due to several reasons. One of the reasons might be that these skills do not necessarily involve a high precision and accuracy like the HEC test. Another reason might be the fatigue of the central nervous system during performance which could lead to a gradual loss of concentration (Enoka & Duchateau, 2016). In addition, peripheral fatigue induced by the large amount of sweat lost and thus the loss of metabolites (vitamins and minerals), could also explain the lack of this relationship in Taekwondo athletes (Lv, 2015). Furthermore, Taekwondo primarily relies on lower body movements rather than upper body movements, making HGS potentially less representative of the strength required for optimal performance in Taekwondo. Our study offers novel insights into the interaction between HES and HGS within Taekwondo, providing a comprehensive analysis by comparing different demographics (age, sex) and considering both dominant and non-dominant hands. This contributes valuable data to the broader understanding of physical attributes in martial arts. Despite including a large number of Taekwondo athletes, the groups were not homogeneous regarding sex (141 men and 25 women), possibly due to cultural and social factors prevalent in Arab countries (Sharara et al., 2018). Future research could explore the correlation between neuromuscular coordination and muscle strength of lower body parts, potentially revealing significant relationships relevant to Taekwondo performance. The findings of this study suggest that training programs for elite Taekwondo athletes might benefit from focusing separately on enhancing hand-eye coordination and hand grip strength, as no significant relationship between these parameters was found.

Conclusion

The main objective of this study was to identify if there is a relationship between HGS and HEC in elite Taekwondo athletes. Even though quick responses and strong kicks are essential when performing in Taekwondo, and despite the several strength and neuromuscular adaptations acquired during practice, HEC and HGS were not significantly correlated. Hence, there is no relationship between HEC and HGS in elite Taekwondo athletes. The results obtained should be taken into consideration by Taekwondo coaches and athletes when developing training strategies.

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Conflicts of interest
The authors certify that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Investigation of Dynamic Balance, Limb Asymmetry and Flexibility in Jiu-Jitsu Athletes: A Preliminary Exploratory Study

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Abstract

This preliminary exploratory study is the first to thoroughly examine dynamic balance, inter-limb asymmetry, and flexibility in jiu-jitsu athletes, focusing on differences by fighting style, competition level, gender, and limb dominance. A total of 25 jiu-jitsu fighters participated in this cross-sectional study, which used repeated measures to ensure the reliability of the findings. Participants underwent measurements for limb lengths and were assessed for dynamic balance using the Lower and Upper Quarter Y Balance Test (LQYBT and UQYBT) and for flexibility with the Sit and Reach Test (SR). The analysis included evaluating the reliability of these tests, as well as comparing outcomes based on gender, fighting styles, competition levels, and limb dominance. The results showed that the LQYBT and UQYBT tests were reliable (ICC=0.74–0.93). There were no significant differences in dynamic balance and flexibility between different fighting styles. No significant differences were found in balance tests between the dominant and non-dominant limbs, but there were notable inter-limb asymmetries in certain measures. This study provides preliminary insights for coaches and researchers by offering reference values for balance, flexibility, and inter-limb asymmetry in jiu-jitsu athletes. However, future research with larger sample sizes is needed to better understand the differences among subgroups in this sport.

Keywords: *athletic training, Y-balance test, physical performance, martial arts, asymmetry*

Introduction

Jiu-jitsu originated from the ancient martial art of jujutsu, which traces its origins back to ancient Japan (Kano, 1994). Today, jiu-jitsu represents a modern self-defense martial art that has gained significant popularity in the last two decades (Andreato, Lara, Andrade, & Branco, 2017). In this sport, there are two styles of fighters – "Pass" and "Guard" fighters. "Pass" fighters aim to break through the guard, earning points or positional advantages, while guard fighters defend the guard and attack from that position (Báez et al., 2014). During matches, fighters utilize various choking, joint-locking, and submission techniques, and a match ends either by the opponent's submission or through scoring by judges (IBJJF, 2016). Match durations vary from 5 minutes for white belt holders (beginners) to 10 minutes for black belt holders (elite athletes; Del Vecchio, Bianchi, Hirata, & Chacon-Mikahili, 2007; IBJJF, 2016), and athletes may have an average of 4 to 6 matches in competitions, requiring good physical preparation (Andreato et al., 2017).

In order to achieve success in jiu-jitsu competitions, fighters must invest various efforts that require motor coordination, strength, flexibility, and good balance abilities (de Paula Lima et al., 2017; James, 2014). For jiu-jitsu fighters, maintaining proper joint stability and postural balance is crucial for achieving successful performances, as it is a factor that can affect the effectiveness of attack and defense techniques among fighters (Sterkowicz, Lech, Jaworski, & Ambrozy, 2012). During each action in combat, the competitor must

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maintain a stable body position that enables them to not lose or disturb their balance (das Graças et al., 2017). Postural stability represents the body's ability to maintain a stable position in a state of balance under the influence of various factors (Krištofič, Malý, & Zahálka, 2018). The level of postural stability reflects an individual's ability to minimize fluctuations in the center of gravity, which means maintaining an upright body position and properly responding to changes in external and internal forces (Angyan, Teczely, & Angyan, 2007). Dynamic balance is defined as maintaining a balanced support base during movement execution. In clinical situations, the Star Excursion Balance Test (SEBT) and its Y-balance version are often used to assess dynamic balance because they are practical tools for identifying deficits in dynamic balance (Gribble, Hertel, & Plisky, 2012). Additionally, the Y-balance test has been used to assess inter-limb asymmetries and related injury risk in both upper and lower extremities (Bubić, & Kozinc, 2023; Fusco et al., 2020; Ribnikar, Maguša, & Kozinc, 2024). Therefore, this test could be very relevant to jiu-jitsu because forced joint loading techniques often increase the incidence of knee, shoulder, and elbow injuries (Kreiswirth, Myer, & Rauh, 2014). Previous research has highlighted athletes showing limb asymmetries in jumping greater than 15% have a higher frequency of injuries (Impellizzeri, Rampinini, Maffiuletti, & Marcora, 2007). However, less is known about the role of inter-limb asymmetries and their impact on physical performance (Bishop, Turner, & Read, 2018). Some studies have shown that inter-limb asymmetries of around 10% result in reduced jump height (Bell, Sanfilippo, Binkley, & Heiderscheit, 2014) and slower change of direction times (Hoffman, Ratamess, Klatt, Faigenbaum, & Kang, 2007), but contradictory results have also been observed (Lockie et al., 2014).

Additionally, flexibility is a relevant physical component of jiu-jitsu, especially in the thoracolumbar spine and posterior chain, which are necessary for performing specific attacking or defensive situations (Andreato et al., 2011, Andreato et al., 2016). High levels of flexibility can assist jiu-jitsu athletes in executing positions and also facilitate the learning of new motor skills (Andreato et al., 2011). A practical and reliable test for assessing the flexibility of hip, back, and posterior lower limb muscles is the sit-and-reach test (Mier, 2011).

While various studies have examined postural balance in fighters such as karatekas (Kocahan, Guraslan, Kabak, Hasanoglu, & Akinoglu, 2022), judokas (Fatih, Çakir, Çavusoglu, Akyurek, & Haykir, 2022; Kilic, Karakoç, & Karakoç, 2022), and taekwondo practitioners (Aldhabi, Albadi, Alnajjar, Fasihudden, & Vincent, 2024; Fatih et al., 2022; Tayshete, Akre, Ladgaonkar, & Kumar, 2020), there is only one study that focused on jiu-jitsu athletes (Leszczak et al., 2022). The authors examined static body balance of fighters according to body mass status, reporting better stability compared to a control group and possible influence of body mass. When it comes to inter-limb asymmetry, research has mainly focused on its impact on injuries in jiu-jitsu athletes (Sarro et al., 2022). On the other hand, there are studies that have investigated flexibility in sit and reach tests among jiu-jitsu senior athletes (Andreato et al., 2011; Del Vecchio et al., 2007), however, differences among groups were not examined; instead, the level of flexibility among jiu-jitsu athletes was determined.

To the best of our knowledge, there are no studies that have examined the dynamic balance and inter-limb asymmetries of jiu-jitsu athletes. This is the first study to investigate differences in dynamic balance, limb asymmetry, and flexibility among fighters. Accordingly, the aim of this study was to preliminarily explore differences in balance, inter-limb asymmetry, and flexibility according to fighting style and competition level, gender differences, asymmetry between dominant and non-dominant legs, and inter-limb asymmetry among jiu-jitsu athletes. Based on these objectives, we can gain a broader insight into dynamic balance, inter-limb asymmetry, and flexibility in jiu-jitsu fighters. This research will have practical significance for fitness coaches in jiu-jitsu sport, providing them with reference values of the given abilities in relation to gender, competition level, and fighting style. It will also be significant for researchers because, as the first study to thoroughly investigate this area, it will contribute to a better understanding of dynamic balance and inter-limb asymmetry in jiu-jitsu athletes.

Methods

Participants

The study involved a convenience sample of 25 jiu-jitsu fighters (8 women, 17 men), representing the maximum feasible recruitment from the available population within the constraints of the study period and resources. The demographic characteristics of the study sample were as follows: the mean height was 177.32 cm (SD=9.46), mean mass was 75.74 kg (SD=11.88), and the mean age was 30 years (SD=8.62). Ten participants (3 women) were "Pass" fighters and 15 (5 women) were "Guard" fighters. The group comprised of 9 participants with purple belts (3 women), 5 with white belts (2 women), 4 with blue belts (2 women), 3 with brown belts (1 women), and 4 with black belts (0 women). The participants reported to perform 4.1±2.2 jiu-jitsu sessions per week (range =2–9) and to have 5.4±4.24 years of jiu-jitsu experience (range =2–15). The dominant arm was right for 22 participants (7 women) and left for 3 participants (1 women). The dominant leg was right for 21 participants (7 women) and left for 4 participants (1 women). To qualify for participation, individuals were required to be free from any upper or lower extremity, or spinal injuries, as well as surgeries in these areas within the previous six months. The exclusion criteria included recent concussions, any conditions that hindered work or sports participation in the last six months, limb amputations, vestibular disorders, active pregnancy, current illnesses or injuries that could affect physical performance or balance, and those receiving treatment for inner ear, sinus, or respiratory infections. Before initiating the study procedures, participants were thoroughly informed about the study's objectives and procedures, provided consent on a voluntary basis, and were assured of their right to discontinue participation at any time. A consent form was signed by each participant prior to the commencement of testing, and they were guaranteed access to their personal results upon completion of the study. Ethical approval for the study was granted by the National Medical Ethics Committee under the reference number 0120-690/2017-8.

Procedures

The study was structured as a cross-sectional investigation incorporating repeated measures to evaluate reliability. Measurements were taken on two separate occasions, with the follow-up visit arranged for 10 to 14 days after the initial session. All assessments were carried out by an examiner with BSc and MSc degrees in kinesiology and ample experience in motor ability testing. To control for potential variations due to the circadian rhythm, all sessions were conducted in the afternoon, and at similar specific time (±1h) for each individual. Participants were instructed to avoid any intense physical activity or training 24 hours prior to their scheduled testing sessions. This precaution was intended to minimize fatigue and other acute effects of exercise that could affect performance and physiological measurements during testing. Additionally, they were advised to maintain their usual dietary habits and ensure adequate hydration to avoid any nutritional imbalances that could impact the study results. Participants completed a structured questionnaire consisting of demographic information and engagement in jiu-jitsu practice. Body weight was assessed using a calibrated digital scale and body height was self-reported.

The warm-up consisted of a 10-minute session involving dynamic stretches and low-intensity cardiovascular exercises, such as jogging and jumping jacks, designed to gradually increase heart rate and muscle temperature, thereby preparing the athletes for the subsequent physical performance tests. This protocol was standardized across all participants to ensure consistency in pre-test conditions. Subsequent measurements involved determining the length of participants' upper and lower limbs to an accuracy of 0.5 cm using a measuring tape. The measurement of the lower limb was conducted in a supine position, extending from the anterior superior iliac spine to the medial malleolus. The upper limb measurement was taken in a standing position, with the arm abducted at 90 degrees, stretching from the seventh cervical vertebra to the tip of the middle finger. Limb measurements were followed by assessment of LQYBT, UQYBT and SR tests, which were conducted sequentially in this order for all participants.

For LQYBT, we followed the procedure was implemented following the methodology outlined by Plisky et al. (2009), where participants executed reaches in anterior, posterolateral, and posteromedial directions from a unilateral stance on a designated device, with allowances for trunk and arm movements and a rest period of 30 seconds between each direction. The YBT-UQ was carried out based on the protocol by Gorman et al. (2012), with participants performing reaches in medial, superolateral, and inferolateral directions from a push-up stance, allowing for pelvic adjustments and a 60-second rest between attempts. Three warm-up attempts were allowed per test, followed by three repetitions. The outcomes of both tests were initially recorded as the absolute lengths to the nearest centimeter. For analysis, the mean of the three reach distances for each direction was calculated. The reach measurements were then normalized to the limb lengths, presented as a percentage.

The SR test was conducted using a measuring tape that was placed and secured on the floor with a starting line marked on it. The test subjects removed their shoes and sat on the floor with the measuring tape positioned between their legs. The soles of their feet were set just behind the marked starting line, with heels approximately 20 cm apart. The palms were touching or slightly overlapping with thumbs interlocked, turned downwards, and placed on the measuring tape. With legs extended, the subjects slowly reached forward along the measuring tape as far as possible. The furthest reach that could be maintained for 2 seconds was measured by the tester. The test was performed twice, with a short break (30 seconds) between attempts. The results were recorded based on the zero-point marked by a line under the heels of the extended legs. If the test subjects did not reach the starting line, the results were recorded as negative values of the distance from the line; if they reached beyond it, the positive values of the reach were recorded in centimeters (Cuberek, Machová, & Lipenská, 2013). The mean value of the two repetitions was taken for further analysis.

Statistical analysis

The collected data were analyzed using the IBM SPSS software (version 25.0). Descriptive statistics are presented as mean and standard deviation. The normality of data distribution was checked with the Shapiro-Wilk test and a visual assessment of histograms. The repeatability of tests across visits was analyzed using the intra-class correlation coefficient (ICC (3,1); absolute agreement model) and the typical error, which we expressed as a percentage of the mean value or the coefficient of variation (CV). ICC values <0.5 were considered indicative of poor repeatability, values between 0.5 and 0.75 as moderate repeatability, values between 0.75 and 0.9 as good repeatability, and values above 0.90 as excellent repeatability (Koo & Li, 2016). Absolute repeatability was considered acceptable if CV was <10% (Hopkins, 2000). The relationships between the test results (both Y-tests and the seated reach test) were examined using Pearson's correlation coefficient. These relationships were interpreted as negligible (<0.1), weak (0.1–0.4), moderate (0.4–0.7), strong (0.7–0.9), or very strong (>0.9) (Akoglu, 2018). The comparison between fight styles was carried out using an independent samples t-test. The comparison between the dominant and non-dominant side was conducted using a paired samples t-test. In the t-tests, the effect size was expressed with Cohen's d, according to which the effect size is interpreted as trivial (<0.2), small (0.2–0.5), medium (0.5–0.8), and large (>0.8) (Cohen, 1988). The differences in inter-limb asymmetry magnitudes across tests and directions were tested with 1-way analysis of variance for repeated measures. Statistically significant differences were accepted at a confidence level of $\alpha<0.05$.

Results

Reliability

Table 1 contains the repeatability analyses of the measurements. The results of LQYBT were found to have good or excellent reliability (ICC=0.84 to 0.95). A similar observation applies to most results of the UQYBT (ICC=0.83 to 0.91), with the exception of moderate reliability in the medial reach with the non-dominant arm (ICC=0.74) and the superolateral reach with the dominant arm (ICC=0.73). The repeatability of the SRT was excellent (ICC=0.93). Absolute repeatability was acceptable for all the tests examined, as the typical error (expressed as CV) was below the 10% threshold for all tests. Despite good relative and absolute repeatability, there were statistically significant systematic errors in 3 out of 8 variables related to the LQYBT (p=0.014 to 0.042) and one variable related to the UQYBT (p=0.038), as well as SRT (p=0.049) (Table 1). In all cases, the test result on the second visit was slightly higher (difference of 1 to 3% of leg length).

Table 1. Reliability of the measurements

Test / Limb	Direction	Visit 1		Visit 2		Reliability		
		Mean	SD	Mean	SD	ICC (95% CI)	CV (95% CI)	Systematic error (p-value)
LQYBT – non-dominant (% leg length)	Anterior	0.71	0.06	0.73	0.05	0.84 (0.59 - 0.93)	3.26 (2.39 - 5.14)	0.014*
	Posterolateral	1.15	0.12	1.16	0.10	0.90 (0.75 - 0.96)	3.11 (2.28 - 4.91)	0.276
	Posteromedial	1.13	0.13	1.14	0.09	0.88 (0.69 - 0.95)	3.67 (2.69 - 5.79)	0.373
	Composite	0.99	0.09	1.00	0.07	0.87 (0.67 - 0.95)	3.18 (2.33 - 5.02)	0.127
LQYBT – dominant (% leg length)	Anterior	0.72	0.07	0.72	0.08	0.93 (0.81 - 0.97)	2.84 (2.08 - 4.48)	0.064
	Posterolateral	1.14	0.11	1.15	0.08	0.92 (0.79 - 0.97)	2.65 (1.94 - 4.17)	0.042*
	Posteromedial	1.12	0.11	1.13	0.09	0.95 (0.87 - 0.98)	2.16 (1.58 - 3.41)	0.079
	Composite	0.99	0.09	1.00	0.07	0.95 (0.88 - 0.98)	1.85 (1.35 - 2.91)	0.014*
UQYBT – non-dominant (% arm length)	Medial	1.06	0.10	1.07	0.08	0.74 (0.37 - 0.89)	4.62 (3.38 - 7.29)	0.884
	Superolateral	0.79	0.13	0.82	0.10	0.88 (0.68 - 0.95)	5.37 (3.93 - 8.47)	0.884
	Inferolateral	0.93	0.11	0.93	0.10	0.88 (0.69 - 0.95)	4.2 (3.08 - 6.63)	0.888
	Composite	0.93	0.09	0.94	0.06	0.85 (0.63 - 0.94)	3.45 (2.52 - 5.44)	0.961
UQYBT – dominant (% arm length)	Medial	1.03	0.10	1.06	0.09	0.83 (0.57 - 0.93)	3.88 (2.84 - 6.11)	0.038*
	Superolateral	0.76	0.12	0.79	0.10	0.85 (0.63 - 0.94)	5.87 (4.3 - 9.26)	0.182
	Inferolateral	0.89	0.10	0.92	0.07	0.73 (0.36 - 0.88)	5.24 (3.84 - 8.27)	0.078
	Composite	0.89	0.08	0.92	0.07	0.91 (0.75 - 0.96)	2.79 (2.04 - 4.39)	0.007*
Sit & Reach test (cm)	/	51.1	10.1	54.5	9.3	0.93 (0.81 - 0.97)	5.21 (3.81 - 8.22)	0.049*

SD – standard deviation; ICC – intra-class correlation coefficient; CV – coefficient of variation; CI – confidence interval; * - significance at the 0.05

Table 2. Fighting styles comparison

Variable		Guard (n = 15)		Pass (n = 10)		Difference		
		Mean	SD	Mean	SD	t	p	d
Body height (cm)		178.0	8.7	176.3	10.9	0.43	0.669	0.15
Body mass (kg)		76.5	10.4	74.5	14.3	0.40	0.691	0.23
Age (yrs)		29.5	9.8	30.7	6.8	-0.33	0.748	0.24
LQYBT – non-dominant (% leg length)	Anterior	0.71	0.05	0.71	0.07	-0.19	0.848	0.05
	Posterolateral	1.15	0.10	1.14	0.15	0.25	0.806	0.10
	Posteromedial	1.14	0.11	1.11	0.15	0.42	0.680	0.19
	Composite	1.00	0.08	0.99	0.11	0.26	0.800	0.12
LQYBT – dominant (% leg length)	Anterior	0.72	0.06	0.71	0.09	0.55	0.589	0.16
	Posterolateral	1.14	0.08	1.14	0.15	0.02	0.988	0.01
	Posteromedial	1.13	0.10	1.10	0.13	0.70	0.493	0.31
	Composite	1.00	0.07	0.98	0.11	0.45	0.658	0.17
UQYBT – non-dominant (% arm length)	Medial	1.09	0.08	1.02	0.12	1.71	0.101	0.57
	Superolateral	0.81	0.10	0.76	0.16	1.00	0.330	0.44
	Inferolateral	0.93	0.10	0.93	0.14	0.09	0.927	0.04
	Composite	0.94	0.06	0.90	0.12	1.13	0.270	0.45
UQYBT – dominant (% arm length)	Medial	1.04	0.08	1.01	0.12	0.73	0.472	0.24
	Superolateral	0.77	0.08	0.75	0.17	0.44	0.667	0.23
	Inferolateral	0.88	0.09	0.90	0.11	-0.46	0.647	0.18
	Composite	0.90	0.06	0.89	0.11	0.31	0.756	0.00
Sit & Reach Test (cm)		53.48	9.26	47.63	10.81	1.45	0.161	0.63

SD – standard deviation.

Differences between fighting styles

We compared the results of 15 "guard" and 10 "pass" fighters. There were no differences between representatives of these fighting styles in terms of body weight (p=0.691), height (p=0.669), and age (p=0.748). There were no differences in styles of fighting in the Y-balance tests (both versions), and likewise, no differences in the seated reach test. The detailed results are displayed in Table 2.

Additionally, we examined the correlations between Y-test results and experience in jiu-jitsu (years of training), as well as the weekly frequency and belt. We found no statistically significant correlations (all r≤0.31; all p≥0.099).

Limb dominance

In Figure 1, the results of the analysis of the difference between the non-dominant and dominant sides are shown. In the analysis of results for the LQYBT, no statistically significant differences were recorded between the non-dominant and dominant leg (p=0.171–0.532) (Figure 1A). Conversely, in the test for the UQYBT, statistically significant higher values were found in the non-dominant arm (p=0.001–0.0015; Figure 1B). The effect size was small for individual directions (d=0.31 for the sideways reach, d=0.21 for the superolateral reach, and d = 0.36 for the inferomedial reach). However, for the overall result, the difference between the non-dominant and dominant arm was large (d=1.48).

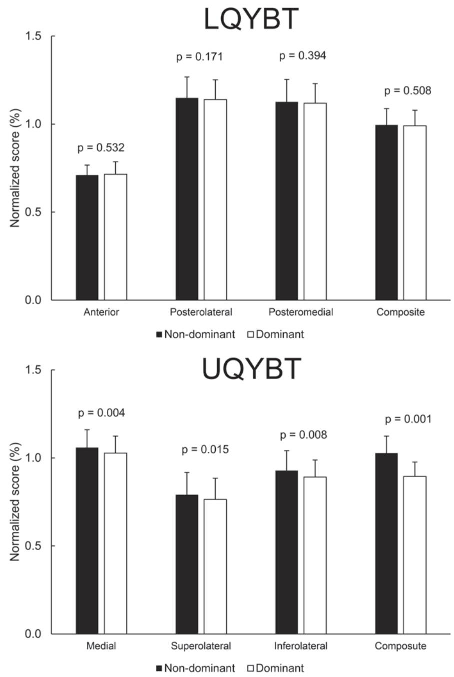

FIGURE 1. Differences between dominant and non-dominant limbs across tests and directions.

Inter-limb asymmetries

We examined whether the inter-limb asymmetries recorded in each test were correlated. We observed a moderate and statistically significant relationship between the asymmetry in the superolateral reach with the arm and the posterolateral reach with the leg (r=0.55; p=0.004), and between the asymmetry in the superolateral reach with the arm and the composite score asymmetry for the leg (r=0.48; p=0.015). No other correlations were found between the two tests. Within the lower body test, there was only a moderate correlation between the posteromedial reach and the posterolateral reach (r=0.46; p=0.019). Within the UQYBT test, there were no correlations between asymmetries in different reach directions.

Table 3 presents the descriptive statistics of the absolute inter-limb asymmetry values for each task and direction of reach. An analysis of variance revealed statistically significant differences in asymmetry values among the three directions of reach in the LQYBT (F=7.19; p=0.002). Post-hoc tests showed that the significant differences were only between the forward reach and the backward-sideways reach (p=0.004). In the UQYBT there were no differences in the size of asymmetries among the reach directions (F=0.55; p=0.578). Additionally, we compared inter-limb asymmetries in the total score between the two tests. It turns out that inter-limb asymmetry was significantly higher in the UQYBT (p=0.022; Table 3).

Lastly, we checked whether the magnitude of inter-limb asymmetries differed between men and women, and between fighters of different styles. A statistically significant difference

Table 3. Mean inter-limb asymmetry values across tests and directions

Test	Direction	Mean	SD	Minimum	Maximum
LQYBT	Anterior	4.52	3.13	0	10.85
	Posterolateral	2.19	1.72	0	5.41
	Posteromedial	3.35	2.29	0	9.52
	Composite	2.29	1.91	0	7.82
UQYBT	Medial	4.34	3.30	0	12.44
	Superolateral	5.69	4.94	0	17.48
	Inferolateral	5.49	5.47	0	24.49
	Composite	3.94	3.01	0.25	11.93

SD – standard deviation

between men and women was observed only in the anterior reach in the LQYBT, where men had significantly lower average asymmetry values (2.79±2.49%) compared to women (5.33±3.13%) (p=0.044; d=0.90). Among fighters of different styles, significant differences were also observed only in one variable, namely the medial reach in the UQYBT. Inter-limb asymmetry was higher in "Guard" fighters (5.43±3.62%) than in "Pass" fighters (2.68±1.89%) (p=0.022; d=0.99).

Discussion

The aim of this study was to thoroughly investigate differences in balance, limb asymmetry, and flexibility according to fighting style, competition level, and gender of Jiu-Jitsu fighters, as well as to determine the asymmetry between the dominant and non-dominant legs and inter-limb asymmetry among them. In this regard, this study had several significant findings: i) there were no differences among groups according to fighting styles in the dynamic balance, flexibility, and body composition; professional athletes had significantly higher results in the inferolateral reach in the UQYBT, both on the non-dominant arm and dominant arm; ii) comparing limb dominance, significantly higher values were found in the UQYBT test on the non-dominant arm, while in LQYBT, there were no differences between the limbs; iii) significant association was found between asymmetry in superolateral reach with the arm and posterolateral reach with the leg, as well as between asymmetry in superolateral reach with the arm and asymmetry in composite result for the leg; iv) within the lower body test, a moderate correlation was found between posteromedial reach and posterolateral reach; significant differences were only between forward reach and backward-lateral reach; v) inter-limb asymmetry was significantly higher in UQYBT; men had significantly lower average asymmetry values than women in anterior reach in LQYBT; inter-limb asymmetry was greater in "Guard" style fighters than in "Pass" style fighters in medial reach in UQYBT.

Reliability of tests

Good to excellent reliability was established for LQYBT (ICC=0.84 to 0.95), and for most UQYBT tests (ICC = 0.83 to 0.91), except for the medial reach with the non-dominant arm (ICC=0.74) and the superolateral reach with the dominant arm (ICC=0.73). In addition, reliability for SRT was excellent (ICC=0.93). These results are consistent with previous studies reporting good to excellent reliability of the LQYBT test (ICC=0.85-0.93; Shaffer et al., 2013), excellent reliability of the UQYBT test (Gorman, Butler, Plisky, & Kiesel, 2012; Westrick, Miller, Carow, & Gerber, 2012), as well as excellent reliability for SRT (ICC=0.95; Ayala, de Baranda, Croix, & Santonja, 2012). In summary, the reliability of all tests on a sample of jiu-jitsu athletes has been confirmed in our study, confirming previous findings. Therefore, LQYBT, UQYBT, and SRT are suitable for use within jiu-jitsu populations, providing a valid foundation for the subsequent analytical processes detailed within this paper.

Differences between groups

There were no differences among groups according to fighting styles in the dynamic balance, flexibility, and body composition. This corresponds to a previous study (de Paula Lima et al., 2017) which also found no differences between styles in jiu-jitsu athletes in dynamic postural balance, flexibility, and isometric handgrip strength, as well as the study (Del Vecchio, Gondim, & Arruda, 2016) in which no differences were found in joint mobility, stability, proprioception, strength, and flexibility. Regarding body composition, in a study (Báez et al., 2014), "Guard" fighters were taller than "Pass" fighters, but there was no difference in body mass and BMI. Based on the recent studies, it is evident that our findings are consistent with previously mentioned studies (Báez et al., 2014; de Paula Lima et al., 2017; Del Vecchio et al., 2016), which can be explained by the fact that jiu-jitsu athletes, regardless of their different fighting styles, participate in the same sport, which imposes relatively similar demands on them. Although, based on the analysis of different styles of jiu-jitsu fighters, it can be observed that "Pass" fighters attack more frequently and require greater strength and power for certain actions (Stock, Mota, Hernandez, & Thompson, 2017), it is important to note that jiu-jitsu is a dynamic sport and jiu-jitsu fighters must rely on other combat strategies depending on various competitive factors. For example, "Guard" fighters sometimes have to fight in a standing position, which is common for "Pass" fighters, and vice versa (Spencer, 2016). Given this information, it is clear why no differences were found between different fighting styles among jiu-jitsu fighters.

Limb dominance, and inter-limb asymmetries

Comparing limb dominance, significantly higher values were found in the UQYBT test on the non-dominant arm. However, although there are no studies with jiu-jitsu athletes, in studies on throwing athletes, there were no differences between the limbs in the UQYBT test (Borms, Maenhout, & Cools, 2016; Bullock et al., 2018; Taylor, Wright, Smoliga, DePew, & Hegedus, 2016). On the other hand, in LQYBT, there were no differences between the limbs, which is in line with previous research (Bressel, Yonker, Kras, & Heath, 2007; Fusco et al., 2020; Gribble & Hertel, 2003). Association was found between asymmetry in superolateral reach with the arm and posterolateral reach with the leg, as well as between asymmetry in superolateral reach with the arm and asymmetry in composite result for the leg. Within the lower body test, a moderate correlation was found between posteromedial reach and posterolateral reach. Therefore, while some correlations were found, asymmetries tend to be test- and outcome-specific, which is consistent with previous studies concerning strength and power asymmetries (Cuthbert et al., 2021; Pérez-Castilla et al., 2021). Therefore, it is crucial to consider a multitude of tests and to compressively assess the presence of inter-limb asymmetries.

Mean asymmetries for our studies were mostly in ~2-5% range, with maximal values at 5–10% in LQYBT and 11-24% in UQYBT. This shows that specific thresholds for meaningful asymmetry for each outcome will likely need to be established. Previous studies have established such benchmarks for other sports. For example, in female cross-country runners, an LQYBT inferolateral reach difference of less than 4 cm was associated with a 75% lower probability of a running-related injury (Ruffe, Sorce, Rosenthal, & Rauh, 2019). In addition, elite football players with high posteromedial asymmetry had a 2.69-fold increased risk of injury (3.26-fold in subgroup with a past injury; Bennett, Chalmers, Milanese, & Fuller, 2022). Future research should focus on investigating inter-limb asymmetries and their correlation with injury risk among jiu-jitsu practitioners. This will lead to the development of reliable thresholds that can be utilized in the development of injury prevention programs. This approach will enable practitioners to design more targeted and effective interventions aimed at reducing the incidence of injuries in this population.

Practical implication

This study is very important as it extensively examined flexibility, dynamic balance, and asymmetries among jiu-jitsu athletes. Based on the findings, reference values for these abilities have been provided, allowing practitioners to adjust training programs according to the gender, level, and style of jiu-jitsu athletes, all aiming to achieve an optimal level of physical preparedness. Furthermore, it will be significant for researchers as the first study to thoroughly investigate this area, contributing to a better understanding of dynamic balance and asymmetry among jiu-jitsu athletes. It can serve as a starting point for further exploration in this field.

Limitations and future suggestion

In the context of this study, one significant limitation is the small sample size, especially regarding the smaller sub-groups within the study. Therefore, future research needs to ensure a large sample of jiu-jitsu athletes, which will also provide a sufficiently large sub-sample to expand on our preliminary findings. Additionally, besides the given abilities, there is a need to examine other motor abilities that are important in jiu-jitsu, primarily muscle strength and muscle endurance, as well as their relationship with dynamic balance and asymmetry, which will provide a broader insight into the motor skills of jiu-jitsu athletes. While intra-rater test-retest reliability was established specifically for the rater in our study to ensure the robustness of the results, inter-rater reliability of the administered tests remains to be established.

Conclusion

This study investigated dynamic-balance (Y-balance test) and flexibility (SR test) among Jiu-Jitsu fighters, focusing on variations by fighting style, competition level, and gender. Key findings include female athletes displaying superior flexibility to males without differences in dynamic balance. No distinctions were observed in dynamic balance, flexibility, or body composition across fighting styles. Professional athletes outperformed in UQYBT for both arms, with notable asymmetry in limb performance. Interestingly, inter-limb asymmetry was more pronounced in the upper body tests, with gender and fighting style influencing specific asymmetry patterns, suggesting tailored training might benefit performance optimization.

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Conflicts of interest
The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Ethical approval
The research adhered to the principles of the Helsinki Declaration and received approval from the National Medical Ethics Committee (approval number: 0120-557/2017/4).

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ORIGINAL SCIENTIFIC PAPER

Innovation, Validity, and Reliability of Modified Dynamic Balance Test for Karate Kata Category

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Abstract

This study aims to create innovation through validity and reliability tests in the form of modification of dynamic balance test to meet the needs of karate athletes in kata category. The sample as a validation test was 11 experts from certified lecturers and coaches. Then, the sample as a reliability test was 50 male athletes with 10.5±8.3 years of karate training experience and 8.3±5.2 years of experience in the kata category, and 50 female athletes with 8.7±9.6 years of karate training experience and 7.7±8.4 years of experience in the kata category. The results of this study included seven aspects which were the suitability of the definition of balance to the test media, the suitability of the test specifically for the kata category, the appearance or design of images that are easy to understand, the test procedure that is easy to do, the distance at each cone, the ease to organize test, then validity value of Aiken V>0.8, reliability value of Cronbach's Alpha 0.778 and ICC value 0.777, the results of the male and female athlete questionnaires through independent t-test ($p<0.05$) of 0.366. In conclusion, the Modified Dynamic Balance Test developed for the Karate Kata category has a high level of validity and reliability in the various aspects tested. The absence of differences in the results between male and female athletes confirms that the test is fair and applicable to all athletes, and it makes the test a proper test tool for improving performance in competition.

Keywords: *dynamic balance, karate test, measuring instrument innovation*

Introduction

Dynamic balance is one of the fundamental components of athletic performance, especially in karate, which demands precise, fast, and varied movements. Dynamic balance is essential in karate because it helps athletes maintain stability when performing complex movements, improve movement efficiency, and maintain correct posture (Pinto-Escalona et al., 2024). Good balance allows athletes to perform techniques with precision, power, and speed without losing control in order to give them an edge in the competition (Mijatović et al., 2023). In the Kata category in karate, dynamic balance is vital in maintaining stability during transitions between movements and correct posture when performing complex techniques. Good balance improves movement efficiency and reduces the risk of injury when performing explosive movements (Kokic et al., 2022).

The balance ability of athletes can be known through balance tests. Balance tests are essential to identify the athlete's level of balance, evaluate the training program's effectiveness, and adjust the training to suit the athlete's specific needs better (Oktavian et al., 2022). In addition, the balance test results can also help design better training strategies and detect potential weaknesses that can interfere with athlete performance (Almas et al., 2023). To date, dynamic balance tests are often performed for athletes such as the Y-Balance Test and Star Excursion Balance Test (Ahern et al., 2021; Picot et al., 2021; Plisky et al., 2021; Ribnikar et al., 2024). Coaches often use balance tests such as Y-Balance Test and

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Star Excursion Balance Test because they effectively measure an athlete's dynamic balance, essential for maintaining body stability during multi-directional movements. These tests help identify asymmetries or weaknesses that could increase the risk of injury and allow coaches to evaluate training progress by comparing results over time (Ahern et al., 2021). In addition, the test is relatively simple and does not require expensive equipment, but provides accurate data on the functional balance of the athlete, making it very useful in optimizing performance and preventing injuries (Plisky et al., 2021; Ribnikar et al., 2024).

Although some balance tests are already designed to measure dynamic balance ability, most of them are vague to the characteristics and needs of karate athletes, particularly in the Kata category. Previous research investigated a balance test on karate using a modified bass test of dynamic balance (Almas et al., 2023; Oktavian et al., 2022), then another research explored the use of Y-balance test to measure balance (Ben Hassen et al., 2022; Pekel et al., 2023). In the results of the novelty study, it was also found that Star Excursion Balance Test was also used for testing balance of karate athletes aged 18-22 (Kindzer et al., 2024), Moreover, the latest study investigated the use of the Stork Standing balance model for karate athletes aged 16-17 (Purba et al., 2024). However, some research results for balance tests are still limited to general fitness tests for the Kumite and Kata categories. Therefore, a more precise and appropriate test is needed to evaluate the dynamic balance of karate athletes in the kata category.

This study aims to develop a modified existing dynamic balance test, adapted specifically to meet the needs of karate athletes in the Kata category. The importance of the results of this study is that it can provide broad insight to coaches and academics in the sport of Karate (Widyastuti et al., 2024). In addition, this study will also evaluate the validity and reliability of the modified test to ensure that it can be used as a reliable and accurate measuring tool in the context of karate sports. This development is expected to improve the quality of training and performance evaluation of karate athletes and prevent injuries related to dynamic balance.

Method

Procedures

This research used quantitative and qualitative approaches. A mixed methods approach is needed to develop measurement tools in innovation research to obtain more profound and measurable study results (Susiono et al., 2024). The first stage in this study was to review articles in English with the topic 'kata' or 'karate' or 'dynamic balance', as well as field observations. The purpose of this stage was to collect material about the physical component of balance so that it can design a form of balance test. The results of the first stage, showed that the karate kata's dynamic balance is the body's ability to remain standing and not fall when changing the direction of each movement or when performing high-speed movements. Then innovations for the testing were made based on the results of other research (Aisyah et al., 2020; Emad et al., 2020; Gaweł & Zwierzchowska, 2024; Guler et al., 2017; Molinaro et al., 2020; Pal et al., 2021; Przybylski et al., 2021; Zago et al., 2015).

After the materials were collected, the second stage was to design a dynamic balance test (Figure 1). The equipment for this test was 1) a large mattress, 2) five cones, 3) a whistle, and 4) a stopwatch. Then, three examiners were needed, with their respective duties: one as an instructor, one as a stopwatch holder, and one as a recorder and reporter of test results.

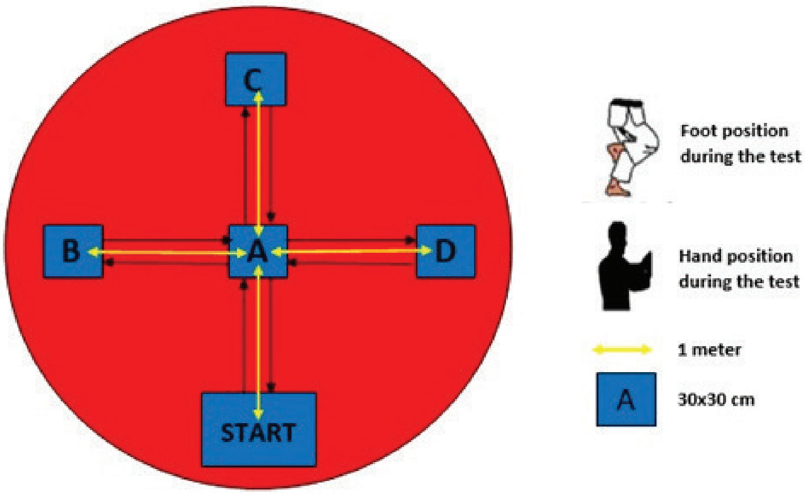

FIGURE 1. Dynamic Balance Test Design

The procedure for this test innovation that must be done by the athlete was as follows: 1) Testee stands with two feet in the Start Position; 2) When the Tester gives the command "Start", Testee steps to Position A with the right foot and maintains the Position for five seconds which is counted by the Tester; 3) From Position A , Testee steps to Position B with the left foot and maintains the position for five seconds; 4) From Position B, Testee steps back to Position A with Right foot as support and maintains position for five seconds; 5) From Position A, Testee steps to Position C with left foot as support and maintains position for five seconds; 6) From Position C, Testee steps back to Position A with Right foot as support and maintains position for five seconds; 7) From Position A, Testee steps to Position D with left foot as support and maintains position for five seconds; 8) From Position D, Testee steps back to Position A with Right foot as support and maintains position for five seconds; 9) From Position A, Testee steps back to the START line with both feet as support; 10) The total time of the test is the time to maintain the body in each position plus one second for locking each position, which is, 42 seconds. The athlete performed three attempts, and the best time of the three attempts was recorded.

After the design, required equipment, and test execution procedures had been determined, the third stage was to conduct validity and reliability tests. The validity test was carried out by academics in sports coaching with a background as karate coaches and karate coaches with a minimum national-level trainer certificate. The Delphi technique was applied to this validity test.

Experienced karate athletes carried out the reliability test in the kata category. Athletes were asked to perform a motor test in the form of balance which had previously been declared valid, then the results were analyzed statistically to determine the reliability of the test.

The validity and reliability test assessment used a questionnaire consisting of seven aspects, and interviews were not structured if needed. The assessment aspect consists of 1) aspects of the suitability of the definition of balance to the test media, 2) aspects of the suitability of the test specific to the Kata category, 3) aspects of the appearance or design of images that are easy to understand, 4) aspects of the test procedure that are easy to do, 5) aspects of the distance on each cone, 6) aspects of the ease of conducting this test at anytime and anywhere.

Participants

The sample of this study involved 11 experts as validators, consisting of four experts from sports coaching education lecturers who are doctoral degree qualified and have karate expertise. Then, seven karate practitioners or trainers already have a national minimum trainer certificate and a bachelor's degree in sports. The 11 validators were identified as having belt of DAN IV-VI levels. This study involved 100 karate athletes (50 male and 50 female) in the Kata category aged 17-23 as a sample for the reliability test. The selection of this sample used purposive sampling, then the athletes also voluntarily became respondents with written consent. Thus, the athletes who were sampled were in the category of having at least been champions in competitions at the provincial level. Sample characteristics in male athletes were as follows: height (mean±SD) 166.43±7.7 cm; weight 67.3±0.3 kg; karate training experience 10.5±8.3 years; experience in the kata category 8.3±5.2 years. The characteristics of the sample in female athletes were as follow: height (mean±SD) 161.6.8 ±3.6 cm, weight 54.6±3.1 kg, karate training experience 8.7±9.6 years, experience in the kata category 7.7±8.4 years. This research has also received ethical approval from the institution 003/S.IP/IKS-UNY-UNRI/A.III/II/2024.

Statistical Analysis

Several tests were conducted to analyze the data in this study. The validity test used the Aiken formula (Aiken, 1985). Then, the reliability test used Cronbach's Alpha and Intraclass Correlation Coefficients (ICC). The significant value reference used in this study was Cronbach's Alpha, which was >0.7, meaning that the innovation of this test has acceptable reliability (Amirrudin et al., 2020). Then, the reference of significance value of Intraclass Correlation Coefficients (ICC) was found between 0.75-0.9 or >0.9, meaning that it has good reliability (Koo & Li, 2016). After the reliability test was carried out on the athlete sample, an independent t-test was conducted on the male and female samples. The independent sample t-test aimed to analyze whether there were differences in assessment between male and female athletes. Analysis of reliability test data and independent sample t-test using SPSS 26 software (George & Mallery, 2019). The scale for the assessment of this test is 1–4,, 1 being 'strongly disagree', 2 being 'disagree', 3 being 'agree', 4 being 'strongly agree'. Level of significance was set at 0.05.

$$\text{V Aiken's: } \frac{\sum S}{n\,(c-1)}$$

S : r – lo

Lo : lowest rating score

C : highest rating score

r : the score given by the assessor

FIGURE 2. Aiken V Validity Formula

Results

The analysis of the results of this study consists of three stages, the first stage is a validity test conducted by nationally certified karate lecturers and trainers. The second stage is a reliability test conducted by athletes by trying this test innovation. The third stage is an independent t-test by comparing the results of the balance test assessment based on gender in this sample.

Validity Test

The first stage is reporting the results of the validity test in table 1 below, which was carried out by 4 doctoral lecturers who have karate trainer expertise and 7 karate trainers who have national trainer certificates. So that the 12 validators will assess six aspects of the indicators in this balance test innovation.

Table 1. Aiken V Validity Test Results

Aspects	ΣS	n(c-1)	Aiken	Description
Suitability of balance definition to test media	28	33	0.848	Valid
Suitability test specific to the Kata category	27	33	0.818	Valid
Easy-to-understand image display or design	28	33	0.848	Valid
Easy test procedure	28	33	0.848	Valid
Distance on each cone	29	33	0.879	Valid
Ease of conducting the test at anytime and anywhere	27	33	0.818	Valid

Based on the results of the Aiken validity test in Table 1, aspect 1), namely the suitability of the definition of balance to the test media was the value of V 0.848, aspect 2), namely the suitability of the test specifically for the kata category was the value of V 0.818, aspect 3). Namely, the appearance or design of images that are easy to understand was the value of V 0.848, aspect 4), namely the easy test procedure was the value of V 0.488, aspect 5), namely the distance at each cone was the value of V 0.879, aspect 6), namely the ease of conducting the test at anytime and anywhere was the value of V 0.818. Therefore, the validity test results based on the seven aspects showed that the Aiken value of V >0.8, which means that the aspects are declared valid (Aiken, 1985).

Reliability Test

The second stage is a reliability test conducted by 50 male athletes and 50 female athletes. This reliability test is, athletes will do a motor test in the form of balance. After the athletes try this innovative form of the test, the athletes are asked to assess six aspects. After the athletes assess, these results will be tested through Cronbach's Alpha and Interclass Correlation Coefficient.

Based on the results in Table 2 below, the value of Cronbach's Alpha for Dynamic Balance Test Modification for Karate Kata was 0.778. The agility test was reliable because it had a Cronbach's Alpha value of >0.7.

Table 2. Cronbach's Alpha Results

Reliability Statistics	
Cronbach's Alpha	N of Items
0.778	6

Based on the results in Table 3 below, the Interclass Correlation Coefficient (ICC) value for the Modified Dynamic Balance Test for the Karate Kata Category, had the value of r=0.777 and had a value of 95% confidence interval (CI) between 0.640–0.807, meaning that this test innovation also has good reliability in addition to Cronbach's Alpha value so that the Modified Dynamic Balance Test can be used to measure and analyze balance for karate kata athletes.

Table 3. Interclass Correlation Coefficient Results

	Intraclass Correlation[b]	**95% Confidence Interval**		**F Test with True Value 0**			
		Lower Bound	**Upper Bound**	**Value**	**df1**	**df2**	**Sig**
Single Measures	0.346[a]	0.387	0.611	36.882	99	255	0.001
Average Measures	0.777[c]	0.640	0.807	36.882	99	255	0.001

Based on the results of table 2 and table 3 above, the Cronbach's Alpha results show 0.778 and the Interclass Correlation Coefficient results show 0.777. Thus, the results of the reliability test are that the balance test innovation for athletes in the word category is reliable.

Independent T-Test

The independent t-test compared the answers of 50 male and 50 female athletes is shown in table 4 below. This assessment was based on the results of the answers from the reliability test that the kata category athletes must answer. The assumption of normality test was also used through Kolmogorov-Smirnov (p>0.05). This normality test showed sig. 0.066 in male athletes and sig. 0.059 in female athletes. Thus, it is feasible to continue the independent t-test because the data is normal distributed.

Based on the results in Table 4 above, the value of sig. showed 0.366, meaning that there is no difference in answers from the questionnaire results between male and female athletes in Karate Kata. Thus, the athlete in the Kata category, supports creating an innovation in the Modification of Dynamic Balance Test for Karate in the Kata category.

Table 4. Independent T-Test Results

		t	**df**	**Sig. (2-tailed)**	**Mean Difference**	**Std. Error Difference**	**95% Confidence Interval of the Difference**	
							Lower	**Upper**
Assessment of Modified Dynamic Balance Test for Karate Kata	Equal variances assumed	0.907	98	0.366	0.400	0.441	-0.475	1.275
	Equal variances not assumed	0.907	97.827	0.366	0.400	0.441	-0.475	1.275

Discussion

Based on the results of this study, the Modified Dynamic Balance Test for Karate Kata has high validity and good reliability. The suitability of the definition of balance to the test media shows that the definition of balance is based on the modified test's form and purpose. Movement balance in Kata requires high body control, both in static and dynamic positions (Younesi et al., 2022). When athletes perform the Kata movement, the athlete must be able to maintain a stable and correct posture while performing various techniques such as kicks, punches, and body position changes (Katanic et al., 2022). The answer or response of the athlete also explains that the movement when performing Kata requires a smooth and controlled weight transfer from one position to another. The athlete also stated that this test modification is more acceptable than other balance tests, such as the standing stork, flamingo, and Y balance tests. This modification requires athletes to move and step with stance (dachi) compared to previous tests. The next aspect of the suitability of specific tests for the Kata category is that they also have a high level of validity and good reliability, which indicates that the modified test has been adjusted to the specific needs of Karate Kata. It is essential because movements in Kata have unique characteristics that require different dynamic balance compared to other sports, such as moving around with a combination of punch movements, fast kicks, then stopping suddenly but with sturdy stance (dachi) or with a balanced body (Yudhistira et al., 2021). With good balance, an athlete may maintain stability while moving, which can reduce the effectiveness of the technique to the detriment of overall performance.

Aspects of the display or image design that test takers easily understand indicate that the visual representation of the test has been well designed, facilitating the athlete's understanding and implementation of the test. In this case, the high level of validity indicates that the image's design supports the implementation of the test in the right way. At the same time, good reliability signifies that this design is reliable in various conditions and by various participants (Navalta et al., 2023). Easy-to-understand drawing design is essential when developing or modifying test forms. It ensures that all users, including athletes, coaches, and evaluators, can quickly follow test instructions and procedures without worrying about confusion. Thus, it will have an impact on the procedural aspects of the test. The results show that the procedural aspects of the test, such as language that is easy to understand and the steps to start and end the test, have a high level of validity and good reliability which confirms that the test developed is not only effective but also practical. It also helps to maintain the accuracy of test results, as all test takers will follow the same procedure with uniform interpretation (Adamakis, 2022). Most of the samples in this study stated that they could easily understand the procedures of this procedure just by looking at the design and reading the implementation procedures. Coaches and athletes also argue that this test can be applied to novice athletes or athletes not from karate because of practicality.

This assessment also corrects the distance aspect of each cone, the results of which have high validity and good reliability, indicating that the spatial arrangement in the test has been optimally arranged to measure dynamic balance. It is important because the distance when the athlete moves can affect the athlete's ability to maintain balance while performing movements similar to those in the Kata competition (Sastra et al., 2022). Trainers and athletes also explained that the distance of each cone along one meter is determined by the stance shape (dachi), where the distance between two feet is one meter. The distance has shown a situation where the athlete must move in various directions with high balance control (Daker et al., 2023). Athletes who move to each cone and stand on one leg for five seconds test their ability to maintain balance after changes in direction and position, which are essential elements in Kata movements. In Karate Kata, athletes often have to stop their movements abruptly and hold certain positions with perfect stability (Elkess et al., 2023).

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Conflit of Interest

The researchers have no conflicts both with the researchers and the results of other studies.

Then, the ease of implementation of this test shows high validity and good reliability, highlighting that this test is practical. The ease of conducting this test is related to several practical factors that make this test very efficient and easy to implement. First, this test does not require high costs, so that various groups can access it without a significant financial burden. Second, the test requires only three people as testers, so the management does not require a lot of human resources, making it simpler and easier to manage. In addition, this test requires little space, so that it can be carried out in various locations such as gyms, school halls, or even outdoors with limited areas. Finally, because this test does not require advanced technology, special equipment, or electronic devices, it further increases the ease of its implementation. This simplicity that has been valid and tested allows the test to be carried out more flexibly and can be widely applied (Purba et al., 2024).

The results and discussion of this study show that the Modified Dynamic Balance Test developed for the Karate Kata category has a high level of validity and reliability in the various aspects tested. The absence of differences in answers from the questionnaire results between male and female athletes shows that this test is fair and can be applied to all athletes regardless of gender. It also supports the fact that athletes have received this innovation in the Modified Dynamic Balance Test, which is relevant for various groups of athletes. The limitation of this study is that specialist karate lecturers and coaches of kata are still very rare in Indonesia, of course it will affect the perspective as an expert in assessing. Then the testing conducted on athletes was done with one trial so that re-testing could not be done. Thus, these tests can be widely adopted in the training and Evaluation of athletes to improve their performance in competitions.

Conclusion

In conclusion, the Modified Dynamic Balance Test developed for the Karate Kata category has a high level of validity and reliability in the various aspects tested. The definition of balance used is based on the test's purpose, which is essential in Kata movements that require heightened body control. This test's good validity and reliability indicate that it fits the specific needs of the Kata category, including the ability to maintain stability and control during complex dynamic movements. The visual design and easy-to-understand test execution procedures support uniform and accurate execution, even by athletes and coaches with diverse backgrounds. The optimal inter-cone spacing helps to simulate the equilibrium conditions encountered in the Kata competition. This practical test does not require high costs, ample space, or advanced technology to be easily implemented in various locations. The absence of differences in the results between male and female athletes confirms that the test is fair and applicable to all athletes. The results of this study explain that this test has proven to be effective and reliable as a dynamic balance measuring tool specific to Karate Kata athletes, making it a valuable tool to improve performance in competition.

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ORIGINAL SCIENTIFIC PAPER

Motivation for Physical Education in Students Aged 12 to 15 – Structure Analysis

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Abstract

The quality of Physical education (PE) largely depends on the degree of students' motivation to participate in various kinesiological activities, whereby the level and structure of the motivational space are important. The aim of this research was to analyse the level and structure of student's motivation to perform tasks in PE classes, as well as gender differences. The research was conducted on 77 female and 61 male primary-school students from Split, Croatia, aged 12 to 15. Motivation assessment was carried out using the Motivation for PE questionnaire. Analysis confirmed that basic metric characteristics of all PE motivation measures are at a satisfactory level. In the total sample, measure of PE motivation is expressed as high, with intrinsic measures of motivation being significantly higher than extrinsic forms of student motivation. In general, no differences were found in measures of motivation according to the sex of the respondents. Differences in several measures of PE motivation were found between respondents of different grades, whereby students of 5th and 6th grades show higher results of PE motivation than students of 7th and 8th grades. Using cluster analysis, three different types of PE motivation were determined for both female and male students. For female student these clusters were defined as very low, high, and extremely high motivated types, while for male students clusters were defined as low, high, and extremely high motivated types. The structure and frequency of types of PE motivation is different for sub-samples of female and male students. Knowledge of the described structure will enable teachers to recognize and intervene in students' motivation, aiming to optimise work effects in kinesiological education.

Keywords: *elementary school students, extrinsic motivation, intrinsic motivation, motivational types, PE motivation*

Introduction

Nowadays, it is of essential importance to give answers to the demands placed on us by society and progress. The aforementioned aspects affect all levels of human life, including movement and exercise. Lack of movement greatly affects the health and psychological aspect of an individual. One such problem is childhood obesity, which represents a distinct problem (Androja et al., 2023). Physical education provides a controlled and guided educational process that fulfils the child's weekly need for movement. Complex interactions between teachers, students, and content within a larger social environment determine the quality of the teaching process (Bavčević, Babin, & Prskalo, 2006; Bavčević, Bavčević, & Androja, 2020). Kinesiological education adds to the complexity since physical education not only imparts new information and skills but also changes students' general anthropological standing. Because of the aforementioned methodology, the impacts of physical education are evident in a larger range of morphological, motor, functional, cognitive, and conative qualities of students, exhibiting both quantitative and qualitative changes (Bavčević, Babin, & Prskalo, 2006; Bavčević, Bavčević, & Androja, 2020; Bavčević, Prskalo, & Bavčević, 2018). Therefore, it is extremely important to determine as many factors as possible that influence the process of physical education as a key segment in the formation of students in growth and development. Growth and development affect numerous aspects related to the educational process, and many parameters change in relation to the age and gender of individuals (Bavčević, 2020; Bavčević, Androja, & Bilić, 2022; Bavčević, & Bavčević, 2015). Student motivation is an import-

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ant factor for the successful implementation of the educational process (de Bruijn, Mombarg, & Timmermans, 2022; Owen, Astell-Burt, & Lonsdale, 2013; Säfvenbom, Haugen, & Bulie, 2015). The degree of motivation also affects cross-curricular outcomes (de Bruijn et al., 2023) and represents an extremely important component in the education process. Among other things, the role of the teacher as the leader of the educational process is important in the degree of student motivation in the course itself (Domville, Watson, Richardson, & Graves, 2019; Navarro-Patón, Lago-Ballesteros, Basanta-Camiño, & Arufe-Giraldez, 2019; Su, Pu, Yadav, & Subramnaiyan, 2022). When we look at motivation as a key factor in the successful implementation of physical exercise, it is important to determine the dynamics of the observed parameter. Some of the previous research dealt with this issue (Chanal, Cheval, Courvoisier, & Paumier, 2019; Bogdan & Babičić, 2015; Burić & Zovko, 2022).

The aforementioned studies dealt with the motivation for sports or PE in different age groups. The joint findings of these studies show a decline in motivation concerning age, especially in the study by authors Chanal, Cheval, Courvoisier and Paumier (2019) who for the first time show a decline in primary school students' motivation for physical education. The findings of other authors provide a strong foundation for a deeper insight into the structure of motivation among students in the period from 12 to 15 years.

The aim of this paper was to examine the motivation structure of seventh and eighth grade elementary school students and the differences in the degree of motivation between female and male students of the specified age.

Methods

Sample of participants

The research was conducted on a sample of 138 subjects, average age of 13.51±0.87 years, of which 77 were female students and 61 were male students from the fifth up to eighth grades of the elementary school in the City of Split, Croatia. All subjects were clinically healthy with no recorded psychophysical aberrations. Participation in the research was voluntary with the regular consent of parents or custodians. The research is part of the research project "The integrative and developmental role of kinesiological education in the educational system – facing the challenges of modern schooling" approved by the Faculty Council of the Faculty of Kinesiology, University of Split (No.: 2181-205-02-01-23-0166, 15 December 2023). The study was conducted in accordance with the principles of the Declaration of Helsinki.

Data collection methods and procedures

Assessment of students' motivation for PE was performed using the adapted Sport Motivation Scale (SMS-28). The original SMS-28 questionnaire (Pelletier et al., 1995) was modified by selecting the items which allow students to assess motivation in relation to PE. In accordance with the recommendations of the authors of the modified questionnaire (Milavić, Milić, Jurko, Grgantov, & Marić, 2015), 20 items measuring 5 subscales were applied for the purposes of this research, as follows: Intrinsic Motivation to Know (IM – Know), Intrinsic Motivation to Accomplish Things (IM – Accomplish), Intrinsic Motivation – Stimulation (IM – Stimulation), Extrinsic Motivation – Identified (EM – Identified), and, Extrinsic Motivation – External Regulation (EM – Ext. Reg.). Total motivation for PE (Σ MOTIVATION) was determined as the mean value of all five variables. All items were evaluated on a Likert scale from 1 (I don't agree) to 5 (I fully agree). The variables were formed by the condensation of items using the item's mean determination method.

Data processing methods

Basic metric characteristics of all motivation scales were determined: homogeneity (by using Principal Component Analyses), reliability (Cronbach alpha coefficients were calculated), and sensitivity (by using more distribution goodness-of-fit indices). As part of the data processing, the following descriptive parameters were calculated: Mean, standard deviation (SD), minimum result (Min), maximum result (Max), Skewness (Skew), Kurtosis, (Kurt). All results of student motivation were interpreted according to the scale: extremely low (1.00-1.74), very low (1.75-2.24), low (2.25-2.74), moderate (2.75-3.25), high (3.26-3.75), very high (3.76-4.25), extremely high (4.26-5.00). The normality of the distribution was tested using the Kolmogorov-Smirnov goodness-of-fit test (K-S D test).

Differences between subsamples of students in the area of motivation were determined using Student t-test and significance level (p) were calculated.

Cluster analysis using the K-means method was applied with the aim of determining groups of students with different types of PE motivation. Descriptive statistics parameters were also calculated for each cluster: Mean and standard deviation (SD). Significance of differences between clusters was determined using One-way ANOVA [F-statistic (F) coefficient, and significance level (p)].

The software STATISTICA v.14.0.1.25 was used for data processing (TIBCO Software Inc, USA).

Results

By analysing the results of the sample of respondents in this study, it was determined that all motivation scales, including the overall measure of PE motivation, have satisfactory measurement characteristics. The scales are homogeneous because all items in each of the scales were projected onto one component, i.e. onto one latent dimension, whereby the percentage of variance explained by that component varies in the scales between 55.1% (extrinsic motivation - identified) to 76.1% (intrinsic motivation – to know). The component consisting of all 5 elements of the motivation measure explains 66.3% of the common variance in the total measure of PE motivation. The scales are at least satisfactorily reliable because the values of all reliability coefficients are above 0.70, with even 5 out of 6 motivation measures having a good level of reliability. Although it was determined for 4 measures of motivation that their distribution of results deviates significantly from the normal distribution (the significance of the K-S D test is $p<0.05$), it is still possible to assess that the sensitivity of all measures of motivation is satisfactory because the values of all other indicators of sensitivity are within acceptable limits. All measures of motivation have a maximum possible range of results (from 1.00 to 5.00). The skewness and kurtosis indices of the distribution of the results are within the range that George and Mallery (2003) stated as acceptable (range of ±2.00).

Descriptive parameters of motivation measures on the total sample vary from low for extrinsic motivation - identified (2.63) to very high for the measure of intrinsic motivation - to accomplish (3.86). The total measure of PE motivation, the

Table 1. Descriptive and metric parameters of PE motivation scales (n=138)

Variable	Mean	SD	Min	Max	Skew	Kurt	K-S D	D test p	Eigen value	Variance %	Cronbach Alpha
IM – Know	3.61	1.17	1.00	5.00	-0.67	-0.61	0.12	p<0.05	3.05	76.1	0.89
IM – Accomplish	3.86	1.10	1.00	5.00	-0.98	0.17	0.15	p<0.01	2.95	73.8	0.88
IM – Stimulation	3.64	1.13	1.00	5.00	-0.73	-0.42	0.15	p<0.01	2.83	70.8	0.86
EM – Identified	2.63	1.20	1.00	5.00	0.42	-0.92	0.13	p<0.05	2.53	63.2	0.81
EM – Ext. Reg.	3.12	1.01	1.00	5.00	-0.04	-0.89	0.09	p>0.20	2.22	55.5	0.73
Σ MOTIVATION	3.37	0.90	1.00	5.00	-0.42	-0.39	0.08	p>0.20	3.32	66.3	0.86

average of 5 elements - facet of motivation, is high (3.37). All three distributions of the results of measures of intrinsic motivation are negatively asymmetric (negative skewness), and the measure of extrinsic motivation has been identified as positively asymmetric (positive skewness).

Results of the Student t-test for independent samples (Table 2) conducted between subsamples of female and male students revealed a difference only in the measure of intrinsic motivation - stimulation, with male students (3.85) having more pronounced results than female students (3.47).

Table 2. Gender differences of PE motivation variables

Variable	Female (n=77)		Male (n=61)		t-test	p
	Mean	SD	Mean	SD		
IM – Know	3.66	1.14	3.56	1.21	0.49	0.62
IM – Accomplish	3.89	1.14	3.83	1.06	0.31	0.76
IM – Stimulation	3.47	1.20	3.85	1.02	1.98	0.05*
EM – Identified	2.62	1.19	2.65	1.21	0.12	0.91
EM – Ext. Reg.	3.04	0.96	3.22	1.08	1.05	0.30
Σ MOTIVATION	3.33	0.91	3.42	0.90	0.55	0.58

Note. *p≤0.05

The Student t-test for independent samples conducted between two subsamples of students belonging to different grades revealed a difference in 4 measures of PE motivation (Table 3): in the measures of intrinsic motivation – to know (p<0.001) and stimulation (p<0.05), in the measure of extrinsic motivation - external regulation (p<0.05) and consequently also in the overall measure of PE motivation (p<0.05). In all these measures of motivation younger students (5th and 6th grade elementary school students) have higher results than older students (7th and 8th grade elementary school students).

Although the findings on the absence of differences between respondents of different genders apparently allow the implementation of further statistical analyses on the entire sample of respondents of this study, the authors of this study decided to conduct a types analysis of students' PE motivation separately for male and female students due to a number of different reasons.

In the implementation of type analysis (K-means clustering), the authors chose a setting in which membership to a particular cluster is determined in such a way as to minimize the differences in the results of members within one cluster (intracluster differences) and to maximize the differences of the arithmetic means of different clusters (intercluster differences) independently about how many members each cluster will have.

Table 3. Age category differences of PE motivation variables

Variable	Grades 5 & 6 (n=73)		Grades 7 & 8 (n=65)		t-test	p
	Mean	SD	Mean	SD		
IM – Know	3.98	1.05	3.20	1.16	4.15	<0.001*
IM – Accomplish	3.92	0.98	3.79	1.22	0.69	0.49
IM – Stimulation	3.82	0.99	3.43	1.26	1.99	0.05*
EM – Identified	2.72	1.20	2.54	1.20	0.88	0.38
EM – Ext. Reg.	3.29	0.96	2.93	1.04	2.13	0.04*
Σ MOTIVATION	3.55	0.83	3.18	0.95	2.42	0.02*

Note. *p≤0.05, ***p<0.001.

Three types of PE motivation were determined for both female and male students (Table 4). According to the characteristics of the expression of motivation measures, the types of PE motivation for female students are named: very low motivated type (22.1% of female students), high motivated type (50.6%), and extremely high motivated type (27.3%).

According to the characteristics of the expression of motivation measures, the types of PE motivation for male students are named: low motivated type (27.8% of male students), high motivated type (37.7%), and extremely high motivated type (34.4%). All established types of PE motivation within the subsamples of female and male students differ significantly from each other in terms of motivation (all differences are at the p<0.001 level).

Table 4. Types of PE motivation among female and male students

Variable	Female students						F	p
	Very low motivated (n=17)		High motivated (n=39)		Extremely high motivated (n=21)			
	Mean	SD	Mean	SD	Mean	SD		
IM – Know	2.18	1.01	3.78	0.73	4.62	0.45	51.79	<0.001
IM – Accomplish	2.12	0.84	4.29	0.60	4.57	0.46	88.55	<0.001
IM – Stimulation	1.78	0.72	3.65	0.79	4.49	0.52	70.72	<0.001
EM – Identified	1.94	0.89	2.18	0.85	4.00	0.77	39.09	<0.001
EM – Ext. Reg.	1.88	0.45	3.03	0.70	3.99	0.58	53.90	<0.001
Σ MOTIVATION	1.98	0.49	3.39	0.35	4.33	0.34	180.42	<0.001
Variable	**Male students**						**F**	**p**
	Low motivated (n=17)		**High motivated (n=23)**		**Extremely high motivated (n=21)**			
	Mean	**SD**	**Mean**	**SD**	**Mean**	**SD**		
IM – Know	2.22	1.08	3.85	0.85	4.32	0.65	30.17	<0.001
IM – Accomplish	2.44	0.69	4.20	0.56	4.55	0.53	66.81	<0.001
IM – Stimulation	2.63	0.89	4.24	0.58	4.40	0.59	37.74	<0.001
EM – Identified	1.87	0.65	1.95	0.68	4.05	0.66	70.87	<0.001
EM – Ext. Reg.	2.09	0.64	3.04	0.66	4.33	0.48	67.44	<0.001
Σ MOTIVATION	2.25	0.39	3.45	0.33	4.33	0.41	144.97	<0.001

Note. F – One-way ANOVA coefficient; p – significance of the One-way ANOVA coefficient

Figure 1 shows the average values of PE motivation types for female and male students. The features of each of the different types are noticeable, as well as their comparative values on individual measures of PE motivation. The very low motivated type of PE motivation of female students differs from the low motivated type of PE motivation of male students in terms of expression and to some extent in terms of the structure of the expression of motivation measures. The two types of higher PE motivation of female students are very similar to the higher types of PE motivation of male students

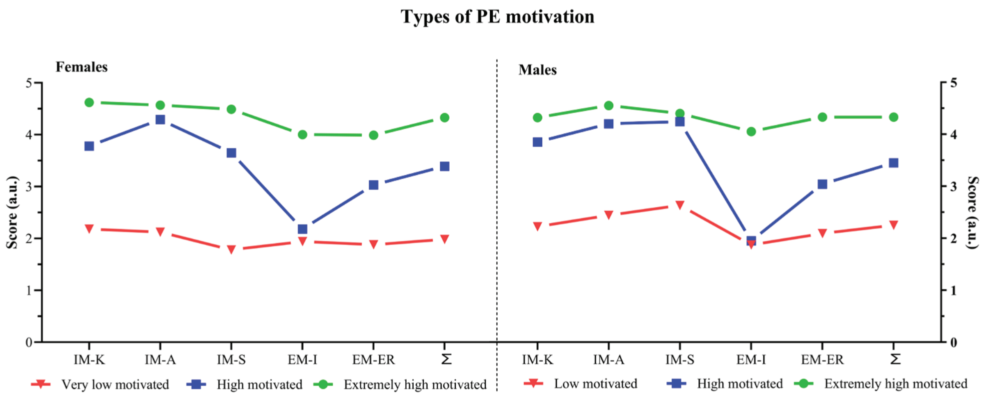

FIGURE 1. Types of PE motivation among female and male students.
Note. IM-K – Intrinsic Motivation to Know, IM-A – Intrinsic Motivation to Accomplish Things, IM-S – Intrinsic Motivation - Stimulation, EM-I – Extrinsic Motivation - Identified, EM-ER – Extrinsic Motivation - External Regulation, Σ – Total Motivation for PE.

Discussion

There are several major findings from this study. First of all, the metric value of all the scales of the PE motivation questionnaire has been confirmed when applied to this sample of respondents. In the total sample, measures of PE motivation vary between low to very high, and the overall measure of PE motivation is expressed as high. Also, intrinsic measures of motivation are significantly higher than extrinsic forms of student motivation. In general, no differences were found in measures of motivation according to the sex of the respondents (a difference was found in one single element - measure of PE motivation). Differences were found in several measures of PE motivation by belonging to different grades, whereby students of lower grades (5th and 6th grades) show higher results of PE motivation than students of higher grades (7th and 8th grades). Furthermore, three different types of PE motivation were determined for female students (named as very low, high, and extremely high motivated types) and for male students (named as low, high, and extremely high motivated types). The structure and frequency of types of PE motivation is different for sub-samples of female and male students. These findings require a more precise and detailed interpretation and will be further presented.

In this study, the PE motivation questionnaire was confirmed as a good instrument. The metric characteristics of all PE motivation measures used are at least at a satisfactory level. These findings are in accordance with the research of the authors Milavić et al. (2015). In the total sample, measures of PE motivation vary between low to very high, and the overall measure of PE motivation is expressed as high. Measures of intrinsic motivation are expressed as high (IM Know; IM Stimulation) or as very high (IM Accomplish). Measures of extrinsic motivation are expressed as low (EM Identified) or moderate (EM External regulation). The total measure of PE motivation is expressed as high. The aforementioned claims were confirmed in the research conducted in the study by Milavić et al. (2015). In the total sample, all three measures of intrinsic motivation are significantly higher than the two measures of students' extrinsic motivation. This shows that students are more intrinsically motivated for PE classes and that they do not expect special external rewards from these classes and PE teachers. These findings are in line with previous research by other authors (Bogdan & Babičić., 2015; Domville, Watson, Richardson, & Graves, 2019; Navarro-Patón, Lago-Ballesteros, Basanta-Camiño, & Arufe-Giraldez, 2019; Su, Pu, Yadav, & Subramnaiyan, 2022).

Additionally, no gender-based variations were observed in the motivation measures taken from the respondents, with the exception of one intrinsic motivation measure called "stimulation," where male student respondents demonstrated a higher degree of "stimulation" in physical education exercises. This phenomenon has been observed in previous research by different authors (de Bruijn, Mombarg, & Timmermans, 2022; Owen, Astell-Burt, & Lonsdale, 2013; Säfvenbom, Haugen, & Bulie, 2015). Differences were found in four of the six measures of PE motivation (IM Know, IM Stimulation, EM Ext. Reg. and total PE motivation) according to belonging to different grades, with students from lower grades (5th and 6th grades) showing higher PE results motivation from students of higher grades (7th and 8th grades). The most pronounced difference is in the IM Know variable, where the prevalence of this variable in the "lower" grades is very high, while in the "higher" grades it is expressed as moderate. These findings suggest that perhaps the transition between the 6th and 7th grade of elementary school is a "key" period for the decline in PE motivation. We cite some possible reasons for this "decline" in PE motivation: entering the 7th grade, students enter a period that is important for students to enrol in high-quality secondary schools, so they pay more attention to subjects important for later enrolment in secondary school, and less to PE. Also, in sports clubs, this period collides with the implementation of a more frequent number of selections for U-16 sports categories, so those who "drop out" of the clubs due to lower sports competences may lose interest in exercise in general (Eliasson & Johansson, 2021). Consequentially, it is possible for children to develop some new interests that at the same time contribute to the reduction of PE motivation, i.e. there is a loss of interest in exercise and in PE classes (Back, Johnson, Svedberg, McCall, & Ivarsson, 2022). The effect of decreasing motivation in a function of time was also observed in other studies (Bogdan & Babičić, 2015; Burić & Zovko, 2022).

Three different types of PE motivation were determined for female students (named as very low, high, and extremely high motivated types) and for male students (named as low, high, and extremely high motivated types). Three types of PE motivation were identified for female and male students. Female students had very low, high, and extremely high motivated types, while male students had low, high, and extremely high motivated types. The structure of these motivational types differed for sub-samples of female and male students, with female students having very low motivation and male students having slightly more PE expressed motivation. Extremely high and high motivational types were similar in their characteristics. The described findings are in accordance with the conclusions of the research of Milavić et al. (2015) in which the complex motivational structure associated with physical activity is also described. The frequency of PE motivation varied for sub-samples of female and male students. Female students had the most numerous type, which contained approximately half of the subsample, and two similar types (extremely high and very low motivated). Male students had two similar types, containing about one-third of the male subsample, and the low motivated type had a slightly lower number. In conclusion, there are similar motivational types in female and male students, but with different frequencies in their subsamples. The obtained findings indicate a complex motivational structure when it comes to gender differences, where the differences were identified only on a partial level. Previous research that has dealt with this topic also does not provide a definitive answer to the issue of gender differences in the field of motivation (Aelterman et al., 2012; Arsani, Maksum, SyamTuasikal, & Kusnanik, 2020; Lee, Fredenburg, Belcher, & Cleveland, 1999).

The general limitation of this research is the relatively small number of the sample which prevents the authors of this study from having a high scientific value of the established scientific findings and conclusions. In this study, a cross-sectional research design was applied, so it is not possible to qualitatively interpret and clarify the possible reasons for the established differences in PE motivation by age (by grade affiliation) of female and male students. The limit of the study is also in the impossibility of qualitatively "describing" the characteristics of each established individual type of PE motivation a sufficiently wide number of socio-demographic variables were not collected, nor variables related to engaging in organized kinesiology (sports) activities (either at school or in sports clubs), which would make it possible. Directions for future research stem from attempts to address the aforementioned limitations of this study. The most important improvement would be to repeat a similar study on a large sample and try to determine an even wider/larger number of types or subtypes (for example 4 types or even 5 types) of PE motivation. It would also be beneficial to use more socio-demographic, academic and kinesiology/sports variables for analysis and possible subsequent description of the characteristics of a particular type of PE motivation. Finally, conducting a longitudinal study on a larger sample of respondents would allow to determine more precisely the reasons for the reduction of motivation or for the lower PE motivation of students in the 7th and 8th grades of elementary school.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

The Effect of Neuromuscular Warm-up on Muscle Contractility of Elite Female Football Players

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Abstract

Neuromuscular warm-up positively affects motor abilities and muscle performance. The purpose of the study was to extend the knowledge about the effect of neuromuscular warm-up on the level of muscle contractility in female football players. The research sample consisted of experimental (EG, n=20) and control (CG, n=16) groups of female football players. The intervention lasted 12 weeks. The neuromuscular warm-up was implemented in the experimental group, while the control group performed a standard warm-up. Muscle contractility was assessed in dominant (DLE) and non-dominant (NDL) legs using Tensiomyograph (TMG-S2) (TMG-BMC Ltd., Ljubljana, Slovenia, 2011) with emphasis on m.biceps femoris, m.gastrocnemius medialis, m.gluteus maximus, m.vastus lateralis, and m.vastus medialis. Muscle contractility was assessed with an emphasis on contraction velocity (Vc). Wilcoxon test was used to determine significant differences. After the intervention, there was a significant improvement in the contraction velocity of m. vastus lateralis DLE (p=0.040, r=0.342) and m. vastus medialis NDL (p=0.048, r=0.330). In contrast to these findings, when the standard warm-up was applied, a significant improvement was observed in the contraction velocity of m. vastus medialis NDL (p=0.011, r=0.422). Additionally, there was a decrease in the contraction velocity of m. biceps femoris, m. gastrocnemius medialis, and m. vastus lateralis in both dominant and non-dominant legs, though the decrease was not significant. Based on the results, it is concluded that NMT warm-up significantly improved the contraction velocity in specific muscles (m. vastus lateralis DLE and m. vastus medialis NDL) in the experimental group. However, the effects on other muscles were not as pronounced, and the standard warm-up showed different results. Therefore, NMT warm-up has an impact on muscle contraction velocity, but the extent and consistency of this effect vary.

Keywords: *tensiomyography, musculoskeletal system, women´s football, contraction velocity*

Introduction

Football performance is a complex of high-level physical performance variables that depend on an individual´s health status, anthropometric and physiological properties and training (Di Salvo et al., 2007). Modern football is characterized by dynamic movements such as impulsive reactions, short and long sprints, and quick changes of direction (Dos'Santos et al., 2021). Neuromuscular abilities play a determinant role in supporting the increasingly high demands of football matches (Loturco et al., 2016).

There is a paradox in the fact that standard football preparation negatively influences a player´s neuromuscular development due to the interference effect between strength and endurance training (Loturco, Pereira, et al., 2015; Noon et al., 2015). Deficiencies in neuromuscular performance heighten the risk of injury (Lehr et al., 2017). Neuromuscular performance factors such as muscle strength in the thigh, trunk, and hip, as well as postural stability, are removable risk factors for lower limb injuries (Read et al., 2016). Injury of the anterior cruciate ligament (ACL) is the most common injury

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in football players and very often causes loss from competition. Commonly purported risk factors of ACL injury include muscular fatigue, decreased core strength, decreased relative hamstring strength, and change in muscle contraction of the hamstrings and quadriceps (Alentorn-Geli et al., 2009; Dai et al., 2012). Hamstring imbalance suggests that players have strength deficiencies that affect their performance and increase ACL injury risk (Ardern et al., 2015).

Repetitive activity consisting of high-velocity contraction movements such as sprints with changes in direction in conjunction with appropriate training methods can lead to positive neuromuscular adaptation (Rusu et al., 2013). Therefore, neuromuscular warm-up appears to be an appropriate means of injury prevention. Reduced technical-tactical training volume and appropriate neuromuscular stimuli can have a positive impact on strength abilities (Loturco, Nakamura et al., 2015; Ramírez-Campillo et al., 2016). Many studies have examined the impact of neuromuscular training on physical fitness (Dhawale, Yeole, & Kargutkar, 2020; Kowalczyk, Tomaszewski, Bartoszek, & Popieluch, 2019; Nunes, Cattuzzo, Faigenbaum, & Mortatti, 2021). The results showed that neuromuscular training positively affects balance, agility, speed, muscular strength, and muscular endurance, essential for a player´s performance (Akbar et al., 2022). Results from several studies have confirmed the efficacy of neuromuscular training to reduce the risk of lower limb injury in football players (Emery & Meeuwisse, 2010; Gilchrist et al., 2008).

However, it is unclear what is the effect of neuromuscular training on muscle contractility. There is also a lack of knowledge about muscle performance and muscle contractility properties in football players that could help prevent injuries related to asymmetry or muscle fatigue. Tensiomyography provides information about muscle stiffness, contraction velocity, and muscle fatigue and assesses the presence of peripheral and central fatigue without requiring additional voluntary effort (García-Manso et al., 2011; Rey et al., 2012). Therefore, we aimed to determine the effect of neuromuscular warm-up on the muscle contractility of lower limbs in elite female football players.

Methods

Participants

The research sample consisted of control (CG, n=16) and experimental (EG, n=20) groups. Twenty female football players from two clubs FC Spartak Myjava and ŠK Slovan Bratislava were assigned to an experimental group and sixteen female football players from 1. FC Tatran Prešov were assigned to a control group. The participation of female football players in the research was voluntary. All participants were informed beforehand about the aim of the study and the procedures to be undertaken and provided written informed consent. The somatic parameters of female football players are presented in Table 1.

Table 1. Description of the research sample

	n	Age (years)	Height (cm)	Weight (kg)	BMI (kg.m-2)	PBF (%)	FFM (kg)
EG	20	17.45±2.63	168.31±6.13	60.21±8.87	21.25±2.73	18.17±5.94	49.04±6.21
CG	16	16.24±1.09	163.84±5.58	55.91±6.87	20.80±2.10	21.89±5.30	43.48±4.63

Note. EG: experimental group; CG: control group; n: sample size; BMI: body mass index; PBF: percent body fat; FFM: fat-free mass; cm: centimeter; kg: kilogram. Data are Arithmetic mean ± Standard deviation.

All the above-mentioned football clubs play in the same league (Slovak Women's First League) comprising 10 teams. The training program was conducted during the season period and comprised 8 hours of training load and 2 hours of match load per week for both groups. The research was conducted in accordance with the Helsinki Declaration and approved by the University Ethics Committee, complying with the Ethics Committee of the University of Prešov, Prešov, Slovakia (Approval no. ECUP032022PO).

Procedures and measurement

Initial testing was carried out at the beginning of the in-season period. Body height was measured using a digital stadiometer SECA 217 (Seca GmbH & Co. KG, Hamburg, Germany). Body weight and body composition were determined by bioelectrical impedance analysis using InBody 270 (InBody Co., Ltd, Seoul, Korea).

The players´ contractile muscle properties were measured using a TMG-S2 (TMG-BMC Ltd., Ljubljana, Slovenia, 2011). Tensiomyography (TMG) is used to assess the mechanical and contractile parameters of skeletal muscles in response to electrical stimuli (Rusu et al., 2013). Various parameters can be obtained from the TMG such as contraction time (Tc), delay time (Td), and maximal radial displacement (Dm). Tc is associated with the speed of force generation, Td relates

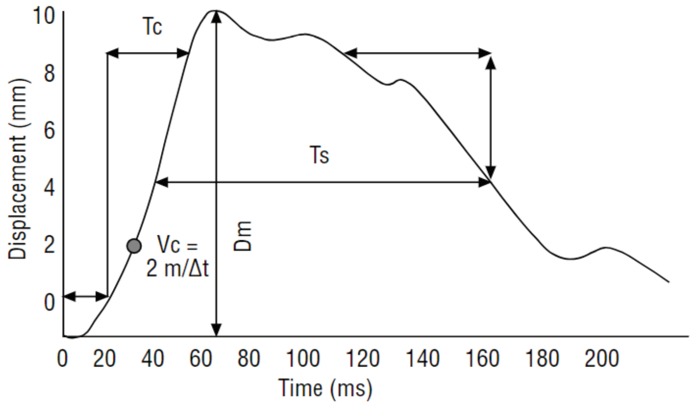

FIGURE 1. TMG parameters (Rodríguez-Matoso et al. 2012)

to the muscle fiber conduction velocity, and Dm reflects the stiffness of the muscle belly (Simunic et al., 2011). The TMG-derived measurements reveal contractile properties of individual muscles, but implementing the TMG outcomes in training sessions is difficult. A possible solution how to make the outcomes useful is to combine individual variables into a single index. In this regard, contraction velocity (Vc) can be determined by dividing maximal radial displacement by the sum of delay time and contraction time: Vc=Dm/(Td+Tc). This index provides a practical means of assessing muscle mechanical functionality. Contraction velocity can be useful for monitoring adverse effects of specific football training that could negatively impact maximum speed (Loturco et al., 2016).

Muscle contractility was assessed in dominant (DLE) and non-dominant (NDL) lower limbs with emphasis on m. biceps femoris, m. gastrocnemius medialis, m. gluteus maximus, m. vastus lateralis, and m. vastus medialis. The above-mentioned assessment procedure was repeated in the research sample after the application of a 12-week intervention program.

Intervention program

Neuromuscular warm-up (NMT warm-up) refers to a training program that includes both fundamental and sport-specific movements. The program includes a variety of activities such as resistance training, dynamic stability exercises, balance work, core strengthening, plyometrics, and agility drills (Davis et al., 2021). NMT warm-up was implemented in the experimental group for 12 weeks, twice a week (15 min each training session), from September 2023 to December 2023. The design of the intervention program was based on research by Hilska et al. (2021) who conducted research on the effectiveness of NMT warm-up intervention in the prevention of non-contact lower limb injuries.

Data analyses

The Shapiro-Wilk test was used to assess the normality of data distribution. Nonparametric methods were used because the data distribution assumptions of parametric tests were not met. Median (ME) as a measure of central tendency, and quartile deviation (QD) as a measure of variation were used. The Wilcoxon test for dependent samples was used to determine significant differences between groups with p<0.05 considered statistically significant. The "effect size" was used to assess the significance of the statistically tested differences, with effect sizes classified as small (0.1), medium (0.3), and large (0.5). Data analyses were conducted using STATISTICA software, version 13.5.0.17 (TIBCO Software Inc., Frankfurt am Main, Germany).

Results

A comprehensive description of the research sample, including descriptive statistics and the Vc values for both research groups, is presented in Table 2.

Table 2. Descriptive results of Vc value

		EG		CG	
M		**Pretest**	**Posttest**	**Pretest**	**Posttest**
BF	DLE	0.087±0.020	0.095±0.015	0.045±0.025	0.064±0.033
	NDL	0.094±0.018	0.091±0.015	0.049±0.023	0.073±0.026
GcM	DLE	0.067±0.011	0.062±0.011	0.048±0.012	0.057±0.016
	NDL	0.074±0.016	0.073±0.012	0.037±0.007	0.058±0.019
GM	DLE	0.106±0.024	0.097±0.021	0.132±0.018	0.118±0.038
	NDL	0.095±0.025	0.095±0.015	0.118±0.025	0.133±0.019
VL	DLE	0.097±0.014	0.086±0.018	0.095±0020	0.096±0.012
	NDL	0.103±0.019	0.103±0.010	0088±0.021	0.098±0.020
VM	DLE	0.153±0.027	0.149±0.018	0.144±0.015	0.124±0.021
	NDL	0.154±0.015	0.134±0.022	0.164±0.029	0.135±0.021

Note. m – muscle; BF – m.biceps femoris; GcM – m.gastrocnemius medialis; GM – m.gluteus maximus, VL – m.vastus lateralis; VM – m.vastus medialis; EG – experimental group; CG – control group; DLE – dominant lower limb; NDL – non-dominant lower limb. Data are Arithmetic mean ± Standard deviation.

The differences in contraction velocity between the control and experimental groups are presented in Table 3. After completion of the 12-week intervention, significant improvements were found in contraction velocity of m. vastus lateralis DLE (p=0.040, r=0.342) and m. vastus medialis NDL (p=0.048, r=0.330) in the experimental group. Contrary to these findings, applying the standard warm-up resulted in statistically significant improvements in contraction velocity of m. vastus medialis NDL (p=0.011, r=0.422) and a decrease in contraction velocity of m. gastrocnemius medialis, m. biceps femoris and m. vastus lateralis of dominant and non-dominant legs, although the decrease was not significant.

Table 4 presents differences between dominant and non-dominant lower limbs of participants. In experimental group, a statistically significant difference between dominant and non-dominant legs was observed in m. vastus medialis (p=0.001; r=0.647). After the intervention based on NMT warm-up, a significant difference was found between dominant and non-dominant legs in m. vastus medialis (p=0.005; r=0.467).

In control group, the initial results showed that there was a statistically significant difference between dominant and non-dominant legs in m. gastrocnemius medialis (p=0.044; r=0.336). After 12 weeks of standard warm-up, the difference between dominant and non-dominant legs in m. gastrocnemius medialis was also significant (p=0.034; r=0.353) (see Table 4).

Table 3. Result of comparison between input and output measurements

	EG					CG				
M	n	T	Z	p	r	n	T	Z	p	r
BF DLE	20	104.000	0.037	0.970	0.006	16	58.000	0.517	0.605	0.086
BF NDL	20	105.000	0.000	1.000	0.000	16	38.000	1.551	0.121	0.259
GcM DLE	20	99.000	0.224	0.823	0.037	16	43.000	1.293	0.196	0.215
GcM NDL	20	98.000	0.261	0.794	0.044	16	47.000	1.086	0.278	0.181
GM DLE	20	97.000	0.299	0.765	0.050	16	44.000	1.241	0.215	0.207
GM NDL	20	102.000	0.112	0.911	0.019	16	65.000	0.155	0.877	0.026
VL DLE	20	50.000	2.053	0.040*	0.342	16	51.000	0.879	0.379	0.147
VL NDL	20	101.000	0.149	0.881	0.025	16	65.000	0.155	0.877	0.026
VM DLE	20	81.000	0.896	0.370	0.149	16	42.000	1.344	0.179	0.224
VM NDL	20	52.000	1.979	0.048*	0.330	16	19.000	2.534	0.011*	0.422

Note. m – muscle; BF – m.biceps femoris; GcM – m.gastrocnemius medialis; GM – m.gluteus maximus, VL – m.vastus lateralis; VM – m.vastus medialis; EG – experimental group; CG – control group; DLE – dominant lower limb; NDL – non-dominant lower limb; n – sample size; T – value of Wilcoxon test; Z – standardized test statistic value; p – significance level; * - significance of p<0.05; r – effect size. Data are Arithmetic mean ± Standard deviation.

Table 4. Differences between dominant and non-dominant lower limbs

		EG					CG				
m		n	T	Z	p	r	n	T	Z	p	r
BF	Pr	20	86.000	0.709	0.478	0.118	16	66.000	0.103	0.918	0.017
	Po	20	102.000	0.112	0.911	0.019	16	40.000	1.448	0.148	0.241
GcM	Pr	20	88.000	0.635	0.526	0.106	16	29.000	2.017	0.044*	0.336
	Po	20	82.000	0.859	0.391	0.143	16	27.000	2.120	0.034*	0.353
GM	Pr	20	70.000	1.307	0.191	0.218	16	66.000	0.103	0.918	0.017
	Po	20	65.000	1.493	0.135	0,249	16	37.000	1.603	0.109	0.267
VL	Pr	20	94.000	0.411	0.681	0.068	16	60.000	0.414	0.679	0.069
	Po	20	40.000	2.427	0.015*	0.404	16	63.000	0.259	0.796	0.043
VM	Pr	20	1.000	3.883	0.001*	0.647	16	42.000	1.344	0.179	0.224
	Po	20	30.000	2.800	0.005*	0.467	16	62.000	0.310	0.756	0.052

Note. m – muscle; BF – m.biceps femoris; GcM – m.gastrocnemius medialis; GM – m.gluteus maximus, VL – m.vastus lateralis; VM – m.vastus medialis; EG – experimental group; CG – control group; DLE – dominant lower limb; NDL – non-dominant lower limb; Pr – pretest; Po – posttest; n – sample size; T – value of Wilcoxon test; Z – standardized test statistic value; p – significance level; * - significance of p<0.05; r – effect size. Data are Arithmetic mean ± Standard deviation.

Discussion

Our study revealed a decrease in muscle contraction velocity in both the dominant and non-dominant legs of female football players, which contrasts with previous studies that found no significant effect of limb laterality on muscle contraction (Gil et al., 2015; Loturco et al., 2016). This finding may suggest that lateral asymmetry can influence muscle contraction under specific conditions or in particular populations, warranting further investigation. Football players often prefer their dominant limb for activities during matches (Daneshjoo et al., 2013), which might lead to visible asymmetry in muscle contraction measurements. Our study supports the theory that bilateral asymmetry may predict injury in football players (Croisier et al., 2008) and that differences in neural responses may affect the loading of the musculoskeletal system (Brophy et al., 2010; Pedersen, Aksdal, & Stalsberg, 2019). This implies that asymmetry could have practical implications for performance and injury prevention.

Our results are at odds with the findings of Loturco et al. (2016), who reported that increases in contraction velocity (Vc) could be influenced by the type of training. De Paula Simola et al. (2016) observed an increase in Vc due to strength training, while Škof and Strojnik (2006) found that higher intensity in warm-ups enhances muscle contraction velocity. Our study indicates that there were no significant improvements in contraction velocity following the application of neuromuscular training (NMT) warm-ups, possibly due to the effects of endurance training on the training outcome.

The NMT warm-up was conducted during the competitive period of the football season, which may have influenced the athletes' performance and recovery due to the high demands and variability of competitive play. Additionally, the warm-up was performed only 2-3 times per week, which may not have been sufficient to fully capture the potential long-term effects of a more frequent or intensive NMT warm-up regimen. These factors could have impacted the effectiveness of the intervention and limited the ability to generalize the findings to different contexts or training schedules. Future studies should consider implementing a more consistent and frequent warm-up protocol and examining its effects in both pre-season and in-season phases to provide a more comprehensive understanding of its benefits and limitations.

Conclusion

The study found that a 12-week neuromuscular warm-up significantly improved contraction velocity in the m. vastus lateralis DLE and m. vastus medialis NDL in the experimental group, indicating enhanced muscle performance. In contrast, the standard warm-up in the control group improved m. vastus medialis NDL but also caused a non-significant decrease in contraction velocity in other muscles. Analysis revealed significant differences in m. vastus medialis between dominant and non-dominant limbs in the experimental group throughout the intervention, while significant differences in m. gastrocnemius medialis were observed in the control group. Based on the results, we conclude that NMT warm-up significantly improved the contraction velocity in specific muscles (m. vastus lateralis DLE and m. vastus medialis NDL) in the experimental group. However, the effects on other muscles were not as pronounced, and the standard warm-up showed different results. Therefore, NMT warm-up does have an impact on muscle contraction velocity, but the extent and consistency of this effect vary.

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Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial and financial relationships that could be construed as a potential conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Family Social Capital as a Predictor of Adolescent Health during the COVID-19 Pandemic

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Abstract

During the COVID-19 pandemic, family social capital may have played a key role in maintaining adolescent health, as they faced challenges such as social isolation and disruptions to their regular activities. Accordingly, the primary aim of this study was to examine the association between family social capital and adolescents' subjective health assessment during the COVID-19 pandemic. The study included 317 final-year high school students from the four largest cities in Croatia (Zagreb, Split, Rijeka, Osijek). The participants were aged between 17 and 20 years (18.24±0.58). Data were collected through survey questionnaires titled Social Capital and Adolescents' Subjective Well-Being During the COVID-19 Pandemic, which included demographic information, assessments of family social capital, and subjective health. Descriptive statistics, the Mann-Whitney U test, the Kruskal-Wallis test, and logistic regression analysis were used to analyze the data. The results of the study indicate that parental support and understanding, as well as quality family time, are key elements in maintaining and improving adolescents' subjective health assessments. These findings emphasize the importance of family social capital in preserving the health of young people during crisis situations. Moreover, these insights have significant implications for the design of interventions and programs aimed at strengthening family bonds and support as a means of improving youth health.

Keywords: *social capital, family, subjective health assessment, adolescents, COVID-19*

Introduction

Social capital represents resources available through social networks and relationships, and it is important for various aspects of human life, including health (Bourdieu, 1983; Coleman, 1990). The theoretical framework of social capital emphasizes interpersonal relationships, trust, and norms of reciprocity as key components of social capital (Putnam, 2000). Within the family, social capital can be particularly significant, as family support, shared activities, and emotional connectedness contribute to better physical and mental health of family members (Furuta et al., 2012).

The COVID-19 pandemic caused numerous changes in daily life, including restrictions on social interactions and physical activities, which may have impacted the well-being of young people (Breaux et al., 2023; Lulić Drenjak et al., 2023). In fact, it has been suggested that adolescents are more affected than adults by the social impact of the pandemic, and one of the most frequently endorsed challenges by parents during the pandemic includes the lack of social interaction for their adolescents (Roy et al., 2022; Unni, 2020). Accordingly, during the COVID-19 pandemic, family social capital may have played a key role in maintaining the health of adolescents, who faced challenges such as social isolation and disruptions to their usual activities (Bai, Jin, & Wan, 2020). Family social capital not only influences adolescent health but also affects their educational outcomes, social skills, and overall sense of well-being (Fraser & Aldrich, 2021; Novak et al., 2016). Parents and close family members serve as primary sources of support, providing emotional security and stability during a developmental period when adolescents are particularly sensitive to external stressors (Fraser & Aldrich, 2021). Understanding the role of family social capital in preserving adolescent health can help in shaping targeted public health measures and support programs (Novak, Doubova, & Kawachi, 2016; Novak, Suzuki, &

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Kawachi, 2015). In line with that, some previous research has established that the COVID-19 pandemic resulted in adolescents spending significantly more time with family members, and the findings indicated a significant improvement in the quality of family relationships and play between siblings, particularly for older adolescents (Campione-Barr, Rote, Killoren, & Rose, 2021; Gadassi Polack et al., 2021).

However, spending significantly more time with family members can provide both benefits and concerns. Therefore, the main research problem was to examine how different aspects of family social capital influence adolescents' subjective health assessment during the COVID-19 pandemic. Specifically, the study focused on understanding the role of family support in maintaining and improving adolescent health in conditions of heightened stress and social isolation caused by the COVID-19 pandemic. The aim was to determine the extent to which family social capital can act as a protective factor against the negative impacts of the COVID-19 pandemic on adolescent health. In line with the stated aim, the hypothesis was that adolescents with higher levels of perceived family support would have a better subjective health assessment, and that shared family activities would be positively associated with better subjective health assessments among adolescents.

Methods

Participants

The sample included 317 adolescents (M=212; F=105) who attended the final year of secondary school from the four largest cities in the Republic of Croatia: Zagreb (n=82), Split (n=80), Rijeka (n=65), and Osijek (n=90). Participants were aged between 17 and 20 years (18.24±0.58). The research was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of the Faculty of Kinesiology, University of Zagreb (number: 33./2021). All participants were informed about the protocol and purpose of the research prior to its commencement and provided written consent to participate. The complete testing protocol was explained to them in detail, with special emphasis on the fact that the research did not require any additional physical effort beyond their regular activities.

Measurements Instruments

Data were collected using a questionnaire titled "Social Capital and Adolescents' Subjective Well-Being During the COVID-19 Pandemic", which was carefully designed to cover all relevant aspects of the research. In the first part of the questionnaire, participants provided demographic information, subjective health assessments, and the educational level of both parents. The measure of subjective health assessment has been frequently used in previous studies on social capital and health, demonstrating good reliability and validity (Hu, Yang, & Luo, 2017; Jiang & Kang, 2019). The second part of the questionnaire assessed family social capital using a social capital questionnaire that also has satisfactory metric characteristics (Carrillo-Álvarez, Villalonga-Olives, Riera-Romaní, & Kawachi, 2019; Novak et al., 2016; Novak et al., 2015).

Variables and Protocol

Before beginning the questionnaire, each participant was given a detailed explanation of the instructions and the completion process, with the opportunity to ask questions if anything was unclear. For the collection of demographic data about the study participants, the following variables were used: age, gender, parents' educational level, the city from which the participants come, and the educational program of secondary school. Participants rated their own health on a Likert scale from 1 (poor health) to 5 (excellent health). The family social capital questionnaire (FSC) consisted of six questions designed to assess family social capital. Responses to the questions in the family social capital segment evaluated the frequency of certain actions addressed by the questions.

Statistical Analysis

The obtained data were processed in the program Statistica 14.0.1.25 (TIBCO Software, Inc.) for the Windows operating system and in Microsoft Excel 2016 (Palo Alto, CA, USA). Basic descriptive parameters (mean and standard deviation) were used to describe each variable. The normality of the distribution was tested with the Kolmogorov-Smirnov test. The relationship between the variable "self-assessed health" and the set of family social capital variables (FSC-1-6) was examined using a sigmoid function in logistic regression analysis. In this analysis, the values of the Omnibus test, Hosmer-Lemeshow test, Cox and Snell R^2, and Nagelkerke R^2 were calculated. The level of statistical significance was set at $p<0.05$.

Results

Table 1 shows the basic statistical parameters for the set of social capital variables. Table 2 presents the coefficients from the logistic regression analysis, while Table 3 displays the coefficients for all three sigmoid functions of the regression equation. The results indicate that family support is significantly positively associated with better subjective health assessments among adolescents.

Table 1. Basic Descriptive Parameters of the Variables for Social Capital

Variables	$\bar{x}$	SD
FSC-1 (Do your parents provide you with support and understanding during your secondary education?)	4.50	0.78
FSC-2 (Do you spend your free time hanging out with your family members?)	3.63	0.95
FSC-3 (Does your family help you with your schoolwork?)	2.60	1.29
FSC-4 (Do you participate in household chores, e.g. washing dishes, vacuuming?)	3.71	1.12
FSC-5 (Do you talk to your parents about what happened during the day at school?)	3.67	1.08
FSC-6 (Are you informed when something important happens in your extended family, e.g. birth, wedding, death?)	4.63	0.74

Note. $\bar{x}$ - mean; SD – standard deviation; FSC – family social capital

Table 2. Coefficients of Logistic Regression Analysis

Variables	Omnibus test of model coefficients (df = 6)	Hosmer and Lemeshow test (df = 8)	Pseudo R2 (Cox and Snell test)	Pseudo R2 (Nagelkerke test)
HE – FSC-1-6	p = 0.00*	p = 0.80	R2 = 0.08	R2 = 0.12

Note. HE – health; FSC – family social capital; df - degrees of freedom; p - significance of the variable; * — statistically significant; R2 - proportion of explained variance.

Table 3. Coefficients of Sigmoid Functions of the Regression Equation

Variables	B	SE	p
FSC-1 (Do your parents provide you with support and understanding during your secondary education?)	0.44	0.20	0.02*
FSC-2 (Do you spend your free time hanging out with your family members?)	0.45	0.19	0.02*
FSC-3 (Does your family help you with your schoolwork?)	-0.10	0.13	0.44
FSC-4 (Do you participate in household chores, e.g. washing dishes, vacuuming?)	-0.08	0.14	0.56
FSC-5 (Do you talk to your parents about what happened during the day at school?)	-0.23	0.16	0.15
FSC-6 (Are you informed when something important happens in your extended family, e.g. birth, wedding, death?)	0.27	0.19	0.15

Note. FSC – family social capital; B – coefficient; SE – standard error; p - significance of the variable; * — statistically significant

Discussion and Conclusion

The analysis of the research results indicates that parental support and understanding, along with quality family time, are key elements in maintaining and improving adolescents' subjective health assessments. Parental support during the COVID-19 pandemic provided emotional stability and a sense of security, which positively impacted the overall health of young people during a time of significant stress and uncertainty. At the same time, shared family activities during this crisis period strengthened family bonds and provided a sense of belonging, which further contributed to a better perception of health. In line with this, it can be concluded that the results of this study clearly point to the important role of social capital within the family, particularly in the context of parental support and shared family activities, in adolescents' subjective health assessments. This also confirms the hypotheses that adolescents with a higher level of perceived parental support will have a better subjective health assessment and that shared family activities will also be positively associated with better subjective health assessments among adolescents.

Also, these results support the findings of previous research, which have shown that family support and healthy family relationships play a crucial role in the psychological and physical health of adolescents (Tuominen & Haanpää, 2022; Zhu, Zhuang & Ip, 2021). Additionally, research confirms that crisis situations such as the COVID-19 pandemic can further highlight the importance of strong family bonds in maintaining the health of children and young people (Bian, 2020; Novak et al., 2018; Zoellner & Maercker, 2006). The results also reinforced existing knowledge by emphasizing that family social capital could play a key role in maintaining the health of adolescents, who faced challenges such as social isolation and the disruption of regular activities (Bai et al., 2020; Fraser & Aldrich, 2021).

In conclusion, this study highlights the crucial role that family social capital plays in adolescents' subjective health assessments. These findings have significant implications for the design of interventions and programs aimed at strengthening family bonds and support to improve youth health. However, this study has certain limitations. First, the sample is restricted to a specific geographic location, which may limit the generalizability of the findings to broader populations. Second, the use of questionnaire surveys may introduce socially desirable response bias, potentially affecting the accuracy of the results. Despite these limitations, the findings provide valuable insights into the role of social capital, not only within families but also in schools, in promoting physical activity and overall well-being. Future research should prioritize longitudinal studies to track changes in social capital and physical activity over time. Furthermore, incorporating qualitative methods would offer a deeper understanding of the nuanced relationship between social capital and adolescent health. Exploring interventions aimed at enhancing social capital within diverse school and family environments could yield practical strategies for improving adolescents' physical activity levels, with broader implications for educational, social, and public health policies. These strategies could be especially relevant in post-crisis settings, where rebuilding social networks is essential for youth development.

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Conflict of Interest
The authors report no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Dietary Supplementation in Competitive Swimming: Analysis of Prevalence and Gender-Differences

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Abstract

Dietary supplementation (DS) is an important factor in contemporary sports, but studies rarely have examined DS practices in high-level swimming. The aim of this study was to investigate the practice of DS in professional swimming, emphasizing the differences between male and female swimmers. The participants were 301 swimmers from Slovenia (148 females; 16.7±2.1 years of age; 9.2±3.1 years of experience in swimming sports), comprising the population of swimmers from Slovenia who competed at the highest level of national competition. Variables were collected via a previously validated questionnaire tool and included DS factors (i.e., knowledge of DS, boundaries, sources of information) and specific usages of the different supplements. The chi-square test and Mann-Whitney test (MW) were used to evaluate differences between genders. Only 30% of the studied swimmers reported not using the DS. Isotonics and vitamins/minerals are mostly used (19% and 18% of regular users, respectively). Males use DS more often than females do (MW=2.90, p=0.03), with a higher consumption of carbohydrates and amino acids in males (MW=2.53 and 2.14, respectively, p<0.05). More than one-third of the swimmers were self-educated at DS, and more than 70% of them declared self-procurement at DS. The results revealed the expected figures of DS in swimming but also highlighted the necessity for objective evaluation of the knowledge of DS in swimming athletes. Further studies are needed to clarify the templates of DS in different age groups.

Keywords: *nutrition, nutritional supplements, athletes, sex-differences*

Introduction

Proper nutrition can enhance training adaptations, improve recovery, and optimize competitive performance; therefore, it is generally accepted that nutrition plays a critical role in sports performance, including swimming (Debnath, Chatterjee, Bandyopadhyay, Datta, & Dey, 2019; Shaw, Boyd, Burke, & Koivisto, 2014). More specifically, most competitive sports, particularly swimming, demand high energy expenditures; therefore, adequate intake of carbohydrates, the primary fuel source for muscles, ensures optimal energy levels during training and competition. Next, intense training causes muscle breakdown, while sufficient protein intake provides the building blocks (amino acids) for muscle repair and growth, leading to increased strength and power. Proper nutrition post-exercise speeds up recovery by replenishing glycogen stores (stored carbohydrates) and facilitating muscle repair, which allows for faster adaptation to training stress and improved performance over time. Maintaining proper fluid balance is crucial for optimal performance and preventing dehydration. Intense training can temporarily suppress the immune system, while adequate intake of vitamins, minerals, and antioxidants supports immune function, reducing the risk of illness and missed training days (Shaw et al., 2014). Nutrition plays a key role in achieving and maintaining optimal body composition, and in

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swimming, this role is particularly important since it includes a balance of muscle mass and a low body fat percentage to enhance performance and reduce drag in the water. Finally, proper nutrition supports cognitive function, improving focus, reaction time, and decision-making during training and competition, which are all important determinants of swimming performance (Royall, 2016).

Dietary supplementation (nutritional supplementation) can play a role in addressing potential gaps in an athlete's nutrition that may arise from inadequate dietary intake or increased nutrient demands due to intense training (AbuMoh'd, Obeidat, & Alsababha, 2021; Rodek, Sekulic, & Kondric, 2012). Although dietary supplementation (DS) should not be seen as a replacement for a well-balanced diet but rather as a complementary tool to optimize performance and support overall health, contemporary swimming training frequently places great overall stress on athletes' bodies; therefore, DS is frequently found to be highly important and necessary (Alfieri, D'Angelo, & Mazzeo, 2023). For example, even with a carefully planned diet, swimmers may fall short on certain vitamins and minerals owing to factors such as limited food variety, poor absorption, or increased needs from training. Therefore, DS can help bridge these gaps and ensure adequate micronutrient levels for optimal health and performance. Swimmers engaging in intense training require a significant amount of energy, and if dietary intake is inadequate, DS with carbohydrates and protein-based ingredients can help meet these increased needs and support optimal recovery (Dhiman & Kapri, 2023). Certain supplements (i.e., creatine, beta-alanine, and caffeine) have been shown to enhance specific aspects of swimming performance, such as power, endurance, and mental focus, and these supplements can be considered if dietary intake alone is not sufficient to achieve performance goals. Finally, swimmers with specific dietary restrictions, such as vegans or those with food allergies, may require DS to ensure adequate intake of nutrients that are challenging to obtain through their diet (Burke, 2007; Correa, 2014).

Not surprisingly, studies have investigated the problem of nutrition and DS in swimming, and reference studies can be clustered into several groups. The first group of investigations examined the nutritional needs of swimmers. In brief, swimmers require high carbohydrate intake (6–12 g/kg/day) to support training demands, whereas protein intake of 2 g/kg/day is recommended, with an emphasis on postexercise consumption (Dominguez et al., 2017; Shaw et al., 2014). Another group of studies examined the effectiveness and appropriateness of DS. In brief, various supplements, including caffeine, creatine, sodium bicarbonate, beta-alanine, and whey protein, have shown potential benefits, while creatine supplementation may improve repeated interval swim performance and power development (Leenders, Lamb, & Nelson, 1999; Mero et al., 2013; Norberto et al., 2020; Peyrebrune, Nevill, Donaldson, & Cosford, 1998). Adequate intake of micronutrients, particularly iron, zinc, and vitamins A, D, E, B6, and B12, is essential for maintaining health and performance, and DS is found to be a convenient way to supplement the lack of micronutrients due to the limited bioavailability of these compounds in a regular diet (Pyne, Verhagen, & Mountjoy, 2014). Furthermore, some studies have investigated problems related to the nutrition of swimmers and have reported inadequate calcium intake and potential risks of eating disorders among swimmers, which clearly support the need for DS to prevent risks related to a lack of this mineral (Paschoal & Amancio, 2004).

While there is no doubt that nutrition is a cornerstone of success in swimming, DS is considered an important factor in reaching optimal nutrient levels, especially in high-level competitive swimming. However, studies in which DS practices were directly examined in high-level swimmers from different teams are lacking. Therefore, this study aimed to investigate the practice of DS in professional swimming, emphasizing the differences between male and female swimmers. It was hypothesized that DS would be more prevalent in male swimmers than in their female peers.

Methods

Participants

The sample in this study included 301 swimmers from Slovenia (148 females; 16.7±2.1 years of age; 9.2±3.1 years of experience in swimming sports). All participants were active swimmers and competed at the highest level of national competition (National Championship). An invitation to participate in the study was sent by the national swimming federation, and none of the athletes refused to participate; therefore, all swimmers who participated in the championship were included, and the tested sample represented the entire population of competitive swimmers from the country. For underage participants, one parent/guardian signed informed consent for study participation, whereas adult swimmers signed consent for their own participation. The study was originally initiated and approved by the national swimming federation, complied with all ethical guidelines and received approval from the Institutional Ethics Review Board of the Faculty of Sport, University of Ljubljana, Slovenia (EBO 10/09/2014-1).

Variables

Variables were collected via a previously validated questionnaire tool (Kondric, Sekulic, Uljevic, Gabrilo, & Zvan, 2013; Rodek et al., 2012). The questionnaire consisted of queries about the following: the swimmers' self-knowledge about DS (self-assessed on a five-point scale ranging from "I have no knowledge at all" to "Excellent"); the primary source of trusted information about DS (possible answers included "I have no knowledge", "Coach, physician", "Formal education", and "Self-education"); and the way of procurement/purchasing of DS (with possible answers: "Medical team", "Coach", "Personally"). Specific DS usage was reported in response to one main question (e.g., "Do you use DS?", and possible responses were "Yes, regularly", "From time to time", and "No, I don't use it") and separate responses regarding the consumption of vitamins and minerals, carbohydrates, proteins, isotonics, recovery supplements, energy bars, and other supplements. For all supplements, four-point scales were offered ("No", "Sporadically", "Often", "Regularly"). Those who did not and/or only sporadically consumed supplements were asked why he/she did not use DS (the answer options were "I don't think it will be useful; I have a proper diet"; "I don't have sufficient knowledge to use DS", "DS is expensive", "I don't think DS is healthy").

Statistics

Descriptive statistics for swimmers' age and experience in competitive swimming included calculations of means and standard deviations. For the remaining study variables, counts (frequencies) and proportions were calculated. Differences between genders for ordinal variables were established via the nonparametric Mann-Whitney test, whereas the chi-square test was applied to identify differences between genders for nominal variables.

A statistical significance level of 95% ($p<0.05$) was applied. Statistical analyses were performed via Statistica (Tibco Inc.) Version 13.5.

Results

Descriptive statistics and differences between genders for DS factors are presented in Table 1. In general, DS usage is relatively common, with only 30% of the studied swimmers declaring no usage of the DS. Males use DS more often than females do (MW=2.90, p=0.03). A relatively small percentage of the participants reported a lack of knowledge of DS, with <4% reporting a total lack of knowledge and <10% reporting poor knowledge of DS. More than one-third of the swimmers were self-educated at DS, and more than 70% of them declared self-procurement at DS.

Table 1. Descriptive statistics (F – frequencies, % - percentages) and differences between genders (MW – Mann—Whitney test, Chi^2—Chi-square test) in dietary-supplementation factors for Slovenian swimmers

	Males		Females		MW/Chi2 (p)	
	F	%	F	%		
Knowledge on dietary supplementation MW					0.56	(0.57)
Very poor	2	1.35	6	3.92		
Poor	12	8.11	16	10.46		
Average	80	54.05	64	41.83		
Good	48	32.43	59	38.56		
Excellent	6	4.05	8	5.23		
Missing	0	0.00	0	0.00		
Main source of information on dietary supplementation Chi					0.35	(0.95)
I have no knowledge	21	14.19	24	15.70		
Coach/physician	33	22.30	33	21.57		
Formal education	38	25.68	42	27.45		
Self-education	56	37.84	54	35.29		
Missing	0	0.00	0	0.00		
Dietary supplement usage MW					2.90	(0.03)
Yes, regularly	34	22.97	13	8.50		
Yes, from time to time	68	45.95	77	50.33		
No, I don't use it	46	31.08	63	41.18		
Missing	0	0.00	0	0.00		
Procurement of dietary supplements Chi					0.32	(0.85)
Medical team	4	2.70	3	1.96		
Coach	11	7.43	8	5.23		
Personally	115	77.70	107	69.93		
Missing	18	12.16	35	22.88		
Limitation/boundary on usage of dietary supplements Chi					7.79	(0.05)
I don't think it will be useful, my nutrition is good	38	25.68	47	30.72		
Lack of knowledge.	25	16.89	37	24.18		
Price	26	17.57	15	9.80		
Health hazard	14	9.46	8	5.23		
Missing/have no boundaries	45	30.41	46	30.07		

Legend: MW – differences calculated via the Mann—Whitney test; Chi – differences calculated via the chi—square test

The overall consumption of the DS is presented in Figure 1.

The specific DS used and the differences between genders are presented in Table 2. Isotonic/electrolyte drinks are the most commonly used DS, with >25% of swimmers using them regularly and an additional 10% using them from time to time. The second most commonly used DS is vitamins/minerals (24% of males and 13% of females who are regular users). Iron supplementation is more common in women (there is no significant difference between genders), with more than 16% of females and <5% of males using it regularly. Significant differences between the genders were detected for carbohydrate supplementation (MW=2.53, p=0.01) and amino acid supplementation (MW=2.14, p=0.03). In both cases, males used it more often than females did.

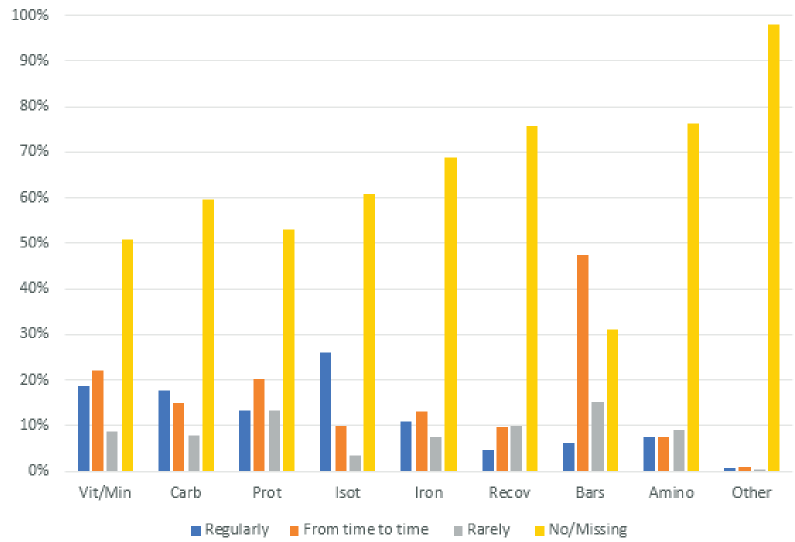

FIGURE 1. Prevalence of dietary supplementation in Slovenian swimming

Table 2. Descriptive statistics (F – frequencies, % - percentages) and differences between genders (MW – Mann–Whitney test) in dietary supplement usage for Slovenian swimmers

	Males		Females		MW (p)	
	F	%	F	%		
Vitamins/Minerals					0.21	(0.83)
Regularly	36	24.32	20	13.07		
From time to time	29	19.59	37	24.18		
Rarely	12	8.11	14	9.15		
No/Missing	71	47.97	82	53.59		
Carbohydrates					2.53	(0.01)
Regularly	33	22.30	20	13.07		
From time to time	29	19.59	16	10.46		
Rarely	10	6.76	13	8.50		
No/Missing	76	51.35	104	67.97		
Proteins					1.51	(0.13)
Regularly	30	20.27	10	6.54		
From time to time	35	23.65	26	16.99		
Rarely	16	10.81	24	15.69		
No/Missing	67	45.27	93	60.78		
Isotonics/Electrolyte drinks					1.90	(0.06)
Regularly	52	35.14	26	16.99		
From time to time	14	9.46	16	10.46		
Rarely	3	2.03	7	4.58		
No/Missing	79	53.38	104	67.97		
Iron based supplements					0.80	(0.41)
Regularly	7	4.73	26	16.99		
From time to time	23	15.54	16	10.46		
Rarely	15	10.14	7	4.58		
No/Missing	103	69.59	104	67.97		
Recovery supplements					1.47	(0.13)
Regularly	14	9.46				
From time to time	15	10.14	14	9.15		

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Table 2. Descriptive statistics (F – frequencies, % - percentages) and differences between genders (MW – Mann–Whitney test) in dietary supplement usage for Slovenian swimmers

	Males		Females		MW (p)	
	F	%	F	%		
Rarely	14	9.46	16	10.46		
No/Missing	105	70.95	123	80.39		
Energy bars					0.19	(0.84)
Regularly	11	7.43	8	5.23		
From time to time	70	47.30	73	47.71		
Rarely	23	15.54	23	15.03		
No/Missing	44	29.73	49	32.03		
Amino-acid supplementation					2.14	(0.03)
Regularly	17	11.49	5	3.27		
From time to time	14	9.46	8	5.23		
Rarely	13	8.78	14	9.15		
No/Missing	104	70.27	126	82.35		
Other					0.04	(0.96)
Regularly	1	0.68	1	0.65		
From time to time	1	0.68	2	1.31		
Rarely	1	0.68				
No/Missing	145	97.97	150	98.04		

Discussion

With respect to the study aims, the results revealed two important findings. First, male swimmers consume DS more often than females do. Second, specific differences between genders are evident in the consumption of protein-based supplements (e.g., amino acids) and carbohydrates. Therefore, the initial study hypothesis can be accepted. Before these findings are discussed, the most important results concerning the prevalence of DS consumption in swimmers are reviewed.

The prevalence of DS in the studied swimmers was relatively high, with only 30% of the studied participants reporting no consumption. However, these results are expected, given that previous studies reported even higher consumption of DS in athletes. Studies suggest that 35% to 100% of athletes report using dietary supplements, with reasons ranging from performance enhancement to recovery support and general health maintenance (Baltazar-Martins et al., 2019; Garthe & Maughan, 2018). Research specifically on swimmers aligns with these general trends. One study on competitive swimmers reported that 79.5% used supplements, with higher usage among national-level athletes (Jimenez-Alfageme et al., 2022). Another investigation into a high-performance swimming club revealed that the majority used supplements, often consuming more than those in development groups did (Newbury, Sparks, Cole, Kelly, & Gough, 2023). However, the results of relatively high consumption of DS are interesting if we consider that the study included not only adult (e.g., senior-level) competitors but also youth categories (please see Methods for details). Despite the background, which is gender- and age specific, the relatively high prevalence of DS in swimmers is relatively understandable.

As discussed in the introduction, DS in swimming are aimed primarily at enhancing performance, supporting recovery, and addressing specific nutritional needs that may arise from the intense training demands of the sport (Alfieri et al., 2023; Burke, 2007; Pyne et al., 2014). Indeed, swimming training and competitions are highly exhaustive. Although the usual volume of training in swimming varies depending on the swimmer's level, age, and goal, competitive swimmers typically train between 5,000 and 10,000 meters per day. This translates to approximately 2–4 hours in the pool per session, often spread across two daily workouts. In addition, high-level swimmers regularly perform additional training in gyms of 4–5 hours per week, which additionally increases the overall volume of training. Therefore, the relatively high prevalence of DS in the studied swimmers could be considered a logical consequence of the high energetic demands of training.

The fact that the majority of swimmers purchase DS individually points to the fact that most of them are actually not properly monitored with regard to the consumption of DS. In other words, even if swimmers are advised to consume specific supplements, individual purchases increase the risk of improper consumption and dosing. This is a particularly dangerous practice because of the two most important issues. First, studies repeatedly highlighted the problem of improper consumption of supplements, which can lead to various health problems, including (i) overdose and toxicity (consuming excessive amounts of certain supplements can lead to toxicity), (ii) nutrient imbalances (taking high doses of one nutrient can interfere with the absorption or utilization of others), (iii) drug interactions (some supplements can interact with prescription or over-the-counter medications, altering their effectiveness or increasing the risk of side effects), and (iv) allergic reactions (certain ingredients in supplements can trigger allergic reactions in susceptible individuals) (Palmer et al., 2003)

Apart from those health-related side effects, many supplements are marketed with exaggerated or misleading claims, creating unrealistic expectations about their benefits, which can lead to disappointment and wasted money. Second, and important is the contamination of supplements with doping agents. In brief, supplements can be contaminated with banned substances (i.e., doping substances). Such contamination poses a serious risk in sports, impacting both athletes' careers and their health. Unfortunately, this issue is prevalent globally and well documented even in the scientific literature (Van Thuyne, Van Eenoo, & Delbeke, 2006). For that reason, monitoring DS in athletes should be highly prioritized.

The consumption of DS is greater in males than in females, and this finding is generally in agreement with reports from other sports (Aguilar-Navarro et al., 2021). Although there are considerable differences in the type of DS used between genders, the generally higher prevalence of DS in males will be briefly discussed, with several potential reasons for such differences. First, regardless of the type of sport athletes are participating in, societal pressures and cultural norms often emphasize muscle mass and strength for men. This can lead to greater pressure on male athletes to achieve a certain physique, leading them to seek out supplements that promote muscle growth or enhanced performance. Second, men generally have greater caloric needs and may require more nutrients, which are difficult to obtain throughout a regular diet. This can lead to a greater reliance on DS to meet these nutritional demands.

Additionally, DS marketing often targets men, focusing on messages of strength, power, and masculinity (Noonan & Patrick Noonan, 2006). This can lead to the perception that DS are primarily for men and may influence their usage patterns. Finally, men may have greater exposure to information about DS through sports media, fitness communities, and peer groups, which can increase their awareness and willingness to try these products. Together, these findings are definitively associated not only with a higher prevalence of DS but also with the types of DS in male and female swimmers.

The most evident difference between genders in DS is the consumption of carbohydrates and proteins (amino acids), with a greater prevalence in men. This finding actually supports previously specified explanations for the higher prevalence of DS in males. However, the background of a specific DS is also interesting. With regard to carbohydrates, the following explanations are possible. First, men typically have a larger body size and muscle mass than women do, leading to a higher metabolic rate and greater energy expenditure during exercise (McMurray, Soares, Caspersen, & McCurdy, 2014). Together, these factors often necessitate increased carbohydrate intake to fuel training and performance. Compared with their female counterparts, male swimmers often engage in training programs with greater volume and intensity, and this increased training load further increases their energy requirements, making greater carbohydrate intake crucial for optimal performance and recovery. Finally, men have greater muscle mass and consequently have a greater capacity to store glycogen, requiring greater carbohydrate intake to replenish these stores after intense training sessions.

While both male and female swimmers need adequate quantities of proteins and amino acids for muscle repair and recovery, a few factors might contribute to increased protein-based DS among male swimmers. First, men generally have a greater percentage of muscle mass than women do (Malina, 2007). The pursuit of increased muscle mass and strength, which is often emphasized in male-dominated sports culture, may lead male swimmers to consume more protein and utilize supplements to support those goals. Furthermore, traditional dietary patterns may lead men to consume more protein-rich foods, such as meat and dairy, than women do, which creates a certain "culture" of protein-based nutrition in male athletes. Male swimmers are regularly engaged in more strength-focused training alongside their swimming routine. Therefore, their protein requirements might be slightly greater to support muscle growth and repair. Finally, in addition to those physiologically based mechanisms, DS marketing often targets men with messages of strength and muscle gain, potentially influencing their DS choices.

The most important limitation of the study is that the DS was self-reported; therefore, there might be concerns about the accuracy of the answers. However, the author believes that the strict anonymity of the questionnaire, as well as the fact that participants were not asked about issues that may be under the influence of "social desirability," decreased the possibility that swimmers did not respond honestly. Additionally, this study investigated athletes from only one country; therefore, the results are generalizable to similar samples (i.e., swimmers from countries with similar cultural backgrounds and cultures).

This is one of the rare studies performed in southeastern Europe where the DS of high-level swimmers were evaluated. The participants were tested with a specific, reliable, and previously validated questionnaire, which is another important strength of the study. Therefore, although results are not the final word on a topic, this study will hopefully broaden the knowledge in a field and initiate further research.

Conclusion

The prevalence of DS in the studied swimmers was in line with previous reports. However, further studies are needed to clarify the templates of DS in different age groups. Specifically, the sample of participants in this study included swimmers of different age categories; therefore, more detailed age-specific analyses are needed.

The fact that the majority of the studied swimmers purchase DS individually, with no proper supervision of educated and certified specialists, is alarming. Seeking guidance from a specialist is crucial to ensure that DS are safe, effective, and tailored to athletes' individual needs and goals. This can help swimmers optimize performance, promote recovery, maintain overall health, and avoid the risk of consuming illegal (banned) substances.

Males consume more DS than females do. While this was expected, attention should be given to the relatively high consumption of protein-based supplements in male swimmers. Specifically, although protein supplementation can be beneficial for swimmers, there can be misconceptions about the amount of protein needed for optimal performance and recovery, leading to unnecessary supplementation. If dietary protein intake is sufficient, additional supplementation might not be necessary. Therefore, it is of utmost importance to educate athletes that the overconsumption of protein can have negative health effects, and it is essential to strike the right balance to support optimal performance and recovery without compromising overall health.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Association of Weight Status with Geographic Region, Residential Status and Sex of 8th Grade School Children: The Prevalence of Overweight and Obesity in Montenegro

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Abstract

The aim of the research was to determine the association between weight status with residential status, geographic region, and sex in younger adolescent age group in Montenegro. The sample was stratified based on gender, settlement type, and geographic region in Montenegro. Anthropometric characteristics were assessed using a battery of three variables: body height (BH), body weight (BW), and body mass index (BMI). BMI was categorized based on the World Health Organisation's (WHO) cut-offs to underweight, normal weight, overweight and obese individuals. The sample comprised 534 early adolescents, specifically 8th grade students, including 283 boys (average age 13.52±0.42 years) and 251 girls (average age 13.51±0.40 years) from primary schools throughout Montenegro. Based on the Chi-square test, an association between weight status and geographic region was found among boys and in the overall sample of children, with the highest prevalence of obesity in the northern region. There is also a significant association between weight status and residential status of adolescents, with obesity being more prevalent among rural children compared to urban children. Additionally, an association between sex and weight status was established, with obesity being more prevalent in boys compared to girls. Therefore, these findings highlight the need for targeted intervention and the promotion of healthy lifestyles among children, taking these demographic factors into account.

Keywords: *anthropometric characteristics, BMI status, nutritional status, prevalence of obesity, adolescents*

Introduction

The substantial changes in people's lifestyles have an impact on children's and adolescents' physical growth as well (Ferreira et al., 2007). The period between the ages of 10 and 19 is termed as adolescence and is characterized by rapid and crucial phases of human development affecting a person's physical, emotional, cognitive, and social development, among other aspects (WHO, 2017). Insufficient physical activity and unhealthy dietary practices are leading to a widespread increase in overweight and obesity, reaching epidemic levels particularly in developed nations (Wijnhoven et al., 2014).

One of the biggest health issues we face today is obesity, which has a number of associated changes that reduce life expectancy (Stankovic et al., 2021). In most economically developed nations as well as some low-income nations, the prevalence of pediatric obesity has doubled or tripled over the past three decades, mostly in metropolitan areas (Wang & Lobstein, 2006). Additionally, obesity has social and psychological repercussions: it is associated with negative feelings of worth, lack of confidence, and sadness (Storch et al., 2007).

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Children are important members of our society, and it has been recognized that monitoring their growth is essential to identifying population health patterns and formulating successful strategies (Sofi & Senthilvelan, 2021). There appears to be a north-south gradient in obesity incidence in Europe, with southern European nations showing the highest frequency (Wijnhoven et al., 2014). The balance between energy intake and expenditure may be affected by previous social and economic changes in Montenegro toward a Western lifestyle and more democratic markets (Telbisz et al., 2014). Despite having limited usefulness in assessing body fat in children and adolescents body mass index is still a practical technique to identify overweight and obesity (Woźniacka et al., 2018). Additionally, the comparison of results between nations is made possible by the availability of data on BMI assessed in sizable population groupings (Must & Anderson, 2006; Reilly, 2006).

In recent decades, there has been a sharp rise in the frequency of overweight and obesity among children and adolescents from Slovenia (Kovač et al., 2014), Poland (Woźniacka et al., 2018), Hungary (Antal et al., 2009), North Macedonia (Morina, Miftari, Georgiev, & Gontarev, 2022), Albania (Jarani, Spahi, Vrenjo, & Ushtelenca, 2022) and Kosovo (Tishukaj et al., 2017). Also, this trend was recorded in Montenegro for children from 6-9 years old (Vasiljevic & Petkovic, 2023), likewise for adolescent (Martinovic et al., 2015; Katanic et al. 2023b). The fact that every third child in Montenegro between the ages of 9 and 13 is overweight or obese (Milasinovic et al., 2019) raises serious concerns among researches. In addition, according to a study in which Hungarian schoolchildren took part, 48% of boys and 28% of girls identified as overweight by BMI were actually obese based on their body fat percentage (Antal et al., 2009). Tishukaj et al. (2017) conducted a study to investigate the physical fitness and anthropometric characteristics of adolescents residing in various regions of Kosovo. The mentioned research highlighted a notable prevalence of overweight and obesity, particularly among boys aged 14 to 15.

Based on the aforementioned information, while numerous studies have focused on the prevalence of overweight and obesity, according to the author's knowledge there are no studies that specifically explore the association between these variables and geographical regions. Therefore, the aim of the research was to determine the association between weight status and residential status, as well as the association between the weight status and the geographic region of boys and girls in the younger adolescent age group in Montenegro. This study will provide a substantial and crucial dataset concerning the weight status and living arrangements of adolescents in the younger age group, specifically within a defined European region.

Methods

Participants

The sample included early adolescents, specifically 8th-grade students, from primary schools throughout Montenegro. A total of 534 students took part in this cross-sectional study (283 boys, with an average age of 13.52±0.42 years, and 251 girls, with an average age of 13.51±0.40 years). The sample was stratified by gender, geographic region, and type of settlement. Montenegro's geographic division comprises northern, central, and coastal regions. Settlement types are categorized according to administrative criteria and population size. Rural areas consist of villages and small towns with populations of 10,000 or fewer, while urban areas include locations with populations exceeding 10,000 (Monstat, 2011). Table 1 provides a detailed breakdown of the sample characteristics. Participation in the study was voluntary, with parental consent, and the research adhered to the principles outlined in the Helsinki Declaration.

Table 1. Characteristics of sample population

Gender	(number, %)
Boys	283 (53.00%)
Girls	251 (47.00%)
Geographical region	(number, %)
Northern	73 (13.67%)
Central	382 (71.54%)
Coastal	79 (14.79%)
Type of settlement	(number, %)
Urban	380 (71.16%)
Rural	154 (28.84%)

Measurements

A standardized international biological method was employed to evaluate morphological traits. Anthropometric measurements were taken using three key variables: body height (BH), body weight (BW), and body mass index (BMI). Height was measured with a portable stadiometer (Seca Ltd., Bonn, Germany), accurate to 0.1 cm. Body weight was recorded using a Tanita body fat scale (Tanita ® model BC-418MA, Tokyo, Japan). The body mass index (BMI) was calculated using the formula: $BMI=BW(kg)/BH(m)^2$. Based on the World Health Organization (WHO) criteria, BMI was categorized into underweight, normal weight, overweight, and obese groups (Onis et al., 2007).

Statistics

The basic descriptive statistics parameters were calculated, including the arithmetic mean, standard deviation, minimum, maximum, and percentages. To examine the relationship between weight status and residential status, the Chi-square (χ^2) test was applied, as well as the association between weight status and residential region, for boys and girls in the younger adolescent age group. For all statistical analyses, a significance level of $p<0.05$ was considered. Data were processed using the Statistical Package for Social Sciences (SPSS), version 26.0 (SPSS Inc., Chicago, IL, USA), along with Microsoft Excel (version 13, Microsoft Corporation, Redmond, WA, USA).

Results

According to Table 1, the total number of participants was 534, with 53% being boys. The largest share of participants comes from the central region, accounting for 71.54%. Additionally, the ratio of urban to rural participants is 71.16% to 28.84%.

For boys, the average age was 13.52±0.42 years, with an average body height of 169.43±8.89 cm, body weight of 60.54±13.47 kg, and a BMI of 20.99±3.78. While for girls, the average age was 13.51±0.40 years, with an average body height of 165.54±6.67 cm, body weight of 55.28±9.27 kg, and a BMI of 20.13±2.99.

Table 2. Descriptive statistics

		Mean	Std. Dev.	Min	Max
Body Height (cm)	Boys	169.43	8.89	145.0	190.0
	Girls	165.54	6.67	151.0	187.0
	Total	167.60	8.15	145.0	190.0
Body Mass (kg)	Boys	60.54	13.47	34.6	110.2
	Girls	55.28	9.27	33.0	82.8
	Total	58.07	11.97	33.0	110.2
BMI	Boys	20.99	3.78	13.63	34.49
	Girls	20.13	2.99	13.91	32.49
	Total	20.58	3.45	13.63	34.49

Association between the weight status and the geographic region

Table 3 presents the distribution of weight status (Underweight, Normal, Overweight, Obese) among three different geographic regions (Northern, Central, Coastal) based on the boys' places of residence. Based on the Chi-square test (χ2) determined a significant association between the weight status and geographic regions (χ2=22.11, p=0.001). The coefficient φ (phi) value of 0.280 signifies a moderate to strong association between these two variables. The highest percentage of obese boys is in the North, with 23.3%, while the Central and Coastal regions have 10% obese boys each. When considering overweight as well, the highest percentage is in the Coastal region at 50%, followed by the North at 39.5%, and the Central zone with 24% of boys classified as overweight.

In this table (Table 3), the results of the Chi-square test indicate that there is no significant association between weight status and geographical region of girls.

Table 3 also illustrates the distribution of weight status based on the geographic region of total children. The results of the Chi-square test indicate significant association between the weight status and geographic regions (χ2=28.19, p=0.000) of children. The coefficient φ (phi) value of 0.230 suggests a moderate association between these two variables. There is a notably high percentage of overweight and obese children combine, totaling nearly 50%, in the central and northern parts, while in the southern part, this percentage is much lower.

Table 3. The association between the weight status and the geographic region of children

Boys	**Underweight**	**Normal**	**Overweight**	**Obese**
Northern	0.0%	37.2%	39.5%	23.3%
Central	3.5%	61.9%	24.3%	10.4%
Coastal	0.0%	39.5%	50.0%	10.5%
χ2(6, n-283)=22.11, p=0.001, φ=0.280				
Girls	Underweight	Normal	Overweight	Obese
Northern	0.0%	66.7%	23.3%	10.0%
Central	1.7%	78.3%	17.8%	2.2%
Coastal	0.0%	65.9%	26.8%	7.3%
χ2(6, n-251)=9.03, p=0.172, φ=0.190				
Total children	Underweight	Normal	Overweight	Obese
Northern	0.0%	49.3%	32.9%	17.8%
Central	2.6%	69.6%	21.2%	6.5%
Coastal	0.0%	53.2%	38.0%	8.9%
χ2(6, n-534)=28.19, p=0.000, φ=0.230				

Legend: χ2 - Coefficient of the Chi square test; p - Coefficient of significance, φ - measure of association.

Association between the weight status and the residential status

Based on the Chi-square test (Table 4), an association between weight status and residential status in boys was determined (χ2=14.22, p=0.003). The coefficient φ (phi) value of 0.224 suggests a moderate association between these two variables. There is a tendency for rural areas to have a higher percentage of overweight and obese boys combined compared to boys from urban areas.

An association between weight status and residential status in girls was determined (χ2=13.17, p=0.004; Table 4). The coefficient φ (phi) value of 0.229 suggests a moderate association between these two variables. Additionally, girls from rural areas have a higher percentage of combined overweight and obesity compared to girls from urban areas.

The association between weight status and the residential status of children was also found in the total sample of children (χ2=24.10, p=0.000; Table 4). The coefficient φ (phi) value of 0.229 suggests a moderate association between these two variables. The table shows that nearly every second child from rural areas is overweight or obese, while this percentage is much lower for urban children, around 28%.

Association between the weight status and the sex

When it comes to the association between the sex and weight status of participants, a significant association has been established (χ2=25.71, p=0.000; Table 5). The coefficient φ (phi) value of 0.219 suggests a moderate association between these two variables. The table indicates a higher frequency of obesity in boys compared to girls.

Table 4. The association between the weight status and the residential status of children.

Boys	**Underweight**	**Normal**	**Overweight**	**Obese**
Urban	3.0%	61.2%	23.9%	11.9%
Rural	1.2%	40.2%	45.1%	13.4%
χ2(3, n-283)=14.22, p=0.003, φ=0.224				
Girls	Underweight	Normal	Overweight	Obese
Urban	0.6%	81.0%	15.1%	3.4%
Rural	2.8%	59.7%	31.9%	5.6%
χ2(3, n-251)=13.17, p=0.004, φ=0.229				
Total children	Underweight	Normal	Overweight	Obese
Urban	1.8%	70.5%	19.7%	7.9%
Rural	1.9%	49.4%	39.0%	9.7%
χ2(3, n-534)=24.10, p=0.000, φ=0.212				

Legend: χ2 - Coefficient of the Chi square test; p - Coefficient of significance, φ - measure of association.

Table 5. The association between the sex and the the weight status.

	Underweight	**Normal**	**Overweight**	**Obese**
Boys	2.5%	55.1%	30.0%	12.4%
Girls	1.2%	74.9%	19.9%	4.0%
χ2(3, n-534)=25.71, p=0.000, φ=0.219				

Legend: χ2 - Coefficient of the Chi square test; p - Coefficient of significance, φ - measure of association.

Discussion

The aim of the study was to determine the relationship between weight status and residential status, geographic regions and sex of children in the younger adolescent age group in Montenegro. The main findings are: i) a significant association between weight status and geographic region was found in boys and in the total sample of children, while this association was absent in the group of girls; the prevalence of obesity is most pronounced in the northern region; ii) a significant association between weight status and residential status was found in both boys and girls groups, as well as in the total sample of children; the prevalence of obesity is more pronounced in rural compared to urban children; iii) a significant association between sex and weight status has been established; higher frequency of obesity was found in boys compared to girls.

Being overweight in adolescent years is a common thing nowadays. Most of this population aren't immune to this disease and if certain actions aren't made in proper time, it can be continued in the adulthood (Kohn & Golden, 2001; Mitić, 2011). Based on the fact that current study have revealed a significant association between weigh status and geographic region, where the obesity prevalence is pronounced in the north of the Montenegro, this result could partially be in the line with study conducted by Gallotta et al. (2022). Although they have revealed that geographic region is influential on the weight status, a higher BMI proportion was also revealed in the center of the Italy, compared to the north and south. In addition, we have also identified a more pronounced obesity prevalence in rural compared to urban children, which is consistent with previously published European studies (Gallotta et al., 2022; Novak et al., 2015), as well as significant association between weight and residential status, which is further confirmed by Katanic et al. (2023a). Since Montenegro is predominantly smaller country, we should take into consideration the fact that bigger and more populated cities are coastally located, while the north of the country have smaller countries with less population, considering that the northern and central parts of the country are dominated by forests. According to the statistical office of Montenegro, 57.64% of the country is covered with forest and 10.02% is forest land (Monstat, 2021) which is mostly spacious in the central and north of the country. Moreover, the population density in central and north part are less pronounced (ranging from 4-9 and 10-37 inhabitans per km2), in regard to coastal (38-78, 79-158 and 159-305 inhabitans per km2) (Monstat, 2023). Although there are studies who have identified that rural area children have lower values of BMI (Chillon et al., 2011; Vasić et al., 2012), which is contrary to our result, the main explaination may lie in the accessibility of physical activities in urban vs rural regions (Davy et al., 2004).

Consequently, different levels of urbanization and density population leads to varying access to sports facilities and opportunity to perform sports (Parks et al., 2003; Reimers et al., 2014), i.e. according to Ilic et al. (2023), adolescents from the urban areas are significantly more involved in sports than those from the rural areas. In addition, current education level of adolescents themselve, but also parental education status and family income status (Andrade et al., 2014) should not be ignored as influential factors on the weight status. Following that, it is most likely that those children will become low-educated themselves, which will accumulate health risks in the area of health behaviour (Viner et al., 2012). Hence, the surroundings in which adolescents live and attend school as well, might influence their healthy habits and contribute to obesity level changes (Levy et al., 2011).

It is a little bit paradox that physical activity level is higher in boys than in girls, but their physical fitness level is the opposite (Strel et al., 2007). In that regard, it is a notable general incosistency among boys and girls weight status as defined by BMI (Kovač et al., 2014; Ogden et al., 2012; Stamatakis et al., 2005; Vuorela et al., 2011; Whelton et al., 2007; Yuca et al., 2010). What's more, there are some recent findings who have emphasizing that boys have higher prevalence of overweight and obesity (Shehzad et al., 2022; Vrevic, 2023), which is in a line with our obtained results. But if we draw a line here, the diverse results may be explained due to biology, culture, or their combination, as well as the genetic conditioning (Banjevic et al., 2022). Although BMI may also vary with age, gender or maturity status (Yoon et al., 2024), there are few things that must be taken into consideration. Weight status could be related to adequate diet and even eating habits, especially while spending time in school (Adams et al., 2014). Likewise, choosing unhealthy habits, such as avoiding physical activities, jung food consumation, being sedentary during leisure time, playing video games, staying up late and sleeping irregularly, can disturb the regular processes of growth and potentially cause health complications (Li et al., 2017; Olson et al., 2019; Zsakai & Bodzsar, 2014). Consequently, it is necessary to reconsider physical activitiy level in physical education classes, as well as a greater emphasis on more frequent weight status evaluations and their external factors influences. In addition, transport to school must not be neglacted. To date, there is little data on the transportation of children and adolescents to places, especially to school. Therefore, future studies should include the mentioned parameters, which would help in gaining a deeper understanding of the factors contributing to childhood obesity.

Therefore, although this study showed that the prevalence of obesity is higher among children from less developed areas, it is clear that numerous potential factors are behind these results. On one hand, additional research is needed to identify specific causes. On the other hand, these findings indicate the need for the involvement of leading sports institutions in Montenegro to promote sports and healthy lifestyles among children from rural areas. This includes organizing sports programs, improving infrastructure for physical activities, and implementing educational programs on nutrition and health, among other initiatives. Only through a comprehensive approach can we expect significant changes in the prevention and reduction of obesity among children in Montenegro.

The strength of the study lies in the fact that this is one of the rare studies that has established an association between weight status with regional and urban-rural status of adolescents. Additionally, it should be emphasized that the study included a large sample of adolescents of both genders from different parts of the country. This allows for relevant and general conclusions to be drawn about the weight status of younger adolescents in Montenegro, as well as the connection between weight status and the regional and urban-rural status of the respondents.

Some of the limitations of the study are reflected in the absence of skinfold and body composition parameters, which would provide a more complete picture of the adolescents' body composition (Đorđević et al., 2024). Future studies could include these parameters, but they could also examine the level of physical activity, dietary habits, and transportation habits of the respondents, which might indicate the reasons for the observed differences in weight status. In this way, a more comprehensive picture of the factors affecting the weight status of younger adolescents in Montenegro would be provided.

Conclusion

This study has shown a significant association between weight status and geographic region, with the highest prevalence of obesity in the northern region. Also, an association between residential status and obesity was found, with children from rural areas being more prone to obesity compared to children from urban areas. Therefore, these findings highlight the need for targeted intervention and the promotion of healthy lifestyles among children, taking these demographic factors into account.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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SportMont

ORIGINAL SCIENTIFIC PAPER

Effect of Post-Activation Potentiation Enhancement Induced Using Elastic Resistance Bands on Female Track and Field Jumpers

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Abstract

Elastic Resistance Bands (ERB) are widely used by athletes separately and in conjunction with heavy loads to activate muscles and improve various athletic abilities. The study aimed to assess the effect of ERB training while warm-up among female track and field jumpers. In the randomized between-group parallel design, 14 athletes were separated into ERB (age: 21±2.16 years, height: 165.6±8.24 cm, body mass: 57.97±4.28 kg) and control (CON) (age: 21.42±2.29 years, height: 164.57±8.54 cm, body mass: 56.57±5.18 kg) groups. A baseline test for CMJ was performed after a general warm-up. Upon completion, both groups performed similar specific pre-competition warm-up drills, with the ERB group performing the exercises using an elastic resistance band. Both groups were tested for follow-up measurements. Significant time x group interaction was observed for jump height ($p=0.024$, $\eta p2=0.36$) and take-off force ($p=0.039$, $\eta p2=0.30$). Significant differences in baseline and follow-up measurements were found in ERB groups for all the dependent variables except peak speed (jump height $p=0.006$, $g=0.48$; take-off force $p=0.009$, $g=0.99$; take-off speed $p=0.046$, $g=0.55$, and max. concentric force $p=0.005$, $g=0.87$). The results suggest that performing running drills with ERB may have the potential to enhance lower limb force generation capacity, as indicated by improvements in CMJ height and take-off force in this study. While these findings are promising, they should be interpreted with caution due to the small subsample size. ERB could be considered as a potential component of warm-up routines for track and field athletes, particularly in situations of low logistical availability. However, further research involving larger sample sizes is necessary to confirm these preliminary findings regarding the efficacy of ERB in warm-up protocol.

Keywords: *resistance bands, counter movement jump, strength and conditioning, take-off force, jumpers, warm-up protocols*

Introduction

Post-activation performance enhancement (PAPE) is a phenomenon characterized by improved muscular power after a "conditioning" contractile activity. Blazevich and Babault (2019) suggested that the mechanism underpinning PAPE may be attributed to increases in neural drive muscle temperature, blood flow cellular water, and muscle-tendon stiffness, although the specific mechanisms require further empirical investigations. The increased force output caused by contractile history has been known to improve sprint ability, acceleration, change of direction speed (COD), and jump performance (Katushabe & Kramer; Turner, Bellhouse, Kilduff, & Russell, 2015; Wallace, Winchester, & McGuigan, 2006; Wyland, Van Dorin, & Reyes, 2015).

Evidential practices causing PAPE include methods such as maximum voluntary isometric contraction (MVIC)

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(Lima et al., 2014; Tsoukos, Bogdanis, Terzis, & Veligekas, 2016), heavy resistance loading (>80% of 1RM) (Linder et al., 2010; Mitchell & Sale, 2011), and loaded and unloaded plyometrics (Aloui et al., 2020; de Villarreal, Izquierdo, & Gonzalez-Badillo, 2011), among soccer, handball, track and field power athletes, and sprinters. Heavy and maximum resistance loading before performance induces PAPE, but as argued by Lum and Chen (2020), it is neither viable to carry heavy weights all the time nor feasible during competitions. Another limitation of heavy resistance is the lack of sports-specific movements (Aandahl, Von Heimburg, & Van den Tillaar, 2018), which are not biomechanically similar to the movements performed, and impede the involvement of muscles for a specific movement. As Turner et al. (2015) emphasized, maximizing the movement velocity results in greater fast-twitch muscle fibre recruitment inducing PAPE. Smith et al. (2014) examined the effects of sledge resistance sprints, utilizing loads ranging from 0% to 30% of body mass as the PAP warm-up protocol. Findings revealed a significant main effect, demonstrating subsequent sprint performance improvement. Chattong (2008) investigated the effect of the weighted vest with 5-20% of the subject's body weight with box jump on vertical jump performance; findings reveal significant post-test results, but no significant difference in jump height between groups was found. Thus, plyometric exercises could be an effective way of performance enhancement other than conventional weight loading and MVIC, inducing less fatigue, shortening the recovery time, and providing movement velocity (de Villarreal et al., 2011; Seitz & Haff, 2016).

Previously used for injury rehabilitation, elastic resistance bands (ERB) are found to be a potential alternative to resistance loading, eliciting explosive performance in athletes; ERB offers Variable resistance (VR) at the range of motion and could substitute conventional weights (dumbbells) offering similar muscle activation (in deltoid and trapezius) (Bergquist, Iversen, Mork, & Fimland, 2018). ERB is also found to be effective when used in conjunction with external resistance, Mina et al. (2019) investigated the potentiating effect of the ERB training (35% load from ERB) against FWR (85% of 1RM) and revealed a significant increase in 1RM Squat in the ERB group (mean=7.7%, $p<0.01$). Dundass (2013) compared the ERB (70% free-weight resistance + 30% ERB) group with the CON group (100% free-weight resistance) on the lower-body power (max. vertical jump) and strength output (1RM) and found significant improvement in both groups, but no significant difference was recorded in the between-group assessment. ERB, in conjunction with contrast strength training (Hammami et al., 2022) and free-weight resistance (FWR) exercises (Wallace et al., 2006), produces PAPE, which substantially improved 30m sprint, CMJ, and peak force, peak power, respectively. However, PAPE-induced performance improvement using ERB with plyometrics is still an area of investigation.

As per the researcher's knowledge and literature review, only a few studies assessed ERB training without free-weight intervention. Aandahl et al. (2018) assessed the efficacy of warm-up with ERB on roundhouse kicking performance in trained martial arts practitioners. 3D motion capture technology and electromyography recorded a 3.3% increase in kicking velocity against the CON group ($p=0.009$, $\eta=0.32$) and 35% and 44% more muscle activation in the vastus-medialis (prime movers) and rectus femoris, respectively, signifying the use of ERB in warm-up can be substantial for kicking performance. However, the study did not record the effect of ERB training on the vertical jump performance of athletes and other performance variables. Another investigation by Lum and Chen (2020) on male national athletes compared unloaded and ERB-loaded CMJ with different resistance levels. Findings revealed a significant increase in peak power, peak velocity, and reduced time to peak torque in the ERB condition. However, no significant change in jump height was recorded between conditions.

Despite the positive effect of ERB-loaded CMJ on subsequent CMJ performance, it is currently not known if the inclusion of ERB when performing running drills would also potentiate lower limb force development capability. As these drills are commonly performed by track and field athletes, such knowledge would have great practical implications for track and field jumpers. Therefore, the study aimed to assess the effect of ERB training with a dynamic warm-up on track and field female athletes. Based on the available literature, the researcher hypothesized that ERB training with dynamic warm-up would induce PAPE to elevate athletes' jump height, take-off force, take-off speed, maximum concentric force, and peak speed.

Material and Methods

Experimental approach to the problem

A between-group parallel experimental design was used to investigate the potentiation effect caused by the elastic resistance band (ERB) during warm-up. The protocol included general and specific warm-ups. After following a similar general warm-up, jumpers were randomly assigned to the ERB or control (CON) group. In the specific warm-up, the ERB group performed plyometrics and jumping movements with resistance bands, whereas the CON group performed identical movements without resistance bands. Pre- and post-testing before and after specific warm-up measured jump height, take-off force, take-off speed, maximum concentric force and peak speed.

Subjects

Female track and field jumpers (long, high, and triple jumpers) with at least university or national participation who suffered no injury in the past six months were admissible for the study. Fulfilling the above criteria, a total of 14 track and field jumpers (6 high, 5 long, and 3 triple jumpers), with a minimum of 1.5 years of competitive experience and past exposure to resistance training, were allocated into ERB (age: 21±2.16 years, height: 165.6±8.24 cm, body mass: 57.97±4.28 kg) and CON group (age: 21.42±2.29 years, height: 164.57±8.54 cm, body mass: 56.57±5.18 kg) as shown in Table 1. Being from the same university and residing on the campus, all the participants had similar conditioning regimes and followed a similar daily routine. Before signing the consent form, participants were well-informed about the associated details, procedure, risks, and benefits. The study adhered to the ethical standards of the Declaration of Helsinki (2013) and was approved by the Departmental Research Committee of Lakshmibai National Institute of Physical Education, Gwalior, India. LNIPE/DRC/MPE/2021-22/816.

Table 1. Demographic, anthropometric and training characteristics of the CON and ERB group

Variables	Mean ± SD		p
	CON Group (n=7)	ERB Group (n=7)	
Age (years)	21.42 ± 2.29	21 ± 2.16	0.72
Training age (years)	6.15 ± 1.57	7.71 ± 1.6	0.09
Height (cm)	164.57 ± 8.54	165.57 ± 8.24	0.82
Body mass (kg)	56.57 ± 5.18	57.97 ± 4.28	0.59

Procedure

After two mandatory weekly familiarization sessions with ERB exercises and CMJ protocols, all the participants were randomly divided into CON and ERB groups. To avoid learning effects during the data collection, familiarization was strictly adhered to and made sure athletes were reaching their best scores. One day before the data collection procedure, basic demographic information and anthropometric measurements (standing height, body mass, fat percentage) were recorded. Athletes were asked to avoid any arduous activity 48 hours before the assessment and refrain from any energy/supplement drinks 4 hours before testing (caffeine, creatine, and other stimulants), which may alter their performance. After 15-minutes of performing the general warm-up, the pre-test measurement and randomization for ERB and CON were done.

Warm-up Protocol

Before commencing the testing procedure, both the CON and ERB groups followed a 15-minute general warm-up consisting of 7-minutes of self-paced joggings, followed by 5 dynamic stretches of 8 repetitions each: knee to chest hugs, butt kicks, lateral lunges, forward deep lunges, and squats, with 3 repetitions of 10-meter sprints. After the general warm-ups and before commencing the specific warm-up, Pre-testing of CMJ was done. After 2-minutes of Pre-testing, athletes went for specific warm-up exercises, 5-7 repetitions each, including High-knee marching, A-skips, straight lead leg-skips, split lunges jump, one leg complete cycle (both legs), single-leg swing and cycle (both legs), piston action, and ABC drills. The ERB group performed the exercises/drills using spiral elastic resistance bands circumscribed on thighs and knees while performing the drills, whereas, in the case of single leg movements, the resistance bands were hooked to the stationary points to perform the exercises, as shown in Figure 1. The colour determined the intensity of resistance bands: yellow, green, red, blue, and black. All the athletes used blue-coloured resistance bands. This induced approximately 12-14 kg of force upon 100% elongation (as per the manual GoFit Pro Power), ensuring an ERB load of at least 10% of body weight (Burkett, Phillips, & Ziuraitis, 2005; Lum & Chen, 2020; Mina et al., 2019).

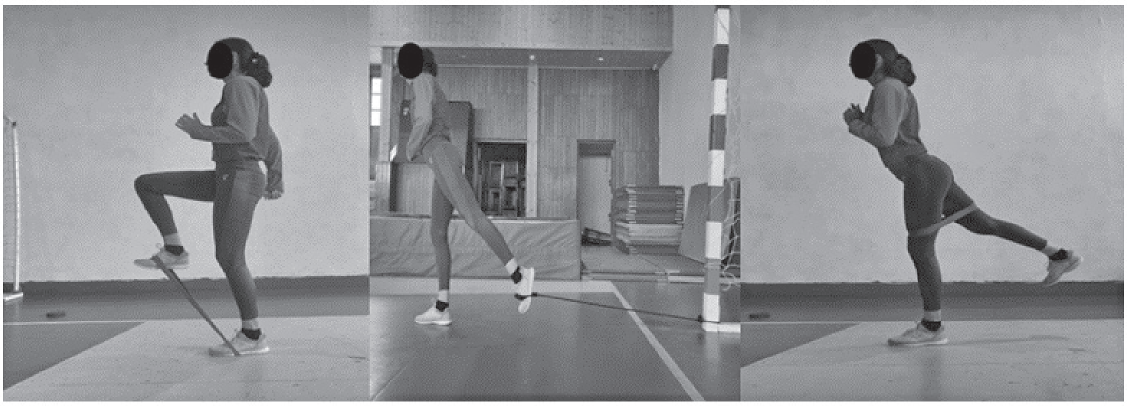

FIGURE 1. Demonstration of exercises performed by the ERB group

Counter Movement Jump

To assess jump height, maximum height, take-off force, take-off speed, maximum concentric force, and peak speed. Countermovement jump performance with arm swing (CMJA) was performed using an inertial moment sensor (BTS G-walk, Italy). The sensor was positioned at the fifth lumbar vertebrae, providing a belt in the centre of the device. Participants were asked to stand with feet slightly apart and were instructed to swing their arms and perform a countermovement to a self-selected depth before taking off and landing with both legs (Domire & Challis, 2007). Knee flexion was not permitted during the flight phase of the jump. The athletes of both groups were consistently motivated throughout the assessment protocols (pre-post attempts) to maximize their CMJ performance. Three trials were performed with 30-40 seconds of rest between jumps, and the best trial was selected for analysis (Rixon, Lamont, & Bemben, 2007). The ICC for test-retest was 0.92–0.96, and CV=11.44-12.52%.

Statistical Analysis

Descriptive analyses were presented as mean and Standard Deviation (SD) for demographic and performance variables. The homoscedasticity was tested using Leven's test of equality of variance followed by an independent (between-group) t-test to compare baseline measurements. Further, the normality assumption for all the data was met using the Shapiro-Wilk test. 2x2 mixed ANOVAs with post hoc analyses using Bonferroni adjustments were used to compare the time, i.e., pre-test and post-test vs. ERB and CON groups. Partial ETA squared was used to calculate the effect size of time and between group testing, and Hegde's g was calculated between baseline and follow-up testing in each group. The magnitude of the intervention was estimated to be classified as trivial (<0.20), small (0.2–0.6), or moderate (>0.60–1.2) based on established criteria (Hopkins, 2007). The magnitude of effects for η2p was interpreted as small (0.6–1.2), large (>1.2–2.0), very large (>2.0–4.0), and extremely large (>4.0) (Hopkins, 2007; Pereira, Horwitz, & Ioannidis, 2012). Statistical significance was set at $p \leq 0.05$. All the analyses were done using IBM SPSS version 25.0 (IBM, New York, USA).

Results

Records of demographic, anthropometric, and training differences for both groups are summarized in Table 1. Further, no statistical baseline differences (p=0.44-0.90) were found between CON and ERB groups when tested for dependent variables, supporting similar physical characteristics. The results comparing the ERB and CON warm-up effect on various CMJ constructs are presented as mean±SD.

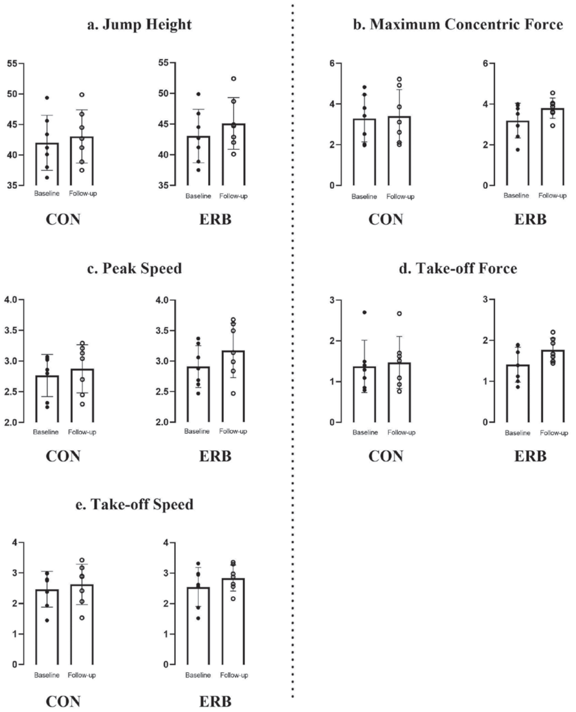

FIGURE 2. Mean (column) and SD for ERB (right) and non-ERB (left) with baseline and follow-up measurements for (a) jump height, (b) maximum concentric force, (c) peak speed, (d) take-off force and (e) take-off speed.

Significant outcomes for the main time effect were concluded for take-off force (p=0.002, ɳp2=0.572); take-off speed (p=0.015, ɳp2=0.404); maximum concentric force (p=0.02, ɳp2=0.374); and peak speed (p=0.003, ɳp2=0.528). However, no significant main time differences were found in jump height (p>0.05). Group-by-time interaction effect was found with a substantial increase in jump height of 4.54% (p=0.024, ɳp2=0.359) and take-off force of 20.45% (p=0.039, ɳp2=0.309) amongst the ERB

Table 2. Counter Movement Jump Variables

Variable	CON (Mean±SD.)				ERB (Mean±SD.)				Main Time Effect	Time x Group	Between-group
	Baseline	Follow-up	(p)	% Δ [g]	Baseline	Follow-up	(p)	% Δ [g]	p-value [ɳp2]	p-value [ɳp2]	p-value [ɳp2]
Jump Height (cm)	42 ± 4.51	42.25 ± 5.01	0.61	0.60 % 0.052	43.05 ± 4.35	45.1 ± 4.2	0.006#	4.54% 0.48	0.06 (0.482)	0.024** (0.359)	0.432 (0.052)
Take off force (N)	1.37 ± 0.64	1.46 ± 0.63	0.17	6.56 % 0.14	1.40 ± 0.42	1.76 ± 0.28	0.009#	20.45% 0.99	0.002* (0.572)	0.039** (0.309)	0.55 (0.039)
Take-off Speed	2.46 ± 0.58	2.62 ± 0.66	0.38	6.50 % 0.25	2.54 ± 0.64	2.84 ± 0.42	0.046#	11.81% 0.55	0.015* (0.404)	0.41 (0.057)	0.64 (0.019)
Max. Concentric Force	3.29 ± 1.15	3.40 ± 1.29	0.22	3.34 % 0.09	3.19 ± 0.85	3.80 ± 0.49	0.005#	19.12% 0.87	0.020* (0.374)	0.90 (221)	0.77 (0.07)
Peak Speed.	2.76 ± 0.34	2.87 ± 0.39	0.12	3.99 % 0.30	2.91 ± 0.34	3.17 ± 0.44	0.068	8.93% 0.66	0.003* (0.528)	0.152 (0.163)	0.282 (0.09)

Note- abbreviations used; % Δ: percentage difference in baseline and follow-up records, [g]: Hedges' g, ɳp2: partial eta squared; SD: standard deviation, #: significant baseline and follow-up measure (with-in group), *: significant main time effect between baseline and follow-up measure; **: significant time by group interaction group when compared to CON. Post-hoc analysis showed a significant increase in peak concentric force in CON but no other CMJ measures. In contrast, all CMJ measures except peak speed increased post-warm-up in the ERB group (p=0.005-0.046).

Discussion

The purpose of the current study was to investigate the effectiveness of including ERB during running drills on the potentiating lower limb force development capability of female track and field jumpers. Our hypothesis was supported by the current results, as CMJ height, take-off, and max concentric force were significantly improved in the ERB group.

The current findings agreed with previous studies that have reported enhancing movement performance after including ERB in movement-specific warm-ups. Aandahl et al. (2018) examined the effect of roundhouse kicking velocity using ERBs during warm-up of trained martial arts practitioners, suggesting a 3.3% increase in linear kicking velocity and higher muscle activation (EMG activity) of associated muscles, confirming PAP. More such studies (Chua et al., 2021; Lum & Chen, 2020) reported that performing CMJ with ERB that provided a resistance level of 10% body mass or more resulted in improved CMJ height and reduced time to take-off. In a recent study, Chua et al. (2021) reported that performing ERB with a resistance level of 10% body mass could also enhance stretch-shortening cycle (SSC) ability. Lum (2019) in a study on judo players with upper and lower body PAP suggests that targeted explosive exercises with ERB can effectively improve peak power. Earlier studies indicated that engaging in ERB-resisted movements during warm-up could enhance subsequent non-resisted movements acutely, particularly when these movements are biomechanically similar, as demonstrated by the significant PAP effects observed in previous research (Baxter, 2014; Stevenson, Warpeha, Dietz, Giveans, & Erdman, 2010). However, the novelty of the current study was that the movement during the warm-up performed with ERB (running drills) was not similar to the movement used for the performance measure (CMJ). Despite that, the inclusion of ERB was still beneficial. The present findings further support a recent study (Singh, Singh, & Sharma, 2023) that highlights the beneficial effects of ERB-induced potentiation during warm-up on CMJ (power) when utilizing lighter loads (less than body weight). This stands in contrast to previous assertions that exercises involving lighter loads do not effectively induce PAP (Hanson, Leigh, & Mynark, 2007; Weber, Brown, Coburn, & Zinder, 2008).

A possible reason for the present observation could be that the inclusion of ERB leads to greater muscle activation during the warm-up due to the increased intensity. It further resulted in a reduction in the recruitment threshold for subsequent movement. Although electromyographic activity was not measured, this phenomenon is reflected in the increased take-off and maximum concentric force, which were observed concurrently with the improved jump height in the ERB group. Furthermore, the findings of this study align with previous research (Chua et al., 2021; Earp, Newton, Cormie, & Blazevich, 2014), indicating that a loaded warm-up can enhance the stiffness of the lower limb tendons. This increase in stiffness facilitates a greater force contribution from the change of direction (COD), which is crucial for improving both take-off force and maximum concentric force, as demonstrated in the current results. Another possible explanation for the significant improvement in the time x group interaction of take-off force can be assigned to the nature of drills performed during the warm-up with ERB and the nature of participants. As the track and field jumpers are conditioned to produce greater take-off force, this may have emulated the take-off mechanics of the female track and field jumpers (Yang, Tang, Liu, & Pandy, 2023). The ergogenic effect of including ERB in running drills has great practical implications for track and field athletes in the practical setting. As athletes are unlikely to have access to heavy resistance equipment during competition, using ERB to enhance their warm-up effect would significantly reduce the logistical requirement. Hence, track and field coaches and athletes may consider using ERB for their competition warm-up regime.

Several limitations should be considered while interpreting the current results. Firstly, important time variables for understanding the mechanism for improving jump performance (Bishop, Jarvis, Turner, & Balsalobre-Fernandez, 2022) were not measured due to limitations in the equipment used. Secondly, it was ensured that the ERB load is around 10% of body mass; however, this parameter was not consistently controlled across all jumpers. This lack of control may have resulted in a greater external load than anticipated. Lastly, the current study used CMJ to indicate lower limb performance. It did not specifically measure track and field-specific performance such as 100-meter sprint, long and high jump, etc. Hence, it is still unknown if running drills with ERB would also directly enhance track and field-specific performance. Future research should aim to explore the impact of ERB-induced PAP on specific track and field performance measures to establish an apparent connection between ERB and track and field performance.

Conclusion

In conclusion, the current study showed that performing running drills with ERB can enhance lower limb force generation capability, as indicated by improved CMJ height and force data. Hence, due to the low logistical requirement, track and field coaches and athletes may consider including ERB when performing running drills as part of their warm-up, specifically during competition settings.

Conflict of Interest
The authors have no relevant financial or non-financial interests to disclose.

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Ethics approval
This study was performed in line with the principles of the Declaration of Helsinki. The Ethics Committee of Lakshmibai National Institute of Physical Education, Gwalior, India. Approved the study. LNIPE/DRC/MPE/2021-22/816

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ORIGINAL SCIENTIFIC PAPER

Assessing the Knowledge Level of Karate Coaches Regarding Physical Training Modalities

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Abstract

Having a certain level of knowledge is always important in doing any task correctly. However, when it comes to being a sports coach, a person needs to have a deep understanding of the sport's rules, the technical aspects involved, and the various physical training modalities. This study aims to identify the knowledge level of karate coaches regarding physical training modalities, on the basics of physical training according to sex, academic qualification, belt degree, classification degree and years of experience. Fifty senior coaches of both sexes were deliberately selected after showing interest in participating into the current study. The study was based on the previously developed and validated questionnaire (Martial Arts Coaches Knowledge Level Questionnaire). The questionnaire is composed of several axes with multiple choice questions to measure the cognitive results of karate coaches in the areas of physical training concerning human exercise physiology, functional sports anatomy, sports nutrition, sports injuries and first aid. The results showed a weak level of knowledge among karate coaches and the correct answers (total level) in all areas scored 45.7%, and coaches over 5 Dan have scored higher levels compared 3 Dan (p=0.033, 0.010; respectively). Based on these findings, we conclude that karate coaches in Jordan have a significant knowledge gap in the field of physical training methods. This gap may impact the effectiveness of training programs and athlete performance, emphasizing the need for targeted educational interventions and continuous professional development programs to enhance coaches' understanding and application of physical training modalities. Such improvements could ultimately lead to better athlete outcomes and a higher level of performance in the sport.

Keywords: *martial arts, performance, injuries, exercise, comprehensive coaching*

Introduction

Sports coaching is an interactive practice that occurs in a dynamic environment which requires the continual development of specific knowledge and skills to guide athletes in an effective way (Cushion et al., 2003; Nash & Collins, 2006; Tan & O'Connor, 2023). Denton and Hasbrouck give a more philosophical definition to the concept of coaching as "different things for different people" (Denton & Hasbrouck, 2009). Coaches must use their expertise to uncover strengths and weaknesses, while also offering support and encouragement and performing a countless number of tasks to drive athletes towards their peak performance (Jones et al., 2004).

Effective athletic trainers and coaches possess key attributes, including a strong level of knowledge, good commu-

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nication skills, effective training techniques, confidence, self-reflection, and risk management (Jones et al., 2010; Kania et al., 2009; Mazerolle et al., 2015; Pitney, 2006). Additionally, a coach may be an administrator, master teacher, curriculum designer, outside expert, classroom teacher, or other professional (Erchul, 2023). However, failure to recognize an athlete's need for improvement can result in ongoing injuries and setbacks (Hegarty et al., 2018; Talpey & Siesmaa, 2017). Therefore, coaches must continually develop their knowledge in their sport to implement a successful coaching process, regardless of the coaching context they belong to.

In karate, there are a wide range of exercises, techniques and training methodologies designed to improve physical abilities such as strength, power, speed, agility, flexibility, endurance and others (García-Hermoso, 2023). The training programs must be designed and modified to meet the unique demands of the practiced type of karate (Pinto et al., 2023). In Kata, which means "form" athletes move in several directions in space applying a series of exhibition movements, while Kumite athletes perform ritualistic rather than actual fights. Although these competitions involve noncontact fighting and symbolic techniques, the athletes must demonstrate the potential force of their movements and execute them as if they were real (Doria et al., 2009). Therefore, it is imperative that karate coaches have a solid understanding of the different physical training modalities to develop training programs that effectively meet the needs of their athletes (Kabadayı et al., 2022).

Unfortunately, the World Karate Federation does not require any level of experience or knowledge for individuals to become coaches (World Karate Federation - WKF, n.d.). Even though, the Jordanian Karate Federation requires at least 3 Dan and being over 25 years old to be able to coach and start a karate academy (Jordan Karate Federation, n.d.). The recent decades have shown a growing interest in exploring the education of coaches (Fullagar et al., 2019; Stoszkowski & Collins, 2016). Studies have analyzed the structure and content of specific coach education programs, highlighting the importance of coaches continually developing their skills to facilitate sports development (Tan & O'Connor, 2023). Olympic committees advise coaches to focus on learning new training methods and to deepening their understanding of the science behind physical training to achieve optimal results (Bompa et al., 2019). The Jordan Olympic Committee has shown considerable interest in developing Jordanian karate, particularly since the 2021 Olympic Games in Tokyo, Japan; by creating the Olympic Preparation Center that focuses on preparing athletes in a scientific and planned way by providing a group of scientists to help Olympic coaches improving their work and supervise the process of the training programs. However, Mesquita et al. (2010) highlight the potential of the academic environment to be considered as a formal coach training agency. Mujika et al. (2018) declared that coaches who have a strong knowledge base of physical training modalities can implement periodized training plans, systematically advance athletes' physical attributes over time, and improve their overall performance. Therefore, coaches must study the science related to physical training more in depth, avoiding the superficial level of information generalities, to be able to achieve the best results (Bompa et al., 2019).

Recent studies have emphasized the importance of structured coach education across various sports. Cushion et al. (2003) found that well-designed coach education programs significantly impact coaching effectiveness and professional development. Similarly, Lyle (2002) demonstrated that coaches with formal education in physical training and coaching methodologies tend to develop more effective training programs, thanks to a deeper understanding of training modalities. This research highlights a gap in formalized coach education in certain sports, such as karate, where federations may not universally require comprehensive training programs, underscoring the need for further research in this area.

The current study aims to fill the gap by investigating the knowledge level of karate coaches regarding physical training modalities, particularly in Jordan, where the Jordan Karate Federation has basic requirements for coaching but lacks a formalized, scientific education structure. While studies specifically focused on karate coaches are limited, research on coaches in martial arts offers valuable insights. For example, Kilic and Ince (2015) explored coaches' access to sport science knowledge in martial arts and highlighted factors such as experience, sex, education level, coaching certification, and remuneration as influencing their access to this knowledge. Understanding these factors is crucial for developing more effective coaching education programs. Additionally, previous research has indicated that martial arts training may enhance certain fitness elements, while neglecting others, such as strength and hand-eye coordination (Hammad et al., 2024). By analyzing Jordanian karate coaches' understanding of training methods and their ability to apply them in practice, this research seeks to contribute to the broader discussion on coach education and the need for more rigorous training standards. The findings of this study can inform sports federations and Olympic committees in enhancing their coach education initiatives.

Materials and Methods

Participants

50 Jordanian karate coaches (46 men, 4 women) working in karate centers across Jordan voluntarily participated in the current study. The number of participants represent (83.33%) of the entire national coaches community, taking into account that they are trainers accredited from the Jordanian Karate Federation and they all own their academies. Participants have different demographic information (table 1). Study procedures followed international ethical guidelines and participants were informed of the confidentiality of the information collected and they provided written consent. To be included in the study the individual must be recognized from the Jordanian Karate Federation as a working coach. The Institutional Review Board (Faculty of Educational Sciences at Al-Ahliyya Amman University) approved this study under protocol (No. FES-18G-274).

Knowledge assessment questionnaire

In this study, the questionnaire that we validated in a previous study on the level of knowledge of martial arts coaches (Questionnaire on the level of knowledge of taekwondo coaches regarding physical training methods) was adopted, as well as a cross-sectional methodology similar to our previous survey to assess the level of knowledge of (martial arts) coaches (Hammad et al., 2022). The questionnaire consisted of two main sections, the first section gathered demographic information and included variables such as the participants' level of education and belt qualification. This section aimed to establish a baseline of the coaches' backgrounds. The second

Table 1. Demographic information.

Variables	Categories	Participants n (%)
sex	Male	(92) 46
	Female	(8) 4
Level of Education	Secondary and below	(20) 10
	Diploma	(16) 8
	Bachelor's	(58) 29
	Postgraduate Studies	(6) 3
Belt degree	3 Dan	(30) 15
	4 Dan	(20) 10
	5 Dan	(32) 16
	6 Dan and above	(18) 9
The degree of classification as a Karate Coach	National	(78) 39
	International	(22) 11
Experience as a Karate Coach	<5 years	12 (24)
	5 – 10 years	(46) 23
	>10 years	(30) 15

section focused on evaluating coaches' knowledge in various areas of the basics of physical training. This section was divided into five distinct areas: physical training, exercise physiology, sports career anatomy, sports nutrition, and sports injuries and first aid.

Within the five areas, a set of 49 multiple-choice questions was developed. Each question offered four options of the answer, with only one option being correct. The participants' responses were used to calculate a score, reflecting their knowledge in each area.

To assess the level of knowledge, we employed a three-category framework. Scores of 75% and above were categorized as demonstrating adequate knowledge. Scores ranging from 50% to 74% were considered indicative of moderate knowledge. Scores below 50% were categorized as reflecting insufficient knowledge (Hammad et al., 2022).

Data Collection

After the study was approved by the Jordanian Karate Federation and a list of recognized coaches' names was obtained, data collection took place during local competitions in the summer of 2023. Each volunteer was given a questionnaire and was monitored while completing it without the use of the internet or mobile phones. On average, it took respondents around 15 minutes to complete the questionnaire. To ensure the reliability of the research tool, the application and re-application method was used. This approach facilitated the extraction of the coefficient of difficulty and discrimination. The obtained values for the discrimination coefficients were deemed acceptable, ranging from 0.30 to 0.90, as confirmed by the Kuder-Richardson method. The tool was also applied to physical education students in their 3rd and 4th years of study, who were able to achieve an adequate level (above 75%) of correct answers.

Data Analysis

The collected data were analyzed using the chefé test and one-way ANOVA, employing SPSS software (version 26, IBM, USA). The statistical analysis was conducted using a significance level of $p \leq 0.05$. Additionally, the effect of guessing was calculated by the equation: FS=R-W/C-1 (FS-corrected score; R-right answers; W-wrong answers; C-number of alternatives). The threshold for statistical significance was set at $p \leq 0.05$ and $F \leq 0.05$.

Results

The results for the cognitive outcome domains of karate coaches are presented in Table2. The scores for each domain indicate the coaches' level of knowledge, calculated as the average number of correct answers. To provide a better understanding, we have also included the corresponding percentage values based on the total number of questions in each area.

According to the findings, the domains, sports injuries and first aid, had the highest average of correct score of 5.63, representing a value of 50.7% (Table 2). This suggests a relatively stronger understanding in this area than other selected ones. In contrast, anatomy of sports area had the lowest average score of 3.96, which represents a percentage of 39.6%. This indicates a lower level of knowledge in this area among coaches. However, physical training comes second, with an average score of 4.82, or in percentage of 48.2%. Sports nutrition follows closely with an average score of 4.80, a percentage of 48%. Following all exercise physiology ranked fourth, with an average score of 4.18, representing a percentage of 41.8% (Table 2). These results indicate a weak level of knowledge in these respective areas.

The average total score for the knowledge of karate coaches in all areas was 23.39, which corresponds to a percentage of 45.7% (Table 2). This overall score suggests a weak level of coach knowledge. It is noteworthy that the values obtained from the percentage values of the cognitive outcome domains indicate that the coaches' knowledge of karate is below the adequate level.

To categorize the knowledge levels of the participants, a grading scale was established. Participants' scores were classified as "weak" if they achieved less than 50% correct answers in each knowledge field. This threshold was determined based on the principle that less than half of the correct responses indicate a lack of sufficient understanding in that area. The average score for each participant was calculated by dividing the total number of correct answers by the total sample size. The

Table 2. The actual correct grades of all participants.

Knowledge fields	Questions Number	Actual Grades Correct answers	Average	Percentage %	Knowledge Levels
Physical Training	10	241	4.82	48.2	Weak
Exercise Physiology	10	209	4.18	41.8	Weak
Sports Career Anatomy	10	198	3.96	39.6	Weak
Sports Nutrition	10	240	4.8	48.0	Weak
Sports Injuries and First Aid	9	228	5.63	50.7	Weak
Total	49	1116	23.39	45.7	Weak

Note. Average - the total number of right answers/the total sample number, Weak - under 50% of the right answers.

overall knowledge level was then assessed based on whether the average percentage of correct answers was above or below 50%, with scores below 50% categorized as "weak."

Table 3. presents the results indicating the presence of statistical differences in the physical training field based on belt degree. The results reveal that the calculated Indication level of P-value for the physical training field reached 0.049, suggesting a statistically significant difference that indicates significant variations in the average scores among different belt degrees within the physical training field. However, in the other fields (exercise physiology, anatomy of sports, sports nutrition, and sports injuries and first aid), no significant differences were observed. This implies that the average scores in these fields did not vary significantly based on belt degree. These findings highlight the importance of belt degree in relation to the knowledge level specifically in the physical training field. It suggests that coaches with different belt degrees may exhibit varying levels of expertise and understanding in this domain.

Table 3. A Variance Analysis of the Average Knowledge Outcomes of Karate Coaches in Different Training Methods Regarding their Belt Degrees.

Fields	The source of the variance	Sum of Squares	Degree of freedom	Mean Square	Probability value (P-value)
Physical training	Between groups	19.983	3	6.661	.049*
	Within groups	108.438	46	2.357	
	Total	128.420	49		
Exercise physiology	Between groups	2.417	3	.806	.798
	Within groups	109.583	46	2.382	
	Total	112.000	49		
Sports functional anatomy	Between groups	.631	3	.210	.921
	Within groups	59.689	46	1.298	
	Total	60.320	49		
Sports nutrition	Between groups	6.230	3	2.077	.221
	Within groups	62.650	46	1.362	
	Total	68.880	49		
Sports injuries and first aids	Between groups	2.507	3	.836	.481
	Within groups	45.993	46	1.000	
	Total	48.500	49		
Overall cognitive achievement	Between groups	41.714	3	13.905	.346
	Within groups	565.406	46	12.291	
	Total	607.120	49		

Note. This calculation determines the ratio of explained variance to unexplained variance. * Significance at $\alpha \leq 0.05$.

Concerning the data analyzes according to the degree of classification as a karate coach, the results show statistical differences (Table 4). The results reveal that in the physical training dimension, a significant difference ($p \leq 0.05$) was observed with a calculated indication level of α value reaching $p \leq 0.002$. This indicates that the average scores in the physical training dimension varied considerably depending on the degree of classification as a karate trainer. Specifically, the average difference was found to be higher for international karate coaches, with an average value of 4.73, compared to the average value of 3.10 for local karate coaches. This suggests that international karate trainers display higher levels of knowledge in the dimension of physical training compared to their local counterparts. However, for the other dimensions, no significant difference was observed. This means that there were no statistically significant variations in mean scores at different degrees of classification as a karate trainer in these dimensions.

Table 4. Mean Differences in Outcome Areas of Karate Coaches' Training Methods Based on Degree of Classification as a Karate Coach (t-test Results)

Fields	Degree of classification	n	Mean ± sd	t	sig
Physical Training	local	39	3.10 ± 1.52	3.20	0.002*
	international	11	4.73 ± 1.35		
Exercise Physiology	local	39	4.46 ± 1.64	0.53	0.593
	international	11	4.18 ± 0.98		
Sports functional anatomy	local	39	1.51 ± 1.02	0.56	0.577
	international	11	1.73 ± 1.42		
Sports Nutrition	local	39	2.18 ± 1.17	1.60	0.115
	international	11	2.82 ± 1.17		
Sports Injuries and First Aid	local	39	1.51 ± 1.10	0.17	0.866
	international	11	1.45 ± 0.52		
Overall cognitive achievement	local	39	12.77 ± 3.65	1.82	0.075
	international	11	14.91 ± 2.51		

Note. * Significance at α≤0.05.

The results of the physical training domain based on the belt level (Table 5) showed significant differences (p≤0.05). The results indicate that there are significant differences in the average scores between different belt degrees within the physical training field. Specifically, 6th Dan and above scored higher levels compared to 5th Dan, and there was also a significant difference favoring 6th Dan and above compared to 3rd Dan. These findings suggest that coaches with higher belt degrees, particularly those at the 5th Dan and above level, exhibited greater knowledge and proficiency in the physical training field compared to those with lower belt degrees. The differences observed highlight the importance of advancing to higher belt degrees as a reflection of increased expertise and understanding in this specific domain.

Table 5. Chefé test of the Belt degree in karate coaches.

Fields	Arithmetic mean	Belt degree	4 Dan	5 Dans	6 Dans and above
Physical training	2.60	3 Dan	0.208	0.033*	0.010*
	3.40	4 Dan		0.508	0.192
	3.81	5 Dan			0.420
	4.33	6 Dan and above			

Note. * Significance at α≤0.05.

Discussion

This study attempted to investigate the knowledge level of karate trainers regarding physical training modalities. The cognitive knowledge assessment of karate coaches in Jordan came within the weak level with an overall percentage of 45.7%, similar data was obtained from the taekwondo coaches (S. Hammad et al., 2022). This finding emphasizes the need for targeted interventions and educational programs to enhance coaches' knowledge which will consequently improve their effectiveness in training athletes. The discrepancy in knowledge levels across the different fields highlights specific areas that require coaches to give attention and develop themselves in (Attardi et al., 2018; Bhadana et al., 2015; Bompa et al., 2019; García-Isidoro et al., 2020; Pope et al., 2015).

Among the different areas assessed, sports injuries and first aid achieved the highest percentage of correct answers. Non-surprisingly the results of Jorge et al., also indicates a relatively higher level of knowledge in first aid area compare to others (Jorge et al., 2009). Conversely, anatomy of sports career recorded the lowest average, those results meets the finding of some previous studies (Attardi et al., 2018; Bhadana et al., 2015; Bompa et al., 2019; García-Isidoro et al., 2020; Pope et al., 2015). This deficiency in knowledge among karate coaches may impede athlete development and increase the risk of injuries and health complications (Bhadana et al., 2015). The study also uncovered findings that are consistent with previous studies and suggest a consistent pattern of insufficient knowledge among karate coaches, which can hinder athlete development and increase the risk of injuries and health complications (Bhadana et al., 2015). One contributing factor to this knowledge gap may be the inadequacy of the training courses offered by the Jordan Karate Federation. The limited duration of the courses, coupled with a lack of practical components, restricts the depth of knowledge that coaches can acquire. The current approach of providing theoretical materials for a short period, typically twice a year, is insufficient in effectively improving coaches' cognitive outcomes and knowledge acquisition. Previous studies have shown that academic training plays an important role in the development of coaches' knowledge, conditioned by the knowledge related to their professional background (Kilic & Ince, 2015; Mesquita et al., 2010). Additionally, the lack of theoretical promotion examinations, except for higher belt degrees such as 6th Dan and above, raises concerns about the quality and standardization of training within the sport (Fullagar et al., 2019; Stoszkowski & Collins, 2016).

To address these challenges, it is crucial to enhance the training programs provided by the Jordan Karate Federation. Increasing the duration and frequency of the courses, incorporating practical elements, and emphasizing evidence-based practices can significantly enhance coaches' understanding and application of physical training fundamentals (S. Hammad et al., 2022; Heath et al., 2012), By providing coaches with comprehensive and up-to-date knowledge, they can better guide athletes in their training, resulting in improved performance and reduced injury rates.

Implementing regular assessments, including theoretical promotion examinations, can ensure that coaches demonstrate an adequate level of training related to the sport. These assessments should not only evaluate theoretical knowledge but also assess coaches' practical application of training methods and their ability to adapt to the individual needs of athletes. By setting higher standards for certification, coaches will be incentivized to continually improve their knowledge and skills (Fabio & Towey, 2018; Moen et al., 2020). This rigorous evaluation process will help maintain high-quality coaching standards and ensure that coaches possess the necessary expertise to effectively train athletes.

Furthermore, collaboration and knowledge exchange platforms should be established to facilitate interaction between coaches within Jordan and international experts. Encouraging coaches to participate in workshops, conferences, and seminars can expose them to innovative training methods and the latest research findings. This exposure to diverse perspectives and best practices can inspire coaches to explore and implement new approaches in their training programs (Boniwell & Osin, 2014; Martindale et al., 2005; Romar et al., 2020).

Additionally, there were some significant differences ($P \leq 0.05$) in the knowledge outcome areas among trainers working in karate centers in Jordan based on the variable of Belt degree. This suggests that international coaches and those with higher Dan ranks have likely undergone advanced training courses, either outside the country or through the Jordanian Karate Federation. These additional training opportunities may have exposed these coaches to different methods and advancements in karate training (García-Isidoro et al., 2020). Consequently, trainers with higher belts and international classification are more likely to have acquired extra knowledge and skills in the field of physical training. This may have motivated other trainers to explore similar training methods through internet research or consulting with other coaches, potentially leading to a higher level of expertise in physical training among trainers with higher belts and international classification (S. Hammad et al., 2022).

The recommendations outlined align with international guidelines and best practices in karate coaching, as emphasized by organizations like the World Karate Federation (WKF) (World Karate Federation, 2021). The WKF recognizes the importance of continuous education and knowledge enhancement for coaches, aiming to elevate the overall coaching standards in karate.

By implementing these strategies in Jordan, a profound transformation in the knowledge and expertise of karate coaches can be achieved, leading to significant benefits for both athletes and their performance. It is worth noting that athletic training has the potential to influence starting from the gene level to the various physiological systems within the body (R. Hammad et al., 2022). Therefore, investing the knowledge and professional levels of coaches will not only contribute to the growth of karate in Jordan but also pave the way for enhanced success in the field (S. Hammad et al., 2022).

Limitations

Despite the results being interesting and providing valuable advice and information regarding the knowledge level of the karate coaches, we only assess the current level, and we cannot predict how this knowledge might evolve or change in the future. Further research is needed to obtain a longitudinal assessment and a broader understanding about the knowledge level improvements.

Conclusion

The study revealed that karate coaches in Jordan have a significant knowledge gap in the field of physical training methods. This knowledge gap was observed across coaches of various belt degrees, indicating a widespread issue within the coaching community. To address this issue, it would be recommended to implement comprehensive educational programs that specifically focus on enhancing coaches' understanding of physical training methods.

These programs should be integrated into coaching curricula and emphasize the importance of continuous professional development for coaches. By incorporating education on the fundamentals of physical training methods, coaches can acquire the necessary knowledge and skills to effectively train athletes and reduce the risk of injuries.

Additionally, it is crucial to establish and enforce regulations that mandate coaches to engage in ongoing education and professional development activities. This will ensure that coaches remain up-to-date with the latest advancements in physical training methods and maintain a high level of competence in their coaching practice.

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Conflicts of interest

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Ethics statement

The studies involving human participants were reviewed and approved by Committee for Scientific Work and Ethics of the Faculty of Educational Sciences at Al-Ahliyya Amman University, under protocol (No. FES-18G-274).

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ORIGINAL SCIENTIFIC PAPER

Differences in the Speed and Power of Elite U12 and U13 Croatian Soccer Players

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Abstract

Identification of talent and continuous evaluation of players' abilities is an indispensable part of the selection process. The main goal of the research was to determine the differences in speed and power of elite U12 and U13 young Croatian soccer players with different roles in the team. Out of the 38 total HNK Hajduk Split players, 18 players were U12 age category while 20 players were U13 age category. Players were divided by the coaches in the groups of higher quality (key players) and lower quality (rotation players). Players were tested with 2 tests to assess explosive power and 3 tests to assess speed. Results obtained with Student's T-test showed that there are no significant differences in the speed and the power between the two observed group of players for each age category. For U12 category key players demonstrated better results than rotation players in the standing long jump and the medicine ball throw (183.89/566.44 vs. 177.22/511.89 cm, respectively). Additionally, key players performed better in 5-, 10- and 20- m sprint (1.11/1.93/3.43 vs. 1.14/1.96/3.52 s). U13 key players performed similar with rotation players in 5-, 10- and 20- m sprint (1.12/1.97/3.44 vs. 1.13/1.97/3.47 s), but interestingly, U13 key players were slightly worse than rotation players in the standing long jump and the medicine ball throw (194.33/587.67 vs. 194.75/606.50 cm). Many variables and abilities may determine soccer success. For the observed participants it seems likely that other abilities could have been more decisive for status differentiation in the team.

Keywords: *young soccer players, key players, rotation players, speed, power*

Introduction

Today's football, which has greatly progressed compared to the football played in the 20th century (Reilly et al., 2000; Wong et al., 2009), differs mainly in terms of the technical-tactical requirements of the game and fitness abilities, which today makes a difference in a large number of cases (Johnson, Farooq & Whiteley, 2017; Morales et al., 2018). Nowadays, football is much faster and more dynamic compared to the football that was played in the past (Cardenas et al., 2019; Vaeyens et al., 2009). Football game is becoming more and more demanding, and players are no longer closely tied to the position they are placed in but are required to participate fully in both the attack and defense phases (Figueiredo et al., 2018; Huertas et al., 2019). Fitness preparation of athletes is a process where various programs are applied to maintain and improve numerous functional and motor abilities, but also morphological characteristics of players (Aquino et al., 2017; Cobley et al., 2009; Costa et al., 2012). It is very important to assess players' abilities continuously during the phases of growth and development and one of the most important things is diagnostics, which plays a big role in planning and programming training, especially for young football players (Cardenas et al., 2019; Wong et al., 2009). Football academies and coaches should regularly conduct player tests in order to be able to identify and select football talents with the most accurate data possible (Philippaerts et al., 2006, Toselli et al., 2022). Today, young football players who are physically more dominant are selected as more successful ones (Duarte et al., 2019; Reilly et al., 2000, Rickesh et al., 2019). Some researchers (Altimari et al., 2021, Duarte et al., 2019) have studied differences in anthropometric characteristics and motor abilities in young soccer players of the same chronological age. The results of their research suggested that first team players performed better when compared to reserve players in motor abilities and were taller and heavier.

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Bidaurrazaga-Letona et al. (2016) investigated the differences in anthropometric characteristics between young soccer players and their research suggested that players who are taller and heavier are selected as more successful and play in the starting line-up in contrast to physically inferior players. Gissis et al. (2006) found that the players of the first team, given their physical dominance, have better results in tests of explosiveness of the lower extremities. Some researchers suggest (Aquino et al., 2017; Deprez et al., 2015) anthropometric characteristics and motor performance without the ball as the main predictors of success that coaches single out for young soccer players. On the other hand, Costa et al. (2012) suggest that selection should not be made with emphasis on morphological characteristics, but primarily on the skills with the ball. Speed and strength are certainly one of the most important and measured abilities in soccer (Gravina et al., 2008). Also, modern football is unimaginable without strong and fast players. In football, one of the most important things is explosive power, which represents the ability of absolute excitation of muscle fibers in a unit of time (Figueiredo et al., 2018). Players who are selected as more successful have better results in manifestations of explosiveness (Gissis et al., 2006). On the other hand, speed represents the ability to overcome a certain way in the shortest possible time. Today's football implies that the player needs to be fast to more easily adapt to the technical and tactical demands placed in front of him (Duarte et al., 2019). As soccer clubs continue to pursue talent in earlier and earlier stages of players' development, it becomes increasingly important to develop efficient ways to detect talent and differences between elite and sub-elite players. Subsequently, it is of great interest for coaches and clubs to make continuous assessment of young players' abilities in order to maintain an appropriate identification and selection of players. Players within U12 and U13 age group are the ones that experience transition from smaller "half-pitch" size fields to "full size" fields. It would be beneficial to determine what abilities differentiate key and rotation players in elite clubs during this sensitive stage of development.

The main goal of the research was to determine the differences in speed and power of elite U12 and U13 young Croatian soccer players with different roles in the team.

Methods

Participants

Out of the 38 total HNK Hajduk Split players, 18 players were U12 age category while 20 players were U13 age category. Players were divided by the coaches in the groups of higher quality (key players) and lower quality (rotation players). Key players were the ones defined by the coach as players that were the most impactful on the match outcome and can dominate most aspects of the game. Rotation players were all other players who were either substitute players or the ones that impact the game in a lesser degree. Ethical Committee of the Faculty of Kinesiology approved the research (approval number: 2181-205-02-05-23-018 Split, Croatia), according to the ethical standards of the 1964 Helsinki Declaration. Inclusion criteria for participating in this study were: player's participation in more than 85% of the training while playing in competitive matches, without reported injury during the time of the research. Players had valid sport identity card signed and were healthy as well as medically examined by a sport specialist doctor. The participants did not consume any caffeine products 24 hours before the test and did not consume any food for two hours prior to the test.

Design and procedures

The research was conducted in the period from October 13, 2022. – October 20, 2022. First, anthropometric characteristics were measured: body height and body mass. After the anthropometric measurements were taken, the players went out on the field with artificial grass to do a warm-up and prepare the organism for the motor tests. The subjects performed medicine ball throw (3kg), standing long jump, the 5 m sprint, the 10 m sprint and the 20 m sprint. For each test, the measurement was performed three times with a break between each measurement. Polarized retroreflective photoelectric Witty timing system – Microgate (16-bit microprocessor, operating temperature 0-45°C, accuracy ±0.4ms) series was used (Bolzano, Italy).

Statistical analysis

The data obtained from the research were entered into the Microsoft Excel (2018 version, Microsoft Corporation, Redmond, Washington) computer program. Furthermore, the software Statistica ver. 13.0 (Dell Inc., Round Rock, TX USA) was used for data analysis. Descriptive statistics parameters were calculated, and T-test for independent samples was used to determine differences between groups ($p<0.05$).

Results

T-test shows that significant differences were not found in the variables of anthropometry and motor abilities between key and rotation players for U12 category. The results obtained in the descriptive statistics show that, on average, key players (N=9) are slightly taller (152.67 vs. 151.67 cm) compared to rotation players (N=9) and that they have quite similar body

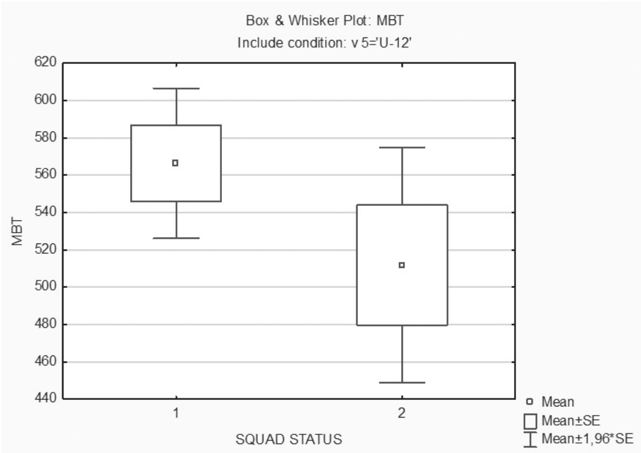
Table 1. T-test of differences between U12 players and range of results.

Variables	M±SD 1	M±SD 2	t-value	p	Range 1	Range 2
BH (cm)	152.67±4.39	151.67±10.74	0.25	0.799	147.00-159.00	41.00-177.00
BM (kg)	38.32±3.51	38.81±7.26	-0.18	0.858	32.00-43.50	30.30-55.80
SLJ (cm)	183.89±9.08	177.22 12.06	1.32	0.204	171.00-197.00	160.00-203.00
MBT (cm)	566.44±61.58	511.89±96.32	1.43	0.171	485.00-670.00	417.00-745.00
S5m (s)	1.11±0.04	1.14±0.06	-1.31	0.208	1.05-1.17	1.07-1.24
S10m (s)	1.93±0.06	1.96±0.07	-1.00	0.331	1.85-2.02	1.82-2.03
S20m (s)	3.43±0.08	3.52±0.12	-1.89	0.077	3.30-3.52	3.23-3.65

Legend: M±SD 1 – mean and standard deviation of key players, M±SD 2- mean and standard deviation of rotation players, t- value, RANGE1- minimum and maximum of key players, RANGE2- minimum and maximum of rotation players, BH - body height, BM- body mass, SLJ - standing long jump, MBT - medicine ball throw, S5m-sprint 5 meters, S10m-sprint 10 meters, S20m-sprint 20 meters.

mass (38.32 vs. 38.81 kg). In the tests for measuring the explosiveness of the lower and upper part of the body (SLJ and MBT) key players achieved better results in contrast to rotation players (183.89/566.44 vs. 177.22/511.89cm). Additionally, key players performed better in S5, S10 and S20 m sprint (Table 1).

In graph 1. it can be seen that average score in medicine ball throw of key players is higher than average score of rotation players that had wider range of results.



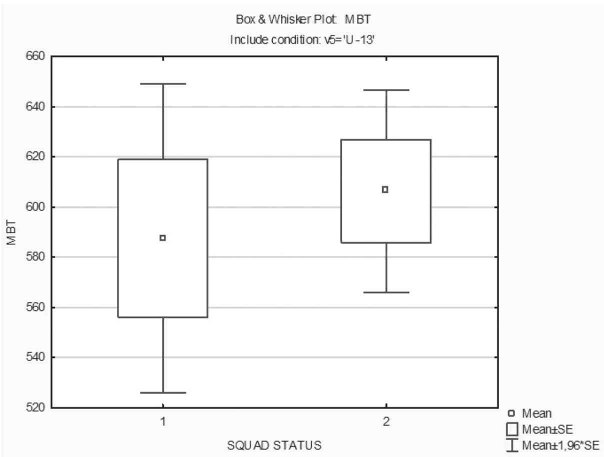
GRAPH 1. Graphic presentation of the results in medicine ball throw for U12 group of subjects

Using the T-test for independent samples, no significant differences were obtained. Rotation players (N=8) are on average taller (159.00 vs. 155.67 cm) and heavier (44.90 vs. 42.48 kg) compared to key players (N=12). Key players and rotation players achieved an almost identical result in the standing long jump (194.33/194.75 cm), while rotation players achieved a better result in the medicine ball throw (606.5 cm) than key players (587.67 cm). U13 key players performed similar with rotation players in S5, S10 and S20 m sprint (Table 2).

Contrast to U12 players, in U13 age category rotation players performed better in variable medicine ball throw while key players had wider range of results.

Table 2. T-test of differences between U13 players and range of results

Variables	M±SD 1	M±SD 2	t-value	p	Range 1	Range 2
BH (cm)	155.67±8.45	159.00±7.35	-0.91	0.380	143.00-172.00	150.00-172.00
BM (kg)	42.48±7.76	44.90±4.63	-0.79	0.440	32.60-58.20	38.50-50.90
SLJ (cm)	194.33±13.37	194.75±17.04	-0.06	0.950	168.00-213.00	168.00-223.00
MBT (cm)	587.67±108.89	606.50±58.22	-0.45	0.660	425.00-795.00	545.00-700.00
S5m (s)	1.12±0.03	1.13±0.04	-0.14	0.890	1.09-1.17	1.08-1.20
S10m (s)	1.97±0.07	1.97±0.08	-0.07	0.940	1.89-2.10	1.88-2.15
S20m (s)	3.44±0.10	3.47±0.12	-0.57	0.570	3.26-3.58	3.31-3.65



GRAPH 1. Graphic presentation of the results in medicine ball throw for U12 group of subjects

Discussion

Results showed that there were no significant differences in the speed and the power between key and rotation players of U12 and U13 age category. For U12 category, key players demonstrated slightly better results than rotation players in the standing long jump and the medicine ball throw (183.89/566.44 vs. 177.22/511.89 cm). U13 key players were slightly worse than rotation players in the standing long jump and medicine ball throw (194.33/587.67 vs. 194.75/606.50 cm). Such results were somewhat expected because U13 rotation players in this research were taller and heavier compared to key players (Table 2). On the other hand, Silva et al. (2010) found that selected players were heavier (F=30.67, p<0.01) and taller (F=35.07, p<0.01); performed better in explosive power (F=21.25, p<0.01), repeated sprints (F=20.04, p<0.01) than non-selected players. Additionally, Gouvea at al. (2017) showed that more skilled young soccer players showed greater performance in two manifestations of explosiveness such as squat jump and countermovement jump. Gissis et al. (2006) also revealed significant effects between elite and sub-elite young soccer players of the group on maximal isometric force (F=5.490, p<0.01), peak force relative to body mass (F=818.728, p<0.01), explosive force at 100 msec (F=140.225, p<0.05), rate of force development (F=7.982, p<0.001), squat jump height (F=127.827, p<0.05). Bidaurrazaga-Letona et al. (2016) showed that elite players were significantly taller than sub-elite across all ages (d=0.30–1.18). In the Under-11 group, he found that elite players were less heavy than sub-elite players (d=0.30–0.54) but, as age increased, elite players became heavier than players in the rest of the groups (d=0.28–0.87). Also, differences were observed by Toselli et al. (2022) in body composition (height, weight, lengths, widths, circumferences, and skinfold thicknesses) between elite and non-elite players and in all performance tests (Yo-Yo test, repeated sprint ability and countermovement jump). The aforementioned research suggests significant differences in the anthropometric characteristics of young players who were selected as more successful which is in contrast to findings in this research. This was possibly one of the reasons that there were not significant differences in manifestations of explosiveness (Table 1 and Table 2).

Key players of this research performed slightly better in S5, S10 and S20 m sprint (1.11/1.93/3.43 vs. 1.14/1.96/3.52 s) in U12 group. U13 key players performed similar with rotation players in the S5, S10 and S20 m sprint (1.12/1.97/3.44 vs. 1.13/1.97 /3.47 s) but in both groups' differences were not significant (Table 1 and Table 2). Similar results with our research were obtained in Gissis et al. (2006) which showed there were no significant differences in variable of speed (10m) between elite and sub-elite players (1.95 m/s vs. 2.14 m/s). Additionally, Rebelo et al. (2012) found no differences between elite and non-elite players in 5m and 30m sprints (p>0.05). Malina et al. (2007) found no differences between young players of different quality in speed, however, players differed in aerobic endurance (p<0.01). On the other hand, Slimani and Nikolaidis (2017) found better results (p<0.01) in speed (10-30 m) and agility of higher-level compared to lower-level young soccer players. Contradictory results amongst number of researchers highlight the complexity of the problem at hand. Given the specific biologic and soccer stages of development, it remains challenging to determine differences in abilities and characteristics between youth players, both at the elite and lower level. Relatively small number of participants was one of the limitations of the study. Further research with higher number of participants with more players from elite clubs and sub-elite clubs would be advised. Also, for the U12 and U13 players, variables that include other dimensions such as soccer "skill", agility and coordination are required in order to obtain more conclusive data.

Conclusion

There were no significant differences in the speed and the power between the key and rotation players of U12 and U13 age category. Looking at the obtained results, in this study with this number of participants, it is likely that some other factors such as football skills were important in the assessment of quality by the coach. In a repeated study it would be advised to increase the number of participants in order to reach more precise conclusions. Nonetheless, speed and power should be evaluated regularly as important predictors of soccer success.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Application of Unilateral and Bilateral Plyometric Exercises on the Ability of Planned Agility and Acceleration; Effectis in Young Soccer Players

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Abstract

The main aim of the study was to determine the difference in the effects between the two applied protocols (Unilateral and Bilateral), on the ability of planned agility and acceleration. For this research, the sample were active soccer players (N=30; 14 years in average). Two equal groups were formed randomly, unilateral group (EG=15) and bilateral group (CG=15). The study included an 8-week intervention of unilateral and bilateral plyometric training, applied as an integral part of soccer training, with three training sessions in one week. Both applied protocols were equalized according to the total load volume, the number of foot contacts with the ground and the character of the jump performance. Variables included tests of planned agility (side step test, and 505 test, arrowhead test), and acceleration tests (5- and 20 meters sprint). T-test for independent samples, and combined analysis of variance (2x2 / time x group) were calculated. The results showed no differences between the treatment groups, but absolute effects were achieved in both groups. The sidestep test, 505 planned agility test, arrowhead test, and 5 and 20-meter sprint test improved equally in both groups ($p<0.05$). In conclusion, unilateral and bilateral plyometric training lasting eight weeks led to significant improvements (pre/post= $p<0.05$) in sprint-type explosive power (acceleration ability) and preplanned agility, but without statistically significant differences in the magnitude of the effects between training groups.

Keywords: *football, conditioning capacity, power, speed, training effects*

Introduction

Soccer is a high-intensity, intermittent sport that requires players to have well-developed abilities to sprint, change direction, jump, and perform repeated duels during a match (Turner & Stewart, 2014; Wagner et al., 2023). Data from the analysis of movement during a soccer match showed that elite soccer players cover an average of 10-11 km, can jump up to 15 times and change direction 1200-1400 times in one match (Bangsboo, 1992; Rampini et al., 2007). Many of these intermittent actions are high-intensity in nature, and it stands to reason that strength and power training serve as useful training methods for soccer players to prepare for the demanding actions of the game of soccer. The capacity to perform fast and explosive movements has a high impact on the quality of performance during the match (Christopher et al., 2016, Mohr et al., 2003, Zamparo et al., 2015).

On the other hand, plyometrics is one of the standard methods that is often used in the development of speed-explosive qualities of athletes in different sports, especially in sports that require a high degree of speed and strength (Idrizovic et al., 2018; Kalata et al., 2023; Ozen et al., 2020). Given that the soccer game contains a large number of short, high-intensity activities, it can be concluded that plyometric training is a valuable training method for improving important motor skills in soccer players (Jaksic et al., 2023). In the physiological background of plyometric training is the cycle of muscle stretching and shortening (Turner & Jeffreys, 2010). According to the very definition, plyometrics include exercises whose goal is to connect the strength and speed of movement to achieve an explosive-reactive movement that will be mani-

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fested in movement activity. Numerous studies performed on soccer players showed that plyometric training significantly improves explosive strength and agility among soccer players (Bedoya, Miltenberger, & Lopez, 2015). Moreover, studies reported enhanced performance in sprinting speed and jumping height among players who engaged in regular plyometric routines (Bedoya et al., 2015).

Considering the above, it could be assumed that both modalities, which include the application of bilateral and unilateral content, can improve sports performance. However, there are still debates and an evident lack of research about the magnitude of the effects achieved by the application of these two modalities. Therefore, the main aim of this study was to determine whether one plyometric training modality is more efficient than the other. More precisely, this study aimed to gain information on how the ability of soccer players to accelerate, change the direction of movement, or jump can be improved with both unilateral and bilateral types of plyometric exercises.

Methods

Participants

This study involved a squad of young soccer players (N=30), 14 years old on average. Players train regularly for a long time in the Football Club „Sloboda" from Tuzla, Bosnia and Herzegovina, which competes at the highest national competitive level. None of the participants of the selected sample reported any chronic diseases, nor did they have medically contraindicated conditions, which are closely and specifically related to the implementation of training and diagnostic procedures. The fitness trainers of the club were informed about the nature and type of the experimental procedure and were instructed not to conduct any additional fitness training during the duration of this study. In this way, it was ensured that the achieved effects for both groups at the end of the experimental procedure can be explained solely by the application of the implemented programs. Before the implementation of the plyometric program, the subjects worked for 15 days on the physiological adaptation of the locomotor system in the fitness center with exercises that included mechanics of jumping and landing, strength development and joint stability.

Procedures and variables

Unilateral and bilateral plyometric training was applied as an integral part of technical-tactical soccer training, and conducted for 10 weeks (two weeks of physiological adaptation and 8 weeks of plyometric program), with three training units during one week. Bilateral protocol implied the application of bilateral contents (i.e. both leg jumps) while unilateral protocol consisted of unilateral plyometric contents (i.e. single leg jumps). Both applied protocols were equalized according to the total volume of the load, the number of contacts with the ground, the character of the performance of the jumps concerning the direction and plane in which they are performed, and the training period in which the contents were performed.

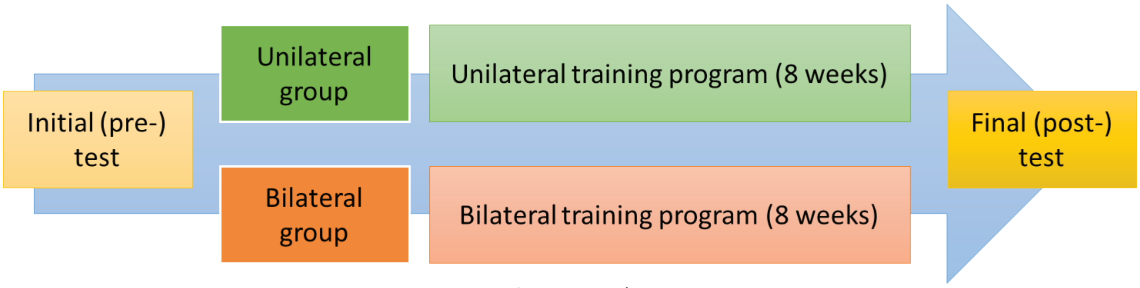

FIGURE 1. Conventional stress test

Initial testing was done 48 hours before the start of the plyometric exercise program. Two equal groups were formed after the test and based on the results. After a standardized warm-up protocol and specific preparation for performing plyometric exercises, the participants of each group performed prescribed plyometric training.

Observed variables consisted of a battery of agility tests that included the Arrowhead Agility Test performed on both sides (ARROW L and ARROW R), Steps to the side test (SS), and 505 Agility Test (A505). Also, acceleration and speed were tested with the sprint on 5 (S5) and 20 (S20) meters.

In the ARROW the participant starts from the high start position with his foot behind the starting line. At the sign from the examiner, the participant runs behind the stand in the middle at a distance of 10 meters, runs right or left to the next stand 5 meters away, and cuts diagonally inward to the final stand 15 meters from the starting line and back to the starting position. In the SS test participants must move laterally between two parallel lines, 1 meter long and 4 meters apart, as quickly as possible without crossing their legs. The test is complete when the participant has crossed the distance of 4 meters six times in this manner. A505 test is performed so that the participant runs 15 meters, turns around at the 15-meter mark, and runs back to the 10-meter mark. The time recorded is from passing the 10-meter gate to returning to the 10-meter gate, with a turn on the 15-meter line. Finally, 5 and 20-meter sprints were performed from the high starting position and participants had to cover the distance as fast as possible. For all the tests, the measurers used Brower Timing System devices (Draper, UT, USA) to read the results.

Statistics

The normality of the distributions was determined using the Kolmogorov-Smirnov test, and descriptive statistic parameters (arithmetic means and standard deviations) were calculated for all variables. An independent samples t-test was used to determine the existence of initial differences between treatment groups, while the differences before and after treatments within the groups were determined using the t-test of paired samples for each group individually. Finally, a combined analysis of variance (2x2 / time x group) was used to determine the differences in the treatments. Statistica 13.0 (TIBCO Software Inc, USA) was used for all calculations with a p-level of 95%.

Results

Mean values and standard deviations of all participants in both groups in observed variables in initial and final test-

Table 1. Descriptive statistics and T-test

Variables	Bilateral group (BIL)					Unilateral group (UNI)				
	Initial		Final		Effects	Initial		Final		Effects
	MEAN	SD	MEAN	SD	pre-post	MEAN	SD	MEAN	SD	pre-post
S5M	1.40	0.17	1.26	0.06	9.9%*	1.37	0.13	1.26	0.06	8.02%*
S20M	3.83	0.26	3.56	0.17	7.04%*	3.87	0.25	3.58	0.18	7.49%*
A505	3.05	0.17	2.42	0.16	20.6%*	3.01	0.26	2.36	0.13	21%*
SS	8.76	0.49	7.93	0.38	9.47%*	8.59	0.36	7.83	0.47	8.84%*
ARROW(L)	9.15	0.45	8.75	0.36	4.87%*	9.29	0.56	8.67	0.47	6.67%*
ARROW(R)	8.78	0.43	8.65	0.39	1.48%*	8.94	0.49	8.62	0.46	3.57%*

Note. S5M – sprint on 5 meters; S20M – sprint on 20 meters; A505 – agility 505 test; SS – side step test; ARROW(L) – Arrow agility test on the left side; ARROW(D) – Arrow agility test on the right side.

ing can be seen in Table 1. Also, the T-test for the analysis of differences between the initial and final state for both groups and indirectly calculated sizes of realized effects by group (%) are presented in Table 1. Results suggest statistically significant differences in all parameters for both groups.

Table 2. shows the results of the 2x2 mixed ANOVA where significant differences can be seen for all variables for the Time factor, but no statistically significant differences for the interaction of the factor (group x time). Results suggest that improvements occurred for all participants after the treatment but with no different effect between each group.

Table 2. Analysis of Variance (2x2 Mixed ANOVA)

Variables	Group			Time			Group x time		
	F	p	η^2	F	p	η^2	F	p	η^2
S5M	0.216	0.646	0.008	20.049	0.001	0.417	0.215	0.645	0.008
S20M	0.195	0.662	0.007	58.674	0.001	0.677	0.090	0.767	0.003
A505	0.602	0.444	0.021	368.157	0.001	0.929	0.073	0.789	0.003
SS	0.904	0.350	0.031	119.657	0.001	0.810	0.602	0.444	0.021
ARROW (L)	0.041	0.840	0.001	52.045	0.001	0.650	2.740	0.109	0.089
ARROW (R)	0.230	0.636	0.008	8.198	0.008	0.226	1.370	0.252	0.047

Note. S5M – sprint 5 meters; S20M – sprint 20 meters; A505 – agility 505 test; SS – side step test; ARROW(L) – Arrow agility test on the left side; ARROW(D) – Arrow agility test on the right side; F – F value for ANOVA; p – statistical significance, η2 – effect size

Discussion

This study aimed to determine the effects of the different plyometric training modalities on agility and acceleration performance among soccer players. Accordingly, two major findings can be highlighted. Firstly, the improvement after the intervention was noted for both groups in all observed variables. Also, there were no significant differences between the effects of treatment for both groups which suggests both types of training provoked similar adaptations.

Both unilateral and bilateral groups had significantly better results after 10 weeks treatment of plyometric training. Plyometric training, which involves explosive exercises that utilize the stretch-shorten cycle, has been consistently shown to improve agility and acceleration in athletes and is generally recognized as one of the most efficient methods for the development of explosive-speed capacities (Čaprić et al., 2022). The significant improvements observed in this study align with existing research. For example, a study on twenty-one youth soccer players showed that even short-term protocols are important and able to give meaningful improvements on power and speed parameters in a specific soccer population (Beato, Bianchi, Coratella, Merlini, & Drust, 2018). Similar findings were found in other sports and other samples. For example, a study on a youth female handball players showed that plyometrics performed two times per week for 10 weeks enhances measures related to game performance, such as change of direction, jump ability, reactive strength index, and power (Gaamouri et al., 2023).

The results of two-factor ANOVA indicate there were no differences in the magnitude of effects between the two groups. More precisely, regardless of whether the players performed unilateral or bilateral type of plyometric training, they enhanced agility and acceleration capacities in similar magnitude. The achieved results are in accordance with the results of an earlier study which dealt with determining the effects of unilateral, bilateral, and combined unilateral and bilateral plyometric training on the performance of sprinting 15 and 30 m in young soccer players (Ramirez-Campillo et al., 2014). Identical to the results presented here, all experimental groups showed a significant increase in the sprint variables. The authors state that a possible reason for the improvement in sprinting at the specified distances in both groups is the use of horizontal jumps (Ramirez-Campillo et al., 2014). Regarding agility performance, our findings are also in accordance with some previous studies. Namely, in the research on young soccer players, participants were divided into three groups - control, and two experimental groups with different types of plyometric training) (Ramirez-Campillo et al., 2015). The results demonstrate that all plyometric groups achieve a significant increase in agility performance, with small to moderate effect sizes. Similar findings were obtained in the study where the effects of horizontal and vertical plyometric modalities were analyzed (Manouras, Papanikolaou, Karatrantou, Kouvarakis, & Gerodimos, 2016). In particular, results showed similar improvements in speed and agility performance in both groups (Manouras et al., 2016).

Conclusion

The results of this study showed a positive effect of plyometric training on speed, acceleration, and agility performance among soccer players. However, more interestingly, no differences in treatment effect were found between the groups that performed unilateral and bilateral jumps. In general, it seems that systematic plyometric training, independent of specific training modalities, can improve explosive capacities in youth athletes. These findings provide valuable information for sports practitioners involved in the development of young athletes, showing that even fundamental types of jumps can gain advancement in players of this age. Therefore, it is necessary that in the process of implementing plyometric training with young athletes, coaches respect the methodical principles of progressiveness and gradualness.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Toward Equitable Sport Opportunities: Addressing Sociodemographic Disparities in Youth Sport Engagement

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Abstract

The aim of this study was to evaluate the associations between sociodemographic factors and different factors of sport participation in early adolescent children. The participants were school-aged children from Kosovo (n=1813; 911 girls) aged 13.7±0.9 years who regularly attended the 8th and 9th grades of elementary school. The variables included sociodemographic data (age, gender, urban/rural living environment, parental education, and socioeconomic status) and sport factors (participation in team and individual sports, experience in sports, and competitive achievement). Differences in sport factors between groups based on sociodemographic variables were established via the Mann-Whitney test (MW) and Kruskal-Wallis ANOVA (KW) The results revealed a positive association between gender and all factors of sport participation, with boys being more involved in sports (MW=7.33 and 13.56, p<0.01 for individual and team sport participation, respectively) and for a longer time (MW=18.23, p<0.001) than girls. The urban living environment was significantly associated with all the observed sport factors (MW=5.04, 2.94, 3.82, all p<0.001). Higher socioeconomic status (KW=13.81, 22.69, and 13.01, all p<0.01), higher paternal education (KW=54.11, 17.11, and 44.83, all p<0.001), and higher maternal education were positively associated with the observed sport factors (KW=85, 11, 28.34, and 108.54, all p<0.001; for team sport, individual sport participation, and experience in sport, respectively). Sociodemographic variables are strong determinants of sport participation in children, and intensive efforts are needed to build similar opportunities for involvement in sports for all children in Kosovo.

Keywords: *puberty, involvement in sports, gender, socioeconomic status, education level*

Introduction

It is globally accepted that physical activity and sport are of utmost importance for children and adolescents (Marinho et al., 2022). The first set of reasons is related to physical health benefits. In brief, activities such as running, jumping, and playing sports help build and maintain strong bones and muscles, setting a good foundation for future growth and development. Regular sports and exercise help children burn calories and maintain a healthy weight, improve overall cardiovascular fitness and reduce the risk of heart disease later in life, whereas activities that involve coordination and balance, such as dancing, swimming, and playing sports, help refine motor skills and improve overall physical coordination (Janssen & Leblanc, 2010; Manojlovic et al., 2023). Furthermore, physical activity in children is related to several social benefits, including opportunities for socialization. Additionally, many sports activities require teamwork and cooperation, teaching children these valuable life skills. Finally, physical activity in childhood has long-term benefits, such as a reduced risk of chronic diseases later in life (i.e., heart disease, type 2 diabetes, and certain types of cancer), whereas encouraging physical activity at a young age helps children develop healthy habits that can last a lifetime (Biddle & Asare, 2011; Yu et al., 2020).

Because of the all previously said, global health authorities

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are concerned with ways to improve physical activity and sport participation in children. Various strategies are employed, aiming to address barriers and create supportive environments that encourage active lifestyles. Depending on the country, region, tradition and resources, different approaches are employed, including the development of national guidelines and recommendations, the implementation of policies that support physical activity, advocacy for funding and investment in sport, school-based interventions (i.e., increasing physical education, providing facilities), and community-based initiatives (developing parks, playgrounds, etc.) (Bassett-Gunter et al., 2017; Hrg et al., 2023). However, it seems that family and individual interventions should be considered the most effective, simply because of their specificity and availability. Similarly, studies have reported the strong influence of various parental and familial factors on the physical activity and sport participation of children (Lloyd et al., 2014; Lopez et al., 2022). While all interventions should be designed and tailored specifically, the knowledge of factors, including sociodemographic indicators associated with physical activity and sports (sociodemographic correlates), is crucial.

In general, sociodemographic correlates of physical activity and sport in children refer to the various social and demographic factors that are associated with a child's level of physical activity and sport participation (Lammle et al., 2012; Sterdt et al., 2014). Understanding these correlates can help us identify groups of children who may be at greater risk of inactivity and design interventions to promote physical activity and sport participation in these populations. To date, studies have explored these problems and reported relatively consistent results. For example, there is a clear association between gender and physical activity, with boys tending to be more physically active than girls are (Sterdt et al., 2014). Sports participation and physical activity levels generally decline with age throughout childhood and adolescence. Children from lower socioeconomic backgrounds are often less involved in sports and less physically active than those from higher socioeconomic backgrounds are, potentially due to limited access to safe spaces for play and organized sports, fewer resources for equipment, or a lack of parental support for physical activity (Stalsberg & Pedersen, 2010). Children of parents with higher levels of education tend to be more involved in sports and more physically active, likely due to greater awareness of the importance of such activities and more resources to support them. Finally, encouragement and support from parents play a vital role in promoting sport participation and physical activity among children (Cadogan et al., 2014; Matijasevic et al., 2023).

Indeed, sport participation plays a crucial role in promoting physical activity among children, often serving as their primary source of exercise. Research has consistently demonstrated a strong positive association between sports participation and overall physical activity levels, with children involved in sports being more likely to meet recommended guidelines (Gilic et al., 2021; Hafsteinsson Ostenberg et al., 2022; Pharr & Lough, 2014). This is not surprising given that organized sports provide structured opportunities for regular physical activity, encouraging children to engage in exercise consistently. However, it is important to recognize that not all children have equal access to sports opportunities and that barriers such as socioeconomic factors and limited resources can hinder participation. Therefore, in addition to physical education (which is mandatory for healthy children and therefore similar for those involved in the scholastic system), observing sports as the most important source of physical activity during childhood is reasonable.

While participation in sports is known to be the main source of physical activity for children, knowledge of the correlates of sport participation could be directly translated into strategies aimed at increasing physical activity in children (Gilic et al., 2021; Hafsteinsson Ostenberg et al., 2022). Therefore, this study aimed to evaluate the associations between sociodemographic factors and different factors of sport participation in early adolescent children from Kosovo. Initially, we hypothesized that the observed sociodemographic factors would be positively associated with sport participation in the studied children.

Methods

The participants in this study were school-aged children from Kosovo (n=1813; 911 girls) aged 13.7±0.9 years. All participants regularly attended the 8th and 9th grades of elementary school in the territory of Kosovo. One week before the study participants were informed of the study aims and procedure, written consent was obtained from their parents. The response rate was greater than 90%. The study fulfilled all ethical guidelines and received the approval of the ethical boards of the University of Split, Faculty of Kinesiology. The survey was strictly anonymous, was administered to groups of at least 12 respondents and contained multiple-choice answers. Each respondent received the questionnaire form and one envelope. After completing the survey, the participants placed the questionnaire form in the envelope, sealed it and then placed it in the closed box.

Variables were collected via previously validated questionnaires (Sekulic et al., 2012) and included sociodemographic data and sport factors. Sports factors consisted of questions asking subjects about their involvement in team and individual sports (two separate questions and answers included never been involved, quit, currently involved) and time of their involvement in sport (never been involved, <1 year, 1--3 years, >3 years). Sociodemographic variables included questions on age (in years), gender (males vs. females), living community (participants were asked about specific places of residence, later grouped into urban vs. rural communities), familial socioeconomic status (below average, average, above average), and paternal and maternal education levels (both answered on a four-point scale: elementary school, high school, college degree, and university degree).

Descriptive statistics included calculations of means and standard deviations (for age) and counts and frequencies (for all remaining variables). To establish the associations between sociodemographic factors and sport factors, we calculated differences between specific groups of participants according to sociodemographic variables. Specifically, for gender and living community (two possible responses), the Mann-Whitney test of differences was performed. For socioeconomic status (three possible responses), paternal education, and maternal education (four possible responses), Kruskal-Wallis ANOVA was employed. Statistica 13.5 (Tibco Inc. Palo Alto, California, USA) was used, and a p-level of 95% was set for all analyses.

Results

The results of the descriptive statistics and differences in sport-related factors according to gender are presented in

Table 1. In general, boys and girls differed in all sport participation factors, with boys being more involved in individual sports (MW=7.33, p<0.001) and in team sports (MW=13.56, p<0.001). Additionally, boys had longer experience in sport (MW=18.23, p<0.001). In general, only 10% of boys reported no participation in sports at all (never participated), whereas more than one-third of girls were never involved in any kind of sport.

Table 1. Descriptive statistics and differences in sport participation factors between genders (MW–Mann–Whitney test of differences)

	Boys		Girls		MW (p)
	F	%	F	%	
Individual sport participation					7.33 (0.001)
No, never	362	40.22	513	56.31	
Yes, but quit	261	29.00	229	25.14	
Yes, still participating	276	30.67	169	18.55	
Missing	1	0.11	0	0.00	
Team sport participation					13.56 (0.001)
No, never	157	17.44	438	48.08	
Yes, but quit	255	28.33	204	22.39	
Yes, still participating	487	54.11	267	29.31	
Missing	0	0.11	2	0.21	
Experience in sport					18.23 (0.001)
Never participated	93	10.33	332	36.44	
< 1 year	259	28.78	369	40.50	
2-5 years	296	32.89	165	18.11	
> 5 years	252	28.00	45	4.94	
Missing	0	0.00	0	0.00	

When children living in urban communities were compared with those living in rural communities, clear differences in sport participation factors were found for all the variables. Children from urban communities were more involved in individual and team sports (MW=5.04, p<0.001, and MW=2.84, p<0.01, respectively) and were involved in sports for a longer period (MW=3.82, p<0.001) than were children from rural communities. For a simple comparison, 21% of the urban children were never involved in sports, whereas this was the case for more than 43% of the rural children (Table 2).

Table 2. Descriptive statistics and differences in sport participation factors between groups based on living community (MW–Mann–Whitney test of differences)

	Urban		Rural		MW (p)
	F	%	F	%	
Individual sport participation					5.04 (0.001)
No, never	770	46.39	107	69.93	
Yes, but quit	468	28.19	22	14.38	
Yes, still participating	421	25.36	24	15.69	
Missing	1	0.06	0	0.00	
Team sport participation					2.84 (0.01)
No, never	525	31.63	70	45.75	
Yes, but quit	431	25.96	29	18.95	
Yes, still participating	701	42.23	54	35.29	
Missing	3	1.81	0	0.00	
Experience in sport					3.82 (0.001)
Never participated	358	21.57	67	43.79	
< 1 year	594	35.78	35	22.88	
2-5 years	438	26.39	24	15.69	
> 5 years	270	16.27	27	17.65	
Missing	0	0.00	0	0.00	

Differences in sport participation factors among groups based on socioeconomic status are presented in Table 3. Once again, significant differences were found for all the study variables. Specifically, the lowest level of participation in individual sports was found for children who declared the "below average" socioeconomic status of the family (KW=13.82, p<0.01). A similar trend was observed for team sport participation (MW=22.69, p<0.001) and experience in sports (MW=13.01, p<0.01). In the group of children who reported "below average" socioeconomic status of the family, 32% never participated in sport, whereas this was the case for 24% of children in the "below average" group and <20% of children who self-declared "above average" financial status.

When participants were grouped on the basis of paternal education and then compared in terms of sport factors, signifi-

Table 3. Descriptive statistics and differences in sport participation factors between groups based on socioeconomic status (KW – Kruskal–Wallis ANOVA test)

	Below average		Average		Above average		KW (p)
	F	%	F	%	F	%	
Individual sport participation							13.82 (0.01)
No, never	28	56.00	665	50.26	184	42.01	
Yes, but quit	12	24.00	347	26.23	131	29.91	
Yes, still participating	10	20.00	310	23.43	123	28.08	
Missing	0	0.00	1	0.08	0	0.00	
Team sport participation							22.69 (0.001)
No, never	19	38.00	471	35.60	105	23.97	
Yes, but quit	9	18.00	335	25.32	115	26.26	
Yes, still participating	22	44.00	515	38.93	217	49.54	
Missing	0	0.00	2	0.16	1	0.23	
Experience in sport							13.01 (0.01)
Never participated	16	32.00	322	24.34	87	19.86	
< 1 year	17	34.00	472	35.68	140	31.96	
2-5 years	12	24.00	324	24.49	124	28.31	
> 5 years	5	10.00	205	15.50	87	19.86	
Missing	0	0.00	0	0.00	0	0.00	

Table 4. Descriptive statistics and differences in sport participation factors between groups based on paternal education (KW – Kruskal–Wallis ANOVA test)

	Elementary		High school		College degree		University degree		KW (p)
	F	%	F	%	F	%	F	%	
Individual sport participation									54.11 (0.001)
No, never	120	62.18	364	54.82	206	45.47	187	37.18	
Yes, but quit	39	20.21	168	25.30	128	28.26	155	30.82	
Yes, still participating	34	17.62	131	19.73	119	26.27	161	32.01	
Missing	0	0.00	1	0.15	0	0.00	0	0.00	
Team sport participation									17.11 (0.001)
No, never	85	44.04	235	35.39	140	30.91	135	26.84	
Yes, but quit	44	22.80	159	23.95	113	24.94	144	28.63	
Yes, still participating	64	33.16	267	40.21	200	44.15	224	44.53	
Missing	0	0.00	3	0.45	0	0.00	0	0.00	
Experience in sport									44.83 (0.001)
Never participated	74	38.34	161	24.25	102	22.52	88	17.50	
< 1 year	74	38.34	237	35.69	152	33.55	166	33.00	
2-5 years	23	11.92	169	25.45	111	24.50	159	31.61	
> 5 years	22	11.40	97	14.61	88	19.43	90	17.89	
Missing	0	0.00	0	0.00	0	0.00	0	0.00	

cant differences among groups were found for all sport factors. Individual sport participation was lowest in children whose fathers were elementary educated only (KW=54.11, p<0.001), and similar results were found for team sport participation (KW=17.11, p<0.001) and experience in sports (KW=44.83, p<0.001). A clear trend of higher sport participation in children whose parents were better educated was obvious from the results of overall sport participation (38%, 24%, 22%, and 17% who declared that they never participated in sports for children whose fathers were elementary educated only, those whose fathers finished high school, college, and university, respectively) (Table 4).

When maternal education was used as a grouping variable, children differed in individual sport participation (KW=85.11, p<0.001), team sport participation (KW=28.34, p<0.001), and experience in sport (KW=108.54, p<0.001), with greater participation and longer experience in children whose mothers were better educated (Table 5).

Table 5. Descriptive statistics and differences in sport participation factors between groups on the basis of maternal education (KW – Kruskal–Wallis ANOVA test)

	Elementary		High school		College degree		University degree		KW (p)
	F	%	F	%	F	%	F	%	
Individual sport participation									85.11 (0.001)
No, never	200	63.69	371	52.92	178	45.29	127	31.44	
Yes, but quit	66	21.02	183	26.11	107	27.23	134	33.17	
Yes, still participating	48	15.29	147	20.97	108	27.48	142	35.15	
Missing	0	0.00	0	0.00	0	0.00	1	0.25	
Team sport participation									28.34 (0.001)
No, never	139	44.27	247	35.24	118	30.03	91	22.52	
Yes, but quit	68	21.66	165	23.54	102	25.95	125	30.94	
Yes, still participating	107	34.08	287	40.94	172	43.77	188	46.53	
Missing	0	0.00	2	0.18	1	0.25	0	0.00	
Experience in sport									108.54 (0.001)
Never participated	121	38.54	182	25.96	77	19.59	45	11.14	
< 1 year	117	37.26	252	35.95	137	34.86	122	30.20	
2-5 years	38	12.10	171	24.39	112	28.50	141	34.90	
> 5 years	38	12.10	96	13.69	67	17.05	96	23.76	
Missing	0	0.00	0	0.00	0	0.00	0	0.00	

Discussion

The results revealed several important findings. First, the significant correlation between gender and various factors of sport participation revealed greater sport participation among boys than among girls. Second, higher sport participation is found in urban children. Third, paternal education and socioeconomic status are strongly positively correlated with sport participation in early adolescent children from Kosovo. Therefore, our initial study hypothesis can be accepted.

Our findings of higher sport participation in boys than in girls are in agreement with the results of previous studies conducted globally. For example, in a U.S. study, boys had greater sport participation than girls did, especially White and Asian students did, which is consistent with the findings of another U.S. study in which boys had higher overall sport participation and physical activity levels than girls did (Hebert et al., 2015; Pharr & Lough, 2014). Asian and European studies also regularly reported that boys are more involved in sports than girls are (Emmonds et al., 2024), while there is conclusive evidence of higher participation in sports among boys than among same-age girls in the territory of southeastern Europe (Sunda et al., 2022; Tahiraj et al., 2016).

However, most of the previous studies have examined this issue in older adolescents; therefore, our findings are relatively novel from the perspective of the participants' age (e.g., we examined younger children than the majority of the previous studies did). Despite differences in sample characteristics between our study and previous studies, there is likely not much difference in factors that actually resulted in our results. First, sports are often perceived as a more masculine domain, and boys are encouraged to be active and competitive from a young age. Girls, on the other hand, may be steered toward less physically demanding activities or face societal pressures to prioritize appearance over athleticism (Plaza et al., 2017). Therefore, girls' lower participation in sports could be at least partially influenced by such gender stereotypes.

Further, girls may have fewer opportunities to participate in sports because of limited funding, fewer female coaches and role models. One specific reason could be the lower emphasis on girls' sports programs in schools and communities. In our experience, this is particularly possible in Kosovo simply because "female-oriented sports" (i.e., gymnastics, dance) are not as common as those known to be more popular in boys (i.e., team-sport games, martial arts, etc.). Next, concerns about safety, harassment, and body image generally discourage girls (and their parents) from participating in sports, especially during childhood and adolescence. Specifically, since parents play a crucial role in encouraging their children to participate in sports, some of them may be less supportive of their daughters participating in sports because of concerns about injury, time commitment, or gender norms (Brackenridge, 1998).

As with the previously discussed association between gender and sport participation, the finding of greater participation in sports among urban children than among their rural peers is also expected and well supported in previous studies (Zenic et al., 2020). In general, the greater degree of participation in sports for urban children than for their rural peers stems from a complex interplay of factors, primarily revolving around access, opportunities, and sociocultural norms. First, and probably most important, urban areas tend to have a denser concentration of sports facilities, organized leagues, and specialized training programs. This greater availability translates into easier access and more opportunities for children to engage in various sports (Zenic et al., 2020). This is related not only to variety programs but also to distance from home to sport facilities, which is another important factor of participation in sports during this time simply because of security issues and organizations.

Socioeconomic factors also play important roles in participation in sport, and urban areas often have a wider range of socioeconomic groups. This logically implies that there will be a significant number of financially better situated families that will be in a position to assure proper support for their children for sport participation (i.e., sport equipment, fees). Additionally, urban environments might place greater emphasis on organized sports as a means of recreation, social development, and even future opportunities. This sociocultural value can lead to increased parental encouragement and peer influence for children to participate in sports.

In contrast, rural areas have specific boundaries with respect to sport participation. First, and most importantly rural areas may have fewer sports facilities, and organized leagues may be less prevalent or cater to a limited range of sports. This is particularly specific for girls' participation in sports (please see the previous discussion on the influence of gender on sport participation and the underrepresentation of "female-oriented sports" in general, which are even more common in rural areas). Furthermore, lower average incomes in rural areas can pose a barrier to sport participation because of the costs involved. Additionally, long distances to sports facilities and programs can create logistical hurdles, particularly for families without reliable transportation (Kellstedt et al., 2021). Finally, in some rural communities, sports might not be as highly prioritized or culturally ingrained compared with other activities such as outdoor recreation or farm work, creating a specific set of limitations for sport participation in rural communities.

Although sociodemographic factors are frequently studied in relation to sport participation in children and adolescents, to the best of our knowledge, this is one of the first studies to examine this issue in the territory of Kosovo, especially considering the (i) relatively large number of tested participants, (ii) participants' young age, and (iii) variety of socioeconomic and sport factors observed. From this perspective, the associations between familial factors, as a specific group of sociodemographic factors, and sport participation are particularly interesting. While the association between (better) familiar socioeconomic status and (higher) prevalence of sport participation in children has been previously discussed, in the following text, we will provide an overview of the theoretical background of the association between parental education and children's sport participation.

In general, studies have confirmed a positive association between parental education and sport participation in children (Kantomaa et al., 2007; Koçak et al., 2002). Therefore, strong positive association between both paternal and maternal education and sport participation in Kosovar children are in line with the findings of previous research. In explaining such findings, several important factors should be elaborated.

First, it is generally accepted that (better) educated parents may be more likely to recognize the importance of sports for their children's physical and mental well-being, encouraging participation. As already stated, the benefits of sport participation in childhood and adolescence are numerous and include both health-related benefits and social benefits (Knight et al., 2017). As a result, better educated parents are more likely to be aware of these benefits and are more interested in supporting and encouraging their children to be involved in sports. Importantly, educated parents are often involved in physical activities and sports themselves (Moore et al., 1991). Therefore, active parents serve as positive role models for their children, fostering an active lifestyle. This is naturally connected to the fact that parents who are well educated are more likely to place special emphasis on education and overall development. In other words, parents with higher education levels may view sports as an integral part of their children's holistic development, promoting participation for its benefits beyond physical fitness.

Despite this greater awareness and value placed on physical activity and sport activities and their benefits for more educated parents, this positive correlation can be interpreted while focusing more on socioeconomic factors that are naturally connected to the educational status of parents. In brief, higher education levels of parents are often associated with higher socioeconomic status of the family. This may simply enable parents to afford sports equipment, fees, and transportation, facilitating their children's involvement in sports. However, it must be emphasized that while higher education is often linked to higher socioeconomic status, this is not always the case. The influence of parental education on sport participation may be intertwined with the family's financial resources and access to opportunities, cultural and social norms (i.e., expectations), individual child factors (i.e., interest, capabilities, health issues, motivation), and other factors. However, in large samples such as the one we observed in this study the general tendency toward greater sport involvement for children who are raised by better educated parents will be identified.

The most important limitation of this study is its cross-sectional design. Therefore, although some causalities can be intuitively interpreted (i.e., male gender or higher parental education are definitively "the causes" of higher sport participation, and not vice versa), for more profound analyses, prospective studies are warranted. Additionally, the participants self-reported all the variables; therefore, a certain tendency toward "socially desirable answers" could be expected. On the other hand, this is one of the first studies examining the factors associated with sport participation in Kosovar children. Additionally, we employed a relatively large number of participants, both from rural and urban communities, which are important strengths of the investigation.

Conclusion

The greater participation of boys in sports is a complex issue rooted in various sociocultural, biological, and psychological factors. Addressing this disparity requires challenging traditional gender norms, providing equal opportunities for girls in sports, and fostering a more inclusive and supportive environment for all children.

Urban children often benefit from a more conducive environment for sports participation, with greater access to facilities, programs, and a cultural emphasis on sports. Rural children, on the other hand, can face significant barriers to participation due to limited resources, transportation challenges, and differing cultural norms. Addressing these disparities through increased investment in rural sports can provide more equitable opportunities for all children to engage in sports.

Higher parental education was found to be associated with increased sport participation in children. However, this relationship is multifaceted and influenced by various socioeconomic, cultural, and individual factors. Regardless of the background, parental involvement and support, regardless of their education level, play crucial roles in fostering children's engagement in sports and promoting a healthy, active lifestyle.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Exercise in the Workplace: A Qualitative-Quantitative Study of Enjoyment

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Abstract

The present study will assess workers' perceptions of the benefits of exercise and their awareness of the usefulness of performing it during working hours. The sample consists of 40 employees, in a small and medium-sized business, equipped with a gymnasium that gives the opportunity to exercise internally for 12 months. Convenience sampling was used, having direct access to employees through contact with the owner. A specially developed questionnaire was then administered to participants to collect detailed information on their fitness habits, perceptions of the benefits of exercise, and awareness of the usefulness of performing exercise activities in the workplace. Post-administration of the questionnaire indicated that 87.5% believed that exercise was important for health, 82.5% believed that having the opportunity to exercise in the workplace would boost mental and physical well-being and productivity. Following the application of the ChiSquare test, a first significant relationship was found between employees' perceptions regarding the usefulness of implementing exercise in the work and non-work context (p=.041). A second relationship was found between the possibility of performing exercise at work and the desire to be more physically fit (p=.002). It is shown that the possibility of performing exercise at workplaces gives way to practice at least the recommended daily amount of movement, making workers perceive the benefits of this practice and its repercussions in the work context.

Keywords: *perception, wellness, questionnaire, company, productivity, sedentariness*

Introduction

The World Health Organization has identified the concept of health as "a state of complete physical, mental and social well-being," the common understanding of health as a disease-free status is bypassed (WHO, 2020). It follows that any program of intervention to safeguard well-being, must attend to the individual as a whole and to the socio-relational and environmental contexts, in which his or her life experience unfolds (Kickbusch, 2010). Scientific studies have highlighted the physical benefits of regular exercise for workers, including a reduction in the risk of cardiovascular disease, obesity and type 2 diabetes (Huang & Novak, 2020). A recent systematic review points out that corporate physical activity programmes help improve mental health, reducing stress and burnout among employees (Foster & Patel, 2021). In particular, regular physical activity is associated with a significant reduction in anxiety and depression symptoms (Lopez & Thompson, 2022). A specific study investigated the effect of exercise on office workers, showing that short active breaks during the working day can improve concentration and reduce mental fatigue (Carter & Brown, 2022). Health and well-being must be framed in an ecosystemic manner, thus on the balance of multiple dimensions social, physical, psychological, environmental, that relate to each other in complex ways (Baum et al., 2014). Work currently represents the activity that adult people engage in for most of their time conditioning various aspects of life even when one is finished performing one's duties (Williams et al., 2016). Thus, well-being depends, not only on the efficient management of health services, but especially on lifestyles and work, leisure time,

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and the condition of the environment and the healthiness of settings (Ducharme et al., 2020). Physical exercise is a crucial tool, considered indispensable by policy to promote shared values, correct lifestyles, social value and the right to well-being, values elevated to Constitutional status recently (Della Repubblica, 2012). However, a trend of decreasing physical activity in adulthood is noted; recent changes in modus vivendi have accelerated this phenomenon (D'Isanto et al., 2023). Most people with active working lives fail to exercise due to lack of time. The digital transition and the increase in smart-working due to corporate downsizing have certainly not been conducive in this regard, as also the great inventions of recent years have considerably reduced the physical effort expended to perform daily work tasks even in purely manual tasks (D'Elia et al., 2023). This situation has caused harm in workers and consequently in the performance of their activities. Based on the available data, between 40% and 60% of the EU population leads a sedentary lifestyle, with a further aggravation of those who are actively working (Lee et al., 2012). Through some everyday changes in business practices and policies, small and medium-sized businesses can become more movement-friendly environments (McEachan et al., 2020). The new industrialization based on digital, has scaled down burdensome work, but at the same time has produced new worker health issues due to low energy expenditure, making it below that recommended for a proper lifestyle (Church et al., 2020; Raiola et al., 2021). This condition is exacerbated by the fact that people, spending most of the day working, have neither time nor desire to exercise (Barr-Anderson et al., 2011). In addition to this, some work tasks can promote the occurrence of various musculoskeletal disorders (Mayer et al., 2013). In this sense, it would be a good practice to introduce programs in companies that allow employees to carry out exercise routines, so as to compensate for the reduced daily energy exertion and use all muscle districts, not only those used in their tasks, also increasing the level of their aerobic threshold (Pedersen et al., 2009). The benefits of such investments extend not only to employees, who experience a better quality of life and greater well-being, but also to the companies themselves, which can observe increased productivity, reduced absenteeism, and improved corporate image (von Thiele Schwarz et al., 2011). The achievement of better mental and physical fitness by employees, would reduce sick leave, reducing risk factors with a better cost-benefit ratio for the company (Altavilla et al., 2018). In the scientific literature, however, there are no studies specifically investigating the perceptions of the main internal stakeholders, i.e. workers, regarding exercise in the workplace. The present study aims to assess the impact in terms of workers' perceptions of the benefits of exercise performed in the work context.

Materials and methods

Study participants

The sample used for the research consists of 40 employees working a Small and Medium Enterprise (SME). Sampling was conducted through a convenience methodology, due to the availability of direct access to employees, facilitated by contact with the employer. Employees ranged in age from 20 to 60 years. All subjects gave their informed consent for inclusion before they participated in the study. The study was conducted in accordance with the Declaration of Helsinki. According to Regulation (EU) 536/2014 and Directive 2001/20/EC, research involving minimal risks to participants may be exempt from formal ethical review, as it does not involve invasive or experimental interventions. Additionally, under Legislative Decree No. 211 of 24 June 2003, research that does not pose significant risks and solely aimed at improving educational practices might be exempt from Institutional Review Board (IRB) or Ethics Committee review and approval.

Study design

This SME has an in-house gymnasium, thus providing employees with the opportunity to exercise within the workplace. Employees for the past 12 months have had the opportunity to use the company gym in a functional manner. An exercise program was set up for them, taking advantage of digital tools such as recorded lectures that allow access to exercise protocols, with the guidance of a personal trainer hired specifically to check that the exercises are being carried out correctly and to motivate people during practice. The questionnaire was developed ad hoc, as there are no previous studies that have addressed this issue. Questions were formulated to investigate the phenomenon in depth and then administered to the participants via individual interviews. The questionnaire is divided into two parts: in the first part, it focuses on collecting information about the sample's habits with respect to exercise; in the second part, it focuses instead, on their perceptions regarding the benefits of exercise in the work context. The questions in the questionnaire provide predefined response options; since they were designed to assess perceptions, no numerical scores were assigned, but the answers given were simply recorded. The whole is represented in Table 1.

Table1. Items of the questionnaire

Questions	Answers
1) How many hours per week do you devote to physical activity outside of work?	(<1)-(1 to 3)-(3 to 6)-(>6)
2) Have you participated in workplace physical activity programs in the past?	Yes-No
3) Do you feel that exercise is important for your overall health?	Very-Fairly-Slightly-Not at all
4) Do you feel that the exercise protocols promoted within your company are helpful?	Very-Fairly-Slightly-Not at all
5) Does having the opportunity to exercise in the workplace boost mental and physical well-being and productivity in your opinion?	Very-Fairly-Slightly-Not at all
6) How many hours per week do you devote to physical activity in the workplace?	Very-Fairly-Slightly-Not at all
7) Obstacles in performing exercise in the workplace concern:	Lack of time-Lack of motivation-Protocols ill-suited to your needs-Fatigue
8) Would you like to be more physically fit?	Very-Fairly-Slightly-Not at all

Statistical analysis

For the statistical analysis, descriptive statistics of the qualitative variables that emerged from the answers to the questionnaire were first calculated, including frequencies, percentages and distributions for each item. Subsequently, the chi-square test was used to examine any significant associations between the variables in order to identify relevant relationships between the participants' perceptions. The statistical analysis was conducted using JASP software (version 0.16.4; University of Amsterdam, Amsterdam, Netherlands), a user-friendly open-source programme for statistical analysis, developed by the research team of the University of Amsterdam and freely available on the official website: https://jasp-stats.org. The software made it possible to perform the analysis accurately and easily. The level of statistical significance was set at $p<0.05$, and any significant relationships between the variables were interpreted in terms of both practical and statistical significance.

Results

After the questionnaire was administered, the results were found to be as shown in Table 2.

Table 2. Results of questionnaire responses

Questions	Answer 1	Answer 2	Answer 3	Answer 4
1. How many hours per week do you devote to physical activity outside of work?	<1	1 to 3	3 to 6	>6
Frequency	30	5	4	1
Percentage	75%	12.5%	10%	2.5%
2. Have you participated in workplace physical activity programs in the past?	YES	NO		
Frequency	2	38		
Percentage	5%	95%		
3. Do you feel that exercise is important for your overall health?	Very	Fairly	Slightly	Not at all
Frequency	35	4	1	0
Percentage	87.5%	10%	2.5%	0%
4. Do you feel that the exercise protocols promoted within your company are helpful?	Very	Fairly	Slightly	Not at all
Frequency	25	10	3	2
Percentage	62.5%	25%	7.5%	5%
5. Does having the opportunity to exercise in the workplace boost mental and physical well-being and productivity in your opinion?	Very	Fairly	Slightly	Not at all
Frequency	33	5	1	1
Percentage	82.5%	12.5%	2.5%	2.5%
6. How many hours per week do you devote to physical activity in the workplace?	<1	1 to 3	3 to 6	>6
Frequency	5	28	7	0
Percentage	12.5%	70%	17.5%	0%
7. Obstacles in performing exercise in the workplace concern:	Lack of time	Lack of motivation	Protocol suited to needs	Fatigue
Frequency	15	10	5	10
Percentage	37.5%	25%	12.5%	25%
8. Would you like to be more physically fit?	Very	Fairly	Slightly	Not at all
Frequency	5	33	2	0
Percentage	12.5%	82.5%	5%	0%

To the first question 'How many hours a week do you devote to physical activity outside of work?' 75% answered less than 1 hour; 12.5% 1 to 3 hours; 10% 3 to 6 hours; 2.5% more than 6 hours. To the second question 'Have you participated in workplace physical activity programs in the past?' 5% answered yes while 95% answered no. Continuing with the third question 'Do you feel that exercise is important for your overall health?' 87.5% responded with a lot; 10% quite a lot and the remaining 2.5% with a little. To the fourth question 'Do you think the exercise protocols promoted within your company are useful?'62.5% answered a lot; 25% quite a lot; 7.5% a little; and the remaining 5% not at all. Next, with to the fifth question 'Does having the opportunity to exercise in the workplace boost mental and physical well-being and productivity in your opinion?' 82.5% answered a lot; 12.5% answered enough; 2.5% with a little. To the sixth question 'How many hours a week do you devote to physical activity in the workplace?'12.5% answered less than 1 hour; 70% 1 to 3 hours; 17.5% 3 to 6 hours. In the seventh question 'Obstacles in performing exercise in the workplace concern:'

37.5% indicated lack of time; 25% lack of motivation; 12.5% poorly fitting protocols and 25% fatigue. Finally, to the eighth question 'Would you like to be more physically fit?' 12.5% answered a lot, 82.5% quite a lot, and 5% a little. Following the Chi-square analysis, two significant relationships were found among the qualitative variables. The first relationship is for question 4, 'Do you think the exercise protocols promoted within your company are helpful?' and question 3, 'Do you think exercise is important for your overall health?' with p=.041. All are shown in Tables 3 and 4.

Table 3. Contingency table between question 4 and question 3

	Do you feel that exercise is important for your overall health?			
Do you feel that the exercise protocols promoted within your company are helpful?	Slightly	Fairly	Very	Total
Not at all	0	0	2	2
Slightly	0	1	9	10
Fairly	0	3	22	25
Very slightly	1	0	2	3
Total	1	4	35	40

Table 4. Chi Square Analysis

	Value	**df**	**p**
X^2	13.126	6	0.041
N	40		

The second significant relationship is between question 5, 'Does having the opportunity to workout in the workplace boost mental and physical well-being and productivity in your opinion?" and question 8, 'Would you like to be more physically fit?" where it was found to be p=.002. The whole can be observed in tables 5 and 6.

Table 5. Contingency table between question 5 and question 8

	Would you like to be more physically fit?			
Does having the opportunity to exercise in the workplace boost mental and physical well-being and productivity in your opinion?	**Slightly**	**Fairly**	**Very**	**Total**
Not at all	1	0	0	1
Slightly	0	1	0	1
Fairly	0	5	0	5
Very slightly	1	27	5	33
Total	2	33	5	40

Table 6. Chi Square Analysis

	Value	**df**	**p**
X^2	20.716	6	0.002
N	40		

Discussion

In recent years, numerous studies have highlighted the importance of exercise for individual well-being, a concept deeply rooted in the bio-psycho-social model (Raiola et al., 2021). The results in this study confirm these inferences, showing that 87.5% of the respondents recognise the importance of exercise for overall well-being and a significant proportion believe that opportunities for physical activity in the workplace can lead to improved mental and physical health. This is in line with Proper's findings that demonstrated how access to workplace exercise facilities enables employees to achieve recommended daily levels of exercise (Proper et al., 2018). Similarly, Raiola emphasised the bio-psycho-social benefits of regular exercise, highlighting how physical activity improves mood, cognitive ability and overall life satisfaction (Raiola et al., 2021). However, the demonstrated evidence deviates slightly from Toker's study, in which it was found that employees who participated in workplace exercise programmes devoted more than 3 hours per week to physical activity (Toker et al., 2011). In this study, only 12.5% of the participants reached this threshold, while 70% devoted between 1 and 3 hours per week. This discrepancy could be attributed to differences in corporate culture, time availability or organisational policies in different sectors and suggests that although employees are generally willing to exercise, various obstacles such as time constraints, lack of motivation and family commitments prevent them from increasing the duration of physical activity. This result is in line with Raiola and his research group's assertion that external pressures, such as family obligations and fatigue, significantly reduce the likelihood of regular physical activity during the working week (Raiola et al., 2020). The questions asked revealed that 95% of employees want to be more physically fit, indicating that the need to move is beginning to take root in people's culture, as it is closely linked to positive implications on several fronts. The questionnaire analysis found a significant correlation between the usefulness of implementing exercise protocols in the company and their relevance to overall health (p=.041), suggesting that the increase in such good practices is positively perceived by staff. A second significant correlation was found between improved mental and physical well-being and productivity, and the desire to be more physically fit (p=.002). It is evident, once again, that good physical fitness produces a positive feedback on the state of workers, improving productivity. Therefore, companies should seriously consider investing in facilities and protocols that facilitate physical exercise, given their potential in improving work performance and reducing health-related absenteeism (D'Isanto et al., 2019). Companies that invest in facilities and programmes to promote exercise can benefit from a healthier, more productive and satisfied workforce. This investment not only improves employees' quality of life, but can also lead to significant economic and reputational benefits for the company (Parks et al., 2022). The demands of the modern work culture, especially in high-pressure environments, can limit employees to make full use of available movement resources. Furthermore, the cultural shift towards prioritising movement and well-being is relatively recent and may take longer for exercise to become a more ingrained habit during the working day. Limitations of this study include the limited sample size, which may not be representative of all work contexts, and the lack of an in-depth analysis of differences between work sectors, which could influence the willingness and propensity to exercise. In terms of practical implications, companies should consider adopting policies that actively encourage exercise, such as creating dedicated exercise spaces and scheduling active breaks during the working day. It would be beneficial for the long-term sustainable success of the company to promote a company policy that values the physical and psychological well-being of employees and adopts accessible equipment, spaces and protocols in the workplace. In addition, providing incentives or support for the adoption of healthier lifestyles, for example through company exercise programmes, can help improve productivity and reduce sick leave.

Conclusion

Using practices such as exercise in the workplace proves to be an effective strategy for improving employees' mental and physical well-being. The study highlights how workers perceive the benefits of exercise not only for physical health, but also for psychological well-being and work productivity. This evidence is supported by the significant correlations arising from the statistical surveys conducted. Companies should seriously consider implementing exercise facilities and programs for their employees, as such initiatives can result in a more efficient and productive workforce. Investing in employee wellness not only improves quality of life, but also brings economic and reputational benefits to the company. It would also be interesting to conduct longitudinal studies to monitor the effects of integrating exercise into the workplace over time. Such studies could examine not only short-term benefits, but also long-term impacts on employee well-being and corporate performance. Promoting physical activity in the workplace proves to be a winning strategy for improving employee well-being and productivity, contributing to sustainable business success. Thus, integrating physical exercise in the workplace has potential multidimensional benefits for employees and organizations. Future research could expand the understanding of these benefits and support the development of evidence-based policies and practices to promote well-being at work.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Effects of Yoga and Combined Yoga with Neuro-Linguistic Programming on Psychological Management in Mothers of Adolescents: A Randomized Controlled Trial

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Abstract

Adolescent parenting presents significant challenges for mothers, often leading to elevated levels of stress and anxiety that can adversely affect their well-being and parenting effectiveness. This study aims to evaluate the efficacy of yoga alone and in combination with Neuro-Linguistic Programming (NLP) in managing stress and anxiety among mothers of adolescent children. In this randomized controlled trial, 90 participants aged 35-55 years (mean age 44.56±4.58 years), each with at least one child aged 13-19 years, were randomly assigned to one of three groups: control, yoga, or yoga with NLP. Interventions were conducted over 12 weeks, with outcome measures assessed pre- and post-intervention by trained research assistants blinded to group allocation. The Depression, Anxiety, and Stress Scale (DASS-21), and Pittsburgh Sleep Quality Index (PSQI), were utilized to evaluate outcomes. Both intervention groups demonstrated significant reductions in depression, anxiety, and stress levels compared to the control group. The yoga with NLP group exhibited superior improvements across all primary outcomes, with statistically significant differences noted in depression (mean difference =7.1, p<0.001), anxiety (mean difference =5.1, p<0.001) and stress levels (mean difference =5.5, p<0.001). Additionally, sleep quality improved significantly in both intervention groups, with the yoga with NLP group showing greater benefits. This study provides evidence that yoga, particularly in combination with NLP, is an effective non-pharmacological approach for reducing stress and anxiety and improving sleep quality among mothers of adolescents. These findings support the integration of mind-body practices into mental health care, highlighting the potential synergistic benefits of combining physical and cognitive interventions. Future research should explore long-term effects and the mechanisms underlying these improvements.

Keywords: *yoga, neuro-linguistic programming, adolescent, depression, stress*

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Introduction

Adolescent parenting presents significant challenges for mothers, often leading to elevated levels of stress and anxiety that can adversely affect their well-being and parenting effectiveness (Charan & Kumar, 2024; Cowling & Van Gordon, 2022; Sharma & Sharma, 2024). Mothers of adolescents experiencing high stress and anxiety may struggle to support their children while managing personal and professional responsibilities (Erfina et al., 2019; Fegert et al., 2020). This can result in ineffective parenting, family conflicts, and strained mother-child relationships (Dotterer et al., 2021). Therefore, there is a critical need for effective interventions to help mothers manage stress and anxiety, thereby improving their quality of life and overall well-being (Shorey et al., 2023).

Yoga is a holistic practice that integrates physical postures, breathing exercises, and mindfulness meditation to enhance physiological and psychological well-being (Charan & Kumar, 2024; Jagadeesan et al., 2022; Mitra, 2023). Multiple studies have demonstrated yoga's ability to reduce stress, anxiety, and depression across various populations, making it a popular choice for stress management (Maheshkumar et al., 2021; Padmavathi et al., 2023; Wang & Szabo, 2020). Beyond physical fitness, yoga promotes mental clarity, emotional stability, and a balanced quality of life by focusing on mindful breathing and providing a natural counterbalance to the stress response (Ong Gaffney et al., 2023; Yan, 2024).

Simultaneously, Neuro-Linguistic Programming (NLP) is a psychological technique that analyzes the relationship between neurological function, language, and behavioral responses (Irsyad & Casmini, 2023; Mehdi et al., 2022). NLP aims to enhance personal efficiency by helping individuals identify their personality traits and alleviate unproductive thoughts and behaviors (Lemieux, 2020). Techniques such as anchoring, cognitive remodeling, and stress management are implemented to promote emotional stability, interpersonal skills, and problem-solving capacity (Gran, 2021; Liff et al., 2024). While NLP has been applied in various contexts like therapy, coaching, and personal development, its potential in combination with yoga remains underexplored (Passmore & Rowson, 2019).

The combined nature of yoga and NLP, where yoga provides physical and mental ease while NLP offers tools for cognitive and emotional restructuring, provides a strong rationale for integrating these practices to improve stress and anxiety management (Akdeniz & Kaştan, 2023; Ybias et al., 2024). Yoga has been shown to alleviate stress and reduce depressive symptoms by positively impacting inflammatory markers, autonomic balance, neurotransmitters, and the HPA axis, making it an ideal complementary therapy for mental health disorders, particularly when used alongside conventional treatments (Martínez-Calderon et al., 2023; Padmavathi et al., 2023). Previous studies also reported that NLP is effective in reducing psychological distress for the patients with various health conditions (Doğan et al., 2022). However, studies on the combined benefits of yoga and NLP, particularly among mothers of adolescents, are sparse. Understanding whether the combination of NLP with yoga can provide additional benefits beyond those achieved through yoga alone is crucial for developing more comprehensive and effective treatments.

This randomized controlled study aims to examine the effects of yoga alone and in combination with NLP on stress and anxiety management in mothers of adolescent children.

Methods

Study design and participants

This randomized controlled trial investigated the effects of yoga and combined yoga with Neuro-Linguistic Programming (NLP) on stress and anxiety management in mothers of adolescent children. The study enrolled 90 participants, aged 35-55 years, (mean age 44.56±4.58 years), each with at least one child between 13-19 years old. Participants were randomly assigned (Figure 1) to one of three groups: control, yoga, or yoga with NLP, with 30 individuals per group, using a lottery method (See Figure 1). The study received approval from the Ethics Committee at Meenakshi Academy of Higher Education & Research in Tamil Nadu (Approval No. MMCH&RI IEC/PhD/02/JAN/2023). All study procedures adhered to the most recent version of the Declaration of Helsinki.

Sample size calculation

The sample size was determined using G*Power software version 3.1.9.2 (Kang, 2021). The calculation was based on a one-way ANOVA with three groups, assuming a medium effect size (f=0.50), an alpha level of 0.05, and a desired power of 0.80. To account for potential dropouts and ensure adequate power for subgroup analyses, the researchers increased the sample size by approximately 20%, resulting in a total target sample of 90 participants, with 30 individuals allocated to each of the three groups.

Inclusion and exclusion criteria

Inclusion criteria required participants to understand

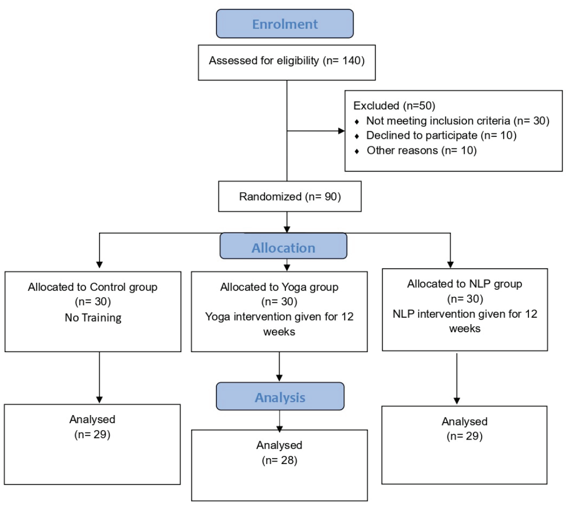

FIGURE 1. CONSORT Flow Chart

and follow instructions in the local language and to be willing to participate for the full study duration. Exclusion criteria included prior experience with yoga or NLP, current participation in other stress management programs, diagnosed psychiatric disorders requiring medication, and physical conditions that would interfere with yoga practice.

Interventions

Yoga Group: Participants in the Yoga Group engaged in 60-minute sessions three times weekly for 12 weeks. The yoga program included a variety of loosening exercises, asanas, pranayama, and meditation techniques. Loosening exercises consisted of Cat-Cow Breathing, Vaksha Sakti Vikasha, and Surya Namaskarasana from the Bihar School of Yoga tradition, followed by Sashangasana (Child Pose), Ustrasana, Matsyasana, Setubandhasana, Bhujangasana, and Makarasana to enhance flexibility and spinal mobility. Pranayama practices focused on Shithali, Bhramari, and Nadi Suddhi, while mental chanting of A-U-M was also incorporated (Jagadeesan et al., 2021; Kuppusamy et al., 2020; Rajagopalan et al., 2023; Thanalakshmi et al., 2020).

Yoga with NLP Group: This group received the same yoga program as above, supplemented with weekly 90-minute NLP sessions for 12 weeks. The NLP program emphasized cognitive restructuring, stress management techniques, and communication skills to enhance parenting competence (See Table 1).

Table 1. Intervention details for Yoga and NLP program

Training Component	Duration	Frequency
Loosening Exercises		
Cat-Cow Breathing	5 minutes	3 times/week
Vaksha Sakti Vikasha	5 minutes	3 times/week
Surya Namaskarasana	10 minutes	3 times/week
Asanas		
Sashangasana (Child	5 minutes	3 times/week
Ustrasana	5 minutes	3 times/week
Matsyasana	5 minutes	3 times/week
Setubandhasana	5 minutes	3 times/week
Bhujangasana	5 minutes	3 times/week
Makarasana	5 minutes	3 times/week
Pranayama|	5 minutes	3 times/week
Shithali Pranayama|	5 minutes	3 times/week
Bhramari Pranayama	5 minutes	3 times/week
Nadi Suddhi	5 minutes	
Meditation		
A-U-M Chanting	10 minutes	3 times/week
NLP Techniques		
Anchoring	30 minutes	1 times/week
De-Busting	30 minutes	1 times/week
Swish	30 minutes	1 times/week

Control Group: Participants in this group received no intervention but were offered the opportunity to participate in yoga and NLP sessions after the study's completion.

Procedure

Prior to the 12-week intervention, participants in both active groups attended two 90-minute familiarization sessions to introduce basic concepts, demonstrate techniques, and address questions. All participants received structured log books to record daily stress levels, significant parenting-related events, and the practice of techniques outside scheduled sessions. The yoga and yoga with NLP groups also logged their session attendance. These log books were collected and reviewed biweekly by the research team to monitor compliance, gather qualitative data, and encourage participant engagement and self-reflection throughout the study period.

Outcome parameters

This study employed three different scales to assess stress, anxiety, sleep quality, and quality of life among participants.

Depression, Anxiety, and Stress Scale (DASS-21): The DASS-21 is a widely used self-report questionnaire designed to assess symptoms of depression, anxiety, and stress over the past week. It consists of 21 items, with scoring calculated by summing the relevant items and multiplying by two to align with the full-scale version. Higher scores indicate increased levels of depression, stress and anxiety (Medvedev, 2023).

Pittsburgh Sleep Quality Index (PSQI): The PSQI is a self-report questionnaire that evaluates sleep quality and disturbances over the past month. It comprises 19 items organized into seven components: sleep duration, sleep disturbances, sleep latency, daytime dysfunction, sleep effectiveness, sleep quality, and frequency of sleep medication use. The global score ranges from 0 to 21, with scores above 3 indicating worse sleep quality and scores above 5 reflecting poor sleep quality (Zitser et al., 2022).

Data collection and analysis

Demographic data were collected at baseline, and all outcome measures were assessed pre- and post-intervention by trained research assistants who were blinded to group allocation. Data analysis was conducted using SPSS version 25.0 (SPSS Inc., Chicago, IL, USA). Descriptive statistics were calculated for all variables, while between-group differences were analyzed using one-way ANOVA for continuous variables followed by post hoc TukeyHSD test. Statistical significance was set at p<0.05.

Results

A total of 90 mothers of adolescent children, aged 35-55 years (mean age 44.56±4.58 years), participated in this study. Baseline demographic characteristics and initial scores on outcome measures were comparable across all groups, with no statistically significant differences observed (Table 2).

Table 2. Baseline demographical details of the study participants

Characteristic	Overall N = 90	Control N = 30	Yoga N = 30	Yoga with NLP N = 30	p-value
Age (yrs)	44.38 (5.14)	44.03 (2.86)	43.90 (3.01)	45.20 (7.92)	>0.9
Height (cm)	153.41 (4.03)	154.03 (4.05)	153.00 (3.84)	153.20 (4.26)	0.5
Weight (kg)	68.58 (8.73)	69.54 (10.13)	67.46 (7.90)	68.73 (8.15)	0.8
Qualification					0.9
Degree and above	54 (60%)	18 (60%)	18 (60%)	18 (60%)	
Graduate	33 (37%)	12 (40%)	11 (37%)	10 (33%)	
HSC	3 (3.3%)	0 (0%)	1 (3.3%)	2 (6.7%)	
Occupation					>0.9
House wife	9 (10%)	2 (6.7%)	4 (13%)	3 (10%)	
Private sector	52 (58%)	19 (63%)	16 (53%)	17 (57%)	
Public sector	10 (11%)	3 (10%)	3 (10%)	4 (13%)	
Self Employed	19 (21%)	6 (20%)	7 (23%)	6 (20%)	

Analysis of the primary outcomes revealed significant improvements in depression, stress and anxiety levels among participants in the intervention groups (Table 2). The yoga group demonstrated a substantial decrease in depression scores (mean difference [MD]=5.8, 95% CI [-7.2 to -4.4], p<0.001) compared to the control group. The yoga with NLP group exhibited an even more pronounced reduction in depressive symptoms (MD=7.7, 95% CI [-9.1 to -6.3], p<0.001) relative to the control group. Moreover, the comparison between the two intervention groups showed that the yoga with NLP group achieved a significantly greater reduction in depression scores than the yoga-only group (MD=1.5, 95% CI [-2.9 to -0.1], p=0.038). Stress levels, as measured by the DASS-21 Stress subscale, showed a marked decrease in both the yoga group (mean difference [MD]=4.1, 95% CI [-5.3 to -2.9], p<0.001) and the yoga with NLP group ([MD]=5.5, 95% CI [-6.7 to -4.3],

Table 3. Comparison of baseline outcome parameters between the groups

	Control Group	Yoga Group	Yoga + NLP Group	P value
DAS 21- Depression				
Pre-intervention	22.4 ± 5.2	23.1 ± 4.9	22.8 ± 5.1	0.23
Post-intervention	21.9 ± 5.3	17.3 ± 4.2	15.1 ± 3.8	0.001
Mean change (95%CI)	0.5 (-0.6 to -0.13)	5.8 (-7.2 to -4.4)	7.7 (-9.1 to -6.3)	
DASS 21- Stress				
Pre-intervention	15.7 ± 4.1	16.2 ± 3.8	15.9 ± 4.0	0.55
Post-intervention	15.3 ± 4.2	12.1 ± 3.3	10.4 ± 2.9	0.001
Mean change	0.4 (-1.5 to 0.7)	4.1 (-5.3 to -2.9)	5.5 (-6.7 to -4.3)	
DASS 21 Anxiety				
Pre-intervention	18.5 ± 4.8	19.0 ± 4.5	18.7 ± 4.6,	0.23
Post-intervention	18.1 ± 4.7	14.6 ± 3.9	12.8 ± 3.5	0.001
Mean change	0.2 (-0.9 to 0.5)	3.7 (-5.3 to -2.1)	5.1 (-6.7 to -3.5)	
PSQI				
Pre-intervention	8.9 ± 2.3	9.1 ± 2.1	9.8 ± 2.2	0.88
Post-intervention	8.7 ± 2.4	7.8 ± 1.9	6.9 ± 1.7	0.001
Mean change	0.5 (-0.7 to -0.4)	2.1 (-3.0 to -1.2)	2.7 (-3.6 to -1.8)	

p<0.001) compared to the control group (Figure 2). The yoga with NLP group demonstrated a greater reduction in stress levels compared to the yoga-only group, and this difference was statistically significant (MD=1.9, 95% CI [-3.7 to -0.1], p=0.03).

Similarly, anxiety levels assessed using the DASS-21 Anxiety subscale significantly decreased in both intervention groups. The yoga group showed a reduction of 3.7 points (95% CI [-5.3 to -2.1], p<0.001), while the yoga with NLP group demonstrated a more pronounced decrease of 5.1 points (95% CI [-6.7 to -3.5], p<0.001) compared to the control group. The difference in anxiety reduction between the two intervention groups was statistically significant (MD=1.4, 95% CI [-2.8 to -0.1], p=0.041) (Table 3).

Secondary outcomes also showed promising results (Figure 3). Sleep quality, as measured by the PSQI, improved significantly in both intervention groups compared to the control group. The yoga group's PSQI scores decreased by 2.1 points (95% CI [-3.0 to -1.2], p<0.01), while the yoga with NLP group showed an even greater improvement with a reduction of 2.7 points (95% CI [-3.6 to -1.8], p<0.001). The difference between the two intervention groups was not statistically significant (MD=0.6, 95% CI [-1.3 to 0.1], p=0.089). Adherence to the interventions was high, with 90% of participants in the yoga group and 93% in the yoga with NLP group attending at least 80% of the scheduled sessions. No serious adverse events were reported during the study period.

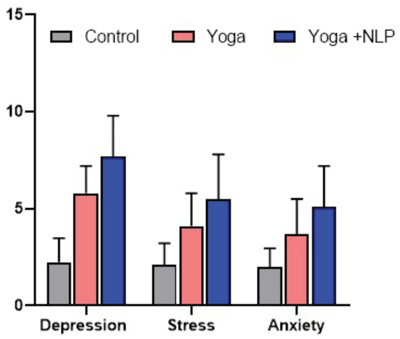

FIGURE 2. Changes in DASS 21 after the intervention among the groups

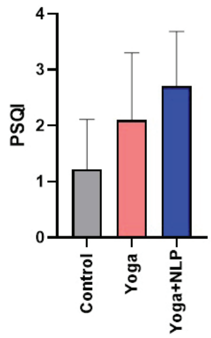

FIGURE 3. Changes in PSQI after the intervention among the groups

Discussion

In this randomized clinical trial involving 90 mothers of adolescent children, we found that both yoga and yoga combined with Neuro-Linguistic Programming (NLP) interventions significantly reduced symptoms of depression, anxiety, and stress compared to a control group. The combined yoga and NLP intervention demonstrated superior efficacy across all primary outcomes. Additionally, both intervention groups showed significant improvements in sleep quality, a key secondary outcome.

These findings suggest that yoga, either alone or in combination with NLP, may be an effective non-pharmacological approach for improving mental health and sleep quality in middle-aged mothers. The more pronounced improvements observed in the yoga with NLP group indicate that the integration of cognitive techniques with physical practice may offer synergistic benefits.

Our results align with previous meta-analytic findings that have demonstrated the efficacy of yoga in reducing symptoms of depression, anxiety, and stress (Breedvelt et al., 2019). They included 23 randomized controlled trials, encompassing 1,373 participants, to examine the effects of meditation, yoga, and mindfulness on depression, anxiety, and stress in tertiary education students. Findings revealed moderate overall effects (g=0.42-0.46) onhese symptoms. However, it's important to note that these effects diminished substantially when compared to active controls (g=0.13). The authors emphasized caution in interpreting these results, as most included studies had a high risk of bias, lacked safety reporting, and rarely assessed academic achievement.

However, to our knowledge, this is the first randomized controlled trial to examine the combined effects of yoga and NLP on mental health outcomes in this specific population. The magnitudes of improvement observed in our study are comparable to or exceed those reported in meta-analyses of yoga interventions for mental health.

Several mechanisms may underlie the observed benefits. Yoga has been shown to modulate the hypothalamic-pituitary-adrenal (HPA) axis and reduce inflammation, both of which are implicated in the pathophysiology of depression and anxiety (Padmavathi et al., 2023). NLP techniques, on the other hand, may enhance cognitive restructuring and coping skills, potentially explaining the additive benefits observed in the combined intervention group (Ybias et al., 2024).

The improvements in sleep quality are particularly noteworthy, given the bidirectional relationship between sleep disturbances and mental health issues. Better sleep may have contributed to the observed improvements in mood and stress levels, and vice versa (Turmel et al., 2022).

A recent quasi-experimental study investigated the impact of hatha yoga on mental health in a group of 52 Iranian women, with an average age of 33.5 years. The participants engaged in a month-long program consisting of three weekly yoga sessions, each lasting 60-70 minutes. Using the DASS-21 scale to measure psychological symptoms, the researchers found significant improvements in depression, anxiety, and stress levels following the intervention ($P<0.001$) (Khunti et al., 2023).

A pilot study evaluated the impact of an 8-week mindful parenting intervention (MPI) on parents of young children with anxiety disorders. The study involved 21 parents whose children were between 3 and 7 years old. Findings revealed significant ($P<0.05$) improvements in mindful parenting practices and notable decreases in problematic parent-child interactions. The results suggest that the MPI may be a promising approach for improving parenting skills and family dynamics in families dealing with childhood anxiety (Farley et al., 2023). This present study extends these findings by demonstrating the enhanced benefits of combining yoga with NLP techniques.

A systematic review by Nompo et al. (2021) found that Neuro-Linguistic Programming (NLP) interventions can effectively reduce anxiety across different populations and contexts (Nompo et al., 2021). Multiple studies showed that NLP techniques, including reframing, anchoring, rapport building, and modeling, helped lessen anxiety symptoms and encourage positive behavioral changes (Kotera et al., 2019). The NLP interventions were typically delivered through multi-session training programs lasting between 2 to 12 weeks. These programs often included components such as identifying sensory preferences, building rapport, reframing negative thoughts, and practicing new cognitive and communication strategies. The benefits of these NLP interventions were not limited to anxiety reduction. Improvements were also noted in participants' self-confidence, communication skills, and overall mental health (Nompo et al., 2021; Shirazi & Kahrazei, 2022).

This study has several limitations. First, the relatively small sample size and short duration of follow-up limit the generalizability of our findings. Second, the use of self-report measures, while validated, may introduce some bias. Third, the lack of an active control group (e.g., exercise only) makes it difficult to isolate the specific effects of yoga and NLP. Finally, the high adherence rates observed may not be representative of real-world settings.

Nevertheless, these findings have important implications for clinical practice and public health. Given the high prevalence of mental health issues among middle-aged women and the potential side effects of pharmacological treatments, yoga and NLP could be considered as adjunctive or alternative therapies in this population. The superior efficacy of the combined intervention suggests that integrating mind-body practices with cognitive techniques may be a promising approach for comprehensive mental health care.

Future research should focus on longer-term follow-up, larger sample sizes, and comparison with other active interventions. Additionally, exploring the underlying neurobiological mechanisms and identifying patient characteristics that predict treatment response would be valuable for personalizing interventions.

Conclusion

In conclusion, this study provides evidence for the efficacy of yoga and yoga combined with NLP in improving mental health and sleep quality among mothers of adolescents. These findings support the integration of mind-body practices into mental health care and highlight the potential synergistic benefits of combining physical and cognitive interventions.

Acknowledgement
All study participants deserve our sincere thanks for committing much of their time and effort to this research.

Conflicts of interest
The authors declare that there are no conflict of interest.

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SportMont

ORIGINAL SCIENTIFIC PAPER

Prevalence of Overweight and Obesity of Preschool Children in Vranje (Serbia) According to the WHO and CDC Recommendations

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Abstract

Overweight and obesity are widespread worldwide among children of all age groups, and the prevalence among preschool children has been steadily increasing over the past twenty years. Tha aim of this study was to determine the prevalence of overweight and obesity of preschool children in Serbia based on the recommendations given by the WHO (World Health Organization) and the CDC (Centers for Disease Control and Prevention). The sample of respondents consisted of children from the preschool institution "Our Child" (serb. Naše dete) from Vranje, Serbia. A total of 380 children participated, of which 195 were boys (51.3%) and 185 were girls (48.7%). The age of the child ranged from 2 to 7 years, that is, from 26 to 84 months. In order to determine the weight status, the children's body weight and height were measured, on the basis of which the Body Mass Index (BMI) was calculated. The obtained BMI values were interpreted based on the recommendations given by the WHO and the recommendations given by the CDC. Research results based on the WHO recommendations showed that 11.1% of the total number of children are overweight and 11.8% are obese, while based on the CDC recommendations, 11.8% of the total number of children are overweight and 7.9% are obese. Also, the research showed that there is no difference in the weight status categories between boys and girls. However, the results show that the proportions of overweight and obese children obtained using the WHO recommendations differ significantly from the proportions of overweight and obese children obtained using the CDC recommendations. High percentages of overweight and obesity among preschool children in Vranje require adequate measures to be taken in order to stop and reduce their growth.

Keywords: *preschoolers, weight status, BMI status, obese children, different recommendations*

Introduction

Overweight and obesity refer to an excessive accumulation of fat that may pose a risk to health (Ilić, 2014). Childhood obesity is a major global public health issue, leading to significant and long-lasting health complications (Liu et al., 2022; Mantzorou et al., 2023). The consequences are in the form of physical and mental health disorders, increased risk of diabetes, cardiovascular diseases, osteoarticular diseases and cancer (Liu et al., 2022). Obesity affects a child's development, often persisting into adulthood and increasing the risk of psychosocial issues, cardiovascular diseases, and cancer (Fäldt et al., 2023).

Overweight and obesity are prevalent worldwide in children of all age groups (Woronko at al., 2023). According to the latest World Health Organization (WHO) reports, it is estimated that 41 million children under the age of five are affected by overweight or obesity (Wang et al., 2022). Nikolić, Gadžić and Stamenković (2022) analyzed 46 studies and determined that the prevalence of overweight and obese preschool children is high on all continents, that it is the highest in North America, that Europe is not far behind them. However, some studies from Asia and Africa indicate that children in Africa have not yet reached the same prevalence of overweight and obesity as their peers in

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North America and Europe. Over the last 30 - 40 years, the prevalence has been steadily increasing and is expected to continue to increase in the coming decades (Miguel-Berges at al., 2023). Between 1980 and 2013, the prevalence of obesity among children and adolescents in developing countries rose from 8% to 13% for both boys and girls. In Africa, the prevalence of obesity in preschool children increased from 4% in 1990 to 8.5% in 2010 (Gebremedhin, 2015). Đurašković at al. (2012) found that the overall prevalence of overweight and obesity in 7-year-old children from Nis (Serbia) increased from 10.63% to 42.05% in the period from 1988 to 2008. Sekhobo at al. (2010) found that the overall prevalence of overweight and obesity in children aged 2 to 5 years from New York (USA) increased from 25.5% to 32.1% between 2002 and 2007. Vuceraković and Mitrović (2021) determined that the overall prevalence of overweight and obesity in children aged 5 to 6 from Podgorica (Montenegro) increased from 33.33% to 45.65% in the period from 2013 to 2020. A national survey conducted in 9 cities in China revealed an increase in the prevalence of overweight and obesity in children from 11.7% to 25.2% between 1991 and 2011 (Wang et al., 2022).

Authors from our region found that children from Serbia are not behind children from Europe in the prevalence of overweight and obesity (Đurašković et al., 2012; Cvetković & Cvetković, 2018; Ilić, 2014; Jovanović et al., 2010; Međedović et al., 2014; Pelemiš et al., 2021; Peulić, Katanic, Jovanović, & Bjelica, 2024; Stupar, Popović & Vujović, 2014). Therefore, childhood obesity is currently viewed as one of the most prevalent public health problems in our country (Kisić-Tepavčević et al., 2008). For these reasons, there is a need to monitor the physical development of children, and one of the most reliable methods for tracking children's weight status is through anthropometric measurements, specifically by calculating the Body Mass Index (BMI).

It should be noted that BMI in adults and children is calculated using the same formula, but the interpretation of the obtained values is not the same because the amount of fat in the body changes with age. Also, the amount of fat in boys and girls is different. Some previous studies support this and show that the prevalence of obesity among preschool children varies depending on whether WHO or CDC (Centers for Disease Control and Prevention) recommendations were used (Al Alawi et al., 2013; Kułaga et al., 2016; Ma et al., 2011; Twells & Newhook, 2011).

Although some international studies have shown that there may be differences in the level of obesity among children depending on the different WHO and CDC cut-off values, there are still no such studies involving our children. In this context, the aim of this paper was to determine the prevalence of overweight and obesity of preschool children in Serbia based on the recommendations given by the WHO and the CDC.

Methods

Participants

The sample of respondents in this research consisted of children from the preschool institution "Our Child" (serb. Naše dete) from Vranje, Serbia. A total of 380 children participated, of which 195 were boys (51.3%) and 185 were girls (48.7%). The criteria for including and selecting subjects were as follows: healthy children (those without any diseases or disorders) of both genders, aged 2 to 7 years, from preschool institution. Therefore the child's age ranged from 2 to 7 years, that is, from 26 to 84 months, and the average age was 4.42 years, that is, 58.63 months. Parental consent was obtained for each child who participated in the research. The research process adhered to the principles outlined in the Helsinki Declaration (World Medical Association, 2011), and the study was approved by the Faculty of Sport and Physical Education from Nis (Ethics Committee).

Measurements

In order to determine the weight status, the children's body weight and height were measured, on the basis of which the Body Mass Index (BMI) was calculated. Body height was measured with an anthropometer (GPM, Zurich, Switzerland) with the subject standing on a horizontal flat surface in an upright position with the back extended and the heels together. The lower side of the arm of the anthropometer is placed on the most prominent part of the crown of the head (vertex). The result of the measurement was read with an accuracy of 0.1 cm, and then they were converted into meters. Body mass was measured with an electronic Tefal 6010 scale (Rumilly, Haute-Savoie, France) in subjects who, minimally dressed, stood calmly on the landing shaft of the scale in an upright position. The measurement result was read from the scale screen with an accuracy of 0.1 kg. BMI was calculated according to the standard formula, i.e. BMI=BM(kg)/BH(m)2.

The obtained BMI values were interpreted based on the recommendations given by the WHO and CDC. These recommendations take into account the gender and age of the child and convert the BMI values into percentiles on the basis of which the position of a certain index value is determined in relation to a group of children of the same gender and age. Therefore, children were classified into four distinct weight categories: underweight, normal weight, overweight, and obese (Table 1).

In order to determine the position of the BMI value, the children's calendar age was taken from the records in the preschool institution, for which there was parental consent. Tables and graphs by which children were categorized according to the level of weight are available on the WHO website (WHO, 2024) and the CDC website (CDC, 2024).

Statistics

Data processing was performed with the SPSS 20 statistics software (SPSS Inc., Chicago, IL, USA). The basic parameters of descriptive statistics were calculated, including the arithmetic mean, standard deviation, frequencies, and percentages. The following was used: Chi-square ($\chi 2$) test of independence to determine the association between the weight status and gender of children; the association between the weight status and the choice of the WHO or CDC recommendations. According to Cohen's (1988) criterion from 0.01 to 0.29 is a small association (effect), from 0.3 to 0.49 is a medium association (effect) and for 0.50 is a large association (effect) (Pallant, 2011). For all statistical analyses, significance was accepted at $p<0.05$.

Table 1. Weight status categories

Underweight	Less than 5th percentile	<5th percentile
Normal weight	From the 5th to less than the 85th percentile	5-85th percentile
Overweight	From the 85th to less than the 95th percentile	85-95 percentile
Obesity	Equal to or greater than the 95th percentile	≥95th percentile

Results

Table 2 presents the descriptive parameters of the analyzed variables, revealing that 70.5% of children were of normal weight, with minor gender differences. The table also shows that the percentage of children with normal weight varied from year to year, without a clear linear trend of increase or decrease. When the variables of body height and body weight were analyzed, both exhibited a linear growth trend with age.

Table 2. Descriptive parameters of the sample of respondents and weight status based on two different classifications

Age	Sex (N)	Height (M±SD)	Weight (M±SD)	BMI (M±SD)	WHO				CDC			
					UW (N/%)	NW (N/%)	OW (N/%)	O (N/%)	UW (N/%)	NW (N/%)	OW (N/%)	O (N/%)
	M (16)	(95.2±3.47)	(14.98±1.89)	(16.52±1.74)	/	(12/75%)	(1/6.3%)	(3/18.8%)	(2/10.5%)	(14/73.7%)	(3/15.8%)	/
2-3	F (19)	(94.2±5.57)	(14.03±1.63)	(15.82±1.52)	(1/5.3%)	(14/73.7%)	(3/15.8%)	(1/5.3%)	(2/10.5%)	(14/73.7%)	(3/15.8%)	/
	T (35)	(94.66±4.69)	(14.46±1.79)	(16.14±1.63)	(1/2.9%)	(26/74.3%)	(4/11.4%)	(4/11.4%)	(2/5.7%)	(27/77.1%)	(3/8.6%)	(3/8.6%)
	M (48)	(101.31±6.61)	(16.58±2.67)	(16.08±1.32)	(2/4.2%)	(31/64.6%)	(7/14.6%)	(8/16.7%)	(3/6.3%)	(34/70.8%)	(7/14.6%)	(4/8.3%)
3-4	F (36)	(101.16±5.6)	(16.53±2.6)	(16.18±2.52)	(3/8.3%)	(22/61.1%)	(5/13.9%)	(6/16.7%)	(4/11.1%)	(21/58.3%)	(6/16.7%)	(5/13.9%)
	T (84)	(101.24±6.16)	(16.56±2.63)	(16.13±1.92)	(5/6%)	(53/63.1%)	(12/14.3%)	(14/16.7%)	(7/8.3%)	(55/65.5%)	(13/15.5%)	(9/10.7%)
	M (31)	(110.11±4.15)	(18.96±2.7)	(15.58±1.41)	(2/6.5%)	(21/67.7%)	(6/19.4%)	(2/6.5%)	(4/12.9%)	(22/71%)	(3/9.7%)	(2/6.5%)
4-5	F (37)	(110.69±5.59)	(18.91±2.79)	(15.4±1.64)	(1/2.7%)	(27/73%)	(6/16.2%)	(3/8.1%)	(5/13.5%)	(23/62.2%)	(6/16.2%)	(3/8.1%)
	T (68)	(110.43±4.96)	(18.93±2.73)	(15.48±1.53)	(3/4.4%)	(48/70.6%)	(12/17.6%)	(5/7.4%)	(9/13.2%)	(45/66.2%)	(9/13.2%)	(5/7.4%)
	M (57)	(116.91±4.85)	(21.15±2.92)	(15.43±1.53)	(5/8.8%)	(44/77.2%)	(4/7%)	(4/7%)	(8/14%)	(43/75.4%)	(3/5.3%)	(3/5.3%)
5-6	F (48)	(115.84±6.22)	(21.45±3.31)	(15.96±2.01)	(1/2.1%)	(37/77.1%)	(2/4.2%)	(8/16.7%)	(2/4.2%)	(36/75%)	(7/14.6%)	(3/6.3%)
	T (105)	(116.42±5.52)	(21.29±3.09)	(15.67±1.78)	(6/5.7%)	(81/77.1%)	(6/5.7%)	(12/11.4%)	(10/9.5%)	(79/75.2%)	(10/9.5%)	(6/5.7%)
	M (43)	(123.82±4.61)	(24.19±4.52)	(15.69±2.15)	(2/4.7%)	(31/72.1%)	(3/7%)	(7/16.3%)	(6/14%)	(28/65.1%)	(4/9.3%)	(5/11.6%)
6-7	F (45)	(123.13±5.0)	(22.94±3.99)	(15.07±2.15)	(8/17.8%)	(29/64.4%)	(5/11.1%)	(3/6.7%)	(12/26.7%)	(25/55.6%)	(6/13.3%)	(2/4.4%)
	T (88)	(123.47±4.8)	(23.55±4.28)	(15.37±2.16)	(10/11.4%)	(60/68.2%)	(8/9.1%)	(10/11.4%)	(18/20.5%)	(53/60.2%)	(10/11.4%)	(7/8%)
SM (195/51.3%)		(111.73±10.75)	(19.84±4.4)	(15.76±1.66)	(11/5.6%)	(139/71.3%)	(21/10.8%)	(24/12.3%)	(21/10.8%)	(140/71.8%)	(17/8.7%)	(17/8.7%)
SF (185/48.7%)		(111.5±11)	(19.58±4.29)	(15.66±2.07)	(14/7.6%)	(129/69.7%)	(21/11.4%)	(21/11.4%)	(25/13.5%)	(119/64.3%)	(28/15.1%)	(13/7%)
SS (380/100%)		(111.62±10.86)	(19.72±4.35)	(15.71±1.87)	(25/6.6%)	(268/70.5%)	(42/11.1%)	(45/11.8%)	(46/12.1%)	(259/68.2%)	(45/11.8%)	(30/7.9%)

Legend: N-number; M- male; F-female; M±SD-mean ± standard deviation; BMI-body mass index; UW-underweight; NW-normal weight; OW-overweight; O-obese; WHO-world health organization; CDC-US centeres for disease control and prevention; SM-sum of male; SF-sum of female; SS-sum of all subjects; T-total.

The results of the research based on WHO recommendations showed that, out of the total number of children, 6.6% were underweight, 70.5% were of normal weight, 11.1% were overweight, and 11.8% were obese (Table 3). To gain a clearer understanding of the weight issue, it was determined that 70.5% of preschool children were of normal weight, while 29.5% had some weight-related problems. According to CDC recommendations, 12.1% of the children were underweight, 68.2% were of normal weight, 11.8% were overweight, and 7.9% were obese. Thus, 68.2% of children were of normal weight, and 31.8% had some weight-related problems.

The X^2 test of independence did not show a significant association between gender and weight status both according to the WHO recommendations (X^2(n=380)=0.670, p=0.880, Cramer's V=0.042) and according to the CDC recommendations (X^2(n=380)=5.013, p=0.171, Cramer's V=0.115) (Table 4). This means that the proportions of boys in all four weight status categories do not differ significantly from the proportions of girls in those same categories. There is no connection between gender and weight status. More precisely, the prevalence of underweight, normal weight, overweight and obesity is the same in boys and girls.

Table 3. Weight status of preschool children in Vranje

Weight status	WHO		CDC	
	N	%	N	%
Underweight	25	6.6%	46	12.1%
Normal weight	268	70.5%	259	68.2%
Overweight	42	11.1%	45	11.8%
Obesity	45	11.8%	30	7.9%
Children who are normally weight	268	70.5%	259	68.2%
Children who are not normally weight	112	29.5%	121	31.8%

Legend: N - number of respondents; % - percentage of respondents; WHO - results according to the recommendations of the World Health Organization; CDC - results according to the recommendations of the Center for Disease Control and Prevention.

Table 4. X^2 test of independence of the association between gender and weight level

WHO							
Gender	Underweight	Normal weight	Overweight	Obesity	X^2	P	Cramer's V
Boys	11 (5.6%)	139 (71.3%)	21 (10.8%)	24 (12.3%)	0.670	0.880	0.042
Girls	14 (7.6%)	129 (69.7%)	21 (11.4%)	21 (11.4%)			
CDC							
Gender	Underweight	Normal weight	Overweight	Obesity	X^2	P	Cramer's V
Boys	21 (10.8%)	140 (71.8%)	17(8.7%)	17 (8.7%)	5.013	0.171	0.115
Girls	25 (13.5%)	119 (64.3%)	28 (15.1%)	13 (7.0%)			

Legend: WHO - results according to the recommendations of the World Health Organization; CDC - results according to the recommendations of the Center for Disease Control and Prevention; X^2 - test result; p - significance level; Cramer's V - effect size.

The X^2 test of independence showed a significant association between weight status and the recommendation used to define it (X^2(n=760)=9.468, p=0.024, Cramer's V=0.112). This indicates that the proportions of children in all four weight status categories based on WHO recommendations differ significantly from those based on CDC recommendations. Specifically, the proportions of underweight, normal weight, overweight, and obese children are not the same when using WHO versus CDC criteria. The prevalence of underweight children is 6.6% according to WHO cut-off values, nearly double under CDC guidelines at 12.1%. The difference in normal weight children is smaller, with 70.5% under WHO and 68.2% under CDC. The smallest discrepancy is for overweight children, with 11.1% under WHO and 11.8% under CDC. However, there is a substantial difference in obesity prevalence, with 11.8% under WHO and 7.9% under CDC guidelines. Based on the Cramer's V coefficient, the differences between the proportions of children classified as underweight, normal weight, overweight, or obese by WHO and CDC standards are considered small.

Table 5. X^2 test of independence - association of weight level with the choice of the WHO or CDC recommendation

Org.	Underweight	Normal weight	Overweight	Obesity	X^2	P	Cramer's V
WHO	25 (6.6%)	268 (70.5%)	42 (11.1%)	45 (11.8%)	9.468	0.024	0.112
CDC	46 (12.1%)	259 (68.2%)	45 (11.8%)	30 (7.9%)			

Legend: Org. - organization; WHO - results according to the recommendations of the World Health Organization; CDC - results according to the recommendations of the Center for Disease Control and Prevention; X^2 - test result; p - significance level; Cramer's V - effect size.

Discussion

The results of our research indicate that the prevalence of overweight and obese children in Vranje is high, regardless of the criteria used to define weight status. According to the results obtained based on the WHO recommendations, as many as 22.9% of children are overweight/obese, while based on the

CDC recommendations, that percentage is 19.7%. The results require adequate measures to be taken in order to reduce the mentioned percentages and save children from the consequences that the state of overweight and obesity can cause.

Overnutrition and obesity according to the WHO recommendation

The percentage of overweight children in our research was 11.1% and 11.8% of obese children. Jovanović et al. (2010) obtained similar results on a sample of 193 children aged 4 and 5 from Pančevo and determined that 13.5% of children were overweight and 15% were obese. Đurašković at al. (2012) obtained even more alarming data. Their research shows that the percentage of overweight and obese children is higher than in our and the previously mentioned research. Based on a sample of 176 children aged 7 years from Niš, the authors determined that 25.6% of children were overweight and 16.5% were obese. Also, Ilić (2014) determined on a sample of 1051 preschool children from Jagodina and Ćuprija that 19.5% of children were overweight and 14.6% were obese. The above shows that the children in our research are slightly less overweight/obese compared to the aforementioned research from Serbia.

However, if we compare our research with the research of Kułaga et al. (2016) we can see that our children from Serbia are more obese than their peers from Poland. The authors, based on a sample of 5026 children aged 2 to 6 years, determined that 11.1% of children are overweight, which is the same as in Vranje, but that only 4.1% of children are obese, which is far less than in Vranje where obesity is 11.8%. It is similar to the data from Madrid obtained by Ortiz-Marrón et al. (2018). On a sample of 2,435 children aged 4 to 6 years, the authors determined that the prevalence of overweight children was 16.5%, which is higher than in Vranje, but that the prevalence of obesity was 5.5%, which is far less than among our children.

According to the research by Ma et al. (2011) children in Northeast China have a similar prevalence of overweight and obesity as children in Serbia. In a sample of 8,653 children aged 2 to 7 years, the authors determined that 10.9% were overweight and 13.8% were obese. Children from Santiago (Chile) have a far higher prevalence of overweight and obesity compared to our and other studies, and this was determined by Kain et al. (2009). Based on a sample of 1089 children aged 5 years, the authors found that 26.6% of children were overweight and 15.6% were obese. That the situation is not better even in Canada is confirmed by Twells and Newhook (2011) who, based on a sample of 1026 preschool children, found that 26% were overweight, which is far more than in Vranje, and that 11.3% were obese, which is approximately as in our research.

According to a study by Gebremedhin (2015) conducted on a sample of 155726 children aged 0 to 5 years in Sub-Saharan Africa, the prevalence of overweight/obese children is far lower than in our study and other mentioned studies. 3.9% of children were overweight and 2.9% were obese. Similar data comes from South India. Kumar et al. (2008) found on a sample of 425 children aged 2 to 5 years that 4.47% of children were overweight and 1.41% were obese, which is far less than our children. Al Alawi at al. (2013) on a sample of 387 children aged 0 to 5 years found that in the Kingdom of Bahrain, the percentage of overweight children is similar to that in Serbia and amounts to 12.1%, and the percentage of obese children is much lower and amounts to 2.6%.

Overnutrition and obesity according to the CDC recommendations

The prevalence of overweight/obesity obtained based on the CDC recommendations is not the same as when using the WHO recommendations. Therefore, the discussion of the obtained results must be done separately. Our research showed that 11.8% of children are overweight and 7.8% are obese. Similar results were obtained by Stupar, Popović & Vujović (2014) on a sample of 206 children aged 6 and 7 years from Novi Sad. The authors found that 12% of children are overweight and 9% are obese. Pelemiš at al. (2021) on a sample of 188 children aged 6 years from Belgrade determined that 9.6% of children were overweight and 7.5% were obese, which is also a similar prevalence as in our research. Somewhat different data comes from Mladenovac. Cvetković & Cvetković (2018) found on a sample of 50 children aged 6 and 7 years that overweight is 6%, which is less than in our research, but that obesity is a surprising 22%. However, the number of respondents in the aforementioned research is small, which may be the reason for different results. According to the data obtained by Đermanović, Miletić & Pavlović (2015) on a sample of 60 children, aged 5 and 6 years, the prevalence of overweight/obesity is higher in the West of Republika Srpska than in Vranje and the mentioned cities in Serbia. The authors determined that 16.7% of children were overweight and 13.3% were obese. The number of respondents in this research is also smaller, so the mentioned data should be viewed with caution.

A rather higher percentage of overweight (16.3%) and obese (15.7%) children compared to our research was found by Manios et al. (2007) on a sample of 2374 children aged 1 to 5 years from Greece. Kułaga at al. (2016) based on a sample of 5026 children aged 2 to 6 years from Poland found that 18.2% of children are overweight, which is much more than in our study, and 8.3% are obese, which is almost the same as in Vranje. Similar results were obtained by Merkiel & Chalcarz (2016) on a sample of 128 children aged 4 to 6 years, also from Poland. Overweight of children was 18.5% and obesity was 5.5%.

Based on the results obtained by Ma at al. (2011) we conclude that children from Northeast China are not behind their peers from Vranje, Serbia and Europe. Based on a sample of 8653 children aged 2 to 7 years, the authors determined that 11.3% of children were overweight and 11.7% were obese.

Nevertheless, although preschool children from Vranje have a pronounced prevalence of overweight and obesity, based on the research conducted by Willows, Johnson & Ball (2007), we see that children in Canada in 2007 had much higher percentages of the mentioned phenomena. In a sample of 1044 children aged 5 years, the authors determined that 27.5% were overweight and 37.4% were obese. Somewhat lower, but still worrying percentages of overweight/obesity of children from Canada were determined by Twells & Newhook (2011) on a sample of 1026 children of preschool age. Overweight of children was represented by 19.1%, and obesity by 16.6%.

The situation is not better in the USA where Lim et al. (2009) based on a sample of 365 children aged 3 to 5 years found that 12.9% of children were overweight and 10.3% were obese. These percentages are similar to those in our research. Sekhobo at al. (2010) arrive at somewhat higher percentages and, based on a sample of 198633 children from New York, aged 2 to 5 years, found that 17.4% were overweight and 14.7% were obese.

Based on the data obtained by Al Alawi at al. (2013) we conclude that the prevalence of overweight/obesity is slightly lower in the Kingdom of Bahrain than in Vranje. The authors, based on a sample of 387 children aged 0 to 5 years, found that 9.8% of children were overweight and 5.2% were obese.

Several studies from the east show that the prevalence of overweight there is similar to that in Vranje, but that obesity is lower. Fatemeh at al. (2012) found on a sample of 500 children aged 2 to 5 years from Birjand (Iran) that 10.6% of children were overweight and 7.6% were obese. Similar results also come from Tehran (Iran), where Gaeini et al. (2011) on a sample of 755 children aged 3 to 6 found that 10.1% of children were overweight and 4.6% were obese, as well as from Basra (Iraq) where Ali Musa & Hassan (2010) on the sample of 550 children aged 0 to 5 years found that 10.5% of children were overweight and 3.3% were obese. A few years later, data came from Iran that overweight was somewhat lower compared to the aforementioned research, and obesity was similar. Namely, Hassanzadeh-Rostami, Kavosi & Nasihatkon (2016) found that 5.7% were overweight and 5.2% obese on a sample of 8821 children aged 2 to 6 years.

Differences in the level of weight between boys and girls

The results of our research showed that there is no statistically significant difference in the prevalence of overweight and obesity between boys and girls, both according to the WHO and CDC recommendations. This means that the proportion of overweight and obese boys is not significantly different from the proportion of overweight and obese girls. However, although there is no significant difference, we should pay attention to the results we obtained following the CDC guidelines, where the percentage of overweight boys is 8.7%, and girls 15.1%, which is almost twice as much. We should repeat this research and use some other statistical analysis or a different procedure to determine if there really is a significant difference.

In contrast to the results we obtained following the WHO guidelines for determining the weight level, Kain et al. (2009) found that Chilean boys were more obese than girls (18.6% vs. 12.6%), but that the percentages of overweight were similar (25.8% vs. 25.5%). Similar data were obtained by Ma et al. (2011) who found that boys from Northeast China are more obese than girls (16.1% vs. 11.3%), while the percentages of overweight are similar (10.3% vs. 11.6%). In Canada, Twells & Newhook (2011) found that the percentage of overweight boys was higher than that of girls (28.5% vs. 23.3%), while the percentages of obesity were similar (11.7% vs. 10.8%). According to research by Đurašković et al. (2012) in Serbia, the percentage of overweight girls is higher than that of boys (28.2% versus 23.1%), while the percentage of obesity is higher among boys than among girls (17.6% versus 15.3%). Ilić (2014) states that in Serbia the percentage of overweight boys is higher than that of girls (21.8% vs. 16.9%), and that boys are more obese than girls (16.3% vs. 12.6%). Kułaga at al. (2016) found that the percentage of overweight boys in Poland is 12.2%, and girls 10%, while the percentage of obese boys is 4.9% and girls 3.4%. Based on the research of Ortiz-Marrón et al. (2018) we can see that in Madrid there is no big difference in overweight of boys (16.8%) and girls (16.2%), nor is there a big difference in obesity between boys (5.4%) and girls (5. 5%).

Following the CDC guidelines for determining weight levels, the authors also came to different results. Willows, Johnson & Ball (2007) in Canada found that girls have higher percentages of overweight than boys (31.2% vs. 23.8%), which is similar to our study, but that boys are more obese than girls (40, 5% against 34.2%). Pelemiš, Mandić, Momčilović, Momčilović & Srdić (2021) in Belgrade found that boys have higher percentages of overweight than girls (12.1% vs. 6.1%), which is different from our and the previous research, while the percentages of obesity are similar (7.5% in boys and 7.4% in girls). Similar data were obtained by Stupar, Popović & Vujović (2014) in Novi Sad and they found that the percentage of overweight boys is higher than the percentage of overweight girls (13.5% vs. 7.6%), while the percentages of obesity are similar (8.6% in boys and 9.7% in girls). Đermanović, Miletić & Pavlović (2015) in Republika Srpska obtained similar data as the previous two groups of authors and found that the percentage of overweight boys is higher than the percentage of overweight girls (18.8% vs. 14.3%), while girls had some higher percentage of obesity than boys (14.3% vs. 12.5%). In contrast to the aforementioned research, Manios et al. (2007) in Greece, Sekhobo at al. (2010) in the USA, Gaeini at al. (2011) in Tehran, Twells & Newhook (2011) in Canada, Fatemeh at al. (2012) in Iran, Kułaga at al. (2016) in Poland and Ma at al. (2011) in Northeast China obtained percentages of overweight and obesity of preschool children that do not differ much between boys and girls.

Differences in weight levels depending on the use of the CDC and WHO recommendations

Results of our research showed that there is a statistically significant relationship between the level of weight and the recommendation that is chosen to define that level. This means that the proportions of overweight and obese children obtained using the WHO recommendations differ significantly from the proportions of overweight and obese children obtained using the CDC recommendations.

The research shows that the prevalence of overweight is higher according to the CDC (11.8%) than according to the WHO (11.1%) recommendations, while the prevalence of obesity is higher according to the WHO (11.8%) than according to the CDC (7.9%) recommendations. Similar results were obtained by Ma et al. (2011) and determined that the prevalence of overweight of preschool children is higher when using the CDC recommendations (11.3%) than when using the WHO recommendations (10.9%), while the prevalence of obesity is higher when using the WHO recommendations (13, 8%) than when using the CDC recommendations (11.7%). However, Twells & Newhook (2011) found that the prevalence of overweight is higher according to the WHO (26%) than according to the CDC recommendations (19.1%), while the prevalence of obesity is higher according to the CDC (16.6%) than according to the WHO recommendations (11.3%). Similar results were obtained by Al Alawi et al. (2013) and they found that the prevalence of overweight is higher according to the WHO (12.1%) than according to the CDC recommendations (9.8%), while the prevalence of obesity is higher according to the CDC (5.2%) than according to the WHO recommendations (2.6%). On the other hand, Kułaga at al. (2016) found that the prevalence of overweight and obesity was higher according to the CDC (18.2% and 8.3%) than according to the WHO recommendations (11.1% and 4.1%).

Research results have shown that the prevalence of obese children in Vranje is high. It is considered that the main contributing factors to obesity are a sedentary lifestyle, lack of physical activity, and improper diet (Mendonça & Anjos, 2004; Planinšec & Matejek, 2004). In response to these challenges, leading health institutions, including the WHO, have developed global strategies for promoting healthy diets and physical activities, which each country should adapt to its specific conditions (Katanic et al., 2023).

This study significantly contributes to a better understanding of the issue of childhood obesity in the country. The results of this study particularly emphasize the importance of regular monitoring of anthropometric parameters and the weight status of preschool-aged children. This monitoring is crucial for the timely diagnosis and prevention of obesity, as well as for reducing the associated health risks. Based on the findings, it is possible to identify children with excess weight and refer them to appropriate treatments, making a significant step towards improving public health measures in the fight against childhood obesity.

Conclusion

The results of the research showed that the prevalence of overweight and obese children in Vranje is high, regardless of whether the WHO or CDC recommendations were used when defining the level of weight. Preschool children from Vranje do not lag behind their peers from Western Europe in terms of overweight and obesity. Also, the research showed that there is no difference in the level of weight between boys and girls, that is, that boys and girls are equally overweight and obese. However, the results also show that the proportions of overweight and obese children obtained using the WHO recommendations differ significantly from the proportions of overweight and obese children obtained using the CDC recommendations. Certainly, high percentages of overweight and obesity of preschool children in Vranje require adequate measures to be taken in order to stop and reduce their growth.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Motor Abilities in Physically Active and Inactive School Adolescents Aged 13-15 Years: A Comparative Study

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Abstract

Accumulating at least 60 min of physical activity every day is health-beneficial and will contribute to development of motor abilities. The objective was to compare some motor abilities of physically active [N=42; (24 boys & 18 girls)] and inactive [N=55; (24 boys & 31 girls)] school adolescents at the age of 13-15 years: (1) abdominal muscle strength; (2) lower back muscle strength; (3) upper limbs muscle strength; (4) lower limbs muscle strength; and (5) explosive leg power. The PAQ-C questionnaire was applied to evaluate the level of physical (in)activity. A cut-off value of 2.73 was set to categorize children as inactive (<2.73) and physically active (≥2.73), as suggested by Benitez-Porres et al. (2016). Student's t-test was applied in order to compare motor abilities between physically active and inactive adolescents. Findings of the present study indicate that physically active adolescents have higher upper limbs muscle strength and explosive leg power than inactive adolescents. No significant differences were observed in abdominal muscle strength, lower back muscle strength and lower limbs muscle strength, between physically active and inactive school adolescents aged 13-15 years. We believe that specifically tailored exercise interventions with an appropriate frequency, intensity, and volume might be necessary in order to induce higher and more significant changes in all segments of the motor abilities spectrum.

Keywords: *adolescents, explosive leg power, muscle strength, physical activity*

Introduction

Physical activity (PA) is any bodily movement that is raising the energy expenditure and is produced by the skeletal muscles (Piggin, 2020). PA is beneficial for the overall health (Wu et al., 2017). More active adolescents display healthier cardio-metabolic profiles and develop higher peak bone masses than their less active counterparts (Boreham & Riddoch, 2001; Brown & Summerbell, 2009; Janssen & LeBlanc, 2010). Also, physically active adolescents have better mental health and psychosocial well-being than those that are inactive (Iannotti, Janssen, et al., 2009; Iannotti, Kogan, et al., 2009). On the other hand, physical inactivity and sedentary behavior are associated with various negative health consequences such as obesity or increased risk for cardio-metabolic diseases (Leung et al., 2012), and may also contribute to a delay of the motor development (Tremblay et al., 2011).

Daily life PA can be categorized into occupational, sports, conditioning, household, or other activities (Dasso, 2019). Nowadays, adolescents have become less physically active with an approximate energy expenditure of 600 kcal per day,

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which is far less than the approximate energy expenditure few years ago (Boreham & Riddoch, 2001). Since PA among children and adolescents has been demonstrated to benefit a number of health issues and diseases through life (Boreham & Riddoch, 2001; Tremblay et al., 2011), assessment and monitoring of the physical (in)activity status is very important. It may contribute to early detection, as well as prevention of the trend of inactivity among children and adolescents by developing and implementing effective interventions that will increase their PA levels. It is generally recommended by the World Health Organization to accumulate at least 60 min of PA with moderate-to-vigorous intensity every day (World Health Organization, 2023).

Promoting PA in early childhood may also contribute to healthier motor development in children, as well as to the development of motor abilities and skills in school adolescents (Timmons et al., 2007; Herodek et al., 2024). Motor abilities are specific abilities that allow performing motor skills, affect performance, and are also very important for the activities of the daily living (Fairclough & Stratton, 2005). Especially strength and explosive leg power, which are essential for the motor tasks during everyday life such as: jumping, running, sprinting and change of direction (Suchomel et al., 2016). Previous evidence reported significant differences in the motor abilities of active adolescents compared to inactive, favoring the active ones (Ahmed et al., 2017; Keiner et al., 2013). Recent systematic reviews have also reported that appropriate levels of motor abilities and skills were distinguishing active from inactive adolescents (Martins et al., 2021; Mateo-Orcajada et al., 2022). However, there are some studies that did not observe significant differences in the motor abilities between active and inactive children and adolescents (Malina & Katzmarzyk, 2006; Martínez-Vizcaíno & Sánchez-López, 2008).

Since there is inconsistency among the scientific literature between the studies that compare motor abilities of active and inactive individuals, this topic requires further research. Therefore, objective of the present study was to compare the motor abilities of physically active and inactive school adolescents at the age of 13-15 years: (1) abdominal muscle strength; (2) lower back muscle strength; (3) upper limbs muscle strength; (4) lower limbs muscle strength; and (5) explosive leg power.

Methods

Participants and procedures

The present study is realized on a sample of 97 participants (52 boys and 45 girls) at the age of 13-15 years, school adolescents from the Elementary School Dimkata Angelov Gaberot-Vatasha, Kavadarci (Macedonia).

All measurements and evaluations were performed during the academic year 2020/2021 at the Elementary School Dimkata Angelov Gaberot-Vatasha, with respect to all prevention and protection protocols due to COVID-19. The study was approved by the principal of the school and realized in accordance with the principles of the Helsinki Declaration. Parents of all adolescents included in the study gave consent for participation.

Instruments

Physical activity

In order to realize the particular aim of the study, the PAQ-C questionnaire was applied during PE class to all participants. The PE teacher was in charge of clarifying concepts and explaining questions if necessary. PAQ-C is a self-reported 7-day recall questionnaire that assesses participation in different types of PA (e.g., walking, bicycling, running, swimming, dancing), as well as the frequency of participation after school and during the weekends (e.g., none; 1; 2 or 3; 4; 5 or more times) (Benítez-Porres et al., 2016). It uses a five-point Likert scale ranging from 1 (low PA level) to 5 (high PA level). The total score of the PAQ-C is calculated as a mean of the scores for the 9 items that form part of the questionnaire. In order to allocate adolescents in the inactive or physically active group, a cut-off value of 2.73 was established as suggested by Benitez-Porres et al., (2016). Finally, two groups were created: (1) inactive adolescents: PAQ-C <2.73 [N=55; (24 boys & 31 girls)]; and (2) physically active adolescents: PAQ-C ≥2.73 [N=42; (24 boys & 18 girls)].

Anthropometric characteristics

Anthropometric characteristics of the adolescents such as body mass, height and BMI were measured barefoot and wearing light clothes, according to the World Health Organization manual (World Health Organization, 2007). Body mass was measured with a calibrated digital scale (TANITA TBF 300; TANITA, Middlesex, UK). Height was measured using a wall mounted stadiometer (SECA SE206, Hamburg, Germany). BMI was calculated from height and body mass as follows: body mass in kg devided by height in m2.

Motor abilities

Motor abilities of the adolescents were assessed by applying the modified EUROFIT (1993) testing battery proposed by Jovanovski (1998), which is cost-effective, practical and easy to apply in school environment (EUROFIT, 1993; Jovanovski, 1998): (1) Abdominal muscle strength test (AMST): abdominal crunches in 1 min; (2) Lower back muscle strength test (LBMST): back extensions in 1 min; (3) Upper limbs muscle strength test (ULMST): push-ups in 1 min; (4) Lower limbs muscle strength test (LLMST): squats in 1 min; (5) Standing long jump (SLJ): standing long jump (cm).

Data analysis

Kolmogorov-Smirnov test was applied to test the normality of the distribution. Appropriate statistical methods were used to calculate descriptive statistical parameters. Student's t-test was applied to compare: abdominal muscle strength, lower back muscle strength, upper limbs muscle strength, lower limbs muscle strength, and explosive leg power, between physically active and inactive adolescents. Statistical analysis was performed with the SPSS 23 statistical package (SPSS Inc, Chicago, IL, United States). Significance level was set to $p<0.05$.

Results

According to what is presented in Table 1, data of adolescents that are physically active are normally distributed, with a normal asymmetry considered when values for Skewness are in range between -1.00 to 1.00, and Kurtosis values that are in range between -3.00 to 3.00 (Kallner, 2013). Exception is the Skewness value for LLMS that indicates on a left-oriented distribution, meaning that more adolescents have shown a higher repetition score in the squat test than the calculated average that is 34 repetitions.

Table 1. Descriptive statistical parameters of physically active school adolescents

	N	Min	Max	X	SD	Skewness	Kurtosis	K-S
Body mass (kg)	42	49	90	65.25	13.46	0.66	0.29	p > 0.20
Height (m)	42	1.58	1.82	1.70	0.07	-0.42	-0.52	p > 0.20
BMI	42	18	29	22.73	4.24	0.34	-1.62	p > 0.20
AMST (rep)	42	35	45	38.75	3.69	0.47	-0.78	p > 0.20
LBMST (rep)	42	11	20	14.75	3.85	0.52	-1.68	p > 0.20
ULMST (rep)	42	8	15	11.38	2.33	0.30	-0.67	p > 0.20
LLMST (rep)	42	25	40	34.25	4.68	-1.18	1.41	p > 0.20
SLJ (cm)	42	165	240	204.38	26.78	0.22	-0.99	p > 0.20

Note. AMST - Abdominal muscle strength test; LBMST - Lower back muscle strength test; ULMST - upper limbs muscle strength test; LLMST - Lower limbs muscle strength test; SLJ - Standing long jump.

Based on Table 2, data of inactive adolescents are normally distributed, except Skewness values for: body mass, height, BMI and LBMST that are above the accepted value, indicating on a right-oriented distribution and meaning that more adolescents have shown lower values for the previously mentioned variables than the calculated average. In addition to this, Skewness values for ULMST and SLJ indicate to a left-oriented distribution, meaning that more adolescents have shown higher repetition score in the push-up test and higher jumping distance than the calculated averages. Furthermore, Kurtosis values for height and LBMST are also higher than the acceptable range, but if in addition to these values we take in consideration standard deviations for the same variables, it is considered as acceptable (Kallner, 2013).

Table 2. Descriptive statistical parameters of inactive school adolescents

	N	Min	Max	X	SD	Skewness	Kurtosis	K-S
Body mass (kg)	55	41	73	52.89	11.48	1.04	-0.21	p > 0.20
Height (m)	55	1.55	1.82	1.68	0.07	2.18	5.72	p > 0.20
BMI	55	15	27	18.8	3.68	1.40	2.13	p > 0.20
AMST (rep)	55	25	47	35.22	7.22	0.07	-0.92	p > 0.20
LBMST (rep)	55	10	22	14.33	3.28	1.64	4.28	p > 0.20
ULMST (rep)	55	0	12	7.22	4.41	-1.08	-0.21	p > 0.20
LLMS (rep)	55	20	45	32.00	8.2	0.17	-0.94	p > 0.20
SLJ (cm)	55	105	210	167.78	36.50	-1.09	0.06	p > 0 .20

Notes. AMST - Abdominal muscle strength test; LBMST - Lower back muscle strength test; ULMST - upper limbs muscle strength test; LLMST - Lower limbs muscle strength test; SLJ - Standing long jump.

Based on what is presented in Figure 1, physically active adolescents have shown better results than inactive adolescents in: abdominal muscle strength, lower back muscle strength, upper limbs muscle strength, lower limbs muscle strength and explosive leg power.

However, according to what is presented in Table 3, we can conclude that there is a significant difference between physically active and inactive adolescents in upper limbs muscle strength and explosive leg power, and no significant difference between them in abdominal muscle strength, lower back muscle strength and lower limbs muscle strength.

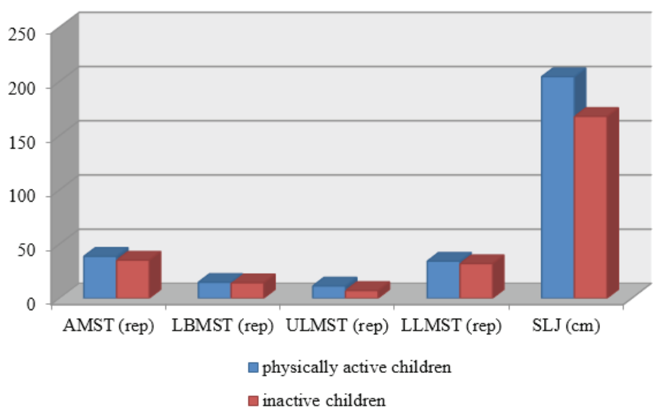

FIGURE 1. Comparison between physically active and inactive school adolescents
Notes. AMST - Abdominal muscle strength test; LBMST - Lower back muscle strength test; ULMST - Upper limbs muscle strength test; LLMST - Lower limbs muscle strength test; SLJ - Standing long jump; rep: repetitions.

Table 3. Comparison of means between inactive and physically active school adolescents, t-test and significance level (p-value)

	Inactive Adolescents	Physically active adolescents	t – test	p level
AMST (rep)	35.22	38.75	1.29	0.22
LBMST (rep)	14.33	14.75	0.24	0.81
ULMST (rep)	7.22	11.38	2.47	0.03*
LLMS (rep)	32.00	34.25	0.70	0.50
SLJ (cm)	167.78	204.38	2.37	0.03*

Notes. AMST - Abdominal muscle strength test; LBMST - Lower back muscle strength test; ULMST - upper limbs muscle strength test; LLMST - Lower limbs muscle strength test; SLJ - Standing long jump; * - p<0.05

Discussion

Results obtained in this study indicate that adolescents that are physically active have higher upper limbs muscle strength and explosive leg power than adolescents that are not physically active. However, there are no differences between physically active and inactive adolescents in terms of abdominal muscle strength, lower back muscle strength and lower limbs muscle strength.

Differences in terms of motor abilities between physically active and inactive adolescents have been reported in the scientific literature so far (Martínez-Vizcaíno & Sánchez-López, 2008). Moreover, it has been suggested that these differences are causal effect of the regular PA (Martínez-Vizcaíno & Sánchez-López, 2008). In this line, there were differences in upper limbs muscle strength and explosive leg power between physically active and inactive adolescents at the present study, but there were no significant differences between them in abdominal muscle strength, lower back muscle strength and lower limbs muscle strength. In support to the present findings, previous evidence also suggested partial and moderate effects on muscle strength induced by PA (Ferreira et al., 2012; Malina & Katzmarzyk, 2006; Martínez-Vizcaíno & Sánchez-López, 2008). PA is usually in context of non-formal sport activities and active games whose primary goal is enjoyment and play, instead of improving motor abilities or specific skills (Coutinho et al., 2016; Ilic et al., 2024). Furthermore, PA seems unlikely to modify all motor abilities to a big extent because it is largely unpredictable and non-systematic, and may occur in relatively short bursts (Martínez-Vizcaíno & Sánchez-López, 2008). We believe that specifically tailored and structured physical exercise interventions with an appropriate frequency, intensity, and volume might be necessary in order to induce higher and more significant changes in all segments of the motor abilities spectrum. According to the concepts of training specificity, specifically tailored physical exercise targeting the specific motor ability will induce better improvement of that ability, rather than any other physical activity that does not directly target that ability (Zhao et al., 2021). In this line, previous study reported that PA based trials induced lower effects on muscle strength than specifically structured resistance exercise trials (Ferreira et al., 2012). Another study has shown that after applying a structured strength-oriented physical exercise intervention, there was a significant improvement in muscle strength after the intervention (Haskell et al., 2007).

However, taking in consideration the small sample size of the present study and the narrow spectrum of motor abilities assessed, larger studies are necessary for further generalization in terms of motor abilities in physically active and inactive adolescents. Inclusion of accelerometer in addition to the PA questionnaire can also be considered in future research, in order to assure that the self-reported PA is habitually maintained and complies with the real PA levels of the adolescents.

Conclusion

Physically active adolescents at the age of 13-15 years have higher upper limbs muscle strength and explosive leg power than the inactive adolescents, but there are no significant differences between them in abdominal muscle strength, lower back muscle strength and lower limbs muscle strength. We assume that structured and specifically designed exercise interventions with an appropriate frequency, intensity, and volume might be necessary to induce significant changes in all segments of the motor abilities spectrum.

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Conflicts of interest

Authors have no relevant financial or non-financial interests to disclose.

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ORIGINAL SCIENTIFIC PAPER

The Effect of Sports Activities on Stress Resilience in Students at Vietnam National University

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Abstract

Exam stress has a significant impact on students' psychological well-being and academic performance, with rising academic pressures and reduced physical activity contributing to adverse health outcomes. This underscores the urgent need for interventions to improve stress resilience. Regular participation in sports is a promising strategy to alleviate these negative effects. The aim of this study was to evaluate the influence of sports activities on students' functional and psychophysiological adaptation to exam-related stress. A total of 186 students from Vietnam National University, Ho Chi Minh City (VNU–HCMC), were divided into two groups: an experimental group (EG), comprising students who had actively participated in sports for at least 2–3 years (including volleyball, football, badminton, table tennis, swimming, and shuttlecock), and a control group (CG), consisting of students not involved in sports. Data collection occurred in two phases—between exams and on exam day. Physiological parameters such as heart rate variability, spectral analysis, and central hemodynamic indices were measured using the MindWare Technologies system. In addition, students' attention distribution was assessed through psychophysiological tests. The study revealed that students in the EG exhibited significantly better cardiovascular and psychophysiological metrics compared to the CG, with $p<0.05$. This suggests that sports participation enhances stress resilience, promotes mental well-being, and supports improved academic performance and daily functioning. These findings are consistent with previous research indicating that physical activity improves cardiovascular reserve, lowers resting heart rate, and stabilizes blood pressure. Additionally, the study showed that students involved in sports had greater attention stability, likely owing to improved autonomic nervous system regulation and reduced central stress. This study offers compelling evidence that incorporating sports activities into higher education can considerably reduce exam-related stress and enhance students' overall health and quality of life. Future research should focus on addressing limitations, such as sample size and assessing long-term effects, to further validate these results.

Keywords: *stress resilience, psychophysiological adaptation, cardiovascular health, autonomic nervous system regulation, higher education interventions*

Introduction

Exam stress is a key factor negatively impacting students' psychological well-being and academic performance. Rising academic demands, the widespread use of information technology, and distance from family create a particularly challenging environment for students. These conditions frequently result in unhealthy behaviors and a deterioration in overall health (Eksterowicz & Napierała, 2020; Gerber et al., 2017; Pengpid et al., 2019). The absence of physical activity in students' lifestyles worsens these problems, weakening the body's regulatory systems and increasing the risk of chronic diseases (Bakiko et al., 2020; Mazin et al., 2021; Syamsudin et al., 2021). Tackling these issues is essential because research shows that elevated stress levels can demotivate students and lead to serious health conditions, including sleep disorders, depression, and even suicidal thoughts (Machová et al., 2020).

With more than 150 million young people enrolled in universities and colleges globally (Santos et al., 2020), the effects of

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exam stress on student well-being and academic performance are a significant worldwide concern. Exam pressure, frequent testing, and psychological overload are key contributors to dysfunctions in the cardiovascular and nervous systems, which impair concentration and performance, ultimately diminishing students' quality of life (Machado & Soares, 2022; Tokaeva & Pavlenkovich, 2012). The transition to university life, marked by new academic challenges, shifting relationships, and financial pressures, often promotes the development of unhealthy habits and declining health. This phase can considerably disrupt students' leisure time, exercise routines, social interactions, and eating habits—especially during exam periods when students tend to consume more sugary, high-fat, and calorie-dense foods while reducing their physical activity (Hernández et al., 2023).

In light of these challenges, it is crucial to explore effective strategies for enhancing students' resilience to stress. Encouraging regular physical exercise is one such approach because it has been proven to positively impact mental health by alleviating symptoms of stress, anxiety, and depression (Riolli et al., 2012). Participation in sports and physical activity not only helps counter the harmful effects of stress but also promotes overall improvements in students' physical and mental well-being.

Regular physical activity has been shown to alleviate the effects of exam stress and promote overall health. Studies indicate that sports not only help the body regulate adverse environmental factors more effectively but also improve psychological stability, increase resilience to stress, aid recovery, and enhance concentration and sleep quality (Bocharin & Guryanov et al., 2023; Yapıcı–Öksüzoğlu, 2020). However, there remains a shortage of in-depth research on the relationship between sports and exam stress among university students, particularly in Vietnam.

Amid growing exam pressure, implementing stress-reduction strategies through sports activities has become essential. These efforts not only enhance the educational experience but also improve students' mental health and overall quality of life. In that regard, the aim of this study was to examine whether sports activities positively influence the adaptability of functional and psychophysiological systems, thereby helping students cope more effectively with stress during exam periods.

Materials and methods

Participants and procedures

A total of 186 students, both male and female, from Vietnam National University, Ho Chi Minh City (VNU–HCMC), with an average age of 19.6±1.4 years, participated in the study. The students were divided into two groups: those who had actively engaged in sports for at least 2–3 years (experimental group (EG), n=88) and those who did not participate in sports (control group (CG), n=98).

The study was performed in two phases: one between exams and the other on exam day. Physiological parameters were collected using the hardware and software system from MindWare Technologies (USA), which included measurements of heart rate variability, spectral analysis of heart rate, and central hemodynamic indices. Additionally, students' attention distribution was evaluated using the psychophysiological "Attention Distribution" test.

This study was performed at VNU–HCMC, Vietnam from January 2023 to February 2024 over two exam periods. Assessments were performed between exams and on exam days. The subsample of interest included 186 full-time students with a mean age of 19.6±1.4 years. Among them, 88 participants in the EG actively engaged in sports such as volleyball, football, badminton, table tennis, swimming, and shuttlecock for 2–3 years. They participated in training sessions 3–4 times a week, with each session lasting 90–120 min, and attended training camps 2–3 times a year. Additionally, they competed at regional, national, and international levels.

The CG, comprising 98 students, had a mean age of 19.4±1.8 years and did not engage in any sports activities. Participants in both groups voluntarily took part in the study and provided written consent. The research adhered to the ethical and moral standards of biomedical research as specified in the 2008 Declaration of Helsinki. The study was approved by the University Ethics Committee (Decision No. 1072, 11.06), and participants voluntarily provided written consent for their involvement.

Exclusion criteria

In this study, students with cardiovascular or nervous system disorders, as well as those with other conditions that could influence the study outcomes, were excluded. Additionally, non-regular students and those who did not consent to participate were also excluded. This approach was taken to ensure the accuracy and objectivity of the results obtained.

Measurements

To evaluate cardiovascular and higher nervous system activities before and after the exams, the study used the MindWare Technologies hardware and software system (MindWare Technologies Ltd., Gahanna, OH, USA). This system includes sensors integrated into a computer mouse for continuous monitoring. The following physiological parameters were assessed.

Chronocardiometry variability: This involved measuring the average heartbeat interval (in milliseconds), the stress index of regulatory systems (SI, in conventional units), the autonomic nervous system balance index (IVE, in conventional units), and the autonomic rhythm index (VRI, in conventional units).

Heart rate spectral analysis: This analysis involved calculating the power ratio of high-frequency (HF) waves to low-frequency (LF) waves, expressed as percentages.

Central hemodynamic parameters: The cardiovascular parameters assessed included heart rate, systolic and diastolic blood pressure, pulse pressure, stroke volume, cardiac output, and the Robinson index, which measures cardiac workload.

Statistical analysis

Various statistical methods were applied to assess the differences between EG and CG. Descriptive statistics, including means and standard deviations (M ± SD), were calculated for all variables. An independent sample t-test was performed to determine the significance of differences between the groups. A paired t-test was used for in-group comparisons (pre-exam vs. exam day). Additionally, repeated measures analysis of variance (ANOVA) analyzed the interaction effects of group (EG vs. CG) and time (pre-exam vs. exam day) on physiological and psychophysiological parameters. A significance threshold of $p<0.05$ was set for all analyses. Data were processed using SPSS software (IBM Corp., Armonk, NY, USA), ensuring a rigorous and robust analysis.

Results

The analysis of cardiovascular parameters, as shown in Table 1, demonstrated a significant positive correlation between sports participation and cardiovascular adaptability in both male and female students.

This study demonstrates that participation in sports activities significantly influences cardiovascular adaptation in students. Table 1 clearly highlights the substantial differences in chronocardiometry indices between the CG and EG.

In the male group specifically, the mean cardiointerval in the CG during the exam was 798.05 ms, decreasing to 701.34 ms during the between-exam period. In contrast, the EG ex-

Table 1. Comparative data on chronocardiometry variability indices between the EG and CG (M ± m)

MALE		**(CG, n = 50)**		**(EG, n = 48)**	
PARAMETER	**Unit**	**On the examination day**	**Intervening period**	**On the examination day**	**Intervening period**
IVE	cu	148.67 ± 7.02	188.42 ± 14.22*	109.43 ± 3.33#	138.67 ± 3.72#*
MC	ms	798.05 ± 10.54	701.34 ± 11.35*	1001.28 ± 10.18#	1042.24 ± 9.88#*
HF	%	45.87 ± 1.66	33.23 ± 1.77*	55.22 ± 1.76#	44.27 ± 1.54#*
LF	%	51.52 ± 1.64	61.78 ± 1.29*	51.25 ± 1.43#	55.24 ± 1.91#*
SI	cu	102.01 ± 4.07	122.75 ± 9.79*	69.45 ± 1.92#	81.56 ± 3.46#*
VRI	cu	5.01 ± 0.11	5.02 ± 0.23	3.52 ± 0.11#	3.51 ± 0.12#
FEMALE		(CG, n = 48)		(EG, n = 40)	
IVE	cu	109.56 ± 3.57	171.45 ± 16.28*	80.31 ± 3.32#	91.72 ± 4.23#*
MC	ms	793.03 ± 37.28	1029.45 ± 22.36*	814.61 ± 10.09#	1141.21 ± 11.81#*
HF	%	51.87 ± 5.45	31.27 ± 4.46*	68.81 ± 7.61#	42.21 ± 1.54#*
LF	%	27.56 ± 4.37	47.1 ± 5.45*	22.31 ± 1.42#	36.21 ± 1.67#*
SI	cu	51.3 ± 4.01	60.65 ± 5.23*	31.38 ± 2.02#	42.35 ± 3.51#*
VRI	cu	3.20 ± 0.43	3.37 ± 0.51*	3.03 ± 0.12#	3.01 ± 0.13#

Note: * Indicates a significant difference in index values on exam day and between exam periods ($p < 0.05$); # indicates a significant difference in index values between the experimental group (EG) and control group (CG) ($p < 0.05$); (IVE): Measure of balance in the autonomic nervous system; (MC): Average interval between heartbeats; (HF): Proportion of high-frequency waves in heart rate variability, associated with parasympathetic activity; (LF): Proportion of low-frequency waves in heart rate variability, reflecting sympathetic and parasympathetic balance; (SI): Indicator of stress on the body's regulatory systems; (VRI): Measure of autonomic nervous system balance

Table 2. Values of central hemodynamic indices in male and female students between EG and CG (M ± m)

PARAMETER		**(CG, n = 50)**			**(EG, n = 48)**		
MALE	**Unit**	**On the examination day**	**Intervening period**	**W%**	**On the examination day**	**Intervening period**	**W%**
Heart rate	bpm	102.34 ± 5.56	74.45 ± 5.04*	31.55	73.57 ± 4.14#	64.37 ± 3.03#*	13.34
Maximal systolic and diastolic blood pressure	mmHg	131.13 ± 10.05	121.62 ± 13.23	7.53	121.02 ± 6.21	118.86 ± 6.48	1.8
Minimal systolic and diastolic blood pressure	mmHg	96.26 ± 9.45	88.62 ± 5.65	8.26	89.61 ± 5.76	84.42 ± 6.01	5.96
Minute volume of blood circulation	l	5.35 ± 1.01	2.09 ± 0.91*	87.63	4.52 ± 1.21#	3.09 ± 1.08#	37.58
Pulse pressure	mmHg	45.39 ± 3.06	41.03 ± 3.51	10.09	36.28 ± 3.13#	39.25 ± 3.42	-7.86
Robinson index	cu	144.81 ± 8.08	97.25 ± 6.09*	39.3	82.28 ± 6.02#	72.17 ± 4.13#*	13.09
Stroke volume of the heart	ml	46.81 ± 4.53	39.05 ± 4.12*	18.08	66.81 ± 4.61#	65.38 ± 4.51#	2.16
FEMALE		(CG, n = 48)			(EG, n = 40)		
Heart rate	bpm	104.07 ± 5.17	74.51 ± 5.21*	33.11	73.03 ± 3.09#	64.25 ± 4.31#*	12.79
Maximal systolic and diastolic blood pressure	mmHg	133.51 ± 7.02	124.03 ± 10.13	7.36	122.01 ± 3.01	126.51 ± 2.81	-3.62
Minimal systolic and diastolic blood pressure	mmHg	92.71 ± 6.14	84.05 ± 5.35	9.8	87.39 ± 5.61	75.81 ± 6.21*	14.19
Minute volume of blood circulation	l	5.21 ± 0.15	3.22 ± 0.54*	47.21	4.32 ± 0.25#	3.44 ± 0.33*	22.68
Pulse pressure	mmHg	46.07 ± 4.13	39.05 ± 4.27*	16.49	39.74 ± 3.54#	43.27 ± 4.26#*	-8.5
Robinson index	cu	141.02 ± 6.41	89.27 ± 6.31*	44.94	94.53 ± 7.18#	80.31 ± 7.21	16.27
Stroke volume of the heart	ml	47.25 ± 4.57	43.25 ± 7.45	8.84	56.45 ± 6.67	53.26 ± 4.28	5.82

Note: * indicates a significant difference in the index values on exam day and between exam periods ($p < 0.05$); # indicates a significant difference in the index values between EG and CG ($p < 0.05$)

hibited a mean cardiointerval of 1001.28 ms, which increased to 1042.24 ms. These results suggest that students in the EG had better cardiovascular regulation, as shown by a lower stress index of the regulatory system compared to the CG.

For female students, the mean cardiointerval in the CG during the exam was 793.03 ms, increasing to 1029.45 ms in the between-exam period, while in the EG, it increased from 814.61 to 1141.21 ms. This further underscores the considerable differences between the two groups, with the EG displaying a notably lower stress index.

Moreover, indices such as HF and LF showed clear distinctions between the groups, with the EG exhibiting better regulation and a more stable balance of the autonomic nervous system (IVE) compared to the CG.

The analysis of central hemodynamic indices between the EG and CG revealed significant differences in several parameters, particularly heart rate, stroke volume, and the Robinson index.

On exam day, the heart rate of male students in the CG averaged 102.34±5.56 bpm, while in the EG, it was considerably lower at 73.57±4.14 bpm, indicating a significant difference between the two groups (p<0.05). After the intervention period, the heart rate in the CG considerably decreased to 74.45±5.04 bpm, corresponding to a 31.55% reduction. Similarly, the EG also showed a substantial decrease, from 73.57±4.14 bpm to 64.37±3.03 bpm, representing a 13.34% reduction. Among female students, the pattern was consistent, with the heart rate in the CG decreasing from 104.07±5.17 bpm to 74.51±5.21 bpm (33.11% reduction), while in the EG, it decreased from 73.03±3.09 bpm to 64.25±4.31 bpm (12.79% reduction).

Both systolic and diastolic blood pressure showed a slight decrease in both male and female groups; however, the changes were not significant, and no statistically meaningful differences were found between the groups (p>0.05). This indicates that the intervention had little clear effect on blood pressure.

In contrast, stroke volume significantly decreased in the CG, from 46.81±4.53 ml to 39.05±4.12 ml (18.08%) in males and from 47.25±4.57 ml to 43.25±7.45 ml (8.84%) in females. However, in the EG, the change was minimal, with only a 2.16% decrease in males and 5.82% in females. These findings suggest that the intervention program helped maintain a more stable stroke volume in the EG compared to the CG.

The Robinson index, a key measure of cardiovascular exertion, also revealed significant differences. In the CG, this index decreased from 144.81±8.08 cu to 97.25±6.09 cu (39.30%) in males and from 141.02±6.41 cu to 89.27±6.31 cu (44.94%) in females. In contrast, the EG showed a more modest decrease of 13.09% in males and 16.27% in females, indicating greater stability and resilience in cardiovascular health in the intervention group.

The results of the psychophysiological tests for EG and CG are presented in Table 3.

Table 3. Values of "Attention distribution" test indices for male and female students in the CG and EG (M ± m).

PARAMETER		(CG, n = 50)		(EG, n = 48)	
MALE	**Unit**	**On the examination day**	**Intervening period**	**On the examination day**	**Intervening period**
Asymmetry	cu	1.34 ± 0.01	1.21 ± 0.01*	1.24 ± 0.01#	1.12 ± 0.02#*
Average reaction time to stimuli	m	2203.35 ± 59.06	2326.04 ± 48.19*	2047.50 ± 43.71#	2109.50 ± 52.21#
Kurtosis	cu	1.79 ± 0.21	2.01 ± 0.34*	1.61 ± 0.24#	1.65 ± 0.41#*
Maximum reaction time	ms	5073.2 ± 135.02	5314.3 ± 93.34*	4302.3 ± 70.23#	4525.3 ± 81.43#*
Minimum reaction time	ms	1003.32 ± 18.45	1004.01 ± 19.34	963.06 ± 16.34#	961.02 ± 15.13#
Mode amplitude	%	18.09 ± 0.31	17.41 ± 0.32*	16.42 ± 0.51#	15.23 ± 0.45#
Mode	ms	1527.31 ± 53.12	1877.12 ± 62.22*	1314.37 ± 51.41#	1391.12 ± 57.42#
Variation range	ms	4052.19 ± 132.04	4266.21 ± 86.15*	3766.55 ± 67.17#	3967.23 ± 62.81#*
FEMALE		(CG, n = 48)		(EG, n = 40)	
Asymmetry	cu	1.01 ± 0.02	2.73 ± 0.80*	0.84 ± 0.04#	1.12 ± 0.73#
Average reaction time to stimuli	m	2256.67 ± 71.21	2454.51 ± 98.52*	1898.51 ± 102.51#	1923.51 ± 118.47#
Kurtosis	cu	0.82 ± 0.22	1.71 ± 0.62*	0.30 ± 0.10#	0.70 ± 0.40#
Maximum reaction time	ms	5132.72 ± 225.71	5784.21 ± 301.23*	3012.34 ± 261.81	4009.46 ± 276.4*
Minimum reaction time	ms	1007.75 ± 26.47	1197.72 ± 51.71*	945.65 ± 54.67#	967.74 ± 52.67#
Mode amplitude	%	16.68 ± 1.05	16.85 ± 1.01	15.31 ± 1.01#	14.79 ± 1.01#
Mode	ms	1698.42 ± 104.21	1887.71 ± 143.13*	1560.17 ± 113.63#	1676.25 ± 151.13#
Variation range	ms	4121.01 ± 338.12	3901.22 ± 332.23	3675.22 ± 297.12	3752.21 ± 312.32

Note: * indicates a significant difference in index values on exam day and between exam periods (p < 0.05); # indicates a significant difference in index values between EG and CG (p < 0.05)

The study included an "Attention Distribution" test, measuring reaction time, standard deviation, and other statistical parameters for both male and female students in the EG and CG. The results revealed significant differences between the groups, highlighting the positive effect of the intervention program on students' attention distribution abilities.

In particular, the findings showed that the average reaction time to stimuli in the EG improved compared to the CG for both male and female students. Among males, the average reaction time in the CG increased from 2203.35±59.06 ms to 2326.04±48.19 ms after the intervention, while in the EG, it increased only slightly from 2047.50±43.71 ms to 2109.50±52.21

ms. This difference was statistically significant ($p<0.05$), indicating that the intervention helped the EG maintain more stable reaction times. Similarly, in the female group, the average reaction time in the CG increased from 2256.67±71.21 ms to 2454.51±98.52 ms, whereas in the EG, it only slightly increased from 1898.51±102.51 ms to 1923.51±118.47 ms, showing a considerable improvement.

Mode amplitude and standard deviation are key indicators of reaction time stability. In the male group, the mode amplitude in the CG decreased from 18.09±0.31% to 17.41±0.32%, whereas in the EG, it decreased from 16.42±0.51% to 15.23±0.45%. This suggests that the EG exhibited better reaction stability than the CG, with a statistically significant difference ($p<0.05$). However, in the female group, the change was not significant, and no statistically meaningful difference was found between the groups, indicating that the intervention's impact on reaction stability in females may be less pronounced.

The range of variation in reaction time (variation range) increased significantly in both males and females in the CG, from 4052.19±132.04 ms to 4266.21±86.15 ms in males, and from 4121.01±338.12 ms to 3901.22±332.23 ms in females. In contrast, the EG showed only a slight increase, from 3766.55±67.17 ms to 3967.23±62.81 ms in males and from 3675.22±297.12 ms to 3752.21±312.32 ms in females. This difference was statistically significant, indicating that the intervention program effectively reduced reaction time variability, particularly in males.

Furthermore, other statistical indices, such as kurtosis and skewness, showed significant improvements in the EG compared to the CG. Among males, kurtosis in the EG decreased from 1.79±0.21 cu to 1.65±0.41 cu, whereas it increased in the CG from 1.79±0.21 cu to 2.01±0.34 cu. Skewness also decreased significantly in the EG, while it increased in the CG. In females, the trend was similar, with both kurtosis and skewness decreasing in the EG and increasing in the CG.

Discussion

Studies have shown that an educational environment can negatively impact students' health, particularly during exams when significant physiological and psychological responses occur (Kolokoltsev et al., 2020; Machado & Soares, 2022; Tokaeva & Pavlenkovich, 2012). The increase in stress associated with exams can disrupt the body's regulatory mechanisms, affecting both the quality of education and overall student health, and may lead to the development of health issues (Junger et al., 2019; Kolokoltsev et al., 2021; Zhang et al., 2019). In this context, it is essential to research and implement strategies to alleviate the effects of exam stress.

The enhancement of cardiovascular adaptability in the EG can be attributed to various physiological mechanisms linked to regular physical activity. Participation in sports improves cardiovascular efficiency through a process known as "cardiovascular conditioning." This process includes an increased stroke volume—the amount of blood the heart pumps per beat—and improved vasodilation, which lowers overall vascular resistance. These changes contribute to a reduced resting heart rate and a more stable cardiovascular system, as evidenced by the lower stress indices and improved Robinson index in the EG. Additionally, physical activity strengthens the autonomic nervous system's capacity to regulate heart rate variability, which is crucial for stress resilience and maintaining cardiovascular homeostasis during stress (Bocharin et al., 2023; Yapıcı-Öksüzoğlu, 2020).

The observed improvements in attention distribution and reaction time can be attributed to sports participation, which stimulates the autonomic nervous system, particularly the parasympathetic branch responsible for calming the body and promoting recovery. Regular physical activity enhances the regulation of this system, reducing stress on the central nervous system and facilitating more stable cognitive performance, as evidenced by decreased variability in reaction times. This increased stability results from improved oxygen delivery to the brain, enhanced neural efficiency, and better cognitive function—benefits that have been documented with consistent physical training (Kamandulis et al., 2020). These findings are consistent with research demonstrating that physical activity positively affects both physiological and psychological parameters, thereby supporting enhanced cognitive regulation and attention control (Zhang et al., 2019; Zurita-Ortega et al., 2019).

In summary, the findings of this study confirm that sports activities not only enhance physical fitness but also significantly improve students' adaptability to exam stress, particularly in cardiovascular and psychological aspects. These results indicate that incorporating sports programs into higher education may be an effective strategy for improving students' overall health and quality of life, ultimately leading to better academic performance and greater resilience to academic stressors.

Limitations and suggestions for further research

The study provides valuable insights into the positive effects of sports activities on cardiovascular adaptability and attention distribution among students; however, it has several limitations. First, the sample size and demographic characteristics may not adequately represent the entire student population, particularly regarding factors such as age, gender, health status, and social background, which may limit the generalizability of the findings. Additionally, the study did not comprehensively control for other variables that might have influenced the outcomes, including nutrition, sleep, external stress levels, and mental health status. The short-term data collection, focused around exam periods, limited the assessment of the long-term effects of sports participation on cardiovascular and psychological health. Furthermore, although the study acknowledged gender differences, it did not investigate the underlying reasons for these disparities or the specific factors influencing males and females. Given these limitations, future research should aim to include larger and more diverse sample sizes while also controlling for additional variables such as lifestyle habits and initial health conditions. Long-term follow-up studies are essential to assess the lasting impact of sports activities on student health and well-being after graduation. Additionally, further research should explore gender differences in more depth and consider incorporating various intervention methods, such as yoga, meditation, or relaxation techniques, to determine the most effective strategies for enhancing student health and resilience to stress.

Conclusion

This study offers compelling evidence of the crucial role that sports activities play in enhancing cardiovascular adaptability and improving attention distribution among VNU–HCMC students. The findings suggest that students engaged in sports exhibit superior cardiovascular regulation, characterized by lower stress indices and a more balanced autonomic nervous system compared to their non-participating peers. Notably, the sports-active group demonstrated greater stability and sustainability in hemodynamic indices, along with improved focus and attention stability during exams.

The discussion highlights that sports activities not only promote physical health but also serve an important function in alleviating the negative impacts of exam stress on students' cardiovascular and psychological well-being. This aligns with previous research and underscores the potential for implementing sports intervention programs in higher education to enhance students' quality of life and academic performance.

However, the study has several limitations, including sample size, short-term follow-up, and a lack of control over extraneous variables, which need to be addressed in future research. Subsequent studies should aim for a larger scale, incorporate long-term follow-up, and consider multiple factors to reach more comprehensive conclusions.

In conclusion, this study confirms that active participation in sports not only enhances cardiovascular adaptability but also improves focus and stress management, ultimately enabling students to achieve better academic results and overall health.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Investigating the Effectiveness of 12 Weeks of Kettlebell Training Compared to Bodyweight Resistance Training on Body Composition in Young Adult Males

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Abstract

This study compared the effects of bodyweight exercises with kettlebell resistance training on important body composition measures in male young adults. A passive control group was also included. Data from 18 young male participants, with an average age of (20.43±2.18) years, were included in the final analysis. The participants were divided into three groups at random: body weight resistance training group (BWRTG, n=6), kettlebell resistance training group (KRTG, n=6), and the control group (CG, n=6), which had no training. Body composition was evaluated through various metrics, including skeletal mass, body fat mass, body mass index, body fat percentage, waist-to-hip ratio, and visceral fat area. The 12-week progressive training programs for KRTG and BWRTG were structured with defined sets, repetitions, and intensity modifications. The results revealed that both the KRTG and BWRTG groups showed similar effects on body fat mass (p=0.032, ES=0.22; p=0.021, ES=0.29), body fat percentage (p=0.040, ES=0.22; p=0.037, ES=0.21), and visceral fat area (p=0.004, ES=0.36; p=0.002, ES=0.60). A significant improvement in the waist-to-hip ratio was observed in the KRTG group (p=0.036, ES=0.36), while no significant change was seen in the BWRTG group (p=0.036, ES=0.52). The results show that the 12-week KRTG and BWRTG regimens had comparable benefits on reductions in visceral fat area, body fat mass, and body fat percentage. These results indicate that both training methods are suitable for individuals aiming to enhance body composition, with kettlebell training providing an extra advantage in decreasing waist-to-hip ratio.

Keywords: *visceral fat reduction, skeletal muscle mass, waist-to-hip ratio, body composition improvement, resistance training benefits*

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Introduction

The rise of sedentary lifestyles in contemporary society has become a global health issue, contributing significantly to the increase in non-communicable diseases (Budreviciute et al., 2020; Singh et al., 2024; Uddin et al., 2020). Sedentarism, characterized by prolonged periods of physical inactivity, is now considered a major risk factor for chronic conditions such as obesity, cardiovascular disease, hypertension, and type 2 diabetes (Bueno-Antequera & Munguía-Izquierdo, 2023; Papry et al., 2024; Park et al., 2020). The World Health Organization (WHO) has repeatedly highlighted the dangers associated with inactivity, and there is growing concern about the long-term consequences of sedentary behaviour on individual health and well-being (World Health Organization, 2024). Identifying exercise strategies that can effectively counteract the adverse effects of sedentarism is paramount for improving public health outcomes (Calcaterra et al., 2023; Faghy et al., 2024).

Body composition, which refers to the proportions of fat mass, lean muscle mass, and other components of the body, plays a crucial role in determining an individual's overall health (Lu et al., 2024; Olshvang et al., 2024; Petřeková et al., 2024). A healthy body composition, characterized by lower fat mass and higher lean mass, has been associated with reduced risk factors for metabolic disorders, improved physical functioning, and enhanced quality of life (Benton & Hutchins, 2024; Chianeh et al., 2024; Lestari et al., 2024). In contrast, excess fat mass, particularly visceral fat, is linked to many health problems, including insulin resistance, inflammation, and a higher likelihood of developing cardiovascular disease (Bosello et al., 2024; Hagberg & Spalding, 2024). For individuals leading a sedentary lifestyle, one of the primary goals of exercise interventions is to shift body composition toward a healthier balance by reducing fat mass and increasing lean muscle mass.

Resistance training has emerged as one of the most effective modalities for altering body composition (Bellicha et al., 2021; X. Chen et al., 2024; Rejeki et al., 2023). In challenging the muscles to work against resistance, this exercise promotes muscle hypertrophy, increases metabolic rate, and contributes to fat loss (Kataoka et al., 2024; Mambrini et al., 2024). Resistance training builds muscular strength, enhances muscle endurance, and promotes long-term weight management (Abou Sawan et al., 2023; Cox, 2024). Various forms of resistance training are available, with kettlebell and bodyweight resistance training being two prominent methods that have gained attention in recent years due to their effectiveness and accessibility.

Kettlebell training, which utilizes a uniquely shaped weight resembling a cannonball with a handle, has grown in popularity due to its multifunctional nature. Kettlebell exercises often involve compound movements that engage multiple muscle groups simultaneously, such as kettlebell swings, snatches, and Turkish get-ups. These dynamic movements improve strength and increase cardiovascular endurance, making kettlebell training a time-efficient option for improving body composition and overall fitness (Meigh et al., 2022). Importantly, the swinging and ballistic motions involved in kettlebell training stimulate core stabilization and full-body coordination, providing functional fitness benefits that translate to daily activities. Studies suggest that kettlebell workouts can significantly improve muscular strength, endurance, and aerobic capacity while also contributing to fat loss and lean muscle gain (H.-T. Chen et al., 2018; Govindasamy et al., 2024; Radhika & Shanmugavalli, 2024). Bodyweight resistance training, on the other hand, has a long history of being a fundamental part of physical fitness routines across various disciplines (Sevilmis et al., 2023). This form of training involves using one's body weight to provide resistance against gravity, incorporating exercises such as push-ups, squats, lunges, and planks (Ernandini & Giovanni Mulyanaga, 2023; Palmer & McCabe, 2023). Bodyweight training requires no equipment and can be performed in virtually any setting, making it one of the most accessible forms of exercise for people at all fitness levels (Spotswood et al., 2023; Wackerhage & Schoenfeld, 2021). For individuals, bodyweight resistance training offers a non-intimidating entry point into physical activity. Research has shown that bodyweight training can be efficient for building muscular strength, enhancing muscular endurance, and improving flexibility, all of which are critical components of physical fitness (Fayazmilani et al., 2022; Koźlenia et al., 2024; Nasrulloh et al., 2022). Moreover, bodyweight training has been associated with improved metabolic health, making it a valuable tool for those seeking to improve body composition through fat loss and muscle gain (Carneiro et al., 2018).

While kettlebell and bodyweight resistance training are effective methods for improving body composition, there must be a more direct comparison between these modalities, especially in young adults populations. Therefore, this study aims to fill this gap by evaluating the effects of kettlebell training versus bodyweight resistance training on body composition in young adults.

Methods

Participants

The analysis revealed that 18 participants were required to identify a significant medium effect with an alpha error probability of 0.05 and a power of 0.80 for the interaction between factors. Figure 1 presents a graphical representation of the study design. Initially, 24 college-age male students consented to participate in the study. Eligibility criteria for the study included: (i) no history of lower limb injuries in the six months preceding the study, (ii) participation in strength training programs for at least three months prior to the study, and (iii) achieving a minimum threshold in the one-repetition maximum (1-RM) test for a major resistance exercise (e.g., squat or bench press) to ensure safe participation in the study's training programs. The top 18 participants who met the 1-RM criteria were selected for inclusion. The assessment of maximal strength was based on the 1-RM test, a commonly used measure in resistance training studies to evaluate muscular strength (Brzycki, 1993). Six participants who did not complete the assessment were excluded. The study comprised 18 participants, categorized into three groups: Body Weight Resistance Training Group (BWRTG, n=6), Kettlebell Resistance Training Group (KRTG, n=6), and a Passive Control Group (CG, n=6). Additionally, throughout the trial period, there were no severe injuries related to testing or training, and both the KRTG and body weight BWRTG demonstrated perfect attendance rates of 100%. All groups of participants shared comparable demographic characteristics, as detailed in Table 1. Before the study commenced, participants were briefed about the potential risks and benefits associated with the research. Subsequently, informed consent was obtained from each participant. The research obtained approval from the local ethics committee (SBU/RHS/SBUEC/20/067/2023) and complied with the ethical standards outlined in the Declaration of Helsinki (1964, revised 2013).

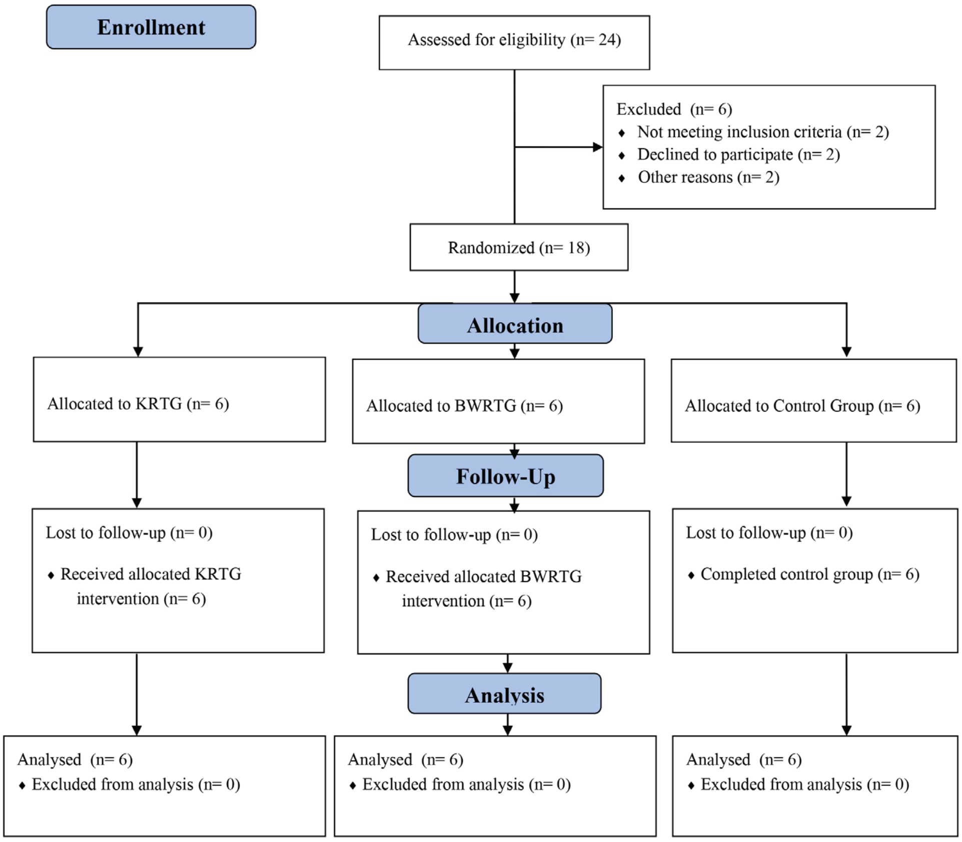

FIGURE 1. CONSORT Flow Chart

Procedure

One week prior to the baseline assessment, participants underwent two familiarization sessions to become familiar with the body composition tests. During these sessions, demographic information was also collected. Participants were instructed to avoid intense physical activity and alcohol consumption for 24 hours before the testing. A randomized longitudinal design was employed to evaluate the effects of the exercise interventions, specifically Kettlebell Resistance Training (KRT) and Body Weight Resistance Training (BWRT) on BM – body mass, SM – skeletal mass, BF- body fat mass, BMI- body mass index, BFP- body fat percentage, WHO- wait hip ratio, VFA-visceral fat area. Measurements were taken after participants completed a 10-minute general warm-up. Evaluations were conducted during the early morning hours, between 7:30 a.m. and 9:30 a.m., at biomedcal lab, under stable room conditions (temperature ranging from (around 20°C to 24°C) and humidity (40-60%)).

Outcome Measures

Body Composition

Body composition was measured using multifrequency bioelectrical impedance analysis with the InBody 720 device (Biospace Co. Ltd., Seoul, Korea). The assessments took place in the morning between 7 a.m. and 8 a.m. Participants were instructed to wear light, standard indoor attire (such as shorts and a T-shirt) to minimize the influence of clothing weight on the measurements. Additionally, all participants were instructed to remove any metallic objects, including rings, watches, bracelets, and other jewelry, to avoid interference with the electrical currents used in the BIA test. This step ensures more accurate body composition readings. They stood barefoot on the platform with their feet placed on the foot electrodes. While maintaining an upright posture, participants abducted their arms slightly from the body and held onto the hand electrodes positioned on the device's handles, holding this position throughout the test for consistent and reliable measurements (Lee & Gallagher, 2008).

Interventions

The training intervention for the KRTG and the BWRTG was designed to be progressive and structured over a 12-week period in Table 1 (Govindasamy et al., 2024). For both groups, participants were instructed to rest for 60-90 seconds between sets, allowing sufficient recovery while maintaining an appropriate intensity level throughout the training sessions. Each group performed their training three times per week, with KRTG sessions scheduled for Monday, Wednesday, and Friday, while BWRTG sessions occurred on Tuesday, Thursday, and Saturday. Each session began with a 10-minute warm-up that included dynamic stretches and mobility exercises to prepare participants for the workout.

For the KRTG, the program was divided into three phases. In the first four weeks, participants completed 2 sets of 10-12 repetitions of exercises such as kettlebell swings, goblet squats, kettlebell deadlifts, kettlebell presses (overhead press), kettlebell lunges, and kettlebell rows, using light to moderate kettlebell weights (20-40% of their one-repetition maximum, or 1RM). The second phase, weeks 5-8, increased the workload to 3 sets of 8-10 repetitions with moderate kettlebell weights (40-60% of 1RM). In the final phase, weeks 9-12, participants performed 4 sets of 6-8 repetitions with challenging weights (60-80% of 1RM), followed by a 10-minute cool-down of static stretches.

The BWRTG followed a similar structure, focusing on bodyweight exercise intensity. Participants began with 2 sets of 10-12 repetitions for the first four weeks, incorporating exercises such as push-ups, squats, lunges, planks, burpees, and mountain climbers at a moderate pace. From weeks 5-8, the program progressed to 3 sets of 8-10 repetitions, increasing the speed and complexity of the movements. In the final weeks, participants completed 4 sets of 6-8 repetitions, incorporating advanced variations, including plyometric exercises, and concluded each session with a 10-minute static stretch cool-down.

Table 1. Training schedule for the KRT group and BWRT group

Week	KRTG	BWRTG
1-4	2 sets of 10-12 reps	2 sets of 10-12 reps
5-8	3 sets of 8-10 reps	3 sets of 8-10 reps
9-12	4 sets of 6-8 reps	4 sets of 6-8 reps

Statistical Analysis

The descriptive statistics for the participants are summarized as mean and standard deviation, calculated utilizing IBM SPSS Statistics version 29.0 (SPSS Inc., Chicago, IL, USA). To assess normality, the Shapiro-Wilk test was employed (Shapiro & Wilk, 1965), Levene's test was employed to assess the homogeneity of variance. The normality of the outcomes was assessed for each group individually, with all results showing p-values greater than 0.05, indicating normal distribution. Homogeneity of variances was evaluated using Levene's test, which also resulted in p-values exceeding 0.05. To analyze baseline data across three groups, one-way ANOVA was conducted (Fisher, 1992), followed by Bonferroni's post hoc tests for pairwise comparisons. A mixed model ANOVA was employed to assess the intervention's effect on body composition outcomes, combining group components (CG, KRTG, BWRTG) and time points (baseline and post-intervention) (Verbeke, 1997). Interaction effects were documented with p-values and partial eta squared (η^2p). Effect sizes were classified as follows: η^2p of 0.01 indicates a small effect, 0.06 denotes a medium effect, and 0.14 represents a large effect (Cohen, 1988). A simple effects test was conducted following the confirmation of the interaction between group and time to ensure accurate analysis. The magnitude of differences between means was assessed using Cohen's d (effect size, ES) (Cohen, 1988), calculated with the formula (mean post-intervention – mean pre-intervention) / pooled standard deviation. Cohen's d values of 0.20, 0.50, and 0.80 are interpreted as representing small, moderate, and large effects, respectively (Cohen, 1988). Statistical significance was assessed using a threshold of $p<0.05$.

Results

Eighteen adults, completed the baseline assessments for the study. Table 2 displays the baseline characteristics of the participants.

Table 2. Participant Demographics in the Experimental Groups (KRTG, BWRTG) and Control Group

Group	KRTG	BWRT	CG	P value
Age	20.62 ± 2.15	20.46 ± 1.74	20.21 ± 2.57	0.542
Height	169.7 ± 4.8	168.1 ± 6.2	169.5 ± 2.4	0.527
Weight	64.5 ± 6.2	62.1 ± 5.4	53.3 ± 7.6	0.634

Note. CG – control group, KRTG – kettlebell resistance training group, BWRTG – body weight resistance training group

Significant differences were observed between the two groups: KRTG and the Control Group (CG) in terms of BM and BMI ($p<0.05$). No other significant differences were found between the groups prior to the exercise intervention ($p>0.05$). Descriptive statistics for body composition before and after the intervention are provided in Table 3. A significant main effect of the group on BM was observed, with a p-value of 0.025 and a partial eta squared (η^2p) of 0.41, as well as on BMI with a p-value of 0.024 and η^2p of 0.35. However, no significant group differences were observed for SM , BF, BFP, VFA, or WHR. A significant main effect of time was found for BF ($p=0.003$, $\eta^2p=0.44$), BFP ($p=0.007$, $\eta^2p=0.40$), and VFA ($p<0.001$, $\eta^2p=0.46$). However, there were no significant changes over time in BM, SM, BMI, and WHR. There was a significant interaction effect between group and time for the WHR ($p=0.036$, $\eta^2p=0.36$). Additionally, a trend toward a significant interaction effect was observed for VFA ($p=0.054$, $\eta^2p=0.37$), although this result did not reach statistical significance. Subsequent analysis indicated no notable differences in WHR between the groups at baseline or following the intervention. Pairwise comparisons within the groups indicated a significant reduction in WHR in the KRTG group ($p=0.036$, effect size=0.52). In contrast, no significant changes were noted in the BWRTG group ($p=0.176$, effect size=0.24) or the control group ($p=0.160$, effect size=0.27). Figure 2 illustrates the variations in body composition both within and between groups. Within-group analyses revealed a significant reduction in BF for the KRTG ($p=0.032$, ES=0.22, small effect) and BWRTG ($p=0.021$, ES=0.29, small effect) groups. In contrast, no significant changes were observed in the CG for BF ($p=0.657$, ES = 0.06, very small effect). Similarly, significant decreases in BFP were noted in the KRTG ($p=0.040$, ES=0.22, small effect) and BWRTG ($p=0.037$, ES=0.21, small effect) groups. However, no significant changes in BFP were found in the CG ($p=0.499$, ES=0.08, very small effect). A significant reduction in VFA was observed in the KRTG ($p=0.004$, ES=0.36, moderate effect) and BWRTG ($p=0.002$, ES=0.60, large effect) groups. No significant changes in VFA were noted in the CG ($p=0.672$, ES=0.06, very small effect). Between-group comparisons showed significant differences in body mass and body mass index post-intervention. Specifically, BM was significantly different between KRTG and CG ($p=0.024$, ES=1.56, large effect), and BMI was significantly different between KRTG and CG (p=0.029, ES=1.67, large effect). It is essential to recognize that there were also significant baseline differences between KRTG and CG in BM (p=0.034, ES=1.69, large effect) and in BMI (p=0.022, ES=1.71, large effect), which may limit the interpretation of these significant post-intervention differences.

Table 3. Pre & Post-test performances of body composition varaibles in young adults following 12 weeks of three groups:KRTG, BWRTG & CG

Outcome Measures	Groups	Mean ± SD		% Change	Main effect group		Main effect time		Interaction group × time	
		Pre Test (before)	Post Test (after)		p	η^2p	p	η^2p	p	η^2p
BM (kg)	CG	53.3 ± 7.6	53.6 ± 6.4	0.56	0.025*	0.41	0.214	0.12	0.541	0.07
	KRTG	64.5 ± 6.2	63.5 ± 3.7	-1.55						
	BWRTG	62.1 ± 5.4	61.3 ± 6.4	-1.28						
SM (kg)	CG	22.4 ± 3.5	22.6 ± 3.7	0.89	0.208	0.21	0.067	0.20	0.620	0.09
	KRTG	25.1 ± 2.3	26.5 ± 3.4	5.57						
	BWRTG	23.2 ± 4.2	23.8 ± 5.1	2.58						
BF (kg)	CG	14.5 ± 4.3	14.2 ± 5.1	-2.06	0.424	0.13	0.003**	0.44	0.341	0.18
	KRTG	17.6 ± 5.1	16.5 ± 4.5	-6.25						
	BWRTG	16.4 ± 3.4	15.3 ± 4.1	-6.70						
BMI (kg/m^2)	CG	21.2 ± 1.4	21.2 ± 1.4	0	0.024*	0.35	0.167	0.19	0.515	0.08
	KRTG	23.4 ± 1.2	23.3 ± 1.2	-0.42						
	BWRTG	22.3 ± 1.5	22.0 ± 1.4	-1.34						
BFP (%)	CG	26.1 ± 7.3	25.5 ± 6.9	-2.29	0.845	0.03	0.007**	0.40	0.541	0.12
	KRTG	27.6 ± 7.6	26.4 ± 1.2	-4.34						
	BWRTG	26.7 ± 6.6	25.4 ± 5.7	-4.86						
WHR	CG	0.83 ± 0.4	0.84 ± 0.4	1.20	0.863	0.04	0.186	0.13	0.036*	0.36
	KRTG	0.85 ± 0.3	0.84 ± 0.3	-1.17						
	BWRTG	0.84 ± 0.4	0.83 ± 0.4	-1.19						
VFA	CG	56.3 ± 27.1	54.6 ± 26.3	-3.01	0.768	0.10	< .001**	0.46	0.054	0.37
	KRTG	75.5 ± 26.3	66.3 ± 24.2	-12.18						
	BWRTG	68.6 ± 19.4	58.3 ± 14.6	-15.01						

*p<0.05; **p<0.01; Abbreviations: CG – control group, KRTG – kettlebell resistance training group, BWRTG – body weight resistance training group, BM – body mass, SM – skeletal mass, BF- body fat mass, BMI- body mass index, BFP- body fat percentage, WHO- wait hip ratio, VFA-visceral fat area.

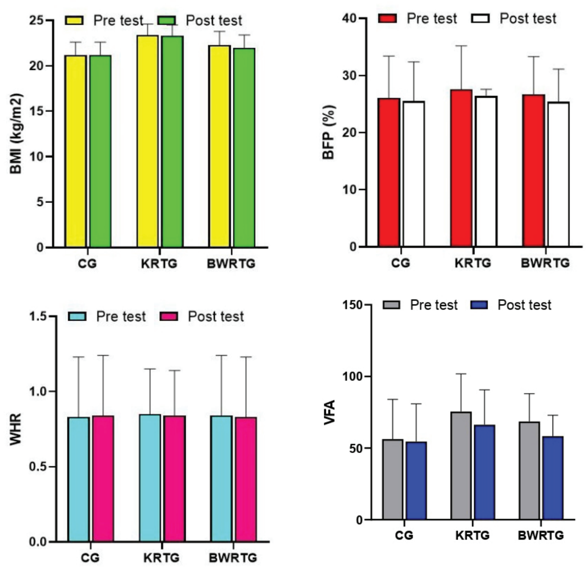

FIGURE 2. Graphical depiction of the percentage change in outcome variables from pre- to post-intervention for each group. BMI- body mass index, BFP- body fat percentage, WHR- wait hip ratio, VFA-visceral fat area.

Discussion

The results of this study provide a comprehensive comparison of the effects of KRTG and BWRTG on body composition among young adult males. KRTG and BWRTG significantly improved key body composition measures, particularly in reducing BFP, BF, and VFA. These findings are consistent with previous studies that support the efficacy of resistance training in fat loss and overall health improvement (Govindasamy et al., 2024; Radhika & Shanmugavalli, 2024). However, KRTG had an even further benefit in decreasing the WHR, indicating that kettlebell exercises' ballistic and compound nature might provide a particular advantage for core stabilization and fat partitioning. These findings offer vital information for elaborating training plans for modifying BM, SM, and other body composition variables, particularly among young adults individuals.

The nature of the two training modalities can account for some of the mechanisms behind the results. Body fat mass, body fat rate, and visceral fat area were decreased in the KRTG and BWRT groups compared with model control groups. The combination of significant muscle mass involvement in a progressive manner and higher energy demands probably caused this. KRTG also showed a particular boon to the WHR through its dynamic and complex actions, unlike static machine-based cross-sectional techniques used in PBL (e.g., kettlebell swings and Turkish getups). These workouts have more impact on core stabilization, making a more significant difference in reducing belly fat and improving fat distribution. The engagement of large muscle groups in KRT might have provided a more significant stimulus for central fat loss, reflected in the superior outcomes in WHR for this group. On the other hand, both training modalities were equally effective in reducing body fat and visceral fat due to the combination of aerobic and resistance training benefits, promoting overall fat loss and muscle hypertrophy.

The young adults group had considerably greater values in variables linked to body fat, particularly in BFP, fat mass, and fat mass index, as compared to the active group (Mateo-Orcajada et al., 2022). Within the scope of this study, a comparison was made between the effects of 12 weeks of BWRT, and KRT on body composition as measured by BM, SM, BF, BMI, BFP, WHR, and VFA in young adults. Some of the research that have previously proven that KRTG has a considerable impact on body composition in various populations. KTRG was found to have a substantial effect on body composition when compared to aerobic exercise on obese male adults (Govindasamy et al., 2022). Similarly, compared to the control group, the research found that volleyball players who undergo with kettlebells and battle ropes saw significant improvements in their body composition (Radhika & Shanmugavalli, 2024). Whereas, BWRT found to have a substantial effect on body composition (Fayazmilani et al., 2022). Several cardiovascular disorders are linked to body composition, particularly fat mass index (Larsson et al., 2020). Prior to the training intervention, the young adults in the current study had BM mean ranges of 53.3 to 64.5 kg, SM of 22.4 to 25.1 kg, BF of 14.5 to 17.6 kg, and BMI of 21.2 to 23.4 kg/m2. Additionally, their BFP ranged from 26.1 to 27.6%, WHR ranged from 0.83 to 0.85m/m, and VFA ranged from 56.3 to 75.5 cm2. All of these values were within the range previously noted in previous research on sedentary adolescents (Sun et al., 2024). In this current study, before and after the 12 weeks of KRT and BWRT the results showed a significant reduction in BF, BFP and VFA for the KRTG and BWRTGs, while no significant changes were observed in the CG. Previous research has also indicated that BWRTG and KRTG are effective in lowering the BFP in obese persons over the course of a 12-week intervention period (Govindasamy et al., 2024). One of the most frequent ways to reduce BFP is through resistance exercise (Poutafkand et al., 2020). Due to the fact that resistance training is not particularly strenuous, the body relies mostly on the aerobic oxidation of sugar and fat as its primary source of energy. This process can be rather intense in order to accomplish the goal of fat reduction (Astorino, 2000; Stisen et al., 2006). Previous research indicates that a resistance training program that does not include the use of any apparatuses and instead makes use of the participant's own body weight as a load appears to be successful in lowering visceral fat and improving metabolic profiles at the same time (Tsuzuku et al., 2007). It has been demonstrated through a systematic review and meta-analysis that resistance training is effective in lowering visceral fat, body fat mass, and BFP in individuals who are in good health (Wewege et al., 2022).

A number of other body composition measurements, including BM, SM, BMI, and WHR, did not exhibit any indication of a significant variation between the pre- and post-assessment periods. Some of the results of the systematic review and meta-analysis are consistent with the findings of the current investigation, which demonstrated that the impact of resistance training on the participant's BMI values was not significant (Liu et al., 2022; Schranz et al., 2013). Similarly, following a period of six weeks of KRTG, there were no discernible changes in BM in healthy 19–26 years age men (Otto et al., 2012) and recreationally active participants (Erbes & Hoar, 2012). KRTG in normal people for 8 weeks showed no significant changes in body composition, including skeletal muscle mass and BMI, according to measurement points (Pre, mid and post) (Jeong et al., 2017). Breukelman et al. (2013) examines the effects on WHR of inactive individuals aged 20-65 years which receive 12 weeks of home-based exercise consisting of aerobic, resistance, and stretching. Following the training procedures, no discernible change was seen WHR (Breukelman et al., 2013). According to the current study, short-term exercise regimens did not have an appreciable improvement in BM, SM, BMI, WHR. The lack of expertise the individuals had with weightlifting and kettlebell activities may have promoted nervous system adaptations more than muscle growth because of their limited resistance training experience. Therefore, brain changes are probably the main mechanism resulting in enhanced performance (Otto et al., 2012). The study by Andersen et al. (2016) show that resistance training for a period of 52 weeks results in substantial improvements in body composition assessments (Andersen et al., 2016).

When we considering the group effect the post-intervention had significant differences in BM and BMI between the KRTG and CG groups. These differences were also observed in baseline BM and BMI between the two groups. However, there are no appreciable variations in any of the body composition metrics between the two intervention groups. Previous studies in line with the current study result, demonstrated no discernible variations in body composition parameters between KRTG and BWRTG (Govindasamy et al., 2024). WHR and BMI are helpful quantifiable indicators for determining overweight and obesity.

The current study has shown that similar outcomes between these two groups suggest that both methods can effectively reduce BMI and improve overall body composition. KRTG was beneficial for central fat distribution regarding WHR improvement, which could be an interesting perspective for many seeking to normalize their central body fat mass. These results indicate that integrating both training approaches could be advantageous for enhancing health outcomes in young adult males.

Limitations

The study has few limitations. This restricted sample size makes it harder to extrapolate the results to a larger population. The objective of next studies ought to encompass a more extensive and heterogeneous cohort to augment the resilience and relevance of the findings. Furthermore, some confounding variables were not controlled, including the participants' nutrition, their level of physical activity outside of the intervention, and lifestyle characteristics. This might have affected the results and made it more difficult to determine the actual benefits of the resistance training interventions. This study may not have given a complete picture of the impact of the training treatments because it only looked at body composition as the dependent variable. The fact that the control group (CG) and the KRTG had notable baseline variations in BM and BMI poses a limitation to this study. These pre-intervention variations might have an impact on how the post-intervention outcomes are interpreted since the changes that are shown may have been impacted by the groups' original circumstances in addition to the intervention itself. To increase the validity of the results, future research should concentrate on employing better randomisation or stratification to provide more balanced baseline circumstances.

Conclusion

In conclusion, both the twelve-week KRTG and BWRTG programs demonstrated similarly effective results in reducing body fat mass, fat percentage, and visceral fat area in young males. These findings highlight that both training approaches are viable options for improving body composition, with kettlebell training offering the added advantage of positively impacting waist-to-hip ratio. For those looking to optimize fat reduction and overall body composition, incorporating either training regimen can be beneficial.

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Conflicts of interest
The authors declare that there are no conflict of interest.

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Revised September 2019

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When preparing the final version of the manuscripts, either NEW or REVISED authors should strictly follow the guidelines. Manuscripts departing substantially from the guidelines will be returned to the authors for revision or, rejected.

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1.1. Overview

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SM honors six-weeks for an initial decision of manuscript submission.

Authors should submit the manuscripts as one Microsoft Word (.doc) file.

Manuscripts must be provided either in standard UK or US English language. English standards should be consistent throughout the manuscripts accordingly.

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Use Times New Roman font, size eleven (11) point.

Number (Arabic numerals) the pages consecutively (centering at the bottom of each page), beginning with the title page as page 1 and ending with the Figure legend page.

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Apart from chapter headings and sub-headings avoid any kind of formatting in the main text of the manuscripts.

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- ✓ In 2007, Duffield and Marino's study of soccer players

Works by three to five authors: cite all the author names the first time the reference occurs and then subsequently include only the first author followed by et al.

- ✓ First citation: Bangsbo, Iaia, and Krustrup (2008) stated that
- ✓ Subséquent citation: Bangsbo et al. (2008) stated that

Works by six or more authors: cite only the name of the first author followed by et al. and the year

- ✓ Krustrup et al. (2003) studied
- ✓ In one study (Krustrup et al., 2003), soccer players

Two or more works in the same parenthetical citation: Citation of two or more works in the same parentheses should be listed in the order they appear in the reference list (i.e., alphabetically, then chronologically)

- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

Nepocatych, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine, 6*(2), 57-65. doi: 10.26773/mjssm.2017.09.008

Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology, 100*(6), 727-735. doi: 10.1007/s00421-007-0468-x

Krustrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise, 35*(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies, 50*(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia, 4*. Retrieved November 15, 2005, from http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html

Conference paper:

Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf

Encyclopedia entry (print, with author):

Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from http://www.bartleby.com/65/et/ethnolog.html

Thesis and dissertation:

Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport.* Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

If you use data from another published or unpublished source, it is the authors' responsibility to obtain permission and acknowledge them fully.

2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See* example:

- ✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

2.5.3. Table footnotes

Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

- ✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. [a, b, c]), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript [a].

- ✓ [a]One participant was diagnosed with heat illness and n = 19.[b]n =20.

Probability notes provide the reader with the results of the texts for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

- ✓ *P<0.05,†p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See* example:

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓ ....in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. SM prefers TIFF, EPS and PNG formats.

If a figure has been published previously, acknowledge the original source and submit a written permission from the copyright holder to reproduce the material. Permission is required irrespective of authorship or publisher except for documents in the public domain. If photographs of people are used, either the subjects must not be identifiable or their pictures must be accompanied by written permission to use the photograph whenever possible permission for publication should be obtained.

Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of SM.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written bellow the figure, in sentence case. *See* example:

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, *p>0.01, †p>0.05.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See* example:

- ✓ Figure 1
    - × figure 1
    - × Figure 1.
    - ✓ ….exhibit greater variance than the year before (Figure 2). Therefore…
    - ✓ ….as shown in Figures 1 and 3. (citing more figures at once)
    - ✓ ….result has shown (Figures 1-3) that... (citing more figures at once)
    - ✓ ….in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See* example:

- ✓ Figure 1a
- ✓ …in Figures 1a and b we can…
- ✓ …data represent (Figures 1a-d)…

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10º	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 º	× 10kg	× 12 : 2	× .056

Signs should be placed immediately preceding the relevant number.		
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age
× 45 ± 3.4	× p < 0.01	× males > 30 years of age

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. *See* example:

- ✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*

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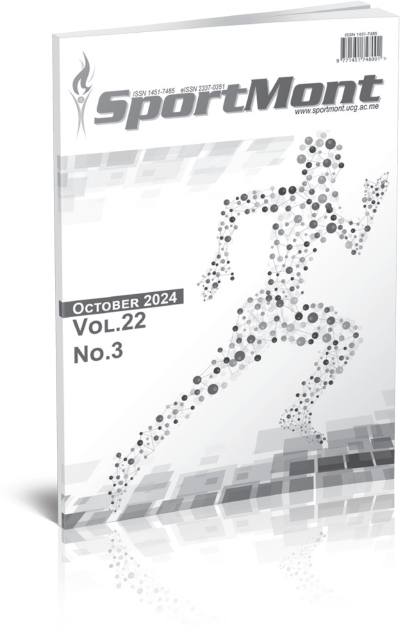

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- Post-publication tools to indicate quality and impact;
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- Worldwide media coverage.

SMJ is published three times a year, in February, June and October of each year. SMJ publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

SMJ covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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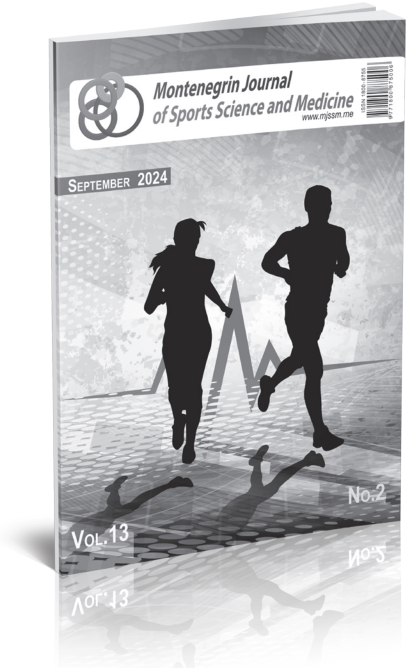

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- Open-access and freely accessible online;
- Fast publication time;
- Peer review by expert, practicing researchers;
- Post-publication tools to indicate quality and impact;
- Community-based dialogue on articles;
- Worldwide media coverage.

MJSSM is published biannually, in September and March of each year. MJSSM publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

MJSSM covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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MONTENEGRIN SPORTS ACADEMY

Founded in 2003 in Podgorica (Montenegro), the Montenegrin Sports Academy (MSA) is a sports scientific society dedicated to the collection, generation and dissemination of scientific knowledge at the Montenegrin level and beyond.

The Montenegrin Sports Academy (MSA) is the leading association of sports scientists at the Montenegrin level, which maintains extensive co-operation with the corresponding associations from abroad. The purpose of the MSA is the promotion of science and research, with special attention to sports science across Montenegro and beyond. Its topics include motivation, attitudes, values and responses, adaptation, performance and health aspects of people engaged in physical activity and the relation of physical activity and lifestyle to health, prevention and aging. These topics are investigated on an interdisciplinary basis and they bring together scientists from all areas of sports science, such as adapted physical activity, biochemistry, biomechanics, chronic disease and exercise, coaching and performance, doping, education, engineering and technology, environmental physiology, ethics, exercise and health, exercise, lifestyle and fitness, gender in sports, growth and development, human performance and aging, management and sports law, molecular biology and genetics, motor control and learning, muscle mechanics and neuromuscular control, muscle metabolism and hemodynamics, nutrition and exercise, overtraining, physiology, physiotherapy, rehabilitation, sports history, sports medicine, sports pedagogy, sports philosophy, sports psychology, sports sociology, training and testing.

The MSA is a non-profit organization. It supports Montenegrin institutions, such as the Ministry of Education and Sports, the Ministry of Science and the Montenegrin Olympic Committee, by offering scientific advice and assistance for carrying out coordinated national and European research projects defined by these bodies. In addition, the MSA serves as the most important Montenegrin and regional network of sports scientists from all relevant subdisciplines.

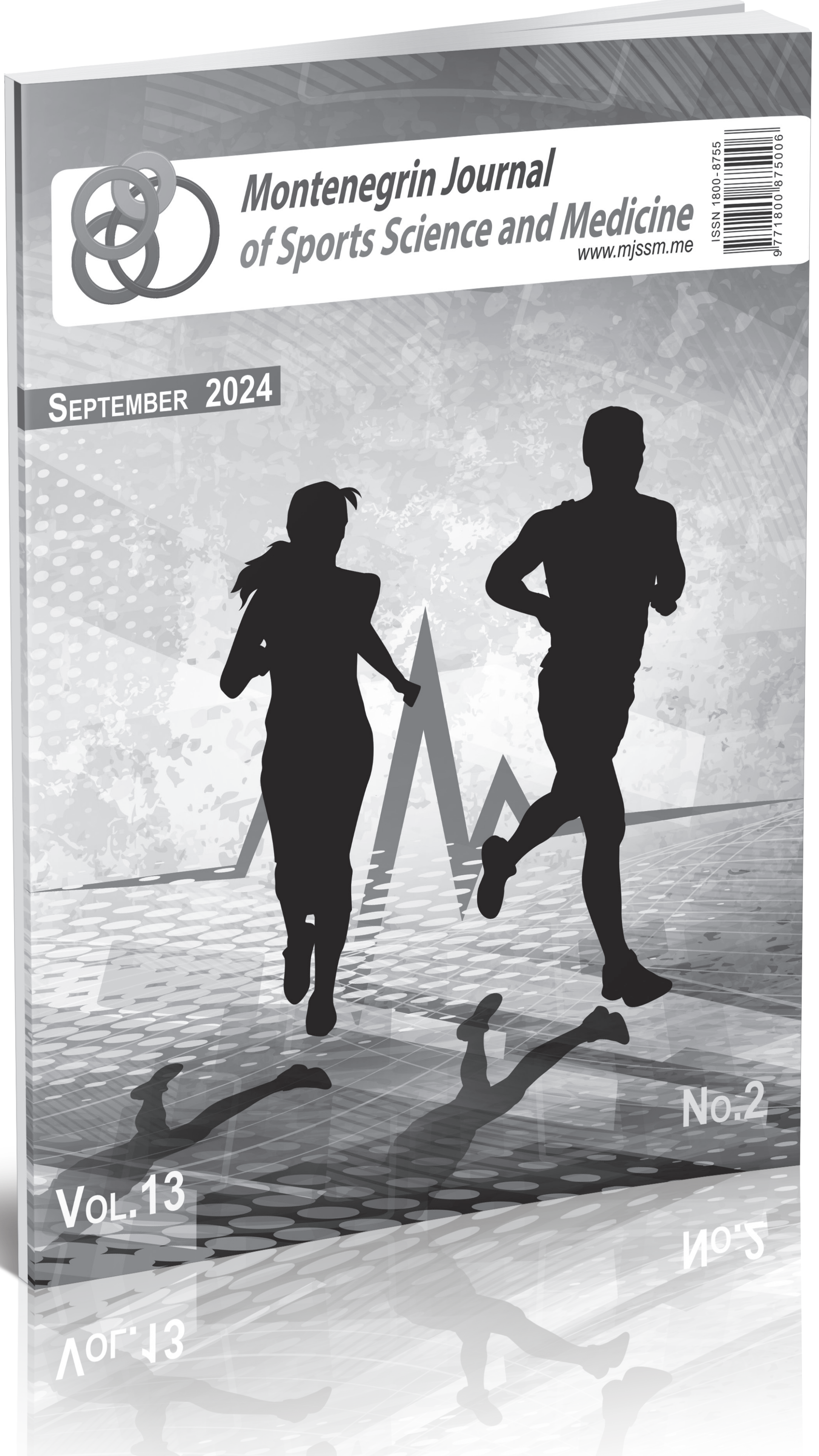
The main scientific event organized by the Montenegrin Sports Academy (MSA) is the annual conference held in the first week of April.

Annual conferences have been organized since the inauguration of the MSA in 2003. Today the MSA conference ranks among the leading sports scientific congresses in the Western Balkans. The conference comprises a range of invited lecturers, oral and poster presentations from multi- and mono-disciplinary areas, as well as various types of workshops. The MSA conference is attended by national, regional and international sports scientists with academic careers. The MSA conference now welcomes up to 200 participants from all over the world.

It is our great pleasure to announce the upcoming 22th Annual Scientific Conference of Montenegrin Sports Academy “Sport, Physical Activity and Health: Contemporary Perspectives” to be held in Dubrovnik, Croatia, from 3 to 6 April, 2025. It is planned to be once again organized by the Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education, University of Montenegro and other international partner institutions (specified in the partner section).

The conference is focused on very current topics from all areas of sports science and sports medicine including physiology and sports medicine, social sciences and humanities, biomechanics and neuromuscular (see Abstract Submission page for more information).

We do believe that the topics offered to our conference participants will serve as a useful forum for the presentation of the latest research, as well as both for the theoretical and applied insight into the field of sports science and sports medicine disciplines.



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